ABSTRACT ALGEBRA

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Preface

These reading notes are a combination of material from the course Math 210A by Prof. Richard Taylor at Stanford University, Dummit and Foote's *Abstract Algebra* [DF04], and Aluffi's *Algebra: Chapter 0* [Alu09]. I have also included my answers to several questions asked in example sheets written by Prof. Taylor for MATH 210A since they seemed very integral to understanding the concepts.

There may be major and minor errors throughout these notes. If you find any, please let me know by sending me an email at jubayer@stanford.edu. Oftentimes, I wrote these notes *after* I had already handwritten them elsewhere, and I was careless or inefficient in rewriting the proofs. I also glossed over various parts of the original texts that these reading notes were trying to follow, despite those parts being important. As such, I take sole responsibility for the errors.

1. Groups

1.1. Definitions and Immediate Properties

Definition 1.1 (Groups)**.** *A **group** is the following data: G is a set and $\bullet$ is a binary operation on G, such that:*

1. *the operation is associative: $(x \bullet y) \bullet z = x \bullet (y \bullet z)$ for all $x, y, z \in G$*
2. *the operation has an identity: there exists an element $1_G \in G$ such that $x \bullet 1_G = 1_G \bullet x = x$ for all $x \in G$*
3. *the operation gives each element an inverse: for all $x \in G$, there exists an inverse $x^{-1} \in G$ such that $x \bullet x^{-1} = x^{-1} \bullet x = 1_G$*

*In particular, the operation $\bullet$ need not be commutative. However, if it is commutative (i.e. $x \bullet y = y \bullet x$ for all $x, y \in G$), then we say that the group is **abelian** or commutative.* ⌟

Observe that a group is always non-empty since it must, at the very least, have the identity, $1_G \in G$. We now look at some immediate examples of groups:

Examples 1.2. *The following are all groups:*

1. *The set of integers ($\mathbb{Z}$), the set of rationals ($\mathbb{Q}$), the set of reals ($\mathbb{R}$), and the set of complex numbers ($\mathbb{C}$) are all groups under addition, and they are all abelian. However, they are not all groups under multiplication. For example, $\mathbb{Z}$ is not a group under multiplication since $x > 1$ and $x < -1$ has no inverse. Furthermore, $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are not groups under multiplication since 0 has no inverse.*
2. *The sets $\mathbb{R} - \{0\}, \mathbb{Q} - \{0\}$, and $\mathbb{C} - \{0\}$ are groups under multiplication (but not under addition since there is no additive identity).*
3. *All vector spaces V are groups under addition. In particular, they are abelian groups.*
4. *The set $\mathbb{Z}/n\mathbb{Z}$ where $n \in \mathbb{Z}_{>0}$ is an abelian group under addition. The identity in the group is $\bar{0}$ and the additive inverse of $\bar{x}$ is $\bar{-x}$ for any $\bar{x} \in \mathbb{Z}/n\mathbb{Z}$.*

Note that $\mathbb{Z}/n\mathbb{Z}$ is not guaranteed to be a group under multiplication since, a priori, all elements are not guaranteed to have a multiplicative inverse. First, we define the subset of $\mathbb{Z}/n\mathbb{Z}$ which admits such inverses to be a group:

Example 1.3. The set $(\mathbb{Z}/n\mathbb{Z})^\times \subseteq \mathbb{Z}/n\mathbb{Z}$ is the set of all equivalence classes $\bar{x}$ such that there exists a multiplicative inverse. This is a group under multiplication where the identity is $\bar{1}$.

It should be clear, from the notation itself, that $\mathbb{Z}/n\mathbb{Z}$ is a group under addition and $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group under multiplication. ⌟

Definition 1.4 (Direct Product of Groups)**.** *Suppose $(X, \bullet)$ and $(Y, \square)$ are groups. Then, the direct product of these groups is the set $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$ under the componentwise operation:*

$$(x_1, y_1)(x_2, y_2) = (x_1 \bullet x_2, y_1 \square y_2).$$

⌟

We now look at some immediate properties that all groups satisfy.

Notation 1.5. *We adopt the shorthand notation where we suppress the group operation $x_1 \bullet x_2$ and write it as $x_1 x_2$. We write multiplicatively even though the group operation need not be.*

Theorem 1.6. *Suppose G is a group. Then,*

1. *the identity 1_G of G is unique*
2. *the inverse x^{-1} of each $x \in G$ is unique*
3. *$(x^{-1})^{-1} = x$ for all $x \in G$*
4. *$(xy)^{-1} = y^{-1}x^{-1}$ for all $x, y \in G$*
5. *If $xy = xz$ for some $x, y, z \in G$, then $y = z$. If $yx = zx$, then $y = z$. In particular, there is a unique solution for $ax = b$ in x and a unique solution for $ya = b$ in y.*

⌟

Finally, we define the *order* of an element.

Definition 1.7 (Order of an Element in a Group)**.** *Let G be a group and let $x \in G$. Then, the order of x is denoted by $|x|$ and is defined as the smallest positive integer n such that $x^n = 1_G$. If there is no such n, then we say that the order is infinite.* ⌟

Examples 1.8.
(1) *The element $1_G \in G$ has order 1. In fact, it is the element with this property, i.e. an element in any group has order 1 if and only if it is the identity.*
(2) *In $\mathbb{R} - \{0\}$, or $\mathbb{Q} - \{0\}$, the element -1 has order 2 under the operation of multiplication. All other elements in these groups have infinite order.*

We briefly also look at the definitions of monoids.

Definition 1.9 (Monoids)**.** *A monoid is a triple $(M, \cdot, 1)$ where M is a non-empty set, $\cdot$ is a binary operation that is associative (i.e. $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ for all $x, y, z \in M$) and $1 \in M$ such that $1 \cdot m = m = m \cdot 1$ for all $m \in M$.* ⌟

In other words, a group is a monoid $G, \cdot, 1)$ where every element in G has an inverse. We also often encounter *monads*, which has the same definition as a monoid except the binary operation need not be associative. On the other hand, a semigroup $(M, \cdot)$ is a non-empty set with an associative binary operation that need not have the identity 1.

1.2. **Symmetric Groups**

The symmetric group is an extraordinarily important and ubiquitous construction.

Definition 1.10 (Symmetric Groups)**.** *Let M be a set and the let S_M be the set of all bijections from M to itself i.e. the set of all permutations of M. Then, S_M is a group under composition i.e. if $\sigma_s, \sigma_t \in S_M$, then $\sigma_s \sigma_t = \sigma_s \circ \sigma_t$. The identitiy of the group is 1 which sends every element in M to itself.*

In particular, when the set is $M = \{1, 2, \cdots, n\}$ of size n, the symmetric group is denoted by S_n and called the symmetric group of degree n. ⌟

Sometimes, the symmetric group of the set M is denoted by $\mathrm{Sym}(M)$. In these notes, however, we simply denote it by S_M.

Proposition 1.11. *The order of the symmetric group S_n is $n!$, i.e. $|S_n| = n!$.* ⌟

Proof. Each permutation in a symmetric group is an injective function from $\{1, \cdots, n\}$ to itself. There are $n!$ such injective functions. □

One can denote the elements of a symmetric group using cycles. A cycle is a string such as $(a_1, a_2, \cdots, a_m)$ which represents a permutation; this cycle represents that the permutation takes a_1 to a_2, a_2 to a_3 and so on until a_m to a_1. The cycle $(3, 4, 1)$ takes 3 to 4, 4 to 1, and 1 to 3. With this notation, we can decompose any $\sigma \in S_n$ as a composition of cycles of the form $(a_1, a_2, \cdots, a_m)(a_p, a_{p+1}, \cdots, a_{p+k})$. The length of a cycle is the number of integers in it. In particular, we have the following method for calculating the order of any element in a symmetric group:

Proposition 1.12. *The order of a permutation in a symmetric group is the l.c.m. of the lengths of the cycles in its cycle decomposition.* ⌟

Proposition 1.13. *S_n is a non-abelian group for all $n \geq 3$.* ⌟

Proof. Consider the set $\{1, 2, 3, \cdots, n\}$. Then, consider the two permutations σ_s defined by $1 \to 3$ and and the permutation σ_t defined by $1 \to 2$. Then, $\sigma_s \sigma_t \neq \sigma_t \sigma_s$. Since these permutations exists in all symmetric groups S_n for $n \geq 3$, we are done. □

1.3. **Dihedral Groups**

Dihedral groups are an extremely important class of groups that define *symmetries* of polygons.

Definition 1.14 (Dihedral groups)**.** *For a positive integer $n \geq 3$, the dihedral group D_{2n} is the set of all symmetries of the regular n-gon. A symmetry is defined as any rigid motion on the object such that the objects look exactly the same. More precisely:*

- *A symmetry of an n-gon is the following: label all n vertices of the polygon first. Then, a symmetry is a permutation σ of the set $\{1, 2, \cdots, n\}$ corresponding to the rigid motion that leaves the polygon unchanged. For e.g., consider the 90 degrees clockwise rotation of a square (i.e. 4-gon) whose vertices are labelled sequentially. Then, the rotation can be described as the permutation σ that sends i to $i+1$, for all $i \in \{1, \cdots, n-1 = 3\}$, and $n = 4$ to 1.*

This set D_{2n} is a group under the operation of composition. In other words, if $s, t \in D_{2n}$ (i.e. s is some symmetry and t is another symmetry), then st is the symmetry obtained by first applying t and then applying s. This can also be defined by looking at the permutations each such symmetry corresponds to: if s corresponds to the perutation σ_s and t corresponds to the permutation σ_t, then st corresponds to the permutation $\sigma_s \circ \sigma_t$.

In D_{2n}, the identity is the identity symmetry which leaves al vertices fixed/unchanged and is denoted by 1. The inverse of $s \in D_{2n}$ is the symmetry that reverses all the rigid motions of s i.e. s^{-1} corresponds to the permutation σ^{-1} if s corresponds to σ. ⌟

Proposition 1.15. *For any dihedral group D_{2n}, the size of the set is $|D_{2n}| = 2n$. We say the dihedral group D_{2n} is of order $2n$.* ⌟

Proof. Consider some sequential numbering of the vertices of the n-gon i.e. 1 is next to 2 which is next to 3 and so on. Now, we can send the vertex number 1 to n possible locations/vertices. Suppose we send 1 to i. Then, we must send 2 to either $i+1$ or $i-1$ because symmetries must be rigid motions. That means we have $n \times 2 = 2n$ possible permutations that define symmetries. □

Definition 1.16 (Rotations and Reflections in D_{2n})**.** *Define $r \in D_{2n}$ to be a clockwise rotation about the origin through $\frac{2\pi}{n}$ radians. Define $s \in D_{2n}$ to be reflection about the line going through the vertex 1 and the origin.* ⌟

Proposition 1.17. *The following are satisfied in a dihedral group:*

1. *$1, r, r^2, \cdots, r^{n-1}$ are distinct and $r^n = 1$. In particular, $|r| = n$*
2. *$|s| = 2$*
3. *$s \neq r^i$ for any i*
4. *$sr^i \neq sr^j$ for $i \neq j$ and $0 \leq i, j \leq n-1$*

As such, we can write out the full group as:

$$D_{2n} = \{1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}\}$$

and any $x \in D_{2n}$ can be written as $x = s^i r^j$ for $i \in \{0, 1\}$ and $0 \leq j \leq n-1$. Furthermore,

1. *$rs = sr^{-1}$. As such, **the dihedral group is not an abelian group**.*
2. *$r^i s = sr^{-i}$ for all $0 \leq i \leq n$.*

⌟

1.4. **Homomorphisms and Isomorphisms**

A recurring, central question we ask in abstract algebra is whether or not two mathematical objects of interest are, in some sense, the same. To be able to ask this question, we define homomorphisms and isomorphisms that help us understand these questions for groups with respect to their operations.

Definition 1.18 (Homomorphisms and Isomorphisms)**.** *Let $(G, \bullet)$ and $(H, \square)$ be two groups. Then, $\varphi : G \to H$ is a **homomorphism** if*

$$\varphi(x \bullet y) = \varphi(x) \square \varphi(y)$$

*for all $x, y \in G$. In other words, a homomorphism is a mapping that respects the structure of the group G and the group H. In particular, if φ is a bijection, then we call it an **isomorphism**, and we write $G \cong H$.* ⌟

We will switch to the more convenient notation and simply write $\varphi(xy) = \varphi(x)\varphi(y)$ by suppressing the group operation.

Recall that S_M is the symmetric group over the set M. We now show that symmetric groups over sets of equal size are isomorphic.

Theorem 1.19. *Let M and M' be two sets. Then, $S_M \cong S_{M'}$ if and only if $|M| = |M'|$.* ⌟

Proof Sketch: The forward direction is simple. First, since both the sets have the same cardinality, define some correspondence of the elements between them via θ i.e. $\theta(m) = m'$ for some $m \in M, m' \in M'$ where θ is bijective. This map, θ, is our way of associating each element in M with an element in M' which we can do since they have the same cardinality. Now, suppose $\sigma(x) = y$ for $x, y \in M$, i.e. σ is an element in S_M. Then, we will map this to a symmetry in $S_{M'}$ by looking at how the corresponding elements in M' get permuted. Define $\varphi : S_M \to S_{M'}$ by $\varphi(\sigma)(\theta(x)) = \theta(y)$. In other words, φ takes σ to a permutation in $S_{M'}$ that moves the elements of M' the same way σ moves the elements of M. One can easily show that this is a homomorphism. Furthermore, this is invertible since θ itself is invertible, i.e. given any $\mu \in S_{M'}$, we can find a corresponding $\sigma \in S_M$ using the map θ. The converse direction is easy to prove for the finite group case (we will prove the infinite case later). Since the two groups are isomorphic, it must be that $|S_M| = |S_{M'}|$. now, suppose $|M| = p$ and $|M'| = q$. Then, $|S_M| = p!$ and $|S_{M'}| = q!$. So, $p = q$ and we are done.

Examples 1.20. *The following are examples of isomorphisms:*

1. *$G \cong G$ for any group G via, at least, the identity isomorphism.*
2. *Any non-abelian group of order 6 is isomorphic to S_3. In particular, $D_6 \cong S_3$. We shall prove this later when we study classification theorems.*
3. *$\varphi : \mathbb{R} \to \mathbb{R}_{>0}$ by $\varphi(x) = \exp(x)$ is an isomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$.*

Proposition 1.21. *If $\varphi : G \to H$ is an isomorphism, then*

1. *$|G| = |H|$*
2. *G is abelian if and only if H is abelian*
3. *$|x| = |\varphi(x)|$ for all $x \in G$.*

⌟

Example 1.22. Note that if $\varphi : G \to H$ is an isomorphism, then $|G| = |H|$, but the converse is not true. For example, S_3 is not isomorphic to $\mathbb{Z}/6\mathbb{Z}$, despite their cardinality being the same, because the latter is abelian while the former is not. ⌟

1.5. **Group Actions**

So far, we have seen homomorphisms/isomorphisms which maps a group G to a group H. Now, we will look at a particular mapping from $G \times A$ to A where G is a group and A is an arbitrary set.

Definition 1.23 (Group Action)**.** *A **group action** of a group G on a set A is a mapping $\varphi : G \times A \to A$ such that:*

1. *$\varphi(g_1, \varphi(g_2, a)) = \varphi(g_1 g_2, a)$ for all $g_1, g_2 \in G, a \in A$*
2. *$\varphi(1, a) = a$ for all $a \in A$*

⌟

We will usually use notation that suppresses the fact that a group action is a mapping and, instead, write ga to mean $\varphi(g, a)$. As such, the requirements can be written as $g_1(g_2a) = (g_1g_2)a$ and $1a = 1$.

Observation. *A group action can be thought of as a permutation of the set. Since each group element maps elements $a \in A$ to $a' \in A$, we can think of each group element as a permutation in the symmetric group, S_A. Concretely, $g \cdot a$ is $g \in G$ is acting on $a \in A$. Now, for each $g \in G$, we can define $\sigma_g : A \to A$ by $\sigma_g(a) = g \cdot a$. Then,*

1. *σ_g is a permutation of A for each g, and*
2. *$\psi : G \to S_A$ via $\psi(g) = \sigma_g$ is a homomorphism. In particular, ψ is called the **permutation representation** associated to the given group action. In other words, for each group action, there exists a permutation representation corresponding to the group action.*

To see the first, one can show that σ_g has an inverse which is $\sigma_{g^{-1}}$; this follows from the first defining property of a group action. To see the second, we can show that $\psi(g_1g_2)(a) = (\psi(g_1) \cdot \psi(g_2))(a)$. This means, a group action of G on a set A is to be understood as a way of turning every element of G into a permutation on the set A.

Observation. *We saw that, given a group action, there is an associated permutation representation i.e. a homomorphism $G \to S_A$. We can reverse this process. Given a homomorphism $\psi : G \to S_A$, we can construct a group action $G \times A \to G$ by setting $g \cdot a = \psi(g)(a)$.*

Examples 1.24. *We now look at a number of examples of group actions.*

1. *The trivial action is defined by setting $g \cdot a = a$ for all $a \in A$ and all $g \in G$ i.e. all elements of G correspond to the identity permutation in S_A. The associated permutation representation sends every element in G to the identity $1 \in S_A$.*
2. *On the other hand, if each element of G corresponds to a unique permutation in S_A, we say that the group action is **faithful**. The associated permutation representation is injective.*
3. *Let our group be a field F and let V be a vector space defined over F. Then, F acts on V via $\alpha\vec{v} = \alpha(v_1, \cdots, v_n) = (\alpha v_1, \cdots, \alpha v_n) \in V$.*
4. *Let A be a nonempty set. Then, the symmetric group of A, S_A, acts on A via $\sigma \cdot a = \sigma(a)$ for any $a \in A, \sigma \in S_A$. The associated permutation representation, $\mathrm{id} : S_A \to S_A$, is identity.*
5. *After fixing some labeling of the vertices of an n-gon, each element of D_{2n} corresponds to a permutation σ in S_n defined by the way the element permutes the corresponding vertices. This gives us a group action $D_{2n} \times \{1, \cdots, n\} \to \{1, \cdots, n\}$ i.e. $\sigma \cdot i = j$ where j is the permutation of i under σ.*

1.6. **Subgroups**

Definition 1.25 (Subgroups)**.** *Let G be a group and let H be a subset of G. Then, H is called a subgroup if*

1. *H is non-empty, i.e. $H \neq \emptyset$*
2. *H is closed under the group operation and inverses, i.e. $xy, x^{-1} \in H$ for all $x, y \in H$*

In symbols, we write $H \leq G$. ⌟

Examples 1.26. *We look at some examples and non-examples of subgroups.*

1. *Any group G has at least two subgroups: $H = G$ and $H = \{1_G\}$.*
2. *$\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R}$.*
3. *Note that the subgroup must be defined with respect to the same operation as that of the group. For example, $(\mathbb{Q} - \{0\}.\times)$ is not a subgroup of $(\mathbb{R}, +)$ even though both are groups.*
4. *D_6 is not a subgroup of D_8 since D_6 is not a subset of D_8 - this is because the symmetries in D_6 are not the same as the symmetries in D_8 (e.g., the rotation is by 60 degrees in D_6 whereas it is by 45 degrees in D_8).*

Alternatively, one can verify whether H is a subgroup of G via the following criterion:

Proposition 1.27 (Subgroup Criterion)**.** *A subet $H \subseteq G$ of a group G is a subgroup if and only if*

(1) $H \neq \emptyset$*, and*
(2) $xy^{-1} \in H$ *for all* $x, y \in H$*.*

If H is finite, an equivalent and sufficient criterion is to simply check that

(1) $H \neq \emptyset$
(2) H *is closed under group operation.*

⌟

We will now look at some special classes of subgroups.

1.6.1. *Centralizers, Normalizers, Centre.* Given a subset A of a group G, we can construct some important classes of subgroups of G defined by how they interact with that specific set A. This leads us to the definitions of centralizers and normalizers. The centre of a group is another important subgroup that is defined by considering the subset $A = G$.

Definition 1.28 (Centralizers)**.** *Let G be a group and let A be a subset of G that is non-empty. Then, the centralizer of A in G is defined as*

$$C_G(A) := \{g \in G \mid gag^{-1} = a, \forall a \in A\}.$$

In other words, the centralizer of A in G are ***all the elements in G that elements of A commute with.*** ⌟

Theorem 1.29. *The centralizer, $C_G(A)$, is a subgroup, i.e. $C_G(A) \leq G$.* ⌟

The proof is straightforward using the subgroup criterion.

Observation. *Note that if G is an abelian group, then for any non-empty subset A of G, $C_G(A) = G$.*

The centralizer is a particularly strict criterion in the sense that $a \in A$ must commute with all the elements in $C_G(A)$ i.e. $ag = ga$ for all $g \in C_G(A)$. Next, we look at the normalizer which relaxes the condtion quite a bit.

Definition 1.30 (Normalizer)**.** *let A be a non-empty subset of the group G. Then, the normalizer of A in G is the set*

$$N_G(A) := \{g \in G \mid gAg^{-1} = A\}$$

where $gAg^{-1} = \{gag^{-1} \mid a \in A\}$. In other words, the normalizer of A in G are all the elements in G that conjugate elements of A (i.e. gag^{-1}) and return some element in A. ⌟

So far, our constructions were with respect to some non-empty subset A of a group G. Next, we look at the centre of a group, which can be seen as the centralizer of the subset G of G.

Theorem 1.31. *The normalizer of a set A in group G is a subgroup, i.e. $N_G(A) \leq G$.* ⌟

Definition 1.32 (Centre)**.** *The centre of a group G is defined as the subset of elements that commute with all elements of G, i.e.*

$$Z(G) := \{x \in G \mid xg = gx, \forall g \in G\}.$$

⌟

Theorem 1.33. *The centre of a group G is a subgroup, i.e. $Z(G) \leq G$.* ⌟

We will soon see an elegant way to understand why the normalizer and the centre are subgroups.

1.6.2. *Stabilizers and Kernels.* Now, we define some classes of subgroups that are defined with respect to a *group action* of G. In particular, again, we consider some non-empty subset $A \subseteq G$. We will also see that the preceding subgroups (i.e. centralizers, normalizers, and centre) can be defined as special cases of the constructions we will see now.

Recall that a group action is a mapping $G \times A \to A$ such that that the mapping is associative and $1 \cdot a = a$.

Definition 1.34 (Stabilizer)**.** *If G is a group acting on a set A, and $a \in A$ is some fixed element, then the stabilizer of a in G is the set*

$$G_a = \{g \in G \mid g \cdot a = a\}$$

where $g \cdot a$ is the group action of G on A. ⌟

Theorem 1.35. *The stabilizer is a subgroup, i.e. for any $a \in A \subseteq A$, $G_a \leq G$* ⌟

Definition 1.36 (Kernel of Group Action)**.** *If the group G is acting on a set A, then the kernel of the group action is defined as*

$$\{g \in G \mid g \cdot a = a, \forall a \in A\}.$$

⌟

Theorem 1.37. *The kernel of a group action of G on a set A is a subgroup of G.* ⌟

Now, we finally come to the main observation where we note that several of these constructions can be unified by the notion of ***conjugation***. Let $X = \mathcal{P}(G)$ be the set of all subsets of the group G. Now, suppose G acts on X by conjugation, i.e. for any $g \in G$ and $M \in X = \mathcal{P}(G)$, we say $g \cdot M = gMg^{-1} = \{gmg^{-1} \mid m \in M\}$. Then, we can note the following:

(1) the normalizer, $N_G(A)$, is the stabilizer of A in G, i.e. $N_G(A) = G_A$ where $A \in \mathcal{P}(G)$
(2) if the normalizer, $N_G(A)$ acts on the set A by conjugation, i.e. $g \cdot a = gag^{-1}, \forall a \in A$ and $g \cdot a \in A$ since $g \in N_G(A)$, then $C_G(A)$ is the kernel of this action. In other words, **the centralizer is the kernel of the action of $N_G(A)$ on A by conjugation.** As such,
$$C_G(A) \leq N_G(A).$$
(3) the center of a group G, $Z(G)$, is the kernel of G acting on G by conjugation. As such,
$$Z(G) \leq G.$$

1.7. Cyclic Groups

Definition 1.38 (Cyclic Groups)**.** *A **cyclic group** is a group H that is generated by a single element, i.e.*

$$H = \{x^n \mid n \in \mathbb{Z}\}.$$

Note that n iterates over all integers, including negative integers. In other words, we product x with both itself and its inverse.

Here, we wrote the group operation as multiplication as just a matter of notation; we could have also written the group operation as addition, which would lead us to write $H = \{nx \mid n \in \mathbb{Z}\}$. In either case, we denote a cyclic group as $H = \langle x \rangle$ and say that H is a group that is generated by x. ⌟

Observation. *The generator in a cyclic group need not be unique. For example, if $H = \langle x \rangle$, we could also write $H = \langle x^{-1} \rangle$. This is because:*

- *In multiplicative notation where $H = \{x^n \mid n \in \mathbb{Z}\}$, any x^n can be written as $(x^{-1})^{-n}$, so $H = \{x^{-1} \mid n \in \mathbb{Z}\}$.*
- *In additive notation where $H = \{nx \mid n \in \mathbb{Z}\}$, any nx can be written as $(-n)(-x)$, so $H = \{n(-x) \mid n \in \mathbb{Z}\}$.*

Examples 1.39. *We take a look at some examples of cyclic groups. They often appear as subgroups of larger groups that may not be cyclic themselves.*
Consider the dihedral group: $G = D_{2n}$. Now, let H be the group of all rotations. Then, $H = \langle r \rangle = \{r^m \mid m \in \mathbb{Z}\}$. Note that $|H| = n$ since $r^n = 1 = r^0$. Also, $r^{-1} = r^{n-1}$.
The group of integers is cyclic i.e. $\mathbb{Z} = \langle 1 \rangle$.

1.7.1. *Cardinality of cyclic group.* We immediately note that there is a simple way of figuring out the cardinality of a cyclic group via its generator:

Proposition 1.40. *Suppose that H is a cyclic group, i.e. $H = \langle x \rangle$. Then, $|H| = |x|$ (note that $|H|$ and $|x|$ are allowed to be infinite).*

- *if $|H| = n < \infty$, then $x^n = 1$ and the distinct elements of H are $\{1, x, \cdots, x^{n-1}\}$.*
- *if $|H| = \infty$, then $x^n \neq 1$ for any $n \neq 0$.*

⌟

Proof. First, suppose $|x| = n$. For any arbitrary $x^k \in H$, we can write using the division algorithm that $k = nq + r$ where $0 \leq r < n$. Then, $x^k = x^{nq+r} = (x^n)^q \cdot x^r = x^r \in \{1, x, \cdots, x^{n-1}\}$. One can easily show that $x^a \neq x^b$ is $a, b \in \{1, \cdots, n-1\}$ and $a \neq b$: if not, then, $x^{b-a} = 1$ but we know that n is the smallest positive integer such that $x^n = 1$. The proof for the infinite order is simple: no two x^a and x^b are equal for $a \neq b$ since, otherwise, we have an n such that $x^n = 1$ where $0 < n < \infty$. □

Lemma 1.41. *Let $x \in G$ where G is a group and let $m, n \in \mathbb{Z}$. If $x^m = 1 = x^n$, then*

$$x^d = 1, \quad \text{where } d = (m, n).$$

In particular, if $x^m = 1$, then $|x|$ divides m. ⌟

Now, we come to an important result:

Theorem 1.42 (Classification of Cyclic Groups)**.** ***Any two cyclic groups of the same order are isomorphic.***

(1) *Suppose the cyclic groups $\langle x \rangle$ and $\langle y \rangle$ are both of order $n \in \mathbb{Z}_{>0}$. Then, the following is an isomorphism: $\varphi(x^k) = y^k$.*
(2) *Suppose $\langle x \rangle$ is a cyclic group, then $\varphi(k) = x^k$ is an isomorphism. This follows from the observation that if $\langle x \rangle$ is of order n, then $\langle x \rangle \cong \mathbb{Z}/n\mathbb{Z}$.*
(3) *In particular, for any $n \in \mathbb{Z}_{>0}$, let Z_n be the cyclic group of order n, then*

$$Z_n \cong \mathbb{Z}/n\mathbb{Z}.$$

⌟

Proof Sketch: First we show that φ is well-defined. Suppose $x^r = x^s$. Then, we show that $\varphi(x^r) = \varphi(x^s)$. Note that $x^{r-s} = 1$, so $n|(r-s)$ by our previous lemma. Write $(r-s) = nt$. Then, $\varphi(x^r) = \varphi(x^{tn+s}) = y^{tn+s} = y^s = \varphi(x^s)$.

1.8. **Formulas for orders and finding generators**

In this section, we try to find the generators of a cyclic group.

First, we try to calculate the order of arbitrary powers of x in an arbitrary group. **Note:** the following result applies to an arbitrary group, not necessarily a cyclic group.

Proposition 1.43. *Let G be a group (not necessarily cyclic) and let $x \in G$. Let $a \in \mathbb{Z} - \{0\}$.*

(1) *if $|x| = \infty$, then $|x^a| = \infty$*
(2) *if $|x| = n < \infty$, then $|x^a| = \frac{n}{(n,a)}$*
(3) *if $|x| = n < \infty$ and $a|n$, then $|x^a| = \frac{n}{a}$.*

⌟

Now, we use this to understand which elements in a cyclic group are generators of the group.

Proposition 1.44. *Let H be a cyclic group, i.e. $H = \langle x \rangle$. Then,*

(1) *If $|x| = \infty$, then $H = \langle x^a \rangle$ if and only if $a = 1$ or $a = -1$.*
(2) *If $|x| = n < \infty$, then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$.*

⌟

Proof Sketch: we know that $\langle x^a \rangle$ generates a group of order $|x^a|$, so if $\langle x^a \rangle = \langle x \rangle$, we require that $|x^a| = |x|$. By 1.43, this is true if and only if $(a, n) = 1$.

Example 1.45. Consider $\mathbb{Z}/12\mathbb{Z}$. Then, we know that $\bar{1}, \bar{5}, \bar{7}$, and $\bar{11}$ are generators, since for each of these $(a, 12) = 1$. ⌟

Theorem 1.46. *Let $H = \langle x \rangle$ be a cyclic group.*

(1) *Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ (i.e. the trivial group) or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.*
(2) *If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m. **In other words, the nontrivial subgroups of H correspond bijectively with the integers** $1, 2, 3, \ldots$.*

(3) *If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$.*

Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{(n,m)} \rangle$, so that ***the subgroups of H correspond bijectively with the positive divisors of n.***

⌟

1.9. Subgroups generated by subsets

The next class of subgroups we will look at are subgroups that are generated by subsets of a group. Let G be a group and let A be a subset of a group. We will define the subgroup generated by A to be the intersection of all subgroups of G that contain A. Naturally, this raises the question: is this intersection guaranteed to be a group? We prove this using the following lemma:

Lemma 1.47. *Let $\mathcal{C} \neq \emptyset$ be a collection of subgroups of a group G. Then, the intersection of all elements of $\mathcal{C}$ is a subgroup of G.* ⌟

Proof. Define $K = \cap_{X \in \mathcal{C}} X$. Given $1_G \in X$ (since X is defined to be a subgroup for each $X \in \mathcal{C}$), $1_G \in K$. If $x \in K$, then $x \in X$ for all $X \in \mathcal{C}$, which implies $x^{-1} \in X$ for all $X \in \mathcal{C}$, which implies $x^{-1} \in K$. This logic can be used to show that if $x, y \in K$, then $xy^{-1} \in K$. □

Definition 1.48 (Subgroup generated by a Subset)**.** *Let G be a group and let A be a subset of G. Then, the subgroup of G generated by A is defined as the intersection of all subgroups of G containing A, i.e.*

$$\langle A \rangle := \cap_{A \subseteq H, H \leq G} H. \tag{1}$$

In particular, for notational clarity, we set the following:

(1) *If $A = \{a_1, \cdots, a_n\}$ is a finite set, we write $\langle a_1, \cdots, a_n \rangle = \langle A \rangle$.*
(2) *If A and B are two subsets, we write $\langle A, B \rangle = \langle A \cup B \rangle$.*

⌟

Observation. *Note that $\langle A \rangle$ is a subgroup by our last lemma if we let $\mathcal{C} = \{H \mid H \leq G, A \subseteq H\}$.*

So far, we have only said that this definition gives us a subgroup that is generated by a subset. However, we do not yet know what the elements of this subgroup look like. We remedy this in the next result.

Definition/Proposition 1.49. *The subgroup of G that is generated by the subset $A \subseteq G$ can be written as:*

$$\langle A \rangle = \{a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_t^{\epsilon_t} \mid a_i \in A, \epsilon \in \{\pm 1\}, t \in \mathbb{Z}\} =: \bar{A}. \tag{2}$$

Proof. It is easy to show that $\bar{A}$ is a subgroup. To show that $\langle A \rangle = \bar{A}$, we first note that $\langle A \rangle \subseteq \bar{A}$ since each $a \in$ can be written as $a^1 \in \bar{A}$. To show the reverse direction, we simply note that products and inverses belong to the group, so $\bar{A} \subseteq \langle A \rangle$. □

1.10. Quotient Groups

Definition 1.50 (Fibers)**.** *Given $\varphi : G \to H$ is a group homomorphism, the fibers of φ are the sets of elements of G that φ maps to single elements. In other words, the fiber of φ above $h \in H$, denoted by X_h, is defined as:*

$$X_h = \{g \in G \mid \varphi(g) = h\}.$$

⌟

Observation. *Note that there is a natural operation we can do on fibers themselves because of the properties of homomorphisms. Let X_h and $X_{h'}$ be two fibers. Then, $X_h X_{h'}$ is the fiber $X_{hh'}$. This is because for any $\alpha \in X_h, \beta \in X_{h'}$, $\varphi(\alpha\beta) = \varphi(a)\varphi(\beta) = hh'$.*

This will allow us to define the quotient group as the group of fibers of a homomorphism.

Definition 1.51 (Kernel and image of a homomorphism)**.** *Let $\varphi : G \to H$ be a group homomorphism. Then, the kernel of φ is*

$$\ker \varphi := \{g \in G \mid \varphi(g) = 1_H\}.$$

On the other hand, the image of the homomorphism is defined as

$$\operatorname{im}\varphi := \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}.$$

⌟

We look at some immediate properties of kernels and images:

Proposition 1.52. *Let $\varphi : G \to H$ be a group homomorphism. Then,*

(1) $\varphi(1_G) = 1_H$, $\varphi(g^{-1}) = \varphi(g)^{-1}$ *and* $\varphi(g^n) = \varphi(g)^n$
(2) $\ker\varphi$ *and* $\operatorname{im}\varphi$ *are subgroups of G and H respectively, i.e.* $\ker\varphi \leq G$ *and* $\operatorname{im}\varphi \leq H$.

⌟

Now, we define the quotient group:

Definition 1.53 (Quotient Group)**.** *Let $\varphi : G \to H$ be a group homomorphism with $K = \ker\varphi$. Then, the quotient group, denoted by G/K, is the group whose elements are fibers X_h for each $h \in H$ and the operation is defined by $X_h X_{h'} = X_{hh'}$. Note that K is also an element in G/K; concretely $K = X_{1_H}$ and K is the identity of the quotient group G/K.* ⌟

The odd thing about this definition is that X_h is very opaque. What we will now see is that each fiber is essentially a translate of the kernel K via *some* representative:

Proposition 1.54. *Let $\varphi : G \to H$ be a group homomorphism and let $K := \ker\varphi$ be its kernel. Then, $X_h \in G/K$ can be written as, for any $g' \in X_h$,*

$$X_h = g'K := \{g'k \mid k \in K\}$$

and

$$X_h = Kg' := \{kg' \mid k \in K\}.$$

⌟

Proof. It is easy to see that $g'K \subseteq X_h$ because $\varphi(g'k) = h$ for any $k \in K$. To see the reverse inclusion, let $g \in X_h$, i.e. $\varphi(g) = h$. Now, define $k = g'^{-1}g$. Then, $\varphi(k) = \varphi(g'^{-1}g) = \varphi(g')^{-1}\varphi(g) = h^{-1}h = 1$. So, $k \in K$. Therefore, $g = g'k \in g'K$. □

We used a particularly general construction in this definition that is worth noting separately:

Definition 1.55 (Left coset/Right coset and Representatives)**.** *For a subgroup N of G and some element $g \in G$, we will use the notation $gN = \{gn \mid n \in N\}$ which is called a **left coset** of N in G. Similarly, $Ng = \{ng \mid n \in N\}$ which is called a **right coset** of N in G. Any element of a coset is called a **representative** of the coset. In particular, $g = g1 = 1g$ itself is a representative of the coset gN or Ng.* ⌟

Note that if G is an additive group, we will write $g + N$ to be the left coset of N in G with representative g (and the equivalent statement for right cosets is true).

Observation. *The quotient group G/K consists of elements that are the left cosets of $K = \ker\varphi$ in G. In other words, the elements of G/K look like gK where g is a representative of a fiber. Furthermore, since different representatives define the same fiber, the choice of the representative does not matter – we will formalize this now:*

Proposition 1.56. *Let G be a grup and let $K = \ker\varphi$ for some group homomorphism φ. Then, consider the set of left cosets of K in G and define the following operation on them: $uK \circ vK = (uv)K$. Then,*

(1) *This operation defined a group, called the quotient group, denoted by G/K*
(2) *The operation is well-defined, i.e. if $u_1 \in uK$ and $v_1 \in vK$, then $u_1v_1 \in uvK$ and $u_1v_1K = uvK$. In other words, the group operation does not depend on the choice of representative.*
(3) *The statements are true after replacing left cosets with right cosets.*

⌟

In light of all this, we write the following as the definition of a quotient group:

Definition 1.57 (Quotient Group)**.** *let $\varphi : G \to H$ be a group homomorphism and let $K = \ker \varphi$. Then, the quotient group G/K is defined as the cosets of the kernel $K = \ker \varphi$ in G, i.e.*

$$G/K := \{\bar{g} := gK \mid g \in G\}$$

with the group operation being $\bar{g} \cdot \bar{g'} = \overline{gg'}$, i.e. $gK \cdot g'K = (gg')K$. Note that $\bar{g} = \bar{g'}$ if $g' \in gK$ or $g \in g'K$. ⌟

Example 1.58. Let $\varphi : G \to H$ be an isomorphism. Then, the kernel is just 1. In particular, the fibres of φ are singleton subsets of G (since φ is injective and subjective), so $G/1 \cong G$. ⌟

Example 1.59. Consider the homomorphism $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$. This has the kernel $K = n\mathbb{Z}$. Then, the cosets of K in $\mathbb{Z}$ are $a + K = a + n\mathbb{Z}$ for each $a \in \mathbb{Z}$. ⌟

Example 1.60. let $G = \mathbb{R}^2$ and let $H = \mathbb{R}$. Define $\varphi : G \to H$ by $\varphi((x, y)) = x$. Then, one can show that φ is a homomorphism and its kernel is $\ker \varphi = \{(0, y) \mid y \in \mathbb{R}\}$. Then, the quotient group G/K consists of the cosets $\overline{(x, 0)} = (x, 0) + \ker \varphi$ for each $x \in \mathbb{R}$. ⌟

1.10.1. *Important properties of cosets and representatives.* Some of what we have seen so far can be generalized. For example, the fact that a coset can be written using any representative and the choice of the representative does not make a material difference in defining the coset can be generalized to cosets of other subgroups as well alongside cosets of a kernel. On the other hand, we will see that the group operation in the quotient group, i.e. $uK \cdot vK = uvK$, is only defined if K satisfies some properties.

Proposition 1.61. *Let N be any subgroup of G, i.e. $N \leq G$. Then, the set of left cosets of N (sets of the form gN for $g \in G$) partition G, i.e. all elements of G belong to some coset of N in G. In particular, for all $u, v \in G$, $uN = vN$ if and only if $v^{-1}u \in N$. In particular, $uN = vN$ if and only if u and v are representatives of the same coset.* ⌟

Proof. Since $1 \in N$ (as N is a subgroup), each element of G can be written as $g = g \cdot 1 \in gN$. As such, g is a representative of the coset gN and, therefore, $G = \cup_{g \in G} gN$. We must now show that distinct cosets have no intersection. In other words, if $uN \cap vN \neq \emptyset$, then $uN = vN$. Let $x \in uN \cap vN$. Then, $x = un = vm$ for some $m, n \in N$. now, $u = vmn^{-1} = vm_1$ where $m_1 := mn^{-1} \in N$. Then, $uN = \{ut \mid t \in N\} = \{vm_1 t \mid t \in N\} = vm_1 N \subseteq vN$. Similarly, $vN \subseteq uN$, which gives us $uN = vN$.

Now, we prove the second statement. We know that $uN = vN$ if and only if $u \in vN$ if and only if $u = vn$ for some $n \in N$ if and only if $v^{-1}u \in N$.

Now, we prove the last statement. Again, $uN = vN$ if and only if $v \in uN$ which is equivalent to saying $v = v \cdot 1_G$ is a representative of uN. Similarly, $uN = vN$ if and only if $u \in vN$ which is equivalent to saying u is a representative of vN. Since $uN = vN$, therefore, u and v are representatives of the same coset. □

Since this makes remembering almost everything so far quite simple, it is worth writing it again: **Two cosets are the same, i.e. $uN = vN$, if and only if u and v are representatives of the same coset. Also, two cosets are the same, i.e. $uN = vN$, if and only if $u \in vN$.**

Note that, in additive notation, we would say that two cosets are the same, i.e. $uN = vN$, if and only if $u \in v + N$

Now, we come to the second part: what are the requirements on the subgroup N that make $uN \circ vN = (uv)N$ a well-defined group operation?

Proposition 1.62. *Let G be a group and let N be a subgroup, $N \leq G$.*

1. *The operation $uN \circ vN = (uv) \circ N$ is well-defined if and only if $gng^{-1} \in N$ for all $g \in G, n \in N$ (as we will soon see - this means, this operation is well-defined if and only if N is a normal subgroup).*
2. *If this condition is satisfied, then the set of left cosets of N in G, i.e. $\{gN \mid g \in G\}$, is a group with the group operation $gN \circ g'N = (gg')N$. In particular, the identity is the coset $1N$ and the inverse of gN is $g^{-1}N$.*

⌟

Proof. We sketch the proof of the first part. Assume that the operation is well-defined. Then, suppose $u, u_1 \in uN$ and $v, v_1 \in vN$. We know that $uN \circ vN = (uv)N$. But choosing different

representatives of uN and vN, we get that $u_1N \circ v_1N = (u_1v_1)N$ must be the same as $(uv)N$, so $(uv)N = (u_1v_1)N$. Now, let $u = 1$, $u_1 = n$ for some $n \in N$ (clearly, then $u, u_1 \in uN$) and let $v = v_1 = g^{-1}$ for some fixed $g \in G$. Then, $1g^{-1}N = ng^{-1}N$ which implies $g^{-1}N = ng^{-1}N$. Clearly, $ng^{-1} \in ng^{-1}N$, so $ng^{-1} \in g^{-1}N$, so $ng^{-1} = g^{-1}n'$ for some $n' \in N$, which gives us that $gng^{-1} \in n' \in N$. Conversely, suppose $gng^{-1} \in N$ for all $g \in G, n \in N$. Again, let $u, u_1 \in uN$ and $v, v_1 \in vN$. Write $u_1 = un$ and $v_1 = vm$ for some $n, m \in N$. We can show that $u_1v_1 = (uv)(n_1m)$ where $n_1 = v^{-1}nv$ which is in N by assumption. Since N is closed under products, $n_1m \in N$, so $u_1v_1 = (uv)n_2$ for some $n_2 \in N$ (precisely $n_2 = n_1m \in N$. Since $u_1v_1 \in uvN$, $u_1v_1N = uvN$. As such, the operation is well-defined. □

Subgroups that satisfy these properties are extremely important in the study of groups and, as such, are given a separate name.

Definition 1.63 (Normal Groups and conjugates)**.** *Let G be a group and let N be a subgroup of G. The element gng^{-1} is called the conjugate of $n \in N$ by $g \in G$. The set $gNg^{-1} = \{gng^{-1} \mid n \in N\}$ is called the conjugate of N by g.*

*In particular, the element g is said to normalize N if $gNg^{-1} = N$. A subgroup N of a group G is called a **normal subgroup** if $gNg^{-1} = N$ for all $g \in G$, i.e. every element of G normalizes N. If N is a normal subgroup of G, we write $N \trianglelefteq G$.* ⌟

Theorem 1.64. *Let N be a subgroup f the group G. Then, the following are equivalent:*

1. *$N \trianglelefteq G$*
2. *$N_G(N) = G$, i.e. the normalizer of N in G is G itself*
3. *$gN = Ng$ for all $g \in G$*
4. *The set of left cosets are a group under the operation $uN \circ vN = (uv)N$*
5. *$gNg^{-1} \subseteq N$ for all $g \in G$.*

⌟

Note that the kernel of a homomorphism satisfies both the requirements in lemma 1.62. One can ask whether all subgroups N that satisfy these properties can be shown to be the kernel of some homomorphisms. This is answered in the next proposition:

Proposition 1.65. *A subgroup N of the group G is normal if and only if it is the kernel of some homomorphism.* ⌟

Proof. The kernel of a homomorphism is normal as we have already shown. We need to now show that if $N \trianglelefteq G$, then there exists a homomorphism whose kernel is N. Define $\varphi : G \to G/N$ by $\varphi(g) = gN$ for all $g \in G$. Note that this is a homomorphism as $\varphi(g_1g_2) = (g_1g_2)N = g_1N \cdot g_2N = \varphi(g_1)\varphi(g_2)$. Now, it is easy to check that $\ker\varphi = \{g \in G \mid \varphi(g) = 1N\} = \{g \in G \mid gN = 1N\} = \{g \in G \mid g \in N\} = N$. □

Definition 1.66. *Let $N \trianglelefteq G$ be a normal subgroup. The homomorphism $\pi : G \to G/N$ defined by $\pi(g) = gN$ is called the **normal projection/homomorphism** of G onto G/N. If $\bar{H} \leq G/N$ is a subgroup of G/N, the complete preimage of $\bar{H}$ in G is the preimage of $\bar{H}$ under the normal projection homomorphism.* ⌟

Example 1.67. For any group G, the subgroups 1 and G are normal in G. In particular, $G/1 \cong G$ and $G/G \cong 1$. ⌟

Example 1.68. Let G be an abelian group, then *any* subgroup N of G is a normal subgroup since $gng^{-1} = gg^{-1}n = n \in N$ for all $g \in G, n \in N$. ⌟

Example 1.69. Let $G = Z_k$ be the cyclic group of order k. Let x be a generator of G and let $N \leq G$. Then, $N = \langle x^d \rangle$ where d is the smallest integer such that $x^d \in N$. Then,

$$G/N = \{gN \mid g \in G\} = \{x^\alpha N \mid \alpha \in \mathbb{Z}\}.$$

Now, since $x^\alpha N = (xN)^\alpha$, $G/N = \langle xN \rangle$, i.e. G/N is cyclic and its generator is xN. Furthermore, $|G/N| = \frac{|G|}{|N|}$. **In other words, the quotient groups of cyclic groups are cyclic**. ⌟

We now prove the following theorem, known as Lagrange's Theorem, which allows us to compute the order of a quotient group.

Theorem 1.70 (Lagrange's Theorem)**.** *If G is a finite group and H is a subgroup of G, then the order of H divides the order of G and the number of left cosets of H in G equals $\frac{|G|}{|H|}$.* ⌟

Proof. Let $|H| = n$ and let the number of left cosets of H in G equal k. Now, we claim that for any fixed $g \in G$, $|gH| = n$. This can be seen by noting that the map $H \to gH$ by $h \to gh$ is (a) surjective by definition and (b) injective because (if $gh_1 = gh_2$, then $h_1 = h_2$), and so $|gH| = |H| = n$.

Now, we have proven previously (see Proposition 1.61) the left cosets of H in G partition G, so $|G| = |$ no. of left cosets $| \cdot |$ cardinality of each left coset $| = kn$. As such, number of left cosets of H in G is $k = \frac{|G|}{n} = \frac{|G|}{|H|}$. □

One point to note and remember from this proof is the idea that $|H| = |gH|$ and that $H \to gH$ is a bijection.

Definition 1.71 (Index of a subgroup)**.** *Let G be a group and let H be a subgroup of G. Then, the index of H in G is the number of left cosets of H in G. We denote the index by $|G : H|$.* ⌟

Note that if G is a finite group, then by Lagrange's theorem, $|G : H| = \frac{|G|}{|H|}$. However, if G is an infinite group, then this is not true. In particular, a subgroup of an infinite group can have finite index or infinite index. As an example, consider the group $\mathbb{Z}$. Then, the trivial subgroup $\{0\}$ has order 1 but has infinite index. On the other hand, the subgroup $\langle n \rangle$ is of infinite order but has index n (assuming $n > 0$).

Corollary 1.72. *let G be a group. We have the following consequences of Lagrange's theorem:*

1. *If G is a finite group and let $x \in G$. Then, $|x|$ divides the order of G. In particular, $x^{|G|} = 1$ for all $x \in G$.*
2. *If G is a group of order p where p is a prime integer. Then, G is cyclic, hence $G \cong Z_p$. (Recall that Z_p is the cyclic group of order p and $Z_p \cong \mathbb{Z}/p\mathbb{Z}$)*

⌟

Example 1.73. Let $H = \langle (1,2,3) \rangle \leq S_3$ and let $G = S_3$. We now show that $H \trianglelefteq S_3$ using Lagrange's theorem. First, one can show that $H \leq N_G(H) \leq G$. Now, by Lagrange's theorem, the order of H divides the order of $N_G(H)$ which divides the order of S_3, which is 6. Now, since G has order 6 and H has order 3, the only possibilities for $N_G(H)$ are that $N_G(H)$ is either the same as H or the same as $G = S_3$. Now, one can show that $(1,2)$ conjugates $(1,2,3)$ to $(1,2,3)^{-1}$. So, $(1,2)$ conjugates a generator of H to another generator of H, which allows us to conclude that $(1,2) \in N_G(H)$. Since $(1,2) \notin H$, so, $N_G(H) \neq H$ and $N_G(H) = G$, which allows us to conclude that $H \trianglelefteq S_3$. ⌟

Definition 1.74 (Simple Groups)**.** *Simple groups are groups whose only normal subgroups are 1 and G.* ⌟

Theorem 1.75 (Cauchy's Theorem)**.** *If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.* ⌟

Theorem 1.76 (Sylow)**.** *If G is a finite group of order $p^\alpha m$, where p is a prime and p does not divide m, then G has a subgroup of order p^α.* ⌟

Proposition 1.77. *If H and K are finite subgroups of a group, then*

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

where $HK = \{hk \mid h \in H, k \in K\}$. ⌟

Proposition 1.78. *If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$.* ⌟

Corollary 1.79. *If H and K are subgroups of G, and $H \leq N_G(K)$, then $HK \leq G$. In particular, if $K \trianglelefteq G$, then $HK \leq G$ for any $H \leq G$.* ⌟

1.11. **Group Isomorphism Theorems**

Theorem 1.80 (First Isomorphism Theorem)**.** *Let $\varphi : G \to H$ be a group homomorphism. Then,* $\ker\varphi \trianglelefteq G$ *and* $G/\ker\varphi \cong \varphi(G)$. ⌟

Proof. Let $x \in \ker\varphi$. Then, for any $g \in G$, $\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} = \varphi(g)1_H\varphi(g)^{-1} = \varphi(g)\varphi(g)^{-1} = 1_H$, so $gxg^{-1} \in \ker\varphi$. We already know that $\ker\varphi$ is a subgroup, so $\ker\varphi \trianglelefteq G$. To prove the second statement, define $\Psi : G/\ker\varphi \to \varphi(G)$ by $\Psi(g\ker\varphi) = \varphi(g)$. Now, with $K = \ker\varphi$,

- Ψ is well-defined. Suppose $gK = hK$. Then, $h^{-1}g \in K$. Then, $\varphi(h^{-1}g) = 1_H$. But $\varphi(h^{-1}g) = \varphi(h)^{-1}\varphi(g) = 1_H$, which implies $\varphi(h) = \varphi(g)$. So, $\Psi(g) = \Psi(h)$.
- Ψ is a homomorphism since $\Psi(gK\circ hK) = \Psi(ghK) = \varphi(gh) = \varphi(g)\varphi(h) = \Psi(gK)\Psi(hK)$.
- Ψ is injective. This is because $\Psi(gK) = \Psi(hK) \implies \varphi(g) = \varphi(h)$. But then, $\varphi(h)^{-1}\varphi(g) = 1_H$, which implies $\varphi(h^{-1}g) = 1_H$. So $h^{-1}g \in K$, so $gK = hK$.
- Ψ is surjective by definition of the morphism.

□

Corollary 1.81. *Let $\varphi : G \to H$ be a group homomorphism. Then, φ is injective if and only if* $\ker\varphi = 1$. *Furthermore,* $|G : \ker\varphi| = |\varphi(G)|$. ⌟

Proof. If φ is injective, then, $\ker\varphi = 1_G$ by definition of an injective map. On the other hand, if $\ker\varphi = 1_G$, then suppose $\varphi(x) = \varphi(y)$. Then, $\varphi(y)^{-1}\varphi(x) = \varphi(y^{-1}x) = 1_H$. Since $y^{-1}x$ is in the kernel and the kernel is simply 1_G, we have that $y^{-1}x = 1_G$, so $x = y$. To prove the second statement, we simply use the first isomorphism theorem (1.80) and Lagrange's theorem (1.70). □

Theorem 1.82 (Second/Diamond Isomorphism Theorem)**.** *Let G be a group and let A and B be subgroups of G. Assume that $A \leq N_G(B)$. Then, $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.* ⌟

Proof. The first statement is the same as 1.79. To show $B \trianglelefteq AB$, we note that $A \leq N_G(B)$ by hypothesis and $B \leq N_G(B)$ trivially, so $AB \leq N_G(B)$. This means, all elements of AB normalize B. As such, B is a normal subgroup of AB considered as a group.

Now, given $B \trianglelefteq AB$, we can define the quotient group AB/B and that allows us to construct the map $\varphi : A \to AB/B$ by $\varphi(a) = aB$. It is straightforward to check that this is well-defined and a homomorphism. From the definition, clearly, φ is surjective. On the other hand, the kernel of φ is $A \cap B$. As such, $\ker\varphi = A \cap B$ which, by the first isomorphism theorem, is a normal subgroup of A. The first isomorphism theorem also allows us to conclude the last statement. □

Theorem 1.83 (Third Isomorphism Theorem)**.** *Let G be a group. Let $H \trianglelefteq G$ and $K \trianglelefteq G$, and let $H \leq K$. Then,*

$$K/H \trianglelefteq G/H$$

and $(G/H)/(K/H) \cong G/K$, which can be equivalently written with the bar notation (i.e. $\bar{J} = J/H$) as:

$$\bar{G}/\bar{K} \cong G/K.$$

⌟

Proof. The fact that $K/H \trianglelefteq G/H$ follows from the fact that $K \trianglelefteq G$ and the group operation in quotient group, i.e. $uK \circ vK = (uv)K$. To prove the second statement, construct the following map: $\varphi : G/H \to G/K$ which sends gH to gK. First, φ is well-defined: if $g_1H = g_2H$, then $g_1 = g_2h$ for some $h \in H$, and since $H \leq K$, $g_1K = g_2K$. So $\varphi(g_1H) = \varphi(g_2H)$. On the other hand, φ is, clearly, a surjective homomorphism. Finally, consider its kernel:

$$\begin{aligned}\ker\varphi &= \{gH \in G/H \mid \varphi(gH) = 1K\}\\ &= \{gH \in G/H \mid gK = 1K\}\\ &= \{gH \in G/H \mid g \in K\}\\ &= K/H.\end{aligned}$$

Using the first isomorphism theorem, we get the final statement in the theorem. □

Finally, we come to the fourth isomorphism theorem, which is arguably more important than the last two:

Theorem 1.84 (Fourth or Lattic Isomorphism Theorem / Correspondence Theorem)**.** *Let G be a group and let $N \trianglelefteq G$. Then, there is the following bijection:*

$$\{ \text{ Subgroups } A \leq G \text{ s.t. } N \subseteq A\} \leftrightarrow \{ \text{ Subgroups } \bar{A} = A/N \leq G/N\}.$$

In particular, any subgroup of G/N is of the form A/N for some subgroup A of G containing N. Furthermore, this bijection is inclusion-preserving (i.e. if $H_1 \subseteq H_2$, then $H_1/N \subseteq H_2/N$).

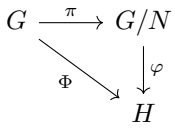
In particular, the bijection has the following properties: for all $A, B \leq G$ with $N \leq A$ and $N \leq B$:

1. *$A \leq B$ if and only if $\bar{A} \leq \bar{B}$ (recall that $\bar{A} = A/N$)*
2. *if $A \leq B$, then $|B : A| = |\bar{B} : \bar{A}|$*
3. *$\overline{\langle A, B\rangle} = \langle \bar{A}, \bar{B}\rangle$*
4. *$\overline{A \cap B} = \bar{A} \cap \bar{B}$*
5. *$A \trianglelefteq G$ if and only if $\bar{A} \trianglelefteq \bar{G}$.*

⌟

Proof Sketch: The mapping is straightforward; it is the normal projection which takes $\pi(x) = xN$ for any $x \in G$. We first show that the map is well-defined. Let $A \leq G$ with $N \subseteq A$. Then, $\pi(A) = A/N$ is a subgroup since homomorphisms map subgroups to subgroups. Conversely, if A/N is a subgroup of G/N, then its preimage, $\pi^{-1}(A/N) = \{k \in G \mid kN \in AN\} = A$. In particular, A must contain N because π maps N to $1N \in AN$. It is straightforward to show that these maps are inverses of each other and preserves inclusion. While (1) to (4) are straightforward, (5) is quite useful. Suppose $A \trianglelefteq G$. Now, consider A/N. For any $xN \cdot aN \cdot (xN)^{-1} = Nx \cdot a \cdot x^{-1}N^{-1} = N \cdot a \cdot N^{-1} = aNN^{-1} = aN$, so $A/N \trianglelefteq G/N$. Converse is proven similarly.

Observation. *In all the theorems and examples, we have constructed homomorphisms on quotient groups, i.e. homomorphisms of the form $\varphi : G/N \to H$. In some of these cases, the value of φ on the coset gN is given in terms of the representative g alone. We then had to prove that φ is well-defined (i.e. the value of φ on the coset was independent of the choice of g). Now, we can get a homomorphism Φ from G to H that makes the following diagram commute:*



In terms of these constructions, we have that

$$\varphi \text{ is well-defined on } G/N \text{ if and only if } N \leq \ker \Phi.$$

1.12. **Group Actions**

Recall: a **group action** of a group G on a set A is a mapping $G \times A \to A$ such that:

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$ for all $g_1, g_2 \in G, a \in A$
2. $1_G \cdot a = a, \forall a \in A$.

With this, for each $g \in G$, we can define a *permutation* of A by

$$\sigma_g : A \to A, \quad \sigma_g(a) = g \cdot a.$$

We can also define a *permutation representation* associated to a given group action by the following homomorphism:

$$\varphi : G \to S_A, \quad \varphi(g) = \sigma_g.$$

Furthermore, we defined the following:

1. The kernel of a group action of G on A is the set $\{g \in G \mid g \cdot a = a, \forall a \in A\}$. The kernel of the group action is the same as the kernel of the corresponding permutation representation. Therefore, the kernel of a group action is a normal subgroup of G.
2. For each $a \in A$, we have the stabilizer of a in G which is the set $G_a := \{g \in G \mid g \cdot a = a\}$.
3. The group action is called faithful if the kernel is the identity 1_G.

Note:

(1) Two group elements g and g' induce the same permutation on A (i.e. $\sigma_g = \sigma_{g'}$) if and only if they are in the same fibre of the permutation representation φ. This happens if and only if they are in the same coset of the kernel.
(2) An action of G on A is a faithful action of $G/\ker\varphi$ on A.

Example 1.85. Consider the action of S_n on $A = \{1, 2, \cdots, n\}$ (where $n \in \mathbb{Z}_{>0}$) by $\sigma \cdot i = \sigma(i)$. In this case, the permutation representation is the identity map, i.e. $\varphi : S_A \to S_A$ by $\varphi(\sigma) = \sigma$. The action is faithful and, for each $i \in \{1, \cdots, n\}$, the stabilizer G_i is isomorphic to S_{n-1}. ⌟

We saw that given a group action of G on A, we can define a permutation representation, which is a homomorphism $\varphi : G \to S_A$. This can be reversed: if $\varphi : G \to S_A$ is a group homomorphism, we can define a group action of G on A by $g \cdot a = \varphi(g)(a)$. This is summarized in the following:

Proposition 1.86. *For any group G and a non-empty set A, there is the following bijection:*

$$\{ \text{ group actions of } G \text{ on } A\} \leftrightarrow \mathrm{Hom}(G, S_A).$$

⌟

Given this result, we can redefine permutation representations as the following:

Definition 1.87 (Permutation Representation)**.** *If G is a group, then a permutation representation of G is a homomorphism*

$$\varphi : G \to S_A$$

where S_A is the symmetric group for some non-empty set A. We say that a given group action of G on A induces a permutation presentation of G. ⌟

Note that in the case where A is a finite set of n elements, we can also say that $S_A \cong S_n$.

Proposition 1.88. *Let G be a group acting on a set $A \neq \emptyset$. The relation on A defined by $a \sim b$ if and only if $a = g \cdot b$ for some $g \in G$ is an equivalent relation. For each $a \in A$, the number of elements in the equivalence class containing a is $|G : G_a|$, the index of the stabilizer of a.* ⌟

Proof. To show that this is an equivalence relation. First, $a \sim a$ since $a = 1_G \cdot a$ by definition of a group action, i.e. the relation is reflexive. Secondly, if $a \sim b$, i.e. $a = g \cdot b$ for some $g \in G$, then $g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = 1_G \cdot b = b$, so $b \sim a$, i.e. the relation is symmetric. Finally, if $a \sim b$ and $b \sim c$, then $a = g \cdot b$ and $b = g' \cdot c$ for some $g, g' \in G$. Then, $a = g \cdot b = g \cdot (g' \cdot c) = \tilde{g} \cdot c$ where $\tilde{g} = g \cdot g' \in G$, so $a \sim c$, i.e. the relation is transitive.

To prove the second statement, we construct a bijection between the left cosets of G_a in G and the elements of the equivalence class of a. Denote the equivalence class of a as

$$\mathcal{C}_a = \{g \cdot a \mid g \in G\}.$$

Now, suppose $b = g \cdot a \in \mathcal{C}_a$. Then, gG_a is a left coset of G_a in G. The map $b = g \cdot a \to gG_a$. Then, gG_a is a left coset of $\mathcal{C}_a$ in G. The map $b = g \cdot a \to gG_a$ is a map from $\mathcal{C}_a$ to the set of left cosets of G_a in G. This is surjective since for any $g \in G$, the element $g \cdot a$ is an element of $\mathcal{C}_a$. Since $g \cdot a = h \cdot a$ if and only if $h^{-1}g \in G_a$ if and only if $gG_a = hG_a$, the map is injective, so this is a bijection. □

1.13. Cycle Decompositions

Definition 1.89 (Orbits)**.** *Let G be a group acting on $A \neq \emptyset$. Then, the equivalence class $\{g \cdot a \mid g \in G\}$ is called the orbit of G containing a.* ⌟

Proposition 1.90. *Every element of the symmetric group S_n has a unique cycle decomposition. In other words, every element can be uniquely expressed as a product of disjoint cycles.* ⌟

Proof. Let $A = \{1, \cdots, n\}$ and let $\sigma \in S_A = S_n$. Now, consider $G = \langle\sigma\rangle$. Then, G acts on A and, so, it partitions A into unique set of disjoint sets, i.e. the orbits.

Now, consider a fixed orbit $\mathcal{O} = \{g \cdot x \mid g \in G\}$ of G containing $x \in \mathcal{O}$. Then, there exists the following bijection:

$$\{ \text{ left cosets of } G_x \text{ in G } \} \leftrightarrow \{ \text{ elements of } \mathcal{O}\}$$

(recall that $G_x = \{g \cdot x \mid g \in G \text{ s.t. } gx = x\}$, so in our case $G_x = \{\sigma^j \mid \sigma^j \in G \text{ s.t. } \sigma^j x = x\}$) by

$$\sigma^i G_x \leftrightarrow \sigma^i x.$$

Since G is cyclic, we know that $G_x \trianglelefteq G$ and G/G_x is cyclic of order d which is the smallest positive integer such that $\sigma^d \in G_x$. On the other hand, because of the bijection, $d = |G : G_x| = |\mathcal{O}|$. Then, there are d distinct cosets of G_x in G which can be written as $\{G_x, \sigma G_x, \sigma^2 G_x, \cdots, \sigma^{d-1} G_x\}$. Using the bijection, the distinct elements of $\mathcal{O}$ are $\{x, \sigma x, \sigma^2 x, \cdots, \sigma^{d-1} x\}$. Therefore, σ is a d-cycle. This allows us to conclude that each σ has a cycle decomposition.

Now, the ordbits of $\langle \sigma \rangle$ are uniquely determined by σ. Within each orbit, $\mathcal{O}$, we can begin with any element as a representative — edifferent choices simply produces a different cyclic permutation of the original list. As such, the cycle decomposition is unique up to a rearrangement of the cycles and up to a cyclic permutation of the integers within each cycle. □

Definition 1.91 (Permutation Groups)**.** *Subgroups of symmetric groups are called permutation groups.* ⌟

For each subgroup G of S_n, the orbits of G will refer to its orbit on $\{1, 2, \cdots, n\}$.

1.14. Cayley's Theorem and the group action of left multiplication

Let G act on itself via left multiplication, i.e. $g \cdot a = ga, \forall g, a \in G$. Note that if G is written additively, then $ga = g + a$. Now, if G is finite, we can label its elements as $\{g_1, \cdots, g_n\}$. Then, each $g \in G$ describes a permutation $\sigma_g \in S_n$ via

$$\sigma_g(i) = j \leftrightarrow g g_i = g_j.$$

This group action, by left multiplication, is both transitive and faithful, as we will see in the next proposition. The stabilizer of any point is the identity subgroup.

Now, let $H \leq G$ be a subgroup. Let

$$A = \{\text{left cosets of } H \text{ in G}\}.$$

Then, G also acts on A by $g \cdot aH = gaH, \forall g \in A, aH \in A$. Note that if $H = \{1\}$, then $aH = \{a\}$ and, so, this action is the same as G acting on itself via left multiplication. On the other hand, if H is of finite index m (i.e. there are m left cosets of H in G), we can label the cosets of H with the integers from 1 to m. We can, then, label the left cosets of H in G as $a_1 H, a_2 H, \cdots, a_m H$. Then, for each $g \in G$, we have a permutation of $\sigma_g \in S_m$, by

$$\sigma_g(i) = j \leftrightarrow g a_i H = a_j H.$$

Proposition 1.92. *Let G be a group and let $H \leq G$ be a subgroup. Let G act on $A = \{$ left cosets of H in $G\}$. Let π_H be the permutation representation of this action:*

$$\pi_H : G \to S_A.$$

Then,

1. *G acts transitively on A*
2. *the stabilizer in G of the point $1_H \in A$ is the subgroup H.*
3. *the kernel of the action, i.e. $\ker \pi_H$, is $\cap_{x \in G} x H x^{-1}$.*
4. *$\ker \pi_H$ is the largest normal subgroup of G contained in H.*

⌟

Proof. To prove (1), we show that there is only one orbit of this group action. Let $aH, bH \in A$. Then, let $g = ba^{-1}$. Then, $g \cdot aH = bH$, so any two arbitrary elements fall into the same orbit.

To prove (2), we note that the stabilizer of 1_H is $\{g \in G \mid g \cdot 1_H = 1_H\} = \{g \in G \mid gH = H\} = H$.

To prove (3), we calculate as follows:

$$\begin{aligned}
\ker \pi_H &= \{g \in G \mid g \cdot xH = xH, \forall x \in G\} \\
&= \{g \in G \mid (x^{-1} g x) H = H, \forall x \in G\} \\
&= \{g \in G \mid x^{-1} g x \in H, \forall x \in G\} \\
&= \{g \in G \mid g \in x H x^{-1}, \forall x \in G\} \\
&= \cap_{x \in G} x H x^{-1}.
\end{aligned}$$

To prove (4), we know already that $\ker \pi_H \trianglelefteq G$ (by first isomorphism theorem) and $\ker \pi_H \trianglelefteq H$. To see the latter, let $x = 1$ and we get that $\ker \pi_H \subset H$. Now, since $\ker \pi_H \trianglelefteq G$, for any $h \in H$, $hkh^{-1} \in \ker \pi_H$ for any $k \in \ker \pi_H$, so $\ker \pi_H \trianglelefteq H$. Now, let $N \trianglelefteq G$ such that $N \subseteq H$. Then, $N = xNx^{-1} \leq xHx^{-1}$ for any $x \in G$. Then, $N \subseteq \cap_{x \in G} xHx^{-1} = \ker \pi_H$. □

Theorem 1.93 (Cayley's Theorem)**.** *Every group is isomorphic to a subgroup of some symmetric group. In particular, if G is a group of order n, then, G is isomorphic to a subgroup of S_n.* ⌟

Proof. Let $H = 1$. Let G act on itself via left multiplication, which can be seen as G acting on $A = \{\text{left cosets of } H \text{ in G}\}$. This gives us a homomorphism $\pi_H : G \to S_G$. Now, by the previous proposition, $\ker \pi_H \trianglelefteq H$, so $\ker \pi_H \subseteq 1_G$. This means, $\ker \pi_H = 1_G$. Therefore, $G \cong \text{im} \pi_H \leq S_G$. □

1.15. **Classification of Finitely Generated Abelian Groups**

We first recall the definition of a direct product of groups:

Definition 1.94 (Direct Product of Groups)**.** *The direct product $G_1 \times G_2 \times \cdots \times G_n$ of the groups $G_1, \cdots, G_n$ is the set*

$$\{(g_1, \cdots, g_n) \mid g_i \in G_i\}$$

with the component-wise operation:

$$(g_1, \cdots, g_n) \cdot (g_1', \cdots, g_n') = (g_1 g_1', \cdots, g_n g_n').$$

Note that one can take the direct product of infinitely many groups as well. ⌟

Example 1.95. Let $G_i = \mathbb{R}$. Then, $\mathbb{R}^\kappa = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$. ⌟

Theorem 1.96. *If $G_1, G_2, \cdots, G_n$ are groups, then*

$$|G_1 \times \cdots \times G_n| = |G_1||G_2| \cdots |G_n|.$$

In particular, the direct product $G_1 \times \cdots \times G_n$ is also a group. ⌟

Proposition 1.97. *Let $G_1, G_2, \cdots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$.*

(1) *For each i,*

$$G_i \cong \{(1, \cdots, 1, g_i, 1, \cdots, 1) \mid g_i \in G_i\}.$$

Identifying the right hand side as G_i, we can say that

$$G_i \trianglelefteq G$$

and

$$G/G_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n.$$

(2) *For each fixed i, define*

$$\pi_i : G \to G_i$$

by

$$\pi_i(g_1, \cdots, g_n) = g_i.$$

Then, π_i is a surjective homomorphism and

$$\ker \pi_i = \{(g_1, \cdots, g_{i-1}, 1, g_{i+1}, \cdots, g_n) \mid g_j \in G_j \forall j \neq i\}.$$

In particular,

$$\ker \pi_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n.$$

(3) *Under the identifications in part (1), for all $x \in G_i$ and for all $y \in G_j$ with $j \neq i$, $xy = yx$. Here we are viewing G_i and G_j as normal subgroups of G.*

⌟

Definition 1.98 (Finitely Generated Groups)**.** *A group G is called finitely generated (f.g.) if there is a finite subset $A \subseteq G$ such that $G = \langle A \rangle$.* ⌟

Definition 1.99 (Free Abelian Group)**.** *For each $r \in \mathbb{Z}$ with $r \geq 0$, we let $\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$ (where we define $\mathbb{Z}^0 = 1$). Then, $\mathbb{Z}^r$ is called the free abelian group of rank r.* ⌟

Theorem 1.100 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let G be a finitely generated abelian group. Then,*

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}$$

where $r, n_1, \cdots, n_s \in \mathbb{Z}$ such that:

1. *$r \geq 0$ and $n_j \geq 2$ for all j*
2. *$n_{i+1} \mid n_i$ for all $1 \leq i \leq s-1$.*

In particular, the expression is unique in the following sense: if $G \cong \mathbb{Z}^q \times Z_{u_1} \times \cdots \times Z_{u_t}$ such that the integers satisfy the conditions above, then $r = q$, $s = t$, and $n_i = u_i$. ⌟

Definition 1.101 (Free Rank, Invariant Factors)**.** *The integer r in Theorem 1.100 is called the free rank of the group G. The integers $n_1, \cdots, n_s$ are called the invariant factors of G.* ⌟

Observation. *Recall that the cardinality of the direct product of a set of groups can be written as $|G_1 \times \cdots \times G_n| = |G_1| \cdot |G_2| \cdots |G_n|$. In light of this, we can see that in Theorem 1.100,*

***G is finite if and only if its free rank is 0, i.e.** $r = 0$.*

.

Observation. *Suppose G is a finite abelian group. Then, we can see that, with the decomposition as in Theorem 1.100,:*

$$|G| = \prod_{i=1}^{s} n_i.$$

Therefore, to classify all finite abelian groups of order n, we must find all sequences $n_1, \cdots, n_s$ of integers such that:

1. *$n_j \geq 2$ for all j*
2. *$n_{i+1} \mid n_i$ for all $i \in \{1, \cdots, s-1\}$*
3. *$n_1 \cdots n_s = n$.*

In particular, we also see that

- *n_1 is the largest invariant factor*
- *each n_i must divide n*
- *if p is a prime factor of n, then it must also divide n_i for some $i \in \{1, \cdots, s\}$*
- *if p divides n_j, then it must also divide all n_i for $i \leq j$*
- ***Every prime divisor of** n **must also divide** n_1.*
- *In particular, if n is a product of **distinct primes**, then $n = n_1$.*

Corollary 1.102. *If G is a finite abelian group of order n such that n is the product of distinct primes, then G is isomorphic to Z_n. In other words, the only abelian group of order n is, up to isomorphism, Z_n.* ⌟

Theorem 1.103 (Elementary Divisor Decomposition of f.g. abelian groups)**.** *Let G be a finitely generated abelian group of order $n > 1$ such that the following is the prime factorization of n:*

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}.$$

Then,

1. *$G \cong A_1 \times A_2 \times \cdots \times A_k$ where $|A_i| = p_i^{\alpha_i}$.*
2. *for each $A \in \{A_1, \cdots, A_k\}$ such that $|A| = p^\alpha$, we can write:*

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_k}}$$

   where $\beta_1 + \cdots + \beta_k = \alpha$ and $\beta_1 \geq \cdots \geq \beta_k \geq 1$.

3. *This decomposition is unique in the sense that if $G \cong B_1 \times \cdots \times B_s$ with $|B_i| = p_i^{\alpha_i}$ for each i, then $B_i \cong A_i$ and $k = s$.*

⌟

2. Rings

2.1. Definitions and Immediate Properties

Definition 2.1 (Rings)**.** *A ring is a set R with two operations, $+$ and $\times$, such that the following are satisfied:*

1. *$(R, +)$ is an abelian group,*
2. *$\times$ is an associative binary operation, i.e. $x \times (y \times z) = (x \times y) \times z$, for all $x, y, z \in R$*
3. *the distributive laws hold in R with respect to the two operations, i.e. $(x + y) \times z = (x \times z) + (y \times z)$ and $x \times (y + z) = (x \times y) + (x \times z)$.*

⌟

Note that a ring is guaranteed to be an abelian group with respect to only the operation of addition, not multiplication. This means that there need not be a multiplicative identity and each $x \in R$ need not have an inverse with respect to multiplication.

Definition 2.2 (Commutative Rings, Unital rings)**.** *A ring such that multiplication is commutative is said to be a commutative ring. A ring with a multiplicative identity, $1_R \in R$ such that $1_R \times x = x \times 1_R = x$ for all $x \in R$, is said to be a unital ring.* ⌟

Observation. *Note that $(R, +)$ is forced to be abelian as a group by the distributive laws. Let R be a unital ring. Consider computing $(1 + 1)(a + b)$ by distributing over the first versus the second term. Then, one can arrive at the conclusion that $a + b = b + a$.*

Definition 2.3 (Division Ring)**.** *A unital ring R with $1_R \neq 0$ such that every $x \in R$ has a multiplicative inverse (i.e. there exists $x^{-1} \in R$ such that $xx^{-1} = x^{-1}x = 1_R$) is called a division ring.* ⌟

Definition 2.4 (Fields)**.** *A commutative division ring is called a field.* ⌟

Example 2.5. The following are examples of rings:

1. The **trivial ring** R is any abelian group $(R, +)$ along with multiplication defined as $x \times y = 0$ for all $x, y \in R$. Note that the trivial ring does not have an identity.
2. The **zero ring** is $R = \{0\}$ with the trivial multiplication. **The zero ring is the only ring where the multiplicative identity is the same as the additive identity**, i.e. $1 = 0$.
3. $\mathbb{Z}$ is a commutative, unital ring with addition and multiplication defined on it as usual.
4. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative, unital rings. In particular, since every $x \in R$ has an inverse, these are fields.
5. $\mathbb{Z}/n\mathbb{Z}$ is a commutative, unital ring. Note that the multiplicative identity is simply $\bar{1}$

⌟

Example 2.6 (Real Hamilton Quaternions). Define the following set:

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}.$$

Define addition in a componentwise manner:

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k.$$

Define multiplication by expanding using the distributive laws and then simplifying using

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

Then, $\mathbb{H}$ is a non-commutative, unital ring where $1 = 1 + 0i + 0j + 0k$. ⌟

Example 2.7. The ring of even integers, $2\mathbb{Z}$, is a commutative ring without identity. ⌟

Proposition 2.8. *Let R be a ring. Then,*

1. *$0r = r0 = 0$, for all $r \in R$*
2. *$(-a)b = -(ab) = a(-b)$ for all $a, b \in R$*
3. *$(-a)(-b) = ab$ for all $a, b \in R$*
4. *if R is a unital ring, then its identity is unique and $-a = (-1) \cdot a$ for all $a \in R$.*

⌟

Definition 2.9 (Zero Divisors)**.** *Let R be a ring. Then, $x \in R$ is a zero divisor if $x \neq 0$ and there exists $y \neq 0$ such that either $xy = 0$ or $yx = 0$.* ⌟

Examples 2.10. *Here are some examples of zero divisors:*

(1) *In $\mathbb{Z}/6\mathbb{Z}$, 3 is a zero divisor, since $3 \cdot 2 = 6 \equiv 0$.*
(2) *In the ring $\{(a, b) \in \mathbb{Z}^2 \mid a \equiv b \bmod 3\}$, $(3, 0)$ is a zero divisor, since $(3, 0) \cdot (0, 3) = (0, 0)$.*

Definition 2.11 (Unit)**.** *Let R be a unital ring with $1 \neq 0$. An element $u \in R$ is called a unit if there exists $v \in R$ such that $uv = vu = 1$. The set of units of the ring R is denoted by $R^\times$.* ⌟

Note that $(R^\times, \times)$ is a group under multiplication. This is called the **group of units of** R. Note that a field is a commutative, unital ring with $1 \neq 0$ where every non-zero element is a unit.

Lemma 2.12. *A zero divisor can not be a unit.* ⌟

Proof. Suppose $x \in R$ is a zero divisor in a unital ring with $1 \neq 0$. Then, $x \neq 0$ and there exists $y \neq 0$ such that $yx = 0$. Now, suppose x is a unit, i.e. there exists $z \in R$ such that $xz = zx = 1$. In particular, if $xz = 1$, then, $y = y \cdot 1 = y \cdot xz = (yx)z = 0 \cdot z = 0$ which is a contradiction. □

Theorem 2.13. *Let $n \geq 2$. Then, in the ring $\mathbb{Z}/n\mathbb{Z}$, the set of units are:*

$$\mathbb{Z}/n\mathbb{Z}^\times = \{u \in \mathbb{Z}/n\mathbb{Z} \mid \quad \text{g.c.d}(u, n) = 1 \quad \}.$$

On the other hand, the elements $v \in \mathbb{Z}/n\mathbb{Z}$ such that $\text{g.c.d}(v, n) > 1$ are zero divisors. As such, every element in $\mathbb{Z}/n\mathbb{Z}$ is either a zero divisor or a unit. ⌟

Proof. To prove the first statement, given $(u, n) = 1$, we know that there exists $x, y \in R$ such that $ux + ny = 1$. Therefore, $ux = 1 - ny$. Now, $\overline{ux} = \overline{1 - ny} = \overline{1}$.

To prove the second statement: let $d = (v, n) > 1$ and let $b = n/d \in (0, n)$, so $\bar{b} \neq \bar{0}$. Now, vb is divisible by n, so $\overline{vb} = 0$. □

Theorem 2.14. *$\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is a prime.* ⌟

Definition 2.15 (Integral Domains)**.** *A commutative, unital ring with $1 \neq 0$ such that it contains no zero divisors is called an integral domain. We will use the term integral domain and domain interchangeably.* ⌟

Proposition 2.16. *Let a, b, c be elements of a ring such that a is not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$. In particular, integral domains have the cancellation property.* ⌟

Corollary 2.17. *Any finite integral domain is a field.* ⌟

Proof. Let R be a finite integral domain. Let $a \neq 0$ be an element of this ring. Now consider the map $\varphi : R \to R$ by $x \to ax$. Since R is finite and this map is injective (because of R being an integral domain), this map is surjective. Therefore, there exists some $b \in R$ such that $ab = 1$. Since a was arbitrarily chosen, R is a field. □

Proposition 2.18 (Wedderburn)**.** *A finite division ring is necessarily commutative. Therefore, a finite division ring is a field.* ⌟

Definition 2.19 (Subring)**.** *A subring of a ring R is a subgroup of R such that it is closed under multiplication.* ⌟

Examples 2.20. *The following are examples of subrings:*

(1) *The set of even integers, $2\mathbb{Z}$, is a subring of $\mathbb{Z}$ which is a subring of $\mathbb{Q}$ which is a subring of $\mathbb{R}$. Any $n\mathbb{Z}$ is a subring of $\mathbb{Z}$ for any $n \in \mathbb{Z}$.*

2.2. **Examples of Rings**

We start with an introduction to polynomial rings. Let R be a commutative, unital ring. Then, a polynomial is a formal sum:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad n \geq 0, a_i \in R.$$

Then, the polynomial ring, $R[x]$, is the set of all such polynomials under the usual definition of addition and multiplication of polynomials.

We have the following vocabulary for describing polynomials:

(1) Leading Term: if $a_n \neq 0$, then $a_n x^n$ is called the leading term.
(2) Leading Coefficient: if $a_n \neq 0$, then a_n is called the leading coefficient.
(3) Degree: the degree of the polynomial is n if its leading coefficient is a_n.
(4) Monic Polynomial: a polynomial such that the leading coefficient is 1.

Observation. *Note that R is a subring of $R[x]$. Also note that $R[x]$ is a commutative, unital ring.*

Proposition 2.21. *Let R be an integral domain (i.e. commutative, unital ring without zero divisors). Let $p(x), q(x) \in R[x]$ such that $p(x) \neq 0$, $q(x) \neq 0$. Then,*

(1) *degree $p(x)q(x)$ = degree $p(x)$ + degree $q(x)$,*
(2) *$R[x]^\times = R^\times$,*
(3) *$R[x]$ is an integral domain.*

⌟

Observation. *If $R[x]$ is an integral domain, then R must be an integral domain, since R is a subring of $R[x]$.*

Proposition 2.22. *Let R be a commutative, unital ring. Then, $p(x) \in R[x]$ is a zero divisor if and only if there exists $c \in R$, $c \neq 0$, such that $cp(x) = 0$.* ⌟

Proof. Suppose $p(x)$ is a zero divisor of degree n:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0.$$

Now, suppose $g(x) \neq 0$ be the polynomial of minimal, non-zero degree such that $g(x)p(x) = 0$. Let

$$g(x) = b_m x^m + \cdots + b_1 x + b_0.$$

After taking the product, we see that $b_m a_n = 0$. As such, we see that $g(x)a_n$ has degree less than m = degree $g(x)$. Furthermore, $g(x)a_n \cdot p(x) = a_n g(x)p(x) = 0$. But this implies $a_n g(x) = 0$.

Since $a_n g(x) = 0$, this also means that $a_n b_{m-1} = 0$, so $a_n x^n b_{m-1} x^{m-1} = 0$. This means that $g(x)p(x) = 0$ implies $a_{n-1}b_m = 0$. This, similarly, implies that $a_{n-1}g(x) = 0$. Continuing this way, we see that $a_i b_m = 0$ for all $i \in \{1, \cdots, n\}$. As such, $b_m p(x) = 0$. □

Proposition 2.23. *Define $R[[x]]$ to be the ring of formal infinite sums in the indeterminate x with coefficients from R:*

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots.$$

Note that

$$\sum_{n=0}^{\infty} a_n x^n \times \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (\sum_{k=0}^{n} a_k b_{n-k}) x^n.$$

Then, $R[[x]]$ is a commutative, unital ring. In particular, $1 - x$ is a unit in $R[[x]]$ with inverse $1 + x + x^2 + \cdots$. Furthermore, $\sum_{n=0}^{\infty} a_n x^n$ is a unit in $R[[x]]$ if and only if a_0 is a unit in R. If R is an integral domain then $R[[x]]$ is also an integral domain. ⌟

Definition 2.24 (Nilpotents)**.** *Let R be a ring. Then, $a \in R$ is called a nilpotent if there exists a positive integer n such that $a^n = 0$.* ⌟

Definition 2.25 (Reduced Rings)**.** *A ring R is called reduced if the only nilpotent element in R is 0 itself.* ⌟

Example 2.26. $\mathbb{Z}/8\mathbb{Z}$ is not a reduced ring as $2^3 = 8$. ⌟

Proposition 2.27. *If R and S are reduced rings, then $R \times S$ is a reduced ring.* ⌟

2.3. **Ring Homomorphisms and Quotient Rings**

Definition 2.28 (Ring homomorphisms, Kernels, and Isomorphisms)**.** *Let R and S be rings. Then, a ring homomorphism from R to S, $\varphi \in \mathrm{Hom}(R, S)$, is a map $\varphi : R \to S$ such that*

$$\varphi(x + y) = \varphi(x) + \varphi(y)$$

and

$$\varphi(xy) = \varphi(x)\varphi(y).$$

Although [DF04] *do not require that ring homomorphisms send 1_R to 1_S, we will adopt that as a part of the definition. We will use the term ring homomorphisms and ring morphisms interchangeably. The kernel of the homomorphism is*

$$\ker \varphi = \{x \in R \mid \varphi(x) = 0\}.$$

If φ is a bijection, we call it an isomorphism. ⌟

Examples 2.29. *The following are examples and non-examples of ring homomorphisms.*

(1) *$\varphi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ which sends evens to 0 and odds to 1.*
(2) *For any $n \in \mathbb{Z}$, define the map*

$$\varphi_n : \mathbb{Z} \to \mathbb{Z}$$

definedby

$$\varphi_n(x) = nx.$$

Then, φ_n is not a ring homomorphism, but is a group homomorphism.
(3) *The following is a ring homomorphism in* $\mathrm{Hom}(\mathbb{Q}[x], \mathbb{Q})$*:*

$$\varphi(p(x)) = p(0).$$

Proposition 2.30. *The kernel and image are subrings. Let $\varphi : R \to S$ be a ring homomorphism. Then,* $\operatorname{im} \varphi \leq S$ *and* $\ker \varphi \leq R$. ⌟

Proposition 2.31. *For any ring R, there exists a unique morphism $R \to \{0\}$. For any ring R, there exists a unique morphism $\mathbb{Z} \to R$. If ring morphisms are required to send 1 to 1, then there is only one ring homomorphism $\{0\} \to X$, and it is the morphism to itself (i.e. $X = \{0\}$).* ⌟

Note that the last statement here comes from the fact that in all other rings $1 \neq 0$.

2.4. **Ideals, Quotient Rings, and Isomorphism Theorems**

Definition 2.32 (Ideals)**.** *Let R be a ring and let $I \subseteq R$ be a subset.*

(1) *I is a left ideal if (i) $I \subseteq R$ is a subring, i.e. $I \leq R$ and (ii) $xI \subseteq I$ for all $x \in R$.*
(2) *I is a right ideal if (i) $I \subseteq R$ is a subring, i.e. $I \leq R$, and (ii) $Ix \subseteq I$ for all $x \in R$.*
(3) *I is an ideal if I is both a left ideal and right ideal. We denote this by $I \trianglelefteq R$.*

⌟

Proposition 2.33. *Let R be a ring and let $I \trianglelefteq R$. Then, the quotient group R/I is a ring under the operations;*

$$(x + I) + (y + I) = (x + y) + I$$

and

$$(x + I) \times (y + I) = (xy) + I.$$

Conversely, for any subgroup I of R such that these operations are well-defined, I is an ideal of R. ⌟

Definition 2.34 (Quotient Ring)**.** *Let R be a ring and let $I \trianglelefteq R$ be an ideal of R. We call R/I a quotient ring.* ⌟

Note that $\overline{x} = \overline{y}$ in R/I if and only if $x + I = y + I$ if and only if $x - y \in I$. This follows from the results we showed for quotient groups.

Example 2.35. In any ring R, the subrings 0 and R are ideals. We call 0 the trivial ideal. ⌟

Proposition 2.36. *Let $\phi : R \to S$ be a ring morphism.*

(1) *If $J \trianglelefteq S$, then $\phi^{-1}(J) \trianglelefteq R$*
(2) *if $I \trianglelefteq R$ and ϕ is surjective, then $\phi(I) \trianglelefteq S$*
(3) *In particular, $\phi^{-1}(0) =: \ker \phi \trianglelefteq R$.*

⌟

It is worth noting that, in (2), if ϕ is not surjective, then $\phi(I)$ need not be an ideal in S. For example, consider the ring morphism $\mathbb{Z} \to \mathbb{Z}[i]$, which maps $(2) \to 2\mathbb{Z}$. But $2\mathbb{Z}$ is not an ideal in $\mathbb{Z}[i]$ (since $i \cdot 2 \notin 2\mathbb{Z}$.

Proposition 2.37 (Universal Property of Quotient Rings)**.** *If $I \trianglelefteq R$ and $\phi : R \to S$ is a ring morphism with $\phi(I) = \{0\}$. Then, there exists a unique morphism $\bar{\phi} : R/I \to S$ such that $\bar{\phi} \circ \pi = \phi$ (where $\pi : R \to R/I$ is the natural morphism):*

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ & \searrow^{\phi} & \downarrow{\scriptstyle \exists! \bar{\phi}} \\ & & S \end{array}$$

In particular, for any ring morphism $\phi : R \to S$, there exists a unique morphism $\bar{\phi} : R/\ker \phi \to S$ such that $\bar{\phi} \circ \pi = \phi$. ⌟

Before we move on to the ring isomorphism theorems, let us look at what ideals inside a quotient ring really look like. In R/I, for each $j \in J$, the element $\overline{j} := j + I$. Let $\overline{J} := \{\overline{j} \mid j \in J\}$. Then the ideal generated by $\overline{J}$ is

$$(\overline{J}) = \left\{ \sum_{k=1}^{n} \overline{r_k} \cdot \overline{j_k} \;\middle|\; \overline{r_k} \in R/I,\ j_k \in J \right\} = \left\{ \sum_{k=1}^{n} r_k j_k + I \;\middle|\; r_k \in R,\ j_k \in J \right\}$$

Since J is an ideal, $\sum_{k=1}^{n} r_k j_k \in J$, so

$$(\overline{J}) = \{j + I \mid j \in J\}.$$

To see that this is the same as saying

$$(\overline{J}) = (J + I)/I = (J, I)/I,$$

first observe that by definition:

$$(J+I)/I := \{a+I \mid a \in J+I\}.$$

But every element in $J+I$ is of the form $j+i$ for some $j \in J$ and $i \in I$. Then, each element in $(J+I)/I$ is of the form

$$j+i+I = j+I.$$

So,

$$(J+I)/I = \{j+I \mid j \in J\} = (\overline{J}).$$

As an example, let's consider the polynomial ring. In $R[x]/(f(x))$, the element $\overline{g(x)} := g(x)+(f(x))$. Then the ideal generated by $\overline{g(x)}$ is $(\overline{g(x)}) = \{\overline{h(x)} \cdot \overline{g(x)} \mid \overline{h(x)} \in R[x]/(f(x))\} = \{h(x)g(x)+(f(x)) \mid h(x) \in R[x]\}$. To see that this is the same as saying $(\overline{g(x)}) = (g(x), f(x))/(f(x))$, first observe that by definition:

$$(g(x), f(x))/(f(x)) := \{a(x)+(f(x)) \mid a(x) \in (g(x), f(x))\}.$$

But every element in $(g(x), f(x))$ is of the form $h(x)g(x)+k(x)f(x)$. Then, each element in $(g(x), f(x))/(f(x))$ is of the form

$$h(x)g(x)+k(x)f(x)+(f(x)) = h(x)g(x)+(f(x)).$$

So,

$$(g(x), f(x))/(f(x)) := \{h(x)g(x)+(f(x)) \mid h(x) \in R[x]\} = (\overline{g(x)}).$$

Note that the quotient ideal on the left-hand side makes sense since $f(x) \subseteq (g(x), f(x))$.

Proposition 2.38 (Ring Isomorphism Theorems)**.** *Let R be a ring.*

(1) *(1st Isomorphism Theorem) Let $\varphi \in \mathrm{Hom}(R, S)$. Then, $\ker\varphi \trianglelefteq R$, $\mathrm{im}\,\varphi \leq S$ (note: image is not necessarily an ideal, it is necessarily a subring), and $R/\ker\varphi \cong \mathrm{im}\,\varphi$. Furthermore, if $I \trianglelefteq R$ is an ideal of R, then the map*

$$R \to R/I$$

defined by

$$x \to x+I$$

is a surjective ring homomorphism with I its kernel. As such, every ideal is the kernel of some ring homomorphism and vice-versa.

(2) *(2nd Isomorphism Theorem) Let $A \leq R$ and $B \trianglelefteq R$. Then, $A+B \leq R$, $A \cap B \trianglelefteq R$, and $(A+B)/B \cong A/(A \cap B)$.*

(3) *(3rd Isomorphism Theorem) Let $I \trianglelefteq R$ and let $J \trianglelefteq R$ such that $I \subseteq J$. Then, $J/I \trianglelefteq R/I$ and $(R/I)/(J/I) \cong R/J$.*

(4) *(4th Isomorphism Theorem) Let $I \trianglelefteq R$. The correspondence*

$$\{A \leq R \text{ s.t. } I \subseteq A\} \leftrightarrow \{A/I \leq R/I\}$$

is an inclusion-preserving bijection. In particular, $A \trianglelefteq R$ with $I \subseteq A$ if and only if $A/I \trianglelefteq R/I$.

⌟

Example 2.39. For any $n \in \mathbb{Z}$, $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, and they are the only ideals of $\mathbb{Z}$.

⌟

Example 2.40. Let $R = \mathbb{Z}[x]$. Let I be the collection of polynomials whose individual terms are of degree greater than or equal to 2. Then, $I \trianglelefteq R$. Even though $R = \mathbb{Z}[x]$ has no zero divisors, R/I has:

$$\bar{x} \cdot \bar{x} = \bar{x^2} = 0.$$

⌟

Example 2.41. For any m and n,

$$m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}, \text{ where } d = \gcd(m, n).$$

⌟

Definition 2.42 (Proper Ideal)**.** *An ideal $I \trianglelefteq R$ is called a proper ideal if $I \neq R$.* ⌟

2.5. **Properties of Ideals**

Definition 2.43 (Ideal Generated by a Subset, Principal Ideal, Finitely Generated Ideal)**.** *Let R be a unital ring with $1 \neq 0$. Let $A \subseteq R$ be a subset.*

(1) *We say (A) is the ideal generated by A, which is defined to be the smallest ideal of R containing A:*
$$(A) = \cap_{I \trianglelefteq R, A \subseteq I} I.$$
In particular,
$$(A) = RAR = \{r_1 a_1 r_1' + \cdots + r_n a_n r_n' \mid r_i, r_i' \in R, a_i \in A, n \in \mathbb{Z}_{>0}\}.$$
If R is also commutative, then
$$(A) = RAR = RA = AR.$$
(2) *An ideal generated by a single element is called a principal ideal. We will use the notation $(a) = (A)$ when $A = \{a\}$.*
(3) *An ideal generated by a finite set is called a finitely generated ideal. We will use the notation $(a_1, \cdots, a_n) := (A)$ where $A = \{a_1, \cdots, a_n\}$.*

⌟

Proposition 2.44. *let $I \trianglelefteq R$ be an ideal and let R be a unital ring with $1 \neq 0$.*

(1) *$I = R$ is and only if I contains a unit.*
(2) *If R is commutative, then R is a field if and only if its only ideals are 0 and R.*
(3) *$R = \{0\}$ if and only if it has only one ideal.*

⌟

Corollary 2.45. *If R is a field, then any non-zero ring homomorphism from R to another ring is an injection.* ⌟

Examples 2.46. *The following are examples of various special ideas.*

(1) *The trivial ideal is a principal ideal: $0 = (0)$.*
(2) *The ideal R is a principal ideal: $R = (1)$.*
(3) *In $\mathbb{Z}$, the ideal $n\mathbb{Z} = \mathbb{Z}n = (n) = (-n)$ for any $n \in \mathbb{Z}$. Therefore, **every ideal of $\mathbb{Z}$ is a principal ideal**. Furthermore, $n\mathbb{Z} \subset m\mathbb{Z}$ if and only if m divides n.*
(4) *In the ring $\mathbb{Z}[x]$, the ideal $(2, x)$ is not a principal ideal.*

Proposition 2.47. *The ideals of $R \times S$ are of the form $I \times J$ where $I \trianglelefteq R$ and $J \trianglelefteq S$.* ⌟

Proof. For any ideal $K \trianglelefteq R \times S$, suppose $(a, b) \in K$. Now,
$$(1, 0) \cdot (a, b) = (a, 0) \in K$$
and
$$(0, 1) \cdot (a, b) = (0, b) \in K$$
since K is an ideal. Now, define $I = \{a \in R \mid (a, 0) \in K\}$ and $J = \{b \in S \mid (0, b) \in K\}$. The properties of ideals forces I to be an ideal in R and J to be an ideal in S. On the other hand, $K = I \times J$. □

2.6. **Maximal Ideals**

Definition 2.48 (Maximal Ideals)**.** *An ideal $M \trianglelefteq S$ is called a maximal ideal if $M \neq S$ and the only ideals containing M are M itself and S.* ⌟

Proposition 2.49. *In a unital ring where $1 \neq 0$, every proper ideal is contained in a maximal ideal.* ⌟

Proof. The proof uses Zorn's lemma. Let I be a proper ideal and define

$$\mathcal{S} = \{J \trianglelefteq R \mid I \subseteq J, J \neq R\}.$$

Now,

$$\mathcal{S} \neq \emptyset$$

since $I \in \mathcal{S}$. Also $\mathcal{S}$ is partially ordered by inclusion. Now, define $\mathcal{C}$ to be a chain inside $\mathcal{S}$ and define $J = \cup_{A \in \mathcal{C}} A$. Then, it is easy to check that J is itself an ideal. Furthermore, J is a proper ideal: if J weren't a proper ideal, then $1 \in J$ which would imply that $1 \in A$ for all $A \in \mathcal{C}$, which is a contradiction since $\mathcal{C}$ is in $\mathcal{S}$, which consists of only proper ideals. As such, every chain $\mathcal{C}$ has an upper bound which belongs in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ has a maximal element which is our maximal ideal. □

Proposition 2.50. *Let R be a commutative, unital ring with $1 \neq 0$. Then, the ideal M is a maximal ideal if and only if R/M is a field.* ⌟

Proof. $M \trianglelefteq R$ is a maximal ideal if and only if there are no ideals I such that $M \subset I \subset R$. By the correspondence theorem, the ideals of R containing M correspond to the ideals of R/M. Therefore, $M \trianglelefteq R$ is a maximal ideal if and only if the only ideals of R/M are 0 and R/M, which is true if and only if R/M is a field. □

Examples 2.51. *The following are examples (and non-examples) of maximal ideals.*

(1) *Let $n \in \mathbb{Z}_{\geq 0}$. Then, $n\mathbb{Z} \trianglelefteq \mathbb{Z}$ is a maximal ideal if and only if n is a prime.*
(2) *$(2, x)$ is a maximal ideal in $\mathbb{Z}[x]$ as $\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z}$.*

    Proof. Inside $\mathbb{Z}[x]/(2, x)$, 2 is 0 since $2 + (2, x) \in (2, x)$. Furthermore, x is 0 since $x + (2, x) \in (2, x)$. Therefore, the only elements in $\mathbb{Z}[x]/(2, x)$ are 0 and 1. □

(3) *(x) is not a maximal ideal in $\mathbb{Z}[x]$ as $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, which is not a field.*

2.7. **Prime Ideals**

Definition 2.52 (Prime Ideals)**.** *Let R be a commutative ring. Then, the ideal $P \trianglelefteq R$ is called a prime ideal if $P \neq R$ and $xy \in P$ implies that either $x \in P$ or $y \in P$.* ⌟

Observation. *Consider the ring $\mathbb{Z}$. Then, for $n \neq 1$, the ideal $n\mathbb{Z}$ is a prime ideal if n is a prime: $n\mathbb{Z}$ is prime is $xy \in n\mathbb{Z}$ implies either $x \in n\mathbb{Z}$ or $y \in n\mathbb{Z}$. So, $n|xy$ implies that $n|x$ or $n|y$. This condition is equivalent to the condition that n is a prime. Therefore,* ***the prime ideals in $\mathbb{Z}$ are the trivial ideal and $p\mathbb{Z}$ for primes p.***

Proposition 2.53. *Suppose R is a commutative ring. Then, the ideal P is a prime ideal in R if and only if R/P is an integral domain.* ⌟

Proof. P is a prime ideal if and only if $xy \in P$ implies $x \in P$ or $y \in P$ and $R \neq P$. This is equivalent to the condition that, in R/P, $\bar{x}\bar{y} = 0$ implies $\bar{x} = 0$ or $\bar{y} = 0$ and $\bar{R} \neq 0$. As such, there are no zero divisors, which, by defintion, means R/P is an integral domain. □

Corollary 2.54. *Suppose R is a commutative ring. Then, any maximal ideal of R is a prime ideal.* ⌟

Examples 2.55. *The following are examples of prime ideals.*

1. *The principal ideals generated by primes in $\mathbb{Z}$ are both prime and maximal ideals. The zero ideal in $\mathbb{Z}$ is prime but not maximal.*
2. *$(x) \trianglelefteq_{\text{prime}} \mathbb{Z}[x]$ as $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. However, (x) is not a maximal ideal.*
3. *$(0) \trianglelefteq_{\text{prime}} \mathbb{Z}[x]$ but (0) is not a maximal ideal.*

Definition 2.56 (Spectrum of a ring)**.** *Define the spectrum of a ring to be the set of all prime ideals of the ring, i.e.*

$$\operatorname{Spec}(R) = \{P \mid P \trianglelefteq_{\text{prime}} R\}.$$

⌟

Theorem 2.57. *Suppose $\phi : R \to S$ is a ring homomorphism. If $J \trianglelefteq S$ is a prime ideal, then $\phi^{-1}(J)$ is a prime ideal in R. In other words,*

$$\phi^{-1} : \operatorname{Spec}(S) \to \operatorname{Spec}(R).$$

If ϕ is an isomorphism, then there is a bijection between the spectrums of R and S are isomorphic. ⌟

Proof. Suppose J is a prime ideal of S. Then, for any $s_1 s_2 \in J$, either $s_1 \in J$ or $s_2 \in J$. Now, consider $r_1 \cdot r_2 \in \phi^{-1}(J)$. Then, if neither r_1 nor r_2 are in $\phi^{-1}(J)$, then we get $\phi(r_1 \cdot r_2) = \phi(r_1)\phi(r_2) \in J$ where neither $\phi(r_1)$ nor $\phi(r_2)$ is in J, which would be a contradiction. So either r_1 or r_2 is in $\phi^{-1}(J)$. □

Observation. *This statement would not hold true for maximal ideals. Consider the homomorphism $\mathbb{Z} \to \mathbb{Q}$. Since $\mathbb{Q}$ is a field, 0 is a maximal ideal, but its preimage is $(0) \subsetneq (2) \subsetneq \mathbb{Z}$.*

Theorem 2.58. *The prime ideals of $R \times S$ are the ideals $I \times S$ for $I \trianglelefteq_{\text{prime}} R$ and $R \times J$ for $J \trianglelefteq_{\text{prime}} S$.* ⌟

Proof. Let P be a prime ideal of $R \times S$. Now, $(1,0) \cdot (0,1) = (0,0) \in P$ so, either $(1,0)$ or $(0,1)$ is in P.
Case 1: $(1,0) \in P$. Then, $(r,0) = (r,0) \cdot (1,0) \in P$, so

$$R \times \{0\} \subseteq P.$$

Now, let $J = \{s \in S \mid (0,s) \in P\}$, which is an ideal of S. Then, $P = R \times J$ since for any $(r,s) \in P$, we have that $(0,s) = (r,s) - (r,0) \in P$, so $s \in J$, which implies $(r,s) \in R \times J$.
Case 2: $(0,1) \in P$. This case, similarly, implies that $P = I \times S$ where $I = \{r \in R \mid (0,r) \in P\}$. □

Theorem 2.59. *There exists the following bijection:*

$$\{\textit{prime ideals of } R/I\} \leftrightarrow \{ \textit{ prime ideals of } R \textit{ containing } I\}.$$

⌟

Proof. Define $\pi : R \to R/I$ by $r \to r + I$. Now, if $\overline{P} \trianglelefteq_{\text{prime}} R/I$, then $\pi^{-1}(\overline{P})$ is a prime ideal in R. Furthermore, $\ker \pi = I \in \pi^{-1}(\overline{J})$, since it maps to 0 which is in $\overline{P}$. So, we have mapped a prime ideal $\overline{P}$ back to a prime ideal in R containing I. Conversely, suppose P is prime in R and $I \subseteq P$. Then, $\pi(P) = P/I$. We claim that P/I is prime: suppose $(a + I) \cdot (b + I) \in P/I \implies (ab + I) \in P/I$. Now, since $ab \in P$, either $a \in I$ or $b \in P$, so either $a + I \in P/I$ or $b + I \in P/I$. □

2.8. **Localisation and Ring of Fractions**

In this section, we will show that a commutative ring R is a subring of a larger ring Q in which every element can become a unit.

Definition 2.60 (Multiplicative Subset)**.** *Let R be a unital ring. Then, D is a multiplicative subset of the ring R if:*

(1) $1 \in D$, *and*
(2) D *is closed under multiplication.*

⌟

Definition 2.61 (Ring of Fractions)**.** *Let R be a unital ring and let D be a multiplicative subset. Then, define*

$$D^{-1}R := \left\{ \frac{r}{d} \mid r \in R, d \in D \right\},$$

and define the equivalence relation:

$$\frac{r_1}{d_1} = \frac{r_2}{d_2} \text{ if and only if } \exists f \in D, \text{ s.t. } f(r_1 d_2 - r_2 d_1) = 0.$$

Then, $D^{-1}R$ is called the ring of fractions or the location of R by D. ⌟

Proposition 2.62. *$0 \in D$ if and only if $D^{-1}R \cong 0$, i.e. the localisation is the zero ring.* ⌟

Proposition 2.63 (Universal property of localizations)**.** *Let $\phi : R \to S$ be a ring homomorphism. Let $D \subset R$ be a multiplicative subset, and suppose $\phi(D) \subset S^\times$. Then, there exists a unique $\tilde{\phi} : D^{-1}R \to S$ such that $\phi = \tilde{\phi} \circ \iota$ (where $\iota : R \to D^{-1}R$ is the natural ring morphism):*

$$\begin{array}{ccc} R & \xrightarrow{\iota} & D^{-1}R \\ & \searrow^{\phi} & \downarrow{\scriptstyle \exists!\, \tilde{\phi}} \\ & & S. \end{array}$$

⌟

As an "application", we will prove the following using the universal property:

Proposition 2.64. *If $D \subset R^\times$, then $D^{-1}R = R$.* ⌟

Proof. Define S to be R in the universal property. Then, $\tilde{\phi} \circ \iota = \mathrm{id}$, for some unique $\tilde{\phi}$, by the commutativity of the diagram. We can check that if we define $\tilde{\phi}(\frac{a}{d}) = ad^{-1}$, we satisfy the diagram. Also, notice that $\tilde{\phi}$ is surjective, since for any $r \in R$, $\tilde{\phi}(\frac{r}{1}) = r$. Now, on the other hand, if we set S to be $D^{-1}R$ in the universal property, then we have that:

$$\begin{array}{ccc} R & \xrightarrow{\iota} & D^{-1}R \\ & \searrow^{\iota} & \downarrow{\scriptstyle \exists!\, \Psi} \\ & & D^{-1}R. \end{array}$$

In particular, since $\tilde{\phi} \circ \iota = \mathrm{id}_R$, we can set $\Psi = \iota \circ \tilde{\phi}$, because then $\iota \circ \tilde{\phi} \circ \iota = \iota \circ \mathrm{id}_R = \iota$. By uniqueness of the universal property, $\iota \circ \tilde{\phi} = \mathrm{id}_{D^{-1}R}$. This shows that $\tilde{\phi}$ is an isomorphism between $D^{-1}R$ and R. □

Definition 2.65 (Total Quotient Ring)**.** *Let R be a ring and let $D := R - \{$ zero divisors and $0\}$. Then, D is a multiplicative subset of R and $Q(R) = D^{-1}R$ is called the total quotient ring of R.* ⌟

Proposition 2.66. *We have the canonical ring morphism $R \to D^{-1}R$ which sends $r \to \frac{r}{1}$. Furthermore, this homomorphism is injective if and only if D contains no zero divisors.* ⌟

Definition 2.67 (Field of Fractions)**.** *Let R be an integral domain. Then $Q(R)$ is a field, and is often denoted by $K(R)$.* ⌟

Example 2.68. $Q(\mathbb{Z}) = \mathbb{Q}$. ⌟

Example 2.69. In $\mathbb{Z} \times \mathbb{Z}$, the zero divisors are $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ such that $a \neq 0, b \neq 0$. Then,

$$Q(\mathbb{Z} \times \mathbb{Z}) = \{\frac{(a_1, a_2)}{(b_1, b_2)} \mid b_1 \neq 0, b_2 \neq 0\}.$$

In particular, $Q(\mathbb{Z} \times \mathbb{Z}) \cong \mathbb{Q} \times \mathbb{Q}$. ⌟

Example 2.70. Let $R = \mathbb{C}[x]/(x^2) = \{a + bx + (x^2) \mid a, b \in \mathbb{C}\}$. The zero divisors in R are $\{bx \mid b \in \mathbb{C}\}$. The units are $\{a + bx \mid a \neq 0\}$. Define $D = \{a + bx + (x^2) \mid a \neq 0\}$ to be the units in R. Then, $Q(R) = R$, since every element of D is already a unit in R. ⌟

Theorem 2.71. *The ring R is an integral domain if and only if R embeds as a subring of a field. Note that this result does not hold if R is not commutative.* ⌟

Next, we look at localizations at an element.

Definition 2.72. *Let $f \in R$. Define $D = \{1, f, f^2, f^3, \cdots\}$. Then, $R_f = D^{-1}R$.* ⌟

Theorem 2.73. $R_f \cong R[x]/(fx - 1) \cong R[\frac{1}{f}]$. ⌟

Example 2.74. $\mathbb{Z}_n = \{\frac{a}{n^k} \mid k \in \mathbb{N}\}$. ⌟

Proposition 2.75. *Let R be a commutative ring with $1 \neq 0$. Then,*

1. *$R[x]/(x - a) \cong R$ for any $a \in R$.*
2. *$R[x]/(fx-1) \cong R_f \cong R[\frac{1}{f}]$, where $f \in R$ is fixed. If f is a unit, then, $R[x]/(fx-1) \cong R$.*
3. *Let A be any commutative, unital ring and let $a \in A$ be fixed. Then, $A[y]/(y - a) \cong A$.*
4. *$R[x, y]/(y - g(x)) \cong R[x]$.*
5. *$R[x_1, \cdots, x_n]/(x_1 - a_1, \cdots, x_n - a_n) \cong R$ for fixed $a_1, \cdots, a_n \in R$.*
6. *If a non-zero integer D is not a perfect square, then $\mathbb{Z}[\sqrt{D}] \cong \mathbb{Z}[x]/(x^2 - D)$.*

⌟

Note: The proof for (2) is easier by directly constructing mutually inverse maps.

Now, we look at localizations in product rings.

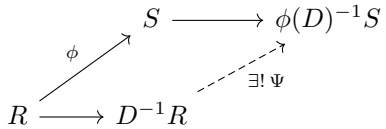
Theorem 2.76. *Let R and S be rings with $D \subset R$ and $E \subset S$ multiplicative subsets. Then, $D \times E \subset R \times S$ is a multiplicative subset, and*

$$(D \times E)^{-1}(R \times S) \cong D^{-1}R \times E^{-1}S,$$

by

$$\frac{(r, s)}{(d, e)} \to (\frac{r}{d}, \frac{s}{e}).$$

Furthermore, if $\phi : R \to S$ is a ring morphism and $D \subset R$ is a multiplicative subset, then $\phi(D) \subset S$ is a multiplicative subset and $\phi(D)^{-1}S$ is a ring with a unique morphism that makes the following diagram commute:



where $\Psi(\frac{a}{d}) = \frac{\phi(a)}{\phi(d)}$. ⌟

Now, some simple results in localizations of polynomial rings:

Theorem 2.77. $D^{-1}(R[x]) \cong D^{-1}R[x]$. ⌟

Theorem 2.78. $\mathbb{Z}[[x]]_2 \subset \mathbb{Z}_2[[x]]$ *but the converse is not true.* ⌟

Proof. The former is simple: map $\frac{a_0+a_1x+\cdots+a_nx^n}{2}$ to $\mathbb{Z}_2[[x]]$ by taking each a_i modulo 2. To show the latter, we can use the counterexample $1 + \frac{1}{2}x + \frac{1}{4}x^2 + \frac{1}{8}x^3 + \cdots$, which cannot be embedded. □

2.9. **Localization and Prime ideals**

Definition 2.79 (Localization at prime ideals)**.** *Let $P \trianglelefteq R$. Then, $D = R - P$ is a multiplicative subset. Then, $R_P := D^{-1}R$.* ⌟

Examples 2.80. *The following are examples of localizations at prime ideals:*

(1) $\mathbb{Z}_{(p)} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, p \text{ does not divide } b\} \subseteq \mathbb{Q}$.
(2) $\mathbb{Z}_{(0)} = Q(\mathbb{Z}) = \mathbb{Q}$.
(3) $\mathbb{C}[x]_{(x)} = \{\frac{f(x)}{g(x)} \mid g(0) \neq 0\}$.

Theorem 2.81. *Let R and S be rings. Recall that the prime ideals of $R \times S$ are of the form $I \times S$ and $R \times J$, where I is a prime ideal in R and J is a prime ideal in S. Let P be a prime ideal of R. Then, $(R \times S)_{P \times S}$ is the localization of $R \times S$ at $\{(a, b) \in R \times S \mid a \notin P\}$. This has the following property:*

$$R_p \cong (R - P)^{-1}R \times S^{-1}S = (R \times S)_{P \times S}.$$

⌟

Proposition 2.82. *Let D be a multiplicative subset and let $I \trianglelefteq R$. Then,*

$$D^{-1}I := \{\frac{i}{d} \in D^{-1}R \mid i \in I\} \trianglelefteq D^{-1}R.$$

Furthermore,

$$D^{-1}I \neq D^{-1}R \iff I \cap D = \emptyset.$$

In other words, an ideal $I \trianglelefteq R$ maps to a proper ideal in $D^{-1}R$ if and only if $I \cap D = \varnothing$. ⌟

Proof. $D^{-1}I$ is an ideal because I is an ideal in R. Now, we have two cases: $I \cap D \neq \emptyset$. In this case, if $s \in I \cap D$, then, $\frac{s}{1} \in S^{-1}I$. But $\frac{s}{1}$ is invertible since $\frac{s}{1} \cdot \frac{1}{s} = 1 \in S^{-1}I$, so $D^{-1}I = D^{-1}R$. The other case is $I \cap D = \emptyset$. Then, we claim $D^{-1}I$ is proper. This is because if $1 = \frac{1}{1} \in D^{-1}I$, then $\frac{1}{1} = \frac{i}{d}$ for some $i \in I, d \in D$. But then, there exists some $d' \in D$ such that $d'(d - i) = 0$. This means $d'd = d'i \in I$. But $dd' \in D$ which contradicts that $I \cap D = \varnothing$. □

Definition 2.83 (Extensions, Contractions)**.** *Consider the canonical map*

$$\iota : R \to D^{-1}R, \quad \iota(r) = \frac{r}{1}.$$

Let $I \trianglelefteq R$. Then, we define the extension of I to be

$$I^e = D^{-1}I := \{\frac{i}{d} \mid i \in I, d \in D\} \trianglelefteq D^{-1}R.$$

On the other hand, if $J \trianglelefteq D^{-1}R$, then its contraction is

$$J^c := \iota^{-1}J = \{r \in R \mid \frac{r}{1} \in J\} \trianglelefteq R,$$

which is an ideal in R since it is the preimage under a homomorphism of an ideal in $D^{-1}R$. ⌟

Definition 2.84 (Saturated Ideals)**.** *We call $I \trianglelefteq R$ saturated with respect to D (or D-saturated) if for all $rd \in I$ with $d \in D$, we have that $r \in I$.* ⌟

Proposition 2.85. *Let P be a prime ideal such that $P \cap D = \varnothing$. Then,*

$$(D^{-1}P)^C := \iota^{-1}(D^{-1}P) = P.$$

⌟

Proof. We prove both inclusions. First, suppose $r \in P$. Then, $\iota(r) = \frac{r}{1} \in D^{-1}P$, so $r \in \iota^{-1}(D^{-1}P)$. Therefore, $P \subseteq \iota^{-1}(D^{-1}P)$. Conversely, suppose $r \in \iota^{-1}(D^{-1}P)$. Then $\iota(r) = \frac{r}{1} \in D^{-1}P$, so there exist $a \in P$ and $d \in D$ such that $\frac{r}{1} = \frac{a}{d}$. So, there exists $t \in D$ such that $t(rd - a) = 0$. Subsequently, $(td)r = ta \in P$, since $a \in P$ and P is an ideal. Now, $td \in D$ because D is multiplicatively closed. Since $P \cap D = \varnothing$, we have $td \notin P$. But P is prime and $(td)r \in P$. Therefore, $r \in P$. Hence, $\iota^{-1}(D^{-1}P) \subseteq P$. Combining both inclusions, we conclude. □

Proposition 2.86. *Let D be a multiplicative subset of ring R. Let $\pi : R \to D^{-1}R$ be the natural morphism:*

$$\pi(r) = \frac{r}{1}.$$

Then,

(1) *If $J \trianglelefteq D^{-1}R$, then*

$$\pi^{-1}(J) \trianglelefteq R$$

is saturated with respect to D, and

$$D^{-1}\pi^{-1}(J) = J, \quad \text{meaning } (J^c)^e = J.$$

(2) *If $I \trianglelefteq R$, then $\pi^{-1}(D^{-1}I) = \{a \in R \mid ad \in I \text{ for some } d \in D\}$ is the smallest ideal saturated with respect to D that contains I. Furthermore,*

$$D^{-1}\pi^{-1}(D^{-1}I) = D^{-1}I, \quad \text{meaning } ((I^e)^c)^e = I^e$$

(3) *There exists the following bijection:*

$$\{\text{ideals } J \trianglelefteq D^{-1}R\} \leftrightarrow \{D\text{-saturated ideals of } R\}$$

via

$$J \to \pi^{-1}(J)$$

and

$$D^{-1}I \leftarrow I.$$

(4) *If $\bar{D}$ is the image of D in R/I, then*

$$\bar{D}^{-1}(R/I) \cong D^{-1}R/D^{-1}I.$$

⌟

Proof. All proof rely on the fact that π maps D to units in $D^{-1}R$.

We start with (1). Suppose $J \trianglelefteq D^{-1}R$. Then, $\pi^{-1}J \trianglelefteq R$ since it is the preimage of an ideal under homomorphism. Now, supose $rd \in \pi^{-1}J$, with $d \in D$. Then, $\pi(rd) = \frac{rd}{1} \in J$. So, $\frac{1}{d}\frac{rd}{1} = \frac{r}{1} \in J$, meaning $r \in \pi^{-1}(J)$. Therefore, $\pi^{-1}J$ is D-saturated. Next, we show that $D^{-1}\pi^{-1}(J) = J$. For any $\frac{j}{d} \in J$, we have that $\frac{j}{1} \in J$ and so $j \in \pi^{-1}J$, which means $\frac{j}{1} \in D^{-1}\pi^{-1}(J)$. Since this is an ideal $\frac{1}{d}\frac{j}{1} = \frac{j}{d} \in D^{-1}\pi^{-1}(J)$. Conversely, suppose $\frac{j}{d} \in D^{-1}\pi^{-1}(J)$. Then, $d \in D$ and $j \in \pi^{-1}(J) \implies \frac{j}{1} \in J$. This means, $\frac{j}{d} = \frac{1}{d}\frac{j}{1} \in J$. With these two inclusions, we conclude.

Now, we prove (2). We construct $\pi^{-1}D^{-1}I$. If $r \in \pi^{-1}D^{-1}I$, then $\frac{r}{1} \in D^{-1}I$. This means, $\frac{r}{1} = \frac{i}{d}$ for some $i \in I, d \in D$. As such, there exists some $d' \in D$ such that $d'(rd - i) = 0 \implies d'dr = d'i \in I$. Conversely, suppose $r \in R$ such that $rd \in I$ for some $d \in D$. Then, $\frac{rd}{1} \in D^{-1}I$, so $\frac{r}{1} \in D^{-1}I$ implying that $r \in \pi^{-1}(D^{-1}I)$. Therefore,

$$\pi^{-1}(D^{-1}I) = \{r \in R \mid rd \in I \text{ for some } d \in D\}.$$

This is an ideal that is D-saturated: if $rd \in \pi^{-1}(D^{-1}I), d \in D$, then $\frac{rd}{1} \in D^{-1}I \implies \frac{r}{1} \in D^{-1}I \implies r \in \pi^{-1}(D^{-1}I)$. Note that $I \subseteq \pi^{-1}D^{-1}I$ clearly. To show that this is the smallest D-saturated ideal containing I, suppose J is another ideal in R that is D-saturated and contains I. Take $r \in \pi^{-1}D^{-1}I \implies rd \in I$ for some $d \in D$. Then, as $I \subseteq J$, we have that $rd \in J$ for some $d \in D$. Since J is D-saturated, this means $r \in J$. So, $\pi^{-1}D^{-1}I \subseteq J$, so we are done.

Next, we how that $D^{-1}\pi^{-1}D^{-1}I = D^{-1}I$.However, this follows from (1) since $D^{-1}I \trianglelefteq D^{-1}R$.

Next, we prove the bijection. Given J is an ideal of $D^{-1}R$, we can map it to a D-saturated ideal of R via π^{-1} as shown in (1). Given $I \trianglelefteq R$, we can map it to $D^{-1}I$ which is an ideal. We have already seen that $D^{-1}\pi^{-1}(J) = J$. To see the other side of the inverse, suppose $I \trianglelefteq R$ is D-saturated. We know that $\pi^{-1}D^{-1}I$ is the smallest D-saturated ideal containing I. But I is already D-saturated, so $\pi^{-1}D^{-1}I = I$. □

Corollary 2.87. *If $P \trianglelefteq_{\text{prime}} R$, and $D \subset R$ is a multiplicative subset, then P is D-saturated if and only if $D \cap P = \emptyset$.* ⌟

Proof. Suppose P is D-saturated. Let $d \in D \cap P$. Since $d \in P$, then $1 \cdot d \in P$. Now, $d \in D$ and P is D-saturated, so $1 \in P$, which is a contradiction (since prime ideals are proper). So, $D \cap P = \emptyset$. Conversely, suppose $D \cap P = \emptyset$. Then, suppose $dr \in P$ for $d \in D$. Then, $r \in P$, so P is D-saturated. □

This leads us to the following observation:

Theorem 2.88. *There exists the following bijection:*

$$\mathrm{Spec}(D^{-1}R) \leftrightarrow \{P \in \mathrm{Spec}(R) \mid D \cap P = \emptyset\}.$$

⌟

Proof. This follows from the correspondence we have already established and that $P \in \mathrm{Spec}(R)$ is D-saturated if and only if $P \cap D = \varnothing$. □

Corollary 2.89. *After localization at a prime ideal, we have the following correspondence:*

$$\mathrm{Spec}(R_P) \leftrightarrow \{Q \in \mathrm{Spec}(R) \mid Q \subseteq P\}.$$

⌟

Corollary 2.90. *There exists the following bijection: for $p \in \mathbb{Z}$ a prime,*

$$\mathrm{Spec}(\mathbb{F}_p[x]) \leftrightarrow \{Q \in \mathrm{Spec}(\mathbb{Z}_{(p)}[x]) \mid p \in Q\}.$$

⌟

We now make some observations:

Proposition 2.91. *If $P \in \mathrm{Spec}(R)$ and $P \cap S = \varnothing$, then*

$$\frac{r}{d} \in D^{-1}P \Leftrightarrow r \in P.$$

⌟

Proof. If $r \in P$, then clearly, $r/d \in D^{-1}P$. Conversely, suppose $r/d \in D^{-1}P$, then, $\frac{r}{1} = \frac{d}{1}\frac{r}{d} \in D^{-1}P$. Since $(D^{-1}P)^c = P$, so $r \in P$. □

Corollary 2.92. *If R is Noetherian, then $D^{-1}R$ is Noetherian.* ⌟

Proposition 2.93. *Let $p \in \mathbb{Z}$ be a prime number. Then,*

$$\mathrm{Spec}(\mathbb{Z}_{(p)}) = \{(0), (p)\}.$$

In particular, $\mathbb{Z}_{(p)}$ is a ring with the unique maximal ideal $p\mathbb{Z}_{(p)}$. ⌟

Proof. By the correspondence theorem for prime ideals under localization,

$$\mathrm{Spec}(S^{-1}\mathbb{Z}) \longleftrightarrow \{P \in \mathrm{Spec}(\mathbb{Z}) \mid P \cap S = \varnothing\}.$$

We know that the prime ideals of $\mathbb{Z}$ are $(0), (q)$, for all prime numbers q.

First, since $0 \notin S$, we have

$$(0) \cap S = \varnothing.$$

So, we get one prime ideal (0) of $\mathbb{Z}_{(p)}$.

Next, consider a prime ideal (q), where q is prime. If $q \neq p$, then $p \nmid q$, so $q \in S$. But $q \in (q)$, and hence

$$(q) \cap S \neq \varnothing.$$

On the other hand, if $q = p$, then by definition of S,

$$(p) \cap S = \varnothing.$$

Hence, we get the prime ideal $S^{-1}(p) = p\mathbb{Z}_{(p)}$. So,

$$\mathrm{Spec}(\mathbb{Z}_{(p)}) = \{(0), p\mathbb{Z}_{(p)}\}.$$

Finally, since $p\mathbb{Z}_{(p)}$ is the only nonzero prime ideal and $(0) \subsetneq p\mathbb{Z}_{(p)}$, it is the unique maximal ideal of $\mathbb{Z}_{(p)}$. □

2.10. **Examples of Classifying Spectrum**

Problem 2.94. *Show that all the prime ideals of $\mathbb{Z}[x]$ are of one of the following forms:*

(1) (0)
(2) (p) *for some prime* p
(3) (f) *for* f *irreducible in* $\mathbb{Q}[x]$ *and not divisible by any prime*
(4) (p, f) *where* p *is prime and* $f \in \mathbb{Z}[x]$ *lifts an irreducible element of* $\mathbb{Z}/p\mathbb{Z}[x]$.

Answer. First, we check that these are indeed prime ideals. (0) is prime and if p is prime, then by definition (p) is a prime ideal (or alternatively we can see this since $\mathbb{Z}[x]/(p) \cong \mathbb{Z}/p\mathbb{Z}[x]$ which is a PID, and therefore (p) is prime). Suppose f is irreducible in $\mathbb{Q}[x]$ and is not divisible by any prime. Then, $f(x)$ is primitive, since otherwise, we can write $f(x) = df'(x)$ where $d \neq 1$ is the gcd of its coefficients, but d can be decomposed into primes which would make $f(x)$ divisible by those primes. An irreducible primitive polynomial in $\mathbb{Q}[x]$ is irreducible in $\mathbb{Z}[x]$ by Gauss's lemma, and irreducibles are primes in $\mathbb{Z}[x]$, which is a UFD. Finally, we show for (p, f): let $ab \in (p, f)$. Then, $ab = \alpha_1 p + \alpha_2 f$. Moving this to $\mathbb{Z}/p\mathbb{Z}[x]$, we see that $\bar{a}\bar{b} \in (\bar{f})$. Since $\bar{f}$ is irreducible in this ring, this is a prime ideal (as the ring is a PID), so one of the two factors is in this ideal. WLOG, suppose $\bar{a} \in (\bar{f})$. Then, $a(x) + (p, f) = f(x)g(x) + (p, f) \implies a(x) \in (p, f)$. Therefore, (p, f) is indeed prime.

Now, consider any arbitrary prime ideal $P \trianglelefteq_{\text{prime}} \mathbb{Z}[x]$. We first classify all prime ideals in $\mathbb{Z}[x]$ in a convenient manner. There are two options: either P contains no non-zero constants in $\mathbb{Z}$ or P contains non-zero constants in $\mathbb{Z}$.

Case 1: P contains no non-zero constants in $\mathbb{Z}$. Let us start with the first: P contains no non-zero constants in $\mathbb{Z}$. If $P = (0)$ we are done. So, suppose $P \neq (0)$. Note that this means $P \cap D = \emptyset$ where $D = \mathbb{Z} - \{0\}$. Then, we can move to the localization $\mathbb{Q}[x] = D^{-1}\mathbb{Z}[x]$. Recall that there exists the following bijection:

$$\text{Spec}(D^{-1}R) \leftrightarrow \{P \in \text{Spec}(R) \mid D \cap P = \emptyset\}.$$

Clearly, $D^{-1}P$ is now a prime ideal in $\mathbb{Q}[x]$. Now, $\mathbb{Q}[x]$ is a polynomial ring over fields, so it is a PID, so $D^{-1}P = \{\frac{a(x)}{d} \mid a(x) \in P, d \in\} = D\} = (h(x))$ for some $h(x)$. Since this is a prime ideal, $h(x)$ is irreducible. However, it need not be primitive. Suppose

$$h(x) = \sum_{i=0}^{n} \frac{u_i}{d_i} x^i.$$

Now, take a common multiple of d_i, which we call d. We get that $dh(x)$ is now a polynomial in $\mathbb{Z}[x]$. Furthermore, d is a unit, so

$$(dh(x)) = (h(x)).$$

Now, let $dh(x) = \sum_{i=0}^{n} \alpha_i x^i$. Let the gcd of the coefficients α_i be a. Then, $\frac{dh(x)}{a}$ is primitive when viewed in $\mathbb{Z}[x]$ and also since this generates the prime ideal, this is irreducible in $\mathbb{Q}[x]$. Call this $g(x)$. We see then that $D^{-1}P = (g(x)) \trianglelefteq_{\text{prime}} \mathbb{Q}[x]$ where $g(x)$ is irreducible and primitive in $\mathbb{Z}[x]$. Then, $g(x)$ is irreducible in $\mathbb{Z}[x]$, and so $(g(x))$ is a prime ideal in $\mathbb{Z}[x]$

Note that by Gauss's lemma, $g(x)$ is irreducible in $\mathbb{Z}[x]$. We claim that $(g(x)) = P$ in $\mathbb{Z}[x]$. On one hand, since $g(x) \in D^{-1}P$, $g(x) = \frac{h(x)}{d}$. Now, $h(x) = dg(x) \in P$. Then, either $d \in P$ or $g(x) \in P$. But d is a non-zero constant which cannot be in P, so $g(x) \in P$. Conversely, suppose $h(x) \in P$. Then, $h(x) \in D^{-1}P = (g(x))$. So, there exists some $q(x)$ such that $h(x) = q(x)g(x)$. Multiplying by a common multiple of the denominators on the right hand side gives us $mh(x) = g(x)a(x)$. Now, $a(x) \in \mathbb{Z}[x]$ and $g(x)a(x) \in (g(x))$. So, $mh(x) \in (g(x))$. We know that $g(x)$ is irreducible so this is a prime ideal and since m is constant, $h(x) \in (g(x))$.

In conclusion, if P is prime ideal that contains no non-zero constant, either $P = (0)$ or $P = (g(x))$ for some $g(x)$ that is irreducible and primitive in $\mathbb{Q}[x]$ (which means $g(x)$ is not divisible by any prime).

Case 2: P contains non-zero constants in $\mathbb{Z}$. Then, $P \cap \mathbb{Z} \neq (0)$. Let $n \in P \cap \mathbb{Z}$ such that $n \neq 0$. Now, let $n = up_1 \cdots p_k$ be its prime factorization in $\mathbb{Z}[x]$ with u a unit, which is a UFD. Since P is prime, $u \notin P$ and so at least one of the prime factors is in P. Furthermore, since n is an integer, all the prime factors are integers. So, we have a prime integer $p_i \in P$. For convenience, let us write $p := p_i$. Note that $(p_i) \subseteq P$. Then, we have two sub-cases: either $(p_i) = P$ or $(p_i) \subsetneq P$. If it is the former, we are done.

Now, suppose $(p) \subsetneq P$. Notice that if q is an integer that is not in (p) but is in P, then since q and p are coprime, we would have that $1 \in P$ (by Bezout's identity), which is a contradiction. So, if $f(x) \in P$ but $f(x) \notin (p)$, the $f(x)$ is not a constant. Now, we can move to the ring

$$\mathbb{Z}[x]/(p) \cong \mathbb{Z}/p\mathbb{Z}[x] = \mathbb{F}_p[x].$$

In particular, since P is a prime ideal containing (p), $P/(p)$ is a prime ideal in $\mathbb{Z}/p\mathbb{Z}[x]$. Now, this is a polynomial ring over a field, so it is a PID, so $P/(p) = (g(x))$ for some $g(x) \in \mathbb{F}_p[x]$. If

$$\pi : \mathbb{Z}[x] \to \mathbb{F}_p[x],$$

then

$$P = \pi^{-1}(P/(p)) = \pi^{-1}((g(x))).$$

Suppose $\pi(f(x)) = g(x)$, which exists since π is surjective. We claim that

$$\pi^{-1}((g(x))) = (p, f(x)).$$

To see this, as $\pi(p) = 0 \in ((g(x)))$ and $\pi(f(x)) = g(x) \in (g(x))$, we have that $(p, f(x)) \subseteq \pi^{-1}((g(x)))$. Conversely, suppose $h(x) \in \pi^{-1}((g(x)))$. Then,

$$\pi(h(x)) = k(x)g(x)$$

for some $k(x)$. Let $\pi(\tilde{k(x)}) = k(x)$. Then, $\pi(h(x) - \tilde{k(x)}f(x)) = k(x)g(x) - k(x)g(x) = 0 \in (g(x))$. So, there $h(x) - \tilde{k(x)}f(x) \in \ker \pi = (p)$, meaning $h(x) \in (p, f(x))$. As such,

$$P = (p, f)$$

where p is prime and f is a polynomials that lift an irreducible elements of $\mathbb{Z}/p\mathbb{Z}[x]$. So, if P is a prime ideal containing non-zero constants in $\mathbb{Z}$, then it is either (p) for a prime p or (p, f) where p is prime and $f(x)$ lifts an irreducible elements in $\mathbb{F}_p[x]$.

Problem 2.95. *Fix a prime number p. Show that $\mathbb{Z}_{(p)}[x] = (\mathbb{Z} - p\mathbb{Z})^{-1}\mathbb{Z}[x]$. Then, deduce that the prime ideals of $\mathbb{Z}_{(p)}[x]$ are of the following forms:*

1. (0)
2. (p)
3. (f) *where f is an irreducible polynomial in $\mathbb{Q}[x]$ that is not divisible by p*
4. (p, f) *where $f \in \mathbb{Z}_{(p)}[x]$ lifts an irreducible polynomial in $\mathbb{Z}/p\mathbb{Z}[x]$.*

Proof. Note that

$$\mathbb{Z}_{(p)} \text{ consists of terms like } \frac{a}{b}, \text{ where } a, b \in \mathbb{Z}, p \nmid b,$$

$$\mathbb{Z}_{(p)}[x] \text{ consists of terms like } \sum_{i=0}^{n} \frac{a_i}{b_i} x^i, \text{ where } a_i, b_i \in \mathbb{Z}, p \nmid b_i$$

which is clearly the same as

$$(\mathbb{Z} - p\mathbb{Z})^{-1}\mathbb{Z}[x] = \{\frac{\sum_{i=0}^{n} a_i x^i}{b} \mid a_i, b \in \mathbb{Z}, p \nmid b\}.$$

We follow a similar strategy in our classification. With $\iota : \mathbb{Z}_{(p)} \to \mathbb{Z}_{(p)}[x]$, if P is a priem ideal of $\mathbb{Z}_{(p)}[x]$, we know that $\iota^{-1}(P)$ is a prime ideal of $\mathbb{Z}_{(p)}$. Now, $\iota^{-1}(P) = \{a \in \mathbb{Z}_{(p)} \mid a \in P\} = P \cap \mathbb{Z}_{(p)}$ and since it is a prime ideal of $\mathbb{Z}_{(p)}$, it must be either (0) or (p) in $\mathbb{Z}_{(p)}$.
Case 1: $P \cap \mathbb{Z}_{(p)} = (0)$**.** Now, if $P = (0)$, we are done. So, now, suppose $P \neq (0)$. Then, P contains no non-zero constants. Now, consider $D = \mathbb{Z}_{(p)} - \{0\}$, the set of non-zero constants in $\mathbb{Z}_{(p)}$. This does not intersect with P and is a multiplicative set. So, we consider the localization $D^{-1}\mathbb{Z}_{(p)}[x]$ and recall that

$$\operatorname{Spec}(D^{-1}\mathbb{Z}_{(p)}[x]) \leftrightarrow \{Q \in \operatorname{Spec}(\mathbb{Z}_{(p)}[x]) \mid Q \cap D = \varnothing\}.$$

Consider the elements of

$$\begin{aligned} D^{-1}\mathbb{Z}_{(p)}[x] &= \{\frac{\sum_{i=0}^{n} \frac{a_i}{b_i}x^i}{\frac{d_1}{d_2}} \mid p \nmid b_i, p \nmid d_2, d_1 \neq 0\} \\ &= \{\sum_{i=0}^{n} \frac{d_2 a_i}{d_1 b_i}x^i \mid p \nmid b_i, p \nmid d_2, d_1 \neq 0\} \\ &= \mathbb{Q}[x] \end{aligned}$$

The last equality follows because any elements $\frac{a}{b}x^i$ for $a, b \in \mathbb{Z}$ can be written as $\frac{a\cdot 1}{b\cdot 1}$ where $a, b \in \mathbb{Z}$ and $p \nmid 1$, so this is in $D^{-1}\mathbb{Z}_{(p)}[x]$. The converse is straightforward, i.e. $D^{-1}\mathbb{Z}_{(p)}[x] \subseteq \mathbb{Q}[x]$. As such, the problem can be reformulated as:

$$\operatorname{Spec}(\mathbb{Q}[x]) \leftrightarrow \{Q \in \operatorname{Spec}(\mathbb{Z}_{(p)}[x]) \mid Q \cap D = \varnothing\}.$$

Now, $\mathbb{Q}[x]$ is a polynomial ring over a field, so it is a PID. Its prime ideals are (0) and $(f(x))$ where $f(x)$ is irreducible. We already took care of the (0) ideal, so now we pay attention to $(f(x))$ where $f(x) \in Q[x]$ is irreducible. Now, if

$$f(x) = \frac{u_0}{d_0} + \frac{u_1}{d_1}x + \cdots + \frac{u_n}{d_n}x^n,$$

we can multiple both sides by a common multiple of d_i to get

$$df(x) = \sum_{i=0}^{n} c_i x^i$$

which can then be turned into a primitive polynomial by dividing the gcd of c_i from both sides:

$$\frac{df(x)}{g} = \sum_{i=0}^{n} a_i x^i.$$

Then, note that

$$(f(x)) = (\frac{df(x)}{g}) = (\sum_{i=0}^{n} a_i x^i) \quad \text{in } \mathbb{Q}[x].$$

So, now, we need to map prime ideals of the form $(f(x))$ for $f(x)$ primitive and irreducible in $\mathbb{Q}[x]$ back to $\mathbb{Z}_{(p)}[x]$. Recall that $\iota : \mathbb{Z}_{(p)}[x] \to \mathbb{Q}[x]$, so

$$\begin{aligned} \iota^{-1}((f(x))) &= \{g(x) \in \mathbb{Z}_{(p)}[x] \mid \iota(g(x)) = \frac{g(x)}{1} \in (f(x))\} \\ &= f(x)\mathbb{Q}[x] \cap \mathbb{Z}_{(p)}[x]. \end{aligned}$$

So, we get that

$$P = f(x)\mathbb{Q}[x] \cap \mathbb{Z}_{(p)}[x].$$

Note that $f(x)$ is primitive and irreducible in $\mathbb{Q}[x]$, so it is also irreducible in $\mathbb{Z}_{(p)}[x]$ by Gauss's Lemma.

Now, we have to show that $f(x)\mathbb{Q}[x] \cap \mathbb{Z}_{(p)}[x] = (f(x)) \trianglelefteq \mathbb{Z}_{(p)}[x]$ To see this, for any $g(x) \in (f(x))$, we have that $g(x) = h(x)f(x)$ where $h(x) \in \mathbb{Z}_{(p)}[x]$. Since this is a product of two polynomials in $\mathbb{Z}_{(p)}[x]$, this is in $\mathbb{Z}_{(p)}[x]$ and since this product contains a factor $f(x)$, this is also in $(f(x))$, so $g(x) \in P$. On the other hand, if $g(x) \in P$, then, $g(x) = h(x)f(x)$ for some $h(x) \in \mathbb{Q}[x]$. Clearing the denominators gives us $g(x) = \frac{1}{d}\tilde{h(x)}f(x)$. So, $dg(x) = \tilde{h(x)}f(x) \in \mathbb{Z}_{(p)}[x]$. Now $dg(x) \in (f(x))$, which is prime since $f(x)$ is irreducible in $\mathbb{Z}_{(p)}[x]$. Then, either $d \in (f(x))$ or $g(x) \in (f(x))$. But d is a constant whereas $f(x)$ is of degree at least 1, so $g(x) \in (f(x))$ as viewed in $\mathbb{Z}_{(p)}[x]$. Therefore,

$$P = (f(x))\mathbb{Z}_{(p)}[x].$$

Case 2: $P \cap \mathbb{Z}_{(p)} = (p)$**.** This has two sub-cases. The first is $P = p\mathbb{Z}_{(p)}[x] = (p)$. In this case, we are done; do note that (p) is a prime ideal since $\mathbb{Z}_{(p)}[x]/(p) = \mathbb{Z}_{(p)}[x]/p\mathbb{Z}_{(p)}[x] \cong \mathbb{F}_p[x]$ which is an integral domain. In particular $\mathbb{Z}_{(p)}[x]/(p) \cong \mathbb{F}_p[x]$ by the morphism:

$$\varphi(\sum_i \frac{a_i}{b_i}x^i) = \sum_i \overline{a_i}\overline{b_i}^{-1}x^i,$$

which is surjective and whose kernel is (p). The other final case we have to deal with is when $(p) = p\mathbb{Z}_{(p)}[x] \subsetneq P$. Then, we can move to $\mathbb{Z}_{(p)}[x]/(p)$. Now, we recall the correspondence (Theorem 2.59):

$$\mathrm{Spec}(\mathbb{Z}_{(p)}[x]/(p)) \leftrightarrow \{Q \in \mathrm{Spec}(\mathbb{Z}_{(p)}[x] \mid (p) \subseteq Q\}.$$

By this correspondence, we see that $P/(p)$ is a prime ideal in $\mathbb{Z}_{(p)}[x]/(p) \cong \mathbb{F}_p[x]$. Now, the prime ideals in $\mathbb{F}_p[x]$, which is a PID, are (0) and (0) for $(g(x))$ for $g(x)$ irreducible in $F_p[x]$. Note that

$$\pi : \mathbb{Z}_{(p)}[x] \to \mathbb{F}_p[x]$$

is surjective, so there exists some $f(x) \in \mathbb{Z}_{(p)}[x]$ such that $\pi(f(x)) = g(x) \in \mathbb{F}_p[x]$. Now, we claim that

$$\pi^{-1}((g(x))) = (p, f(x)).$$

Firstly, we know that $\pi(p) = 0 \in (g(x))$ and $\pi(f(x)) = g(x) \in (g(x))$, so $(p, f(x)) \subseteq \pi^{-1}((g(x)))$. Conversely, suppose $h(x) \in \pi^{-1}((g(x)))$. Then, $\pi(h(x)) = k(x)g(x)$ for some $k(x) \in \mathbb{F}_p[x]$ whose preimage is $\tilde{k}(x)$. Now, $\pi(h(x) - \tilde{k}(x)f(x)) = k(x)g(x) - k(x)g(x) = 0$. So, $h(x) - \tilde{k}(x)f(x) \in (p) = p\mathbb{Z}_{(p)}[x]$. Then, $h(x) \in p\mathbb{Z}_{(p)}[x] + (f(x))$, so $h(x) \in (p, f(x))$. With this, we conclude that in this case

$$P = (p, f)$$

for some $f(x)$ that maps to an irreducible polynomial in $\mathbb{F}_p[x]$. □

2.11. **Chinese Remainder Theorem**

In this section, we let R be a commutative, unital ring with $1 \neq 0$.

Recall the definition of a direct product of rings:

Definition 2.96 (Direct Product of Rings)**.** *Let R and S be rings. Then, their direct product is:*

$$R \times S = \{(r, s) \mid r \in R, s \in S\}.$$

⌟

Definition 2.97 (Comaximal Ideals)**.** *The ideals $A \trianglelefteq R$ and $B \trianglelefteq R$ are comaximal if $A + B = R$.* ⌟

Theorem 2.98 (Chinese Remainder Theorem)**.** *Let R be a commutative, unital ring with $1 \neq 0$. Let $A_1, A_2, \cdots, A_k$ be ideals in the ring R. Then, the map*

$$\varphi : R \to R/A_1 \times R/A_2 \times \cdots \times R/A_k$$

is a ring homomorphism. In particular:

1. *$\ker \varphi = A_1 \cap A_2 \cap \cdots \cap A_k$ is the kernel.*
2. *If $\forall i, j \in [k]$ such that $i \neq j$, A_i and A_j are comaximal, then φ is surjective, and $A_1 \cap A_2 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$. So,*

$$R/A_1 \cdots A_k = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k.$$

⌟

Proof. We prove this via induction, starting with $k = 2$. The first part is obvious, via the map $\varphi(r) = (r + A, r + B)$. Now, suppose A and B are comaximal. Then, there exists $x \in A, y \in B$ such that $x + y = 1$. Now, $\varphi(x) = (0, 1)$ as $x \in A$ and $x = 1 - y \in 1 + B$. On the other hand, $\varphi(y) = (1, 0)$. This creates our *basis*: for any $(r_1 + A, r_2 + B)$, we have that:

$$\begin{aligned}
\varphi(r_2 x + r_1 y) &= \varphi(r_2 x) + \varphi(r_1 y) \\
&= \varphi(r_2)\varphi(x) + \varphi(r_1)\varphi(y) \\
&= (r_2 + A, r_2 + B)(0, 1) + (r_1 + A, r_1 + B)(1, 0) \\
&= (r_1 + A, r_2 + B).
\end{aligned}$$

So, φ is surjective. One can easily show that $AB = A \cap B$.

Now, for any general $k - 1$, suppose the statement is true. Then, for k: define $A = A_1$ and $B = A_2 \cdots A_k$. Then, we claim that A and B are comaximal. To prove this, note that inductive hypothesis says that A_1 and A_i for $i \neq 1$ are comaximal, so there exists $x_i \in A_1$ and $y_i \in A_i$ such that $x_i + y_i = 1$. Now, $x_i + y_i \equiv y_i \bmod A_1 \in R/A_1$. Then,

$$\begin{aligned}
1 &= (x_2 + y_2) \cdots (x_k + y_k) \\
&= \text{ terms containing } x_j + y_2 \cdots y_k \\
&\in A_1 + A_2 \cdots A_k.
\end{aligned}$$

So, A and B are comaximal. □

Corollary 2.99. *Let n be a positive integer such that its prime factorization is $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$. Then,*

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z} \times \cdots \mathbb{Z}/p_k^{\alpha_k}\mathbb{Z},$$

and

$$\mathbb{Z}/n\mathbb{Z}^\times \cong \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}^\times \times \cdots \mathbb{Z}/p_k^{\alpha_k}\mathbb{Z}^\times.$$

⌟

Observation (Constructing basis elements for CRT.)**.** *This is simply for note-keeping of a technique we already used in the proof of CRT. Consider the CRT isomorphism:*

$$R/A_1 \cdots A_k \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k.$$

We can construct basis elements $e_1, \cdots, e_k \in R$ *such that* $e_i \equiv 1 \pmod{A_i}$ *and* $e_i \equiv 0 \pmod{A_j}$ *for* $j \neq i$*. To do so, we first observe that for any* A_i *and* A_j*, we can find* $u_{ij} \in A_i, v_{ij} \in A_j$ *such that* $u_{ij} + v_{ij} = 1$ *because of pairwise comaximality. Then, define*

$$e_i = \prod_{j \neq i} v_{ij}.$$

This works as $e_{ij} \equiv 0 \pmod{A_j}$ *for* $j \neq i$ *and, since* $v_{ij} = 1 - u_{ij} \in 1 + A_i$ *which is* $\overline{1} \in R/A_i$*, therefore,* $e_{ij} \equiv 1 \pmod{A_i}$*. Then, for any specific choice of*

$$(b_1, \cdots, b_k) \in R/A_1 \times R/A_2 \times \cdots \times R/A_k,$$

we can find a corresponding elements in R *via setting*

$$b_1 e_1 + \cdots + b_k e_k.$$

Proposition 2.100 (CRT and Idempotents)**.** *An element* $x \in R$ *is called an idempotent if* $x^2 = x$*. Suppose* x *is an idempotent and assume that* $xr = rx$ *for all* $r \in R$*. Then,*

1. Rx *and* $R(1-x)$ *are ideals of* R*,*
2. $R \cong Rx \times R(1-x)$*, and*
3. x *is the identity of* Rx *and* $(1-x)$ *is the identity of* $R(1-x)$*.*

⌟

Problem 2.101. *Find* $a \in \mathbb{Z}$ *such that* $a \equiv 1 \pmod{19}$ *and* $a \equiv 2 \pmod{17}$*.*

Answer. Note that 17 and 19 are coprime, so they are comaximal. Then, we can use CRT: $\mathbb{Z}/(17 \cdot 19) \cong \mathbb{Z}/(17) \times \mathbb{Z}/(19)$. Now, $a \equiv 1 \pmod{19}$, then $a = 19k + 1$ for some integer k. On the other hand,

$$\begin{aligned} a = 19k + 1 &\equiv 2 \pmod{17} \\ 2k + 1 &\equiv 2 \pmod{17} \\ 2k &\equiv 1 \pmod{17}. \end{aligned}$$

Now, in $\mathbb{Z}/17\mathbb{Z}$, the multiplicative inverse of 2 is $\overline{9}$. Note that taking $k = 9$, we get $a = 19(9)+1 = 172$ which is 1 modulo 19.

Problem 2.102. *Find the smallest integer* $m > 0$ *suh that* $a^m \equiv 1 \pmod{105}$ *for all* a *coprime to 105.*

Answer. Note that $105 = 3 \times 5 \times 7$. Using CRT, we get that $(\mathbb{Z}/(105))^\times \cong (\mathbb{Z}/(3))^\times \times (\mathbb{Z}/(5))^\times \times (\mathbb{Z}/(7))^\times$. If a is coprime to 105, then $\gcd(a, 105) = 1$, so a is a unit on the left hand side, and so it maps to units in the three rings on the right hand side. In particular, $a^2 \equiv 1 \pmod{3}$, $a^4 \equiv 1 \pmod{5}$ and $a^6 \equiv 1 \pmod{7}$. Then, the smallest integer m that is a multiple of 2, 4, and 6 is 12.

2.12. **Some Example Computations**

We will go through a series of computations here that tie together various parts of the notes so far.

(1) $\mathbb{Z}[\sqrt{-3}]/(2) \cong \mathbb{Z}[x]/(2, x^2+3)$.

Proof. First, note that $\mathbb{Z}[\sqrt{-3}]/(2) \cong \mathbb{Z}/2\mathbb{Z}[\sqrt{-3}]$. This is simply because in $\mathbb{Z}[\sqrt{-3}]/(2)$, each element is of the form $a+b\sqrt{-3}+(2)$, and so $a, b \in \{0, 1\}$. Now, consider the morphism $\varphi : \mathbb{Z}[x] \to \mathbb{Z}/2\mathbb{Z}[\sqrt{-3}]$ using $\varphi(a) = \bar{a} \in \mathbb{Z}/2\mathbb{Z}$ and $\varphi(x) = \sqrt{-3}$. Then, this is surjective and its kernel is the ideal $(2, x^2+3) \subset \mathbb{Z}[x]$. □

(2) $\mathbb{Z}[x]/(2, x^2+3) \cong \mathbb{F}_2[x]/(x^2+1)$.

Proof. First, we know from the definitions that $\mathbb{Z}[x]/(2, x^2+3) \cong \mathbb{F}_2[x]/(x^2+3)$. Now, consider the ideal $(x^2+3) \subset \mathbb{F}_2[x]$. But in $\mathbb{F}_2[x]$, $\bar{3} = \bar{1}$, so $(x^2+3) = (x^2+1)$. □

(3) $\mathbb{F}_2[x]/(x^2+1) \cong \mathbb{F}_2[x]/((x+1)^2)$.

Proof. Consider the ideal $((x+1)^2) \subset \mathbb{F}_2[x]$. Now, in $\mathbb{F}_2[x]$, $(x+1)^2 = x^2+2x+1 \equiv x^2+1$, which implies the statement. □

(4) $\mathbb{Z}[\sqrt{-3}]/(2)$ is not reduced, since it is isomorphic to $\mathbb{F}_2[x]/((x+1)^2)$ in which $(x+1) \neq 0$ is nilpotent.
(5) $\mathbb{Z}[\sqrt{-3}]/(2)$ is not an integral domain or a field.
(6) The only maximal ideal of $\mathbb{Z}[\sqrt{-3}]/(2)$ is $(\overline{1+\sqrt{-3}})$.

Proof. We compute the maximal ideals of $\mathbb{F}_2[x]/((x+1)^2)$. A maximal ideal, M, must contain $0 \equiv (x+1)^2$. Since R is commutative and unital, the maximal ideal is also prime, and therefore, $\overline{x+1} \in M$. Now, the ideal generated by this element must be in M, i.e. $(\overline{x+1}) \subset M$. However, $(\overline{x+1})$ itself is a maximal ideal in $\mathbb{F}_2[x]/((x+1)^2)$.:

$$\begin{aligned}\left(\mathbb{F}_2[x]/((x+1)^2)\right) / \left(\overline{x+1}\right) &= \left(\mathbb{F}_2[x]/((x+1)^2)\right) / \left((x+1)/((x+1)^2)\right) \\ &= \mathbb{F}_2[x]/(x+1) \qquad \text{(by 3rd isomorphism theorem)} \\ &= \mathbb{F}_2,\end{aligned}$$

which is a field. So, we have that $(\overline{x+1})$ is a maximal ideal contained inside M, so M itself must be $(\overline{x+1})$.

Now, we aim to find the ideal in $\mathbb{Z}[\sqrt{-3}]/(2)$ that corresponds to $(\overline{x+1})$ in $\mathbb{F}_2[x]/((x+1)^2)$. First, we find the corresponding ideal in $\mathbb{Z}[x]/(2, x^2+3)$. Recall the map:

$$\mathbb{Z}[x]/(2, x^2+3) \to \mathbb{F}_2[x]/((x+1)^2)$$

via taking every integer modulo 2. Then, $f(x) + (2, x^2+3) \to \overline{f(x)} + ((x+1)^2)$ (here, $\overline{f(x)}$ is simply $f(x)$ modulo 2). So if $\overline{f(x)} + ((x+1)^2) \in (\overline{x+1}) = (x+1+((x+1)^2))$, then $\overline{f(x)} \in (x+1)$ (given $((x+1)^2) \subset (x+1)$). So, $\overline{f(x)} = \overline{q(x)} \cdot (x+1) \in \mathbb{F}_2[x]$, which implies

$$f(x) - q(x)(x+1) \in (2),$$

so

$$f(x) \in (2, x+1).$$

Conversely, we claim that if $f(x) \in (2, x+1)$, then its image lies in $(\overline{x+1}) = (x+1+((x+1)^2))$. Indeed, if $f(x) = 2a(x) + (x+1)b(x)$, then it maps to (after taking modulo 2) $\overline{f(x)} = \overline{b(x)} \cdot (\overline{x+1})$, which proves the statement.

So, the corresponding maximal ideal in $\mathbb{Z}[x]/(2, x^2+3)$ is $(\overline{2, x+1})$. In $\mathbb{Z}[\sqrt{-3}]/(2)$, this ideal corresponds to $(\overline{1+\sqrt{-3}})$. □

Problem 2.103. *For each of the following rings, determine whether they are (a) reduced, (b) integral domains, and (c) fields:*
$\mathbb{Z}[x]/(x^2-1)$; $\mathbb{Z}[x]/(x-1)^2$; $\mathbb{Z}[x]/(x^2+1)$; $\mathbb{Z}[\sqrt{-5}]/(2)$; $\mathbb{Z}[\sqrt{-5}]/(3)$; $\mathbb{Z}[\sqrt{-5}]/(1+\sqrt{-5})$; $\mathbb{Z}[\sqrt{-5}]/(11)$.

Answer. (1) $\mathbb{Z}[x]/(x^2-1) = \mathbb{Z}[x]/((x+1)(x-1))$. This ring is clearly not an integral domain since $x+1$ and $x-1$ are nonzero elements but their product is 0. Therefore, it is also not a field. Now, suppose $f(x) \neq 0$ in this ring and $f(x)^n \equiv 0$. Then, $f(x)^n \in ((x+1)(x-1))$. Now, $x+1$ is prime (since it is irreducible in the PID $\mathbb{Z}[x]$), and since $x+1$ divides $f(x)^n$, it must divide $f(x)$. Similarly, $x-1$ is also prime and therefore it must divide $f(x)$. Since $x+1$ is not associate to $x-1$, $f(x) = r(x)(x+1)(x-1)$ for some $r(x) \in \mathbb{Z}[x]$, so $f(x) \equiv 0$. Therefore, the only nilpotent element is 0 itself and so, the ring is reduced.

(2) $\mathbb{Z}[x]/(x-1)^2$ has a non-zero nilpotent element, namely $(x-1)$. So, this ring is not reduced. Therefore, it is also not an integral domain and, therefore, not a field.

(3) $\mathbb{Z}[x]/(x^2+1)$. Note that, in $\mathbb{Z}[x]$, x^2+1 is an irreducible polynomial since -1 is not a square. $\mathbb{Z}[x]$ is a PID, so (x^2+1) is a prime ideal, and so $\mathbb{Z}[x]/(x^2+1)$ is an integral domain. However, it is not a maximal ideal since $(x^2+1) \subsetneq (2, x^2+1) \subsetneq \mathbb{Z}[x]$. So, the ring is not a field. Finally, we claim that the ring is reduced bcause we have shown that x^2+1 is irreducible: suppose $f(x)^n \equiv 0$. Then, $f(x) \in (x^2+1)$. But x^2+1 is prime, so $f(x) \in (x^2+1)$.

(4) Consider the ring: $\mathbb{Z}[\sqrt{-5}]/(2)$. Consider $\mathbb{Z}[x]/(2, x^2+5) \cong \mathbb{Z}/2\mathbb{Z}[x]/(x^2+5) = \mathbb{F}_2[x]/(x^2+5) = \mathbb{F}_2[x]/(x^2+1) = \mathbb{F}_2[x]/(x+1)^2$, where the last equality follows from $(x+1)^2 = x^2 + 2x + 1 \equiv x^2+1$. In this ring, $(x+1)$ is nonzero but nilpotent, so the ring is not reduced. Therefore, it is also not an integral domain and not a field. Therefore, these statements hold for $\mathbb{Z}[\sqrt{-5}]/(2) \cong (\mathbb{Z}[x]/(x^2+5))/(\bar{2}) \cong \mathbb{Z}[x]/(2, x^2+5)$, where the last isomorphism follows from the third isomorphism theorem: In $(\mathbb{Z}[x]/(x^2+5))$, the ideal $(\bar{2}) = \{2f(x) + (x^2+5) \mid f(x) \in \mathbb{Z}[x]\}$. As an ideal in $\mathbb{Z}[x]$, this is $(2, x^2+5)$. Since we are quotienting by (x^2+5), this becomes

$$(\mathbb{Z}[x]/(x^2+5))/(2, x^2+5)/(x^2+5) \cong \mathbb{Z}[x]/(2, x^2+5).$$

(5) Consider the ring $\mathbb{Z}[\sqrt{-5}]/(3)$. Then, similar to the previous problem, $\mathbb{Z}[\sqrt{-5}]/(3) \cong (\mathbb{Z}[x]/(x^2+5))/(\bar{3}) \cong \mathbb{Z}[x]/(3, x^2+5) \cong \mathbb{F}_3[x]/(x^2+5) = \mathbb{F}_3[x]/(x^2+2) = \mathbb{F}_3[x]/(x^2-1) = \mathbb{F}_3[x]/((x+1)(x-1))$. Now, $(x+1)$ and $(x-1)$ are nonzero but their product is, so we have zero divisors, implying that this is neither an integral domain nor a field. Suppose $f(x)^n \equiv 0$, i.e. $f(x)^n = g(x)(x+1)(x-1)$ for some $g(x) \in \mathbb{F}_3[x]$. Now, $(x+1)$ is irreducible and $\mathbb{F}_3[x]$ is a UFD, so it is also prime, so $(x+1)$ must divide $f(x)$. Similarly, $(x-1)$ must also divide $f(x)$. But these two factors are not associate, so $f(x)$ must be divisible by $(x+1)(x-1)$, therefore, $f(x) \equiv 0$ in our ring, meaning there are no non-zero nilpotent elements. As such, our ring is reduced.

(6) Consider the ring, $\mathbb{Z}[\sqrt{-5}]/(1+\sqrt{-5})$. Firstly, again, $\mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}[x]/(x^2+5)$. Then, $(\mathbb{Z}[x]/(x^2+5))/\overline{(1+\sqrt{-5})} \cong \mathbb{Z}[x]/(x^2+5, 1+\sqrt{-5}) = \mathbb{Z}[x]/(x^2+5, 1+x)$.
Now,

$$\begin{aligned}
\mathbb{Z}[x]/(x^2+5, 1+x) &\cong (\mathbb{Z}[x]/(1+x))/(\overline{x^2+5}) \qquad \text{(3rd isomorphism theorem)} \\
&\cong \mathbb{Z}/(\overline{x^2+5}) \qquad \text{(using } \mathbb{Z}[x]/(1+x) \cong \mathbb{Z} \text{ via evaluating } x=-1) \\
&\cong \mathbb{Z}/(\bar{6}) \\
&= \mathbb{Z}/6\mathbb{Z}.
\end{aligned}$$

Since 6 is not a prime, this is not a field and also not an integral domain, since $\bar{3} \cdot \bar{2} = 0$. This ring is reduced: $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ by Chinese Remainder Theorem. Both $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are fields, so they are reduced. The product of reduced rings is reduced (from lecture 1), so our ring is reduced.

(7) Consider the ring $\mathbb{Z}[\sqrt{-5}]/(11)$. Again, we simplify the ring:

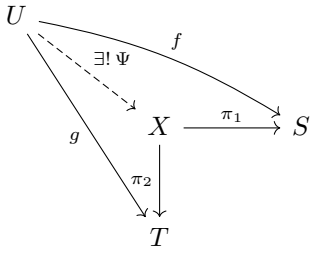
$$\begin{aligned}\mathbb{Z}[\sqrt{-5}]/(11) &\cong (\mathbb{Z}[x]/(x^2+5))/(\overline{11}) \\ &\cong \mathbb{Z}[x]/(11, x^2+5) \\ &\cong \mathbb{F}_{11}[x]/(x^2+5) \\ &\cong \mathbb{F}_{11}[x]/(x^2-6).\end{aligned}$$

Now, 6 is not a square modulo 11, so x^2-6 is irreducible and, therefore, prime (as $\mathbb{F}_{11}[x]$ is a PID). Then, this ring is reduced since $f(x)^n \in (x^2-6)$ implies $f(x) \in (x^2-6)$. Since this is a prime ideal, $\mathbb{F}_{11}[x]/(x^2-6)$ is an integral domain. In PIDs, prime ideals are maximal ideals, so this is also a field.

2.13. **Products and Fibre Products**

We will, first, define the product of rings more abstractly via a universal property.

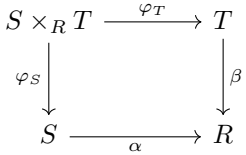
Definition 2.104. *Given rings S and T, the product of the rings is a ring X equipped with morphisms $\pi_S : X \to S$ and $\pi_T : X \to T$ that satisfies the following universal property: if U is another such ring with morphisms $f : U \to S$ and $g : U \to T$, then there exists a unique morphisms $\Psi : U \to X$:*



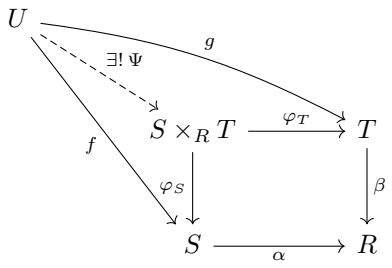
⌟

Note that $X \cong S \times T$. Firstly, observe that by the universal property of X, there exists a unique morphism $\Psi_{\to X} : S \times T \to X$. By the universal property of $\Psi_{X\to} : X \to S \times T$. Now, we can show that $\Psi_{\to X} \circ \Psi_{X\to} = \mathrm{id}_X$. In the diagram above, replace U with X, and replace Ψ with $\Psi_{\to X} \circ \Psi_{X\to}$. However, since the morphism must be unique while id_X also makes the diagram commute, it must be so that $\Psi_{\to X} \circ \Psi_{X\to} = \mathrm{id}_X$. Similarly, we can show that $\Psi_{X\to} \circ \Psi_{\to X} = \mathrm{id}_{S\times T}$. As such, $X \cong S \times T$.

Definition 2.105 (Fibre Products)**.** *Let R, S, and T be rings, and let $\alpha : S \to R$ and $\beta : T \to R$ be ring morphisms. Then, $S \times_R T$ is a ring, called the fibre product, equipped with two morphisms $\varphi_S : S \times_R T \to S$ and $\varphi_T : S \times_R T \to T$, such that $\alpha \circ \varphi_S = \beta \circ \varphi_T$ (as shown in the diagram below which commutes):*



The fibre product also, by definition, satisfies the following universal property: if U is another ring with $f : U \to S$ and $g : U \to T$ such that $\alpha \circ f = \beta \circ g$, then there exists a unique ring morphisms $\Psi : U \to S \times_R T$:



⌟

Note that the fibre product is defined with respect to not only the rings but also the morphisms. Therefore, a more precise notation would be $S \times_{R,\alpha,\beta} T$, since the fibre product is not independent of $\alpha : S \to R$ and $\beta : T \to R$.

Example 2.106. Consider the ring $\mathbb{Z} \times \mathbb{Z}$. Then, $\mathbb{Z} \times_{\mathbb{Z}/3\mathbb{Z}} \mathbb{Z}$ is a subring of $\mathbb{Z}/3\mathbb{Z}$. The elements in the fibre product are:

$$\mathbb{Z} \times_{\mathbb{Z}/3\mathbb{Z}} \mathbb{Z} = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \equiv b \bmod 3\}.$$

⌟

Proposition 2.107. *Let I and J be ideals in R. Then,*

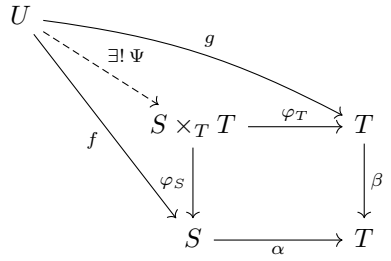
$$R/(I \cap J) \cong (R/I) \times_{R/(I+J)} (R/J).$$

Note that $(R/I) \times_{R/(I+J)} (R/J) = \{(r+I, s+I) \in R/I \times S/I : r+I+J = s+I+J.\}$ ⌟

As a generalization of the fibre product, we have the *limit* in category theory:

Problem 2.108. *Suppose that $\mathcal{R}$ is a set of rings and that $\mathcal{M}$ is a set of homomorphisms between elements of $\mathcal{R}$. We call a ring R together with a homomorphism $\phi_S : R \to S$ for all $S \in \mathcal{R}$ a limit (or inverse limit) if: (1) for each $\alpha : S \to T$ in $\mathcal{M}$, $\phi_T = \alpha \circ \phi_S$, and (2) for any other $(R', \{\phi_{S'} : R' \to S'\}_{S' \in \mathcal{R}})$ that satisfies these requirements, there exists a unique $\Psi : R' \to R$ such that $\phi_{S'} = \phi_S \circ \Psi$ for all $S \in \mathcal{R}$. Then, $(\mathcal{R}, \mathcal{M})$ has a limit.*

Proof. First, observe that if S and T are rings with $\alpha : S \to T$, then the fibre product $S \times_T T$ satiesfies:



Now, we explicitly construct the fibre product. Note that since $\alpha \circ \phi_S = \mathrm{id} \circ \phi_T = \phi_T$, we define the fibre product with projection maps such that after composing with α and id, the morphisms agree:

$$S \times_T T = \{(s,t) \in S \times T \mid \alpha(s) = t\},$$

where

$$\phi_S(s,t) = s, \phi_T(s,t) = t.$$

Now, for an arbitrary set of rings and morphisms between the rings, we can define the fibre product:

$$X := \{(x_S)_{S \in \mathcal{R}} \mid \forall \alpha_{ST} : S \to T, \alpha_{ST}(x_S) = x_T\}.$$

Now, we must verify all the important requirements.
X is a ring. Clearly, $0 = (0, \cdots, 0) \in X$ since 0 is in each ring $S \in \mathcal{R}$ and every ring morphism maps 0 to 0. Similarly, $1 = (1, \cdots, 1) \in X$ since 1 is in each ring. We can define addition and multiplication as:

$$(x_S)_{S \in \mathcal{R}} + (x'_S)_{S \in \mathcal{R}} = (x_S + x'_S)_{S \in \mathcal{R}}$$

and

$$(x_S)_{S \in \mathcal{R}} \cdot (x'_S)_{S \in \mathcal{R}} = (x_S \cdot x'_S)_{S \in \mathcal{R}}.$$

In particular, the elements on the right hand side are also in X, since if $\alpha_{ST}(x_S) = x_T \in T$ and $\alpha_{ST}(x'_S) = x'_T$, then, $\alpha(x_S + x'_S) = \alpha(x_S) + \alpha(x_{S'}) = x_T + x'_T$. Similarly, $\alpha(x_S \cdot x'_S) = \alpha(x_S)\alpha(x'_S) = x_T \cdot x'_T$. Commutativity of addition and multiplication follows from them being respected in each ring $S \in \mathcal{R}$. Associativity and distributivity also follows from the fact that the morphisms preserve these properties: given $x_p + (x'_p + x''_p) = (x_p + x'_p) + x''_p$ in the ring P. Then, $\alpha : P \to Q$ satisfies:

$$\begin{aligned}
\alpha(x_p + (x'_p + x''_p)) &= \alpha(x_p) + \alpha(x'_p + x''_p) \\
&= \alpha(x_p) + \alpha(x'_p) + \alpha(x''_p) \\
&= \alpha(x_p + x'_p) + \alpha(x''_p) \\
&= \alpha((x_p + x'_p) + x''_p).
\end{aligned}$$

Distributivity is also preserved:

$$\begin{aligned}\alpha(x_p \cdot (x'_p + x''_p)) &= \alpha(x_p) \cdot \alpha(x'_p + x''_p) \\ &= \alpha(x_p) \cdot (\alpha(x'_p) + \alpha(x''_p)) \\ &= \alpha(x_p) \cdot \alpha(x'_p) + \alpha(x_p) \cdot \alpha(x''_p) \\ &= \alpha(x_p \cdot x'_p) + \alpha(x_p \cdot x''_p) \\ &= \alpha(x_p \cdot x'_p + x_p \cdot x''_p).\end{aligned}$$

Constructing each morphism $\phi_S : X \to S$. We define these morphisms to be simply the projections:

$$\phi_S((x_P)_{P\in\mathcal{R}}) = x_S.$$

Then, for any $\alpha : S \to T$ such that $\alpha(x_S) = x_T$, with $(x_P)_{P\in\mathcal{R}}$, $(\alpha \circ \phi_S)(x_P)_{P\in\mathcal{R}} = \alpha(x_S) = x_T$. On the other hand, $\phi_T((x_P)_{P\in\mathcal{R}}) = x_T$, so $\alpha \circ \phi_S = \phi_T$. Finally, suppose R' is another ring with morphisms $\phi'_P : R' \to P$ satisfying these properties, then construct the morphism

$$\Psi : R' \to X$$

by

$$\Psi(r') = ((\phi'_P(r'))_{P\in\mathcal{R}}).$$

This is a well-defined map because each ϕ'_P is well-defined and it is a ring morphism since each ϕ'_P is a ring morphism. Then,

$$\phi_S \circ \Psi(r') = \phi_S(((\phi'_P(r'))_{P\in\mathcal{R}})) = \phi'_S(r')$$

for all $r' \in R'$. So, $\phi'_S = \phi_S \circ \Psi$.

Finally, we show uniqueness: suppose $\varphi : R' \to X$ is another such morphism. Then,

$$(\phi_S \circ \varphi)(r') = \phi'_S(r')$$

for each $r' \in R'$ and each $S \in \mathcal{R}$. Since ϕ_S is a projection, $\varphi(r')$ must have $\phi'_S(r')$ in the S-th coordinate. Since this is true for each ring S in $\mathcal{R}$, therefore

$$\varphi(\cdot) = ((\phi'_S(\cdot))_{S\in\mathcal{R}}) = \Psi(\cdot).$$

□

2.14. **Tensor Products of Rings**

Now, we look at another way of constructing a new ring out of two existing rings. Suppose we have a ring R with morphisms to S and T. We construct a new ring, $S \otimes_R T$ which gets maps from S and T that are compatible with respect to R. Furthermore, it will satisfy the universal property that that if A is another ring with maps from S and T that are compatible with respect to R as well, then there exists a unique pushout morphism from $S \otimes_R T \to A$.

2.14.1. *Constructing $S \otimes_R T$ as a quotient polynomial ring.* We first construct this ring via explicitly defining the elements in it. Suppose R is a ring with morphisms to rings S and T as follows:

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\tau}\downarrow & & \\ T & & \end{array}$$

Now, define the polynomial ring in variables x_s and y_t for each $s \in S$ and $t \in T$:

$$\begin{aligned} P &= R[x_s, y_t]_{s\in S, t\in T} \\ &= \{\sum_i r_i x_0^{a_0} x_1^{a_1} \cdots x_s^{a_s} \cdots y_0^{b_0} y_1^{b_1} \cdots y_t^{b_t} \cdots \mid r_i \in R \text{ is 0 for all but finitely many } i, a_n, b_m \in \mathbb{Z}\}. \end{aligned}$$

This is a very large ring that has none of the structure that S and T have, and does not demonstrate how the image of ring R behaves in S or T. For example, in S, $s_1 + s_2 = s_3$, but in our ring P, $x_{s_1} + x_{s_2}$ has no relation to x_{s_3}. Similarly, in $\phi(r)s = s'$ but $x_{\phi(r)s}$ has no relation to $x_{s'}$. To impose these structures, we define the ideal to be generated by some equations:

$$\begin{aligned} I = (&\{x_{s_1} + x_{s_2} - x_{s_1+s_2}, x_{s_1} \cdot x_{s_2} - x_{s_1 s_2}\}_{s_i \in S}, \\ &\{y_{t_1} + y_{t_2} - y_{t_1+t_2}, y_{t_1} \cdot y_{t_2} - y_{t_1 t_2}\}_{t_i \in T},) \\ &\{x_{\phi(r)} - r, y_{\Psi(r)} - r\}_{r\in R}\big). \end{aligned}$$

Then, the tensor product is defined as

$$S \otimes_R T := P/I = R[x_s, y_t]_{s\in S, t\in T}/I.$$

Note that the elements of $S \otimes_R T$ can be written in the form:

$$\sum_i \overline{r_i} \cdot \overline{x_{s_i}} \cdot \overline{y_{t_i}} = \sum_i \overline{x_{\phi(r_i)s_i}} \cdot \overline{y_{t_i}} = \sum_i \overline{x_{s_i}} \cdot \overline{y_{\Psi(r_i)t_i}}$$

since $\overline{x_{s_i}} \cdot \overline{x_{s_j}} = \overline{x_{s_i s_j}}$ and $\overline{r} \cdot \overline{x_s} = \overline{x_{\phi(r)}} \cdot \overline{x_s} = \overline{x_{\phi(r)s}}$.

Since this is a quotient ring, we already have that the tensor product is a ring. Furthermore, we can explicitly write down the identities in this ring:

Proposition 2.109. *$S \otimes_R T$ is a ring with*

$$\begin{aligned} \overline{0} &= \overline{x_0} = \overline{y_0}, \\ \overline{1} &= \overline{x_1} = \overline{y_1}. \end{aligned}$$

The operations are defined via the operations in polynomial rings with the following "simplifications":

$$\overline{x_{s_1}} + \overline{x_{s_2}} = \overline{x_{s_1+s_2}}, \qquad \overline{x_{s_1}} \cdot \overline{x_{s_2}} = \overline{x_{s_1 \cdot s_2}}, \qquad \textit{(similarly for } y_t\textit{'s too)}$$

In particular,

$$\overline{x_{\phi(r)}} = r = \overline{y_{\Psi(r)}}.$$

⌟

Proof. Since $I \trianglelefteq P$, the quotient P/I is a ring whose additive and multiplicative identities are $0_{P/I} = 0 + I$ and $1_{P/I} = 1 + I$, respectively. Indeed, for any $p + I \in P/I$, $(0 + I) + (p + I) = p + I$ and $(1 + I)(p + I) = p + I$. In particular, $\overline{x_{0_S}} = \overline{y_{0_T}} = \overline{0} = 0 + I$. Indeed, since $x_{0_S+0_S} - x_{0_S} - x_{0_S} = -x_{0_S} \in I$, we have $\overline{x_{0_S}} = \overline{0}$, and similarly $\overline{y_{0_T}} = \overline{0}$.

Moreover, $\overline{x_{1_S}} = \overline{y_{1_T}} = \overline{1} = 1 + I$. Since ϕ and Ψ are unital ring homomorphisms, $\phi(1_R) = 1_S$ and $\Psi(1_R) = 1_T$. Hence, by the defining relations of I, $x_{1_S} - 1 = x_{\phi(1_R)} - 1 \in I \implies \overline{x_1} = \overline{1}$ and $y_{1_T} - 1 = y_{\Psi(1_R)} - 1 \in I \implies \overline{y_1} = \overline{1}$ □

Proposition 2.110. *We get the canonical morphisms*

$$\iota_S : S \to S \otimes_R T, \qquad s \to \overline{x_s}$$

and

$$\iota_T : T \to S \otimes_R T, \qquad t \to \overline{y_t}.$$

As such, the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\Psi}\downarrow & & \downarrow{\scriptstyle\iota_S} \\ T & \xrightarrow[\iota_T]{} & S \otimes_R T \end{array}$$

⌟

Changing notation to tensors: Recall that every element in $S \otimes_R T$ is of the form $\sum_i \overline{r_i} \cdot \overline{x_{s_i}} \cdot \overline{y_{t_i}} = \sum_i \overline{x_{\phi(r_i)s_i}} \cdot \overline{y_{t_i}} = \sum_i \overline{x_{s_i}} \cdot \overline{y_{\Psi(r_i)t_i}}$. Now, we switch to the standard notation by first definiting pure tensors and then defining every element in the tensor product ring to be finite sums of pure tensors.

Definition 2.111 (Pure Tensors)**.** *We define a pure tensor to be the following:*

$$s \otimes t := \overline{x_s} \cdot \overline{y_t}.$$

In particular,

$$s \otimes 1 = \overline{x_s}, \qquad 1 \otimes t = \overline{y_t}.$$

Note that

$$s \otimes t = (s \otimes 1)(1 \otimes t).$$

⌟

Now, we rewrite the definition of the tensor product ring in terms of this notation.

Definition 2.112 (Tensor Product Ring)**.** *Given rings R, S, T and ring morphisms $\phi : R \to S$ and $\Psi : R \to T$, we define the tensor product ring $S \otimes_R T$. Every element of $S \otimes_R T$ can be expressed as a finite sum*

$$\sum_i s_i \otimes t_i, \qquad s_i \in S,\ t_i \in T.$$

The following relations hold:

$$(s_1 + s_2) \otimes t = s_1 \otimes t + s_2 \otimes t,$$

$$s \otimes (t_1 + t_2) = s \otimes t_1 + s \otimes t_2,$$

$$(s_1 \otimes t_1)(s_2 \otimes t_2) = s_1 s_2 \otimes t_1 t_2,$$

and, for every $r \in R$,

$$r(s \otimes t) = \phi(r)s \otimes t = s \otimes \Psi(r)t.$$

Moreover, the ring has the following additive and multiplicative identities:

$$0 \otimes 1 = 1 \otimes 0 = 0_{S \otimes_R T},$$

and

$$1 \otimes 1 = 1_{S \otimes_R T}.$$

⌟

Now, we look at the definition in terms of the universal property, which turns out to be significantly more useful in practice:

Proposition 2.113 (Universal Property of Tensor Products $S \otimes_R T$)**.** *Let R, S, T be rings with morphisms*

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\Psi}\downarrow & & \\ T & & \end{array}$$

Then, the tensor product $S \otimes_R T$ comes with canonical morphisms $\iota_S : S \to S \otimes_R T$ by

$$\iota_S(s) = s \otimes 1$$

and $\iota_T : T \to S \otimes_R T$ by

$$\iota_T(t) = 1 \otimes t,$$

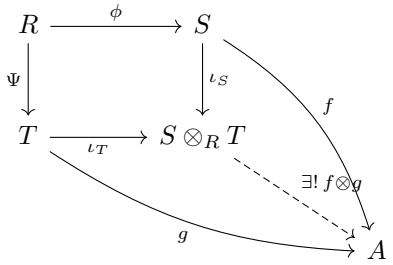
such that

$$\iota_S \circ \phi = \iota_T \circ \Psi,$$

i.e. the following commutes:

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\Psi}\downarrow & & \downarrow{\scriptstyle\iota_S} \\ T & \xrightarrow[\iota_T]{} & S \otimes_R T \end{array}$$

This satisfies the universal property that if A is a ring with $f : S \to A$ and $g : T \to A$ such that $f \circ \phi = g \circ \Psi$, then there exists a unique morphism $f \otimes g : S \otimes_R T \to A$, i.e.



In particular,

$$(f \otimes g)(s \otimes t) = f(s)g(t),$$

and

$$(f \otimes g)\left(\sum_{i=1}^{n} s_i \otimes t_i\right) = \sum_{i=1}^{n} f(s_i)g(t_i).$$

If $S \otimes'_R T$ satisfies the same universal property, then $S \otimes_R T \cong S \otimes'_R T$. In other words, $S \otimes_R T$ is unique up to unique isomorphism. ⌟

Proof. Recall that $s \otimes t = \overline{x_s}\,\overline{y_t} = \iota_S(s)\iota_T(t)$.

(1) Check that the canonical maps $\iota_S : S \to S \otimes_R T$ and $\iota_T : T \to S \otimes_R T$ are ring morphisms. We check this for ι_S; the same argument works for ι_T. For addition,

$$\begin{aligned} \iota_S(s_1 + s_2) &= (s_1 + s_2) \otimes 1 \\ &= s_1 \otimes 1 + s_2 \otimes 1 \\ &= \iota_S(s_1) + \iota_S(s_2). \end{aligned}$$

For multiplication,

$$\begin{aligned} \iota_S(s_1 s_2) &= (s_1 s_2) \otimes 1 \\ &= (s_1 \otimes 1)(s_2 \otimes 1) \\ &= \iota_S(s_1)\iota_S(s_2). \end{aligned}$$

Moreover, $\iota_S(1) = 1 \otimes 1$, which is the multiplicative identity in $S \otimes_R T$, since for every $s \otimes t$, $(s \otimes t)(1 \otimes 1) = s \otimes t$. Finally, $\iota_S(0) = 0 \otimes 1$. Since

$$\begin{aligned} 0 \otimes 1 &= (0 + 0) \otimes 1 \\ &= 0 \otimes 1 + 0 \otimes 1, \end{aligned}$$

cancellation gives $0 \otimes 1 = 0$.

(2) Check that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ {\scriptstyle\Psi}\downarrow & & \downarrow{\scriptstyle\iota_S} \\ T & \xrightarrow[\iota_T]{} & S \otimes_R T. \end{array}$$

For every $r \in R$,

$$\begin{aligned} \iota_S(\phi(r)) &= \phi(r) \otimes 1 \\ &= 1 \otimes \Psi(r) \\ &= \iota_T(\Psi(r)). \end{aligned}$$

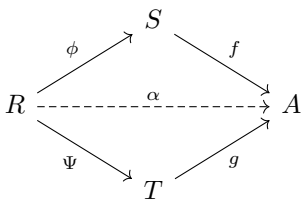
Hence $\iota_S \circ \phi = \iota_T \circ \Psi$.

(3) We now construct the morphism $f \otimes g : S \otimes_R T \to A$.

Suppose we are given ring morphisms $f : S \to A$ and $g : T \to A$ satisfying $f \circ \phi = g \circ \Psi$.

Define $\alpha : R \to A$ by $\alpha(r) = f(\phi(r)) = g(\Psi(r))$.

Thus we have the diagram



Recall that $P = R[x_s, y_t]_{s \in S, t \in T}$. By the universal property of polynomial rings in Proposition 2.172, there exists a unique ring morphism $F : P \to A$ such that $F|_R = \alpha$, $F(x_s) = f(s)$ for every $s \in S$, and $F(y_t) = g(t)$ for every $t \in T$.

We now show that the ideal I defining $S \otimes_R T$ satisfies $I \subseteq \ker F$. For the additive relations in S,

$$\begin{aligned} F(x_{s_1} + x_{s_2} - x_{s_1+s_2}) &= f(s_1) + f(s_2) - f(s_1 + s_2) \\ &= 0. \end{aligned}$$

For the multiplicative relations in S,

$$\begin{aligned} F(x_{s_1} x_{s_2} - x_{s_1 s_2}) &= f(s_1) f(s_2) - f(s_1 s_2) \\ &= 0. \end{aligned}$$

Similarly, $F(y_{t_1} + y_{t_2} - y_{t_1+t_2}) = 0$ and $F(y_{t_1} y_{t_2} - y_{t_1 t_2}) = 0$. Finally, for every $r \in R$,

$$F(x_{\phi(r)} - r) = f(\phi(r)) - \alpha(r) = 0,$$

and

$$F(y_{\Psi(r)} - r) = g(\Psi(r)) - \alpha(r) = 0.$$

Thus every generator of I lies in $\ker F$, so $I \subseteq \ker F$. Therefore, by the universal property of quotient rings, there exists a unique ring morphism

$$f \otimes g : \frac{R[x_s, y_t]_{s \in S, t \in T}}{I} \longrightarrow A.$$

Since $S \otimes_R T = P/I$, this gives $f \otimes g : S \otimes_R T \to A$. The map $f \otimes g$ is simply the map induced by F after passing to the quotient modulo I. We check that it gives the desired maps from S and T:

$$\begin{aligned} (f \otimes g)(s \otimes 1) &= (f \otimes g)(\overline{x_s}) \\ &= F(x_s) \\ &= f(s), \end{aligned}$$

and similarly,

$$\begin{aligned}(f \otimes g)(1 \otimes t) &= (f \otimes g)(\overline{y_t}) \\ &= F(y_t) \\ &= g(t).\end{aligned}$$

Consequently,

$$\begin{aligned}(f \otimes g)(s \otimes t) &= (f \otimes g)\big((s \otimes 1)(1 \otimes t)\big) \\ &= (f \otimes g)(s \otimes 1)(f \otimes g)(1 \otimes t) \\ &= f(s)g(t).\end{aligned}$$

Hence, more generally, $(f \otimes g)\left(\sum_{i=1}^n s_i \otimes t_i\right) = \sum_{i=1}^n f(s_i)g(t_i)$.

(4) We now prove uniqueness.

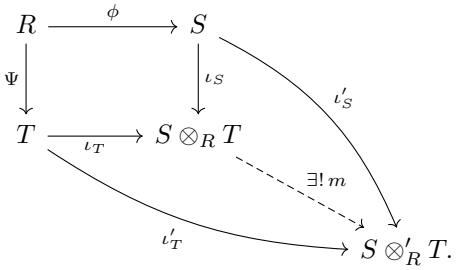
Suppose $h : S \otimes_R T \to A$ is another ring morphism satisfying $h \circ \iota_S = f$ and $h \circ \iota_T = g$. Equivalently, $h(s \otimes 1) = f(s)$ and $h(1 \otimes t) = g(t)$. Then, for every pure tensor $s \otimes t$,

$$\begin{aligned}h(s \otimes t) &= h\big((s \otimes 1)(1 \otimes t)\big) \\ &= h(s \otimes 1)h(1 \otimes t) \\ &= f(s)g(t) \\ &= (f \otimes g)(s \otimes t).\end{aligned}$$

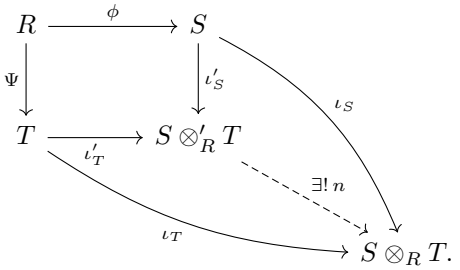
Since every element of $S \otimes_R T$ is a finite sum of pure tensors, it follows that $h = f \otimes g$. So, $f \otimes g$ is unique.

(5) Finally, we show that $S \otimes_R T$ is unique up to unique isomorphism. Suppose $S \otimes'_R T$ is another ring satisfying the same universal property, with canonical maps $\iota'_S : S \to S \otimes'_R T$ and $\iota'_T : T \to S \otimes'_R T$.

Since $S \otimes'_R T$ receives compatible maps from S and T, the universal property of $S \otimes_R T$ gives a unique map $m : S \otimes_R T \to S \otimes'_R T$ such that the following diagram commutes:



Conversely, since $S \otimes_R T$ receives compatible maps from S and T, the universal property of $S \otimes'_R T$ gives a unique map $n : S \otimes'_R T \to S \otimes_R T$ such that



Now consider $n \circ m : S \otimes_R T \to S \otimes_R T$. This map satisfies $(n \circ m) \circ \iota_S = \iota_S$ and $(n \circ m) \circ \iota_T = \iota_T$.

But $\mathrm{id}_{S \otimes_R T}$ also has these properties. By uniqueness in the universal property, $n \circ m = \mathrm{id}_{S \otimes_R T}$.

Similarly, $m \circ n = \mathrm{id}_{S \otimes'_R T}$.

Therefore m and n are inverse isomorphisms, and hence $S \otimes_R T \cong S \otimes'_R T$.

In fact, this isomorphism is the unique isomorphism compatible with the canonical maps from S and T. Thus $S \otimes_R T$ is unique up to unique isomorphism.

□

Examples 2.114. *Here are some examples of tensor products.*

(1) *We have* $\mathbb{Z}/(2) \otimes_{\mathbb{Z}} \mathbb{Z}/(3) \cong (0)$.
Proof using elements. *Proof 1. Consider any pure tensor* $\overline{a} \otimes \overline{b}$ *in* $\mathbb{Z}/(2) \otimes_{\mathbb{Z}} \mathbb{Z}/(3)$. *Then*

$$2(\overline{a} \otimes \overline{b}) = \overline{2a} \otimes \overline{b} = \overline{0} \otimes \overline{b} = 0,$$

and

$$3(\overline{a} \otimes \overline{b}) = \overline{a} \otimes \overline{3b} = \overline{a} \otimes \overline{0} = 0.$$

Therefore,

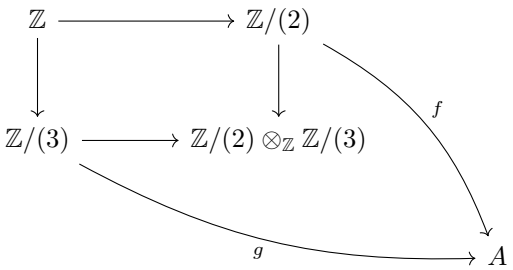
$$\begin{aligned} \overline{a} \otimes \overline{b} &= 3(\overline{a} \otimes \overline{b}) - 2(\overline{a} \otimes \overline{b}) \\ &= 0. \end{aligned}$$

Thus every pure tensor is zero. Since every element of $\mathbb{Z}/(2) \otimes_{\mathbb{Z}} \mathbb{Z}/(3)$ *is a finite sum of pure tensors, the entire tensor product is zero.*

Proof 2. Every pure tensor is generated additively by $\overline{1} \otimes \overline{1}$. *Indeed,* $\overline{a} \otimes \overline{b} = ab(\overline{1} \otimes \overline{1})$. *But* $2(\overline{1} \otimes \overline{1}) = 0$ *and* $3(\overline{1} \otimes \overline{1}) = 0$. *Hence*

$$\begin{aligned} \overline{1} \otimes \overline{1} &= 3(\overline{1} \otimes \overline{1}) - 2(\overline{1} \otimes \overline{1}) \\ &= 0. \end{aligned}$$

Therefore every element of the tensor product is zero.
Proof using the universal property. *Suppose* A *is a ring with ring morphisms* $f : \mathbb{Z}/(2) \to A$ *and* $g : \mathbb{Z}/(3) \to A$ *compatible with the canonical maps from* $\mathbb{Z}$, *as in the following diagram:*



Since f *and* g *are unital ring morphisms,* $f(\overline{1}) = 1_A$ *and* $g(\overline{1}) = 1_A$.
Now,

$$\begin{aligned} f(2\overline{1}) &= f(\overline{1} + \overline{1}) \\ &= f(\overline{1}) + f(\overline{1}) \\ &= 2 \cdot 1_A. \end{aligned}$$

But $2\overline{1} = \overline{0}$ *in* $\mathbb{Z}/(2)$, *so* $f(2\overline{1}) = f(\overline{0}) = 0$. *Thus* $2 \cdot 1_A = 0$.
Similarly, since $3\overline{1} = \overline{0}$ *in* $\mathbb{Z}/(3)$, *we have* $3 \cdot 1_A = 0$.
Therefore,

$$\begin{aligned} 1_A &= 3 \cdot 1_A - 2 \cdot 1_A \\ &= 0. \end{aligned}$$

Hence $A = (0)$.
Thus any ring receiving compatible morphisms from $\mathbb{Z}/(2)$ *and* $\mathbb{Z}/(3)$ *is necessarily the zero ring. Moreover, the zero ring itself receives the canonical morphisms from both* $\mathbb{Z}/(2)$ *and* $\mathbb{Z}/(3)$, *and there is a unique morphism from* (0) *to any such ring* $A = (0)$. *Hence* (0) *satisfies the universal property of the tensor product. By uniqueness up to unique isomorphism,* $\mathbb{Z}/(2) \otimes_{\mathbb{Z}} \mathbb{Z}/(3) \cong (0)$.
(2) *More generally,* $\mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n) \cong \mathbb{Z}/(\gcd(m, n))$.

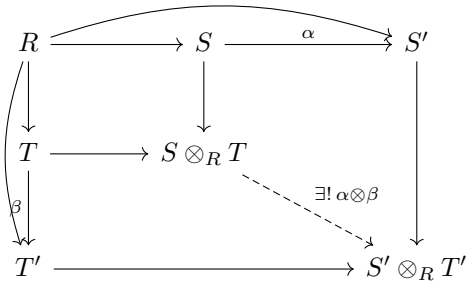
Proposition 2.115. *Suppose we have commuting diagrams*

$$\begin{array}{ccc} S & \xrightarrow{\alpha} & S' \\ & \nwarrow \quad \nearrow & \\ & R & \end{array} \qquad \text{and} \qquad \begin{array}{ccc} T & \longrightarrow & T' \\ & \nwarrow \quad \nearrow & \\ & R & \end{array}$$

so that $\alpha : S \to S'$ and $\beta : T \to T'$ are compatible with the given morphisms from R. Then there exists a unique ring morphism $\alpha \otimes \beta : S \otimes_R T \to S' \otimes_R T'$ such that $(\alpha \otimes \beta)(s \otimes t) = \alpha(s) \otimes \beta(t)$. ⌟

Proof sketch. Consider the canonical morphisms $S' \to S' \otimes_R T'$ and $T' \to S' \otimes_R T'$. Composing these with α and β, respectively, gives compatible morphisms $S \to S' \otimes_R T'$ and $T \to S' \otimes_R T'$.

Thus, by the universal property of $S \otimes_R T$, there exists a unique morphism $\alpha \otimes \beta : S \otimes_R T \to S' \otimes_R T'$ making the following diagram commute:



In particular,

$$\begin{aligned} (\alpha \otimes \beta)(s \otimes t) &= (\alpha \otimes \beta)\big((s \otimes 1)(1 \otimes t)\big) \\ &= (\alpha \otimes \beta)(s \otimes 1)(\alpha \otimes \beta)(1 \otimes t) \\ &= (\alpha(s) \otimes 1)(1 \otimes \beta(t)) \\ &= \alpha(s) \otimes \beta(t). \end{aligned}$$

□

Proposition 2.116. *Let $\Psi : R \to T$ be a ring morphism. Then $R \otimes_R T \cong T$.* ⌟

Proposition 2.117. *$S \otimes_R T \cong T \otimes_R S$ by $s \otimes t \to t \otimes s$.* ⌟

Proposition 2.118. *Let $I \trianglelefteq R$. Then*

$$(R/I) \otimes_R T \cong T/\Psi(I)T,$$

where $\Psi : R \to T$ is a ring morphism and $\Psi(I)T = \left\{ \sum_{j=1}^{n} \Psi(i_j)t_j \;\middle|\; i_j \in I,\ t_j \in T \right\}$. ⌟

Proof. First, construct a morphism $\Phi : (R/I) \otimes_R T \to T/\Psi(I)T$.

Note that for any $(r + I) \otimes t$,

$$\begin{aligned} (r + I) \otimes t &= r(1 + I) \otimes t \\ &= (1 + I) \otimes \Psi(r)t. \end{aligned}$$

Thus, any element of $(R/I) \otimes_R T$ can be written as

$$\begin{aligned} \sum_j (r_j + I) \otimes t_j &= \sum_j (1 + I) \otimes \Psi(r_j)t_j \\ &= (1 + I) \otimes \left(\sum_j \Psi(r_j)t_j \right). \end{aligned}$$

Now, in R/I, we have $i + I = 0$ for every $i \in I$. Thus, for any $t \in T$,

$$\begin{aligned} (1 + I) \otimes \Psi(i)t &= (i + I) \otimes t \\ &= 0 \otimes t \\ &= 0. \end{aligned}$$

Hence every element of the form $\Psi(i)t$ is killed in the tensor product. Therefore, we quotient T by the ideal $\Psi(I)T$.

Define $\Phi((r+I)\otimes t) = \Psi(r)t + \Psi(I)T$.

Conversely, define $\Phi^{-1}(t + \Psi(I)T) = (1+I)\otimes t$.

These maps are mutually inverse, and hence $(R/I)\otimes_R T \cong T/\Psi(I)T$. □

Proposition 2.119. *Let $D \subseteq S$ be a multiplicative set. Then:*

(1) *The set $D \otimes 1 := \{d \otimes 1 \in S \otimes_R T \mid d \in D\}$ is a multiplicative subset of $S \otimes_R T$.*
(2) *There is an isomorphism $D^{-1}S \otimes_R T \cong (D\otimes 1)^{-1}(S \otimes_R T)$, given by $\frac{s}{d} \otimes t \longmapsto \frac{s\otimes t}{d\otimes 1}$.*

⌟

Proposition 2.120. *Let $D \subseteq R$ be a multiplicative set. Then $(D^{-1}R)\otimes_R T \cong (D\otimes 1)^{-1}(R\otimes_R T) \cong \Psi(D)^{-1}T$.* ⌟

Proposition 2.121. *Suppose we have ring morphisms*

$$\begin{array}{ccc} R & \longrightarrow & S \\ & \searrow & \\ & & T. \end{array}$$

Then $S \otimes_R T[x] \cong (S\otimes_R T)[x]$. ⌟

Proposition 2.122. *The following natural isomorphisms hold:*

(1) $S \otimes_R (T \otimes_R U) \cong (S\otimes_R T)\otimes_R U$.
(2) $S \otimes_R (T \times U) \cong (S\otimes_R T) \times (S \otimes_R U)$.

⌟

Proposition 2.123. $S \otimes_R (T[x]) \cong (S\otimes_R T)[x]$. ⌟

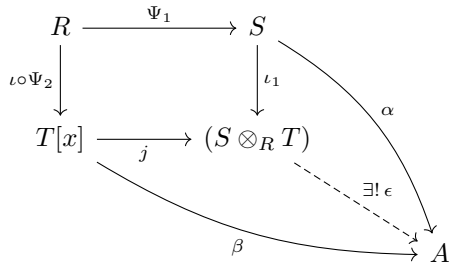
Proof. We show that $(S\otimes_R T)[x]$ satisfies the same universal property as $S\otimes_R (T[x])$. First, we know, by the existence of $S\otimes_R T$, that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\Psi_1} & S \\ {\scriptstyle\Psi_2}\downarrow & & \downarrow{\scriptstyle\iota_1} \\ T & \xrightarrow[\iota_2]{} & S\otimes_R T \\ {\scriptstyle\iota}\downarrow & & \\ T[x] & & \end{array}$$

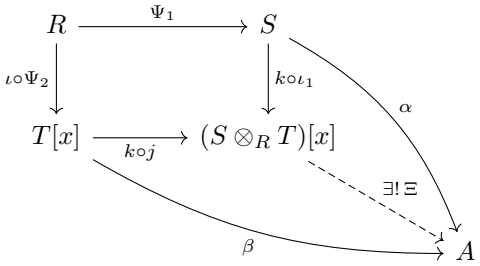
By the universal property of polynomial rings, we then have a unique morphism $j : T[x] \to S\otimes_R T$ such that $\iota_2 = j\circ\iota$. In other words, the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\Psi_1} & S \\ {\scriptstyle\Psi_2}\downarrow & & \downarrow{\scriptstyle\iota_1} \\ T & \overset{\iota_2}{\searrow} & \\ {\scriptstyle\iota}\downarrow & & \\ T[x] & \xrightarrow[j]{} & S\otimes_R T \end{array}$$

Now, suppose A is a ring with morphisms $\beta : T[x] \to A$ and $\alpha : S \to A$ such that $\alpha\circ\Psi_1 = \beta\circ(\iota\circ\Psi_2)$. Then, we have morphisms $S \to A$ and $T \to A$ that are compatible with R. By the universal property of $S\otimes_R T$, there exists a unique morphism $\epsilon : S\otimes_R T \to A$ that makes the following diagram commute:



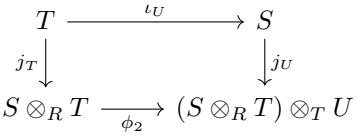
Now, by the universal property of polynomial rings, with $k : S \otimes_R T \to (S \otimes_R T)[x]$ being the inclusion morphism, there exists a unique morphism $\Xi : (S \otimes_R T)[x] \to A$ such that $\Xi \circ k = \epsilon$. So, we have that



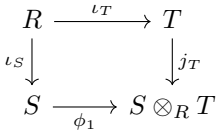
As such, $(S \otimes_R T)[x]$ satisfies the same universal property as $S \otimes_R (T[x])$. Since the tensor product $S \otimes_R (T[x])$ is unique up to a unique isomorphism, therefore, $S \otimes_R (T[x]) \cong (S \otimes_R T)[x]$. □

Proposition 2.124. $(S \otimes_R T) \otimes_T U \cong S \otimes_R U$. ⌟

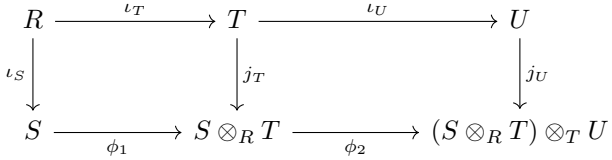
Proof. Given $(S \otimes_R T) \otimes_T U$ exists, we have that the following diagram commutes:



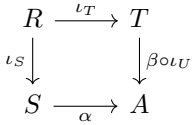
By the existence of $S \otimes_R T$, the following diagram commutes:



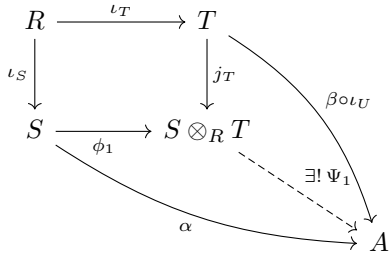
As such, the following diagram commutes too:



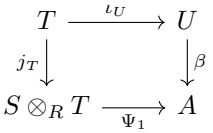
Consider a ring A with morphisms $\alpha : S \to A$ and $\beta : U \to A$. Then, we see that the following diagram commutes:



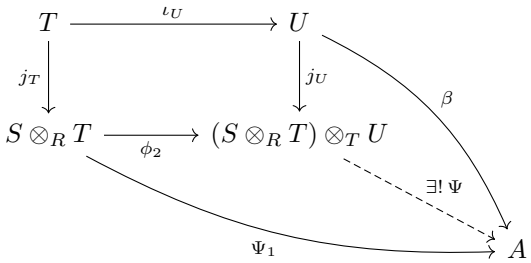
As such, by the universal property of $S \otimes_R T$, there exists a unique morphism $\Psi_1 : S \otimes_R T \to A$ such that



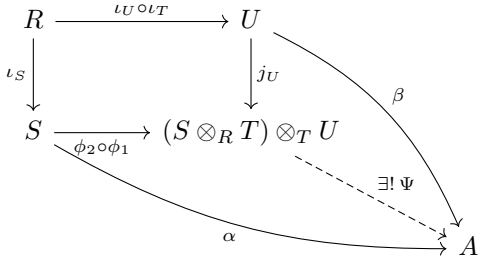
This diagram already has the following information in it:



Then, by the universal property of $(S \otimes_R T) \otimes_T U$, there exists a unique ring morphism $\Psi : (S \otimes_R T) \otimes_T U \to A$, such that



Now, we see that $(S \otimes_R T) \otimes_T U$ satisfies the same universal property as $S \otimes_R U$ which we pictorially show as:



As such, $S \otimes_R U \cong (S \otimes_R T) \otimes_T U$. □

Proposition 2.125. *If I and J are ideals of R, then $R/I \otimes_R R/J \cong R/(I+J)$.* ⌟

Proof. Before proving this, we will require the following:

Claim: $\iota : R \to R/J$ is an epimorphism in the category of commutative, unital rings. To show this, Suppose $\iota : R \to R/J$ and $\phi : R \to S$ for some ring S are ring morphisms. Now, suppose there are two morphisms $\alpha_1 : R/J \to S$ and $\alpha_2 : R/J \to S$ such that $\phi = \alpha_1 \circ \iota = \alpha_2 \circ \iota$. Note that α_1 and α_2 are ring morphisms, so they send 0 to 0. Therefore, for any $r \in J$, $(\alpha \circ \iota)(r) = 0$, so $\phi(r) = 0$ for all $r \in J$. By the universal property of quotient rings, there exists a unique morphism $p : R/J \to S$ such that $\phi = p \circ \iota$. Since the morphism must be unique, $\alpha_1 = \alpha_2$. This shows that ι is an epimorphism. Alternative wayo of seeing this is that ι is surjective, so for any $\bar{r} \in R/J$, there exists some $r \in R$ such that $\iota(r) = \bar{r}$. Then, $\alpha_1(\bar{r}) = \alpha_1(\iota(r)) = \alpha_2(\iota(r)) = \alpha_2(\bar{r})$ for all $\bar{r} \in R/J$ (and r being its preimage under the surjective morphism that is ι).

Now, we show that $R/(I+J) = R/(I,J)$ has the same universal property as $R/I \otimes_R R/J$. Note that we have the natural morphism $\pi : R/J \to R/(J,I)$ by sending $\pi(r+J) \to r+I+J$. We also have the natural morphism $\pi' : R/I \to R/(J,I)$ by $\pi'(r+I) = r+I+J$. Then, with $\iota : R \to R/J$ (which is $\iota(r) = r+J$) and $\iota' : R \to R/I$ (which is $\iota'(r) = r+I$), we see that

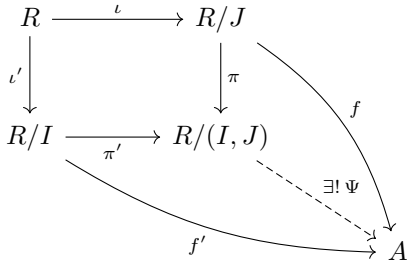
$(\pi \circ \iota)(r) = \pi(r + J) = r + I + J = \pi'(r + I) = \pi'(\iota'(r))$ for all $r \in R$. As such, the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\iota} & R/J \\ {\scriptstyle \iota'}\downarrow & & \downarrow{\scriptstyle \pi} \\ R/I & \xrightarrow[\pi']{} & R/(I, J) \end{array}$$

Now, let A be an arbitary ring and suppose we have morphisms $f : R/J \to A$ and $f' : R/I \to A$ such that $f' \circ \iota' = f \circ \iota$. Now, f is a ring morphism, so it sends 0 to 0. This means for any $x \in J \trianglelefteq R$, $f(\iota(x)) = 0$. Then, $f'(\iota'(x)) = 0$ for all $x \in J$. Since $f' \circ \iota' = f \circ \iota$, $f'(r + I) = 0$ for all $r \in J$. Then, by the universal property of $R/(I, J)$, there exists a unique ring morphism $\Psi : R/(I, J) \to A$ such that

$$\Psi \circ \pi' = f'.$$

We now observe that $\Psi(\pi'(\iota'(r))) = f'(\iota'(r))$ for all $r \in R$. Now, we know that ι' is an epimorphism, so $\Psi \circ \pi' = f'$. On the other hand, $\Psi(\pi(\iota(r))) = \Psi(\pi(r + J)) = \Psi(r + I + J) = \Psi(\pi'(r+I)) = f'(r+I) = f'(\iota'(r)) = f(\iota(r))$. Same argument shows that $\Psi \circ \pi = f$. So, we have the following commutative diagram (note that we already argued that Ψ is a unique morphism by the universal property of quotient rings):



This means, $R/(I, J)$ satisfies the same universal property as $R/I \otimes_R R/J$, which proves the statement. □

2.15. **Euclidean Domains**

Definition 2.126 (Norm)**.** *Let R be an integral domain. Then, any function $N : R \to \mathbb{Z}_{\geq 0}$ with $N(0) = 0$ is called a norm on the integral domain R. If $N(x) > 0$ for all $x \neq 0$, then N is called a positive norm.* ⌟

Definition 2.127 (Euclidean Domains)**.** *Let R be an integral domain. R is called a Euclidean Domain if there exists a norm N on R such that for all $x, y \in R$ such that $y \neq 0$, there exists $q, r \in R$ such that*

$$x = qy + r, \text{ where } r = 0 \text{ or } N(r) < N(y).$$

We say q is the quotient and r is the remainder. ⌟

Euclidean Domains allow us to use the Euclidean algorithm.
Euclidean Algorithm. Let $a, b \in R$. Then, we can apply successive "divisions":

$$a = q_0 b + r_0 \tag{3}$$

$$b = q_1 r_0 + r_1 \tag{4}$$

$$r_0 = q_2 r_1 + r_2 \tag{5}$$

$$r_1 = q_3 r_2 + r_3 \tag{6}$$

$$\vdots$$

$$r_{n-2} = q_n r_{n-1} + r_n \tag{7}$$

$$r_{n-1} = q_{n+1} r_n. \tag{8}$$

Here r_n is the last nonzero remainder. Such an r_n must exist since

$$N(b) > N(r_0) > N(r_1) > \cdots > N(r_n)$$

is a strictly decreasing sequence of positive integers whenever the remainders are nonzero, and therefore the process cannot continue indefinitely.

Proposition 2.128 (All ideals in Euclidean Domains are principal)**.** *Every ideal in a Euclidean Domain is a principal ideal, i.e. if $I \neq 0$ is an ideal in the Euclidean Domain R, then $I = (x)$ where $x \neq 0$ and x is the element of I with the minimum norm.* ⌟

Proof. If $I = (0)$, we are done, by setting $x = 0$. Assume that $I \neq 0$. Then, consider the set $\{N(r) \mid r \in I\} \subseteq \mathbb{Z}$. By the well-ordering of $\mathbb{Z}$, this set has an element x that has the minimum norm. Now, $(x) \subseteq I$ since $x \in I$. On the other hand, consider any $r \in I$. Then, $r = xq + d$ for some q and d in R. Now, $d = r - xq \in I$ since $r \in I$ and $qx \in (x) \in I$. Then, $d \in I$ with $N(d) < N(x)$ if $d \neq 0$, which is impossible since x is the element of I with minimum norm. Therefore, $d = 0$, which shows $r = xq \in (x)$, so $I \subseteq (x)$. This shows $I = (x)$. □

Examples 2.129. *Here are examples and non-examples of Euclidean Domains.*

1. *All fields are Euclidean Domains, since for all $x, y \in F$ such that $y \neq 0$, we have that $x = qy$ for $q = xy^{-1}$. We can use the trivial zero norm: $N(r) = 0$ for all r.*
2. *$\mathbb{Z}$ is a Euclidean Domain with $N(x) = |x|$.*
3. *If F is a field, then $F[x]$ is a Euclidean Domain, with the norm $N(p(x)) = \deg p(x)$. So, $\mathbb{Q}[x]$ is a Euclidean Domain.*
4. *$\mathbb{Z}[x]$ is not a Euclidean Domain. We have previously shown that the ideal $(2, x) \trianglelefteq \mathbb{Z}[x]$ is not a principal ideal.*

Definition 2.130 (Divisors and GCD)**.** *Let R be a commutative ring and let $x, y \in R$ such that $y \neq 0$.*

1. *x is a multiple of y if there exists $m \in R$ such that $x = my$. We say y divides x, i.e. $y \mid x$.*
2. *A greatest common divisor of x and y is a non-zero $d \in R$ such that (i) $d \mid x$ and $d \mid y$, and (ii) if $d' \mid x$ and $d' \mid y$, then $d' \mid d$.*

⌟

Proposition 2.131. *In a commutative ring R, if $x, y \in R$ are nonzero and $(x, y) = (d)$ for some $d \in R$, then d is a greatest common divisor of x and y.* ⌟

Proof. Since $(x, y) = (d)$, we have $x, y \in (d)$. Thus, $d \mid x$ and $d \mid y$, so d is a common divisor of x and y.

Now, suppose $c \in R$ is any common divisor of x and y. Then $c \mid x$ and $c \mid y$. Since $d \in (x, y)$, there exist $r, s \in R$ such that

$$d = rx + sy.$$

Since c divides both x and y, it follows that $c \mid d$. Therefore, d is a greatest common divisor of x and y. □

Proposition 2.132. *Let R be an integral domain. Then, if $x, y \in R$ generate the same principal ideal, i.e. $(x) = (y)$, then $x = yu$ for some unit $u \in R^\times$.*

In particular, if x and y are both greatest common divisors of some $a, b \in R$, then $x = yu$ for some unit $u \in R^\times$. ⌟

Proof. Since $(x) = (y)$, we have $x \in (y)$ and $y \in (x)$. Thus, there exist $r, s \in R$ such that

$$x = ry, \qquad y = sx.$$

Substituting the second equation into the first gives

$$x = r(sx) = (rs)x.$$

If $x = 0$, then $y = sx = 0$, so $x = yu$ with $u = 1 \in R^\times$.

Otherwise, since R is an integral domain and $x \neq 0$, cancellation gives $rs = 1$. Hence, r is a unit with inverse s. Taking $u = r$, we obtain $x = yu$, as desired.

Now, suppose x and y are both greatest common divisors of a and b. By the defining property of a greatest common divisor, we have $x \mid y$ and $y \mid x$. Consequently, $(x) = (y)$. By the first part of the proposition, there exists a unit $u \in R^\times$ such that $x = yu$. □

Proposition 2.133 (Euclidean Domains' ideals are generated by GCDs)**.** *Let R be a Euclidean Domain. Let $x \neq 0$ and $y \neq 0$ in R. Let r_n be the last non-zero remainder of the Euclidean algorithm on x and y. Then,*

$$r_n = \gcd(x, y),$$
$$(r_n) = (x, y),$$

and

$$\exists \alpha, \beta \in R, \ \text{s.t.} \ \alpha x + \beta y = r_n.$$

⌟

2.16. **Principal Ideal Domains (PID)**

Definition 2.134 (Principal Ideal Domains)**.** *A principal ideal domain (PID) is an integral domain in which every ideal is a principal ideal.* ⌟

Examples 2.135. (1) *$\mathbb{Z}$ is a PID, as we have already shown before.*
(2) *$\mathbb{Z}[x]$ is not a PID since the ideal $(2, x)$ can not be a principal ideal.*
(3) *All Euclidean Domains are PIDs.*
(4) *If F is a field, then $F[x]$ is a Euclidean domain, so $F[x]$ is a PID.*
(5) *The quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is not a PID as the ideal $(3, 1 + \sqrt{-5})$ is a non-principal ideal.*

Observation. *It is possible for two non-principal ideals' product to be a principal ideal. For example, in $\mathbb{Z}[\sqrt{-5}]$, the ideals $(3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5})$ is principal even though neither of the two individual ideals is principal.*

Observation. *All Euclidean Domains are PIDs. All PIDs are not Euclidean Domains. For example, the ring $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ is a PID but not a Euclidean Domain.*

Proposition 2.136 (GCD generates ideals in PIDs)**.** *Let R be a PID. Let $a \neq 0$ and $b \neq 0$ in R. Let $(d) = (a, b)$ as ideals. Then, $d = \gcd(a, b)$ and there exists $x, y \in R$ such that $d = ax + by$. Furthermore, d is unique up to a unit of R, i.e. if $(d) = (a, b) = (d')$, then $d = ud'$ for some unit $u \in R^\times$.* ⌟

Proposition 2.137. *Every non-zero prime ideal in a PID is a maximal ideal.* ⌟

Proof. Suppose (p) is a prime ideal in a PID and $(p) \neq 0$. Suppose, for contradiction, $(p) \subset (q) \subset R$. Then, since $q \mid p$, i.e. there exists $s \in R$ such that $p = qs$. Then, either $q \in (p)$ or $s \in (p)$.

Case 1: $q \in (p)$. Then, $(q) \subset (p)$, so $(p) = (q)$, which is a contradiction.

Case 2: $s \in (p)$, so $(s) \subset (p)$. Consider any $r \in (p)$, i.e. there exists $\tilde{r} \in R$ such that $r = \tilde{r}p = \tilde{r}qs \in (s)$, so $(p) \subset (s)$, so $(s) = (p)$. Then, $s = pu$ for some unit u, which suggests that $p = qs = qpu$, so $p(qu - 1) = 0$. Since $p \neq 0$, and an integral domain cannot have zero divisors, therefore, $qu = 1$, so $(q) = R$, which is also a contradiction. As such (p) is a maximal ideal. □

Proposition 2.138. *If R is a commutative ring such that $R[x]$ is a PID or a Euclidean Domain, then R is a field.* ⌟

Proof. Let $R[x]$ be a PID or Euclidean Domain. Then, $R[x]$ is an integral domain, and so R is an integral domain. As such $(x - 1)$ is a prime ideal in $R[x]$ as one can easily verify. Since $R[x]$ is a PID (or Euclidean domain), then $(x - 1)$ is a maximal ideal. Therefore, $R[x]/(x - 1)$ is a field. However, $R[x]/(x - 1) \cong R$, so R is a field.

Note that this last isomorphism follows by explicitly constructing the map: $\varphi : R[x] \to R$ by $\varphi(x) = 1$. Then, this map is clearly surjective and its kernel is $x - 1$, so $R[x]/(x - 1) \cong R$. □

2.17. **Prime Elements and Irreducible Elements**

Definition 2.139 (Irreducible elements, prime elements, and associates)**.** *Let R be an integral domain.*

(1) *Suppose $r \neq 0$ and r is not a unit. Then, r is called an irreducible element if $r = xy$ implies x is a unit and/or y is a unit. Otherwise, r is called a reducible.*
(2) *$p \neq 0$ is called a prime if (p) is a prime ideal. Equivalently, $p \neq 0$ is called a prime if $p \mid xy$ implies $p \mid x$ and/or $p \mid y$.*
(3) *The elements x and y are called associate if there exists a unit u such that $x = uy$.*

⌟

Proposition 2.140 (In IDs, prime $\implies$ irreducible)**.** *In an integral domain, a prime element is always irreducible.* ⌟

Proof. Suppose p is a prime element in an integral domain. Now, suppose $p = xy$ for some $x, y \in R$. Then, $xy \in (p)$, so either $x \in (p)$ or $y \in (p)$. WLOG, suppose the former. Then, $x = px'$ for some $x' \in R$. Then, $p = xy = px'y = x'yp$. Since p is not 0 and there is no zero divisor in the integral domain, $x'y = 1$, so y is a unit. □

Observation. *In an integral domain, primes are irreducible, but irreducibles are not necessarily prime. An example of an irreducible that is not a prime is 3 in $\mathbb{Z}[\sqrt{-5}]$.*

Proposition 2.141. *In a PID, $x \neq 0$ is a prime if and only if it is irreducible.* ⌟

Proof. We already know that in integral domains, primes are irreducible. To show the converse, suppose x is irreducible. Now, we show that (x) is a prime ideal. Suppose, we can write $(x) \subset (m) \subset R$ where each inclusion is proper. Then, $x = mx'$ for some x'. Since x is irreducible, either m is a unit or x' is a unit. If m is a unit, then $(m) = R$, which is not possible. So x' is a unit. Then, $m = x \cdot x'^{-1}$, so $m \in (x)$, and $(m) = (x)$, which is also not possible. Therefore, (x) is a maximal ideal. We know that in any commutative ring, maximal ideals are prime ideals, so x is prime. □

Example 2.142. In $\mathbb{Z}$, prime elements are irreducible. ⌟

2.18. **Unique Factorization Domains (UFDs)**

Definition 2.143. *A Unique Factorization Domain (UFD) is an integral domain R such that, for any non-zero $r \in R$ that is not a unit satisfies the following conditions:*

1. *r can be decomposed as a product of finitely many irreducible elements: $r = r_1 r_2 \cdots r_n$, where the r_i are not necessarily distinct.*
2. *This decomposition is unique, i.e. if there is another decomposition $r = q_1 q_2 \cdots q_m$, then $m = n$, and after some renumbering, $r_i = q_i$.*

(An equivalent definition can also require that any non-zero $r \in R$ (not necessarily non-unit) be uniquely factorized by allowing n, the number of irreducibles in the factorization, to be 0.) ⌟

Examples 2.144.

1. *Fields are trivially a UFD (since there is no non-zero element that is not a unit).*
2. *Every PID is a UFD, as we will prove next.*
3. *If R is a UFD, then $R[x]$ is a UFD, as we will see later.*
4. *$\mathbb{Z}$ is a UFD and $\mathbb{Z}[x]$ is a UFD.*
5. *$\mathbb{Z}[2i]$ is an integral domain but not a UFD. $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD.*

Proposition 2.145. *In a UFD, $x \neq 0$ is a prime if and only if it is irreducible.* ⌟

Proof. We already know that in integral domains, primes are irreducibles, so we show the converse. Suppose $x \neq 0$ is irreducible. Now, suppose $ab \in (x)$. We can write $ab = xy$ for some $y \in R$. Now, after decomposition of a, b and y into irreducibles, suppose

$$a_1 \cdots a_n \cdot b_1 \cdots b_m = x \cdot y_1 \cdots y_k.$$

Given the decomposition of ab must be unique, that means x is associate to some a_i or b_i. WLOG, suppose x is associate to a_1. Then, $a_1 = ux$ for some unit x. Then, $a = a_1 \cdots a_n = uxa_2 \cdots a_n \in (x)$, so x is prime. □

There is a converse of this in some sense:

Proposition 2.146. *Suppose R is an integral domain that satisfies the first condition of a UFD (i.e. any non-zero element has an irreducible factorization). Then, the following are equivalent:*

1. *(π) is a prime if and only if π is irreducible.*
2. *R is a UFD.*

⌟

Proof. We also showed that if R is a UFD, then an element in R is a prime if and only if it is irreducible. We now show the other direction. Since by hypothesis, we already have factorization, we now need to show uniqueness only. Suppose

$$a = u\pi_1 \cdots \pi_n = v\pi_1' \cdots \pi_m'$$

where $u, v \in R^\times$ and π_i, π_j' are irreducible for all i, j. We prove that this factorization is unique by induction. Suppose $m = 0$. Then, $u\pi_1 \cdots \pi_n = v$. But then, $v^{-1}u\pi_1 \cdots \pi_n = 1$ which shows each π_i is a unit despite being irreducible - a contradiction. Therefore, $n = 0$. Now, suppose the statement is true for some $m - 1$. Since $u\pi_1 \cdots \pi_n = v\pi_1' \cdots \pi_m'$, we have that $\pi_m' \mid u\pi_1 \cdots \pi_n$. Since π_m' is irreducible/prime, it must divide either u or $\pi_1 \cdots \pi_n$. If it divides u, then $\pi_m u' = u \implies \pi_m u' u^{-1} = 1 \implies \pi_m$ itself becomes a unit - a contradiction. So, π_m' divides $\pi_1 \cdots \pi_n$. Again, since π_m' is prime, it must divide one of the irreducibles $\pi_i, \forall i \in [n]$. WLOG, suppose π_m' divides π_n. So, $\pi_n = \pi_m w$. But π_n is irreducible, and therefore, $w \in R^\times$. This means π_m and π_n are associates. So, our decompositions become (after absorbing the unit w):

$$u\pi_1 \cdots \pi_{n-1}\pi_m' = v\pi_1' \cdots \pi_{m-1}'\pi_m' \implies u\pi_1 \cdots \pi_{n-1} = v\pi_1' \cdots \pi_{m-1}'.$$

But, by inductive hypothesis, this decomposition is already unique so $m - 1 = n - 1$, which implies $m = n$ and we have that $\pi_i \sim \pi_i'$ (after renumbering) for all $i \in [n]$. □

Proposition 2.147 (Existence and Uniqueness of GCDs in a UFD)**.** *Let R be a UFD. Then, for any $a, b \in R$, there exists an element $d \in R$ such that:*

(1) $d \mid a$ *and* $d \mid b$.
(2) *If* $e \in R$ *satisfies* $e \mid a$ *and* $e \mid b$, *then* $e \mid d$.

Such an element d *is called a greatest common divisor of* a *and* b, *denoted* $\gcd(a, b)$. *Furthermore, the gcd is unique up to multiplication by a unit of* R. ⌟

Proof. We first prove existence. Suppose $a, b \neq 0$. Since R is a UFD, we can write

$$a = u\pi_1^{\alpha_1} \cdots \pi_n^{\alpha_n},$$
$$b = v\pi_1^{\beta_1} \cdots \pi_n^{\beta_n},$$

where $u, v \in R^\times$, the π_i are pairwise non-associate irreducibles, and $\alpha_i, \beta_i \geq 0$. Here we have included all irreducible factors appearing in either factorization, allowing exponents to be zero.

Define

$$d := \prod_{i=1}^{n} \pi_i^{\min(\alpha_i, \beta_i)}.$$

We claim that d is a gcd of a and b. First, since $\min(\alpha_i, \beta_i) \leq \alpha_i$ and $\min(\alpha_i, \beta_i) \leq \beta_i$ for every i, we have $d \mid a$ and $d \mid b$. Next, suppose $e \in R$ satisfies $e \mid a$ and $e \mid b$. Since R is a UFD, we can write

$$e = w\pi_1^{\gamma_1} \cdots \pi_n^{\gamma_n},$$

where $w \in R^\times$ and $\gamma_i \geq 0$. Indeed, every irreducible factor of e must be associated to some π_i, since $e \mid a$. By uniqueness of factorization, $e \mid a$ implies $\gamma_i \leq \alpha_i$, and $e \mid b$ implies $\gamma_i \leq \beta_i$. Consequently,

$$\gamma_i \leq \min(\alpha_i, \beta_i)$$

for every i. Therefore, $e \mid d$, proving that d is a gcd.

If $a = 0$ and $b \neq 0$, we may take $d = b$, since every divisor of b also divides 0. Similarly, if $b = 0$ and $a \neq 0$, we may take $d = a$. If $a = b = 0$, we may take $d = 0$. Thus, a gcd exists in all cases.

Finally, we prove uniqueness up to associates. Suppose d and d' are both gcds of a and b. Since d divides both a and b, the defining property of d' implies $d \mid d'$. Similarly, $d' \mid d$.

If $d = d' = 0$, they are trivially associates. Otherwise, there exist $r, s \in R$ such that

$$d' = rd, \qquad d = sd'.$$

Substituting gives

$$d = srd.$$

Since R is an integral domain and $d \neq 0$, cancellation yields $sr = 1$. Hence r and s are units, so d and d' are associates. Therefore, the gcd exists and is unique up to multiplication by a unit. □

Proposition 2.148 (GCD's in UFDs)**.** *Let* $a \neq 0$ *and* $b \neq 0$ *in a UFD* R. *Suppose their prime factorizations are*

$$a = up_1^{e_1} \cdots p_n^{e_n}$$

and

$$b = vp_1^{f_1} \cdots p_n^{f_n}$$

where u *and* v *are units, the primes* p_i *are distinct, and* $e_i, f_i \geq 0$. *Then,*

$$d := p_1^{\min e_1, f_1} \cdots p_n^{\min e_n, f_n} = \gcd(a, b).$$

⌟

We will need the following lemma for the main result in this section:

Lemma 2.149. *In a PID, any ascending chain of ideals*

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq R$$

must eventually becomes stationary, i.e. there exists n *such that for all* $k \geq n$, $I_k = I_n$. ⌟

Proof. Consider the union, $I = \cup_{i=1}^{\infty} I_i$. Then, we can easily show that I is itself an ideal. Given this is a PID, suppose $(x) = I$. Then, $x \in I_n$ for some n. Since this is a PID, $I_n = (a)$, and as $x \in (a) = I_n$, so $I \subseteq I_n$ which shows $I = I_n$, and so the chain becomes stationary at n. □

Now, we prove the main result:

Theorem 2.150. *Every principal ideal domain is a unique factorization domain. Therefore, every Euclidean domain is also a unique factorization domain.* ⌟

Proof. Let R be a PID and let $r \in R$ be a non-zero, non-unit element. Now, if r is an irreducible, we are done. If r is not irreducible, then we can write $r = r_1 \cdot r_2$ where neither r_1 nor r_2 is a unit. Now, if both r_1 and r_2 are irreducible, we have found the irreducible decomposition and we are done. Otherwise, suppose r_1 is not irreducible. Now, we can write $r_1 = r_{11}r_{12}$. Continuing this way, we see that

$$(r) \subset (r_1) \subset (r_{11}) \subset \cdots \subset R,$$

where each ideal is generated by a reducible. Note that each inclusion must be proper. Now, this process must terminate by the previous lemma. Then, renaming the generators of each ideal, we get the decomposition $r = r_1 r_2 \cdots r_n$.

Now, we must show that this decomposition is unique. If $n = 0$, then r is, by definition, a unit. Now, if we have another decomposition $r = qs$, then q divides a unit, so it must be a unit itself, which is not possible. Therefore, the decomposition with $n = 0$ is unique. Now, suppose that $n \geq 1$. Suppose, we have another decomposition:

$$r = r_1 \cdots r_n = s_1 \cdots s_m$$

where $m \geq n$ and r_i and s_i are (not necessarily distinct) irreducible. Now, r_1 divides $s_1 \cdots s_m$, and since primes and irreducibles are equivalent in PIDs, therefore, r_1 must divide some s_i. WLOG, suppose r_1 divides s_1, i.e. $s_1 = ur_1$. Since s_1 is irreducible, u must be a unit. Now, we cancel r_1 from both sides. We continue this way, and show that $m = n$ and each r_i is associate to s_i. □

Corollary 2.151 (Fundamental Theorem of Arithmetic)**.** *$\mathbb{Z}$ is a unique factorization domain.* ⌟

Proposition 2.152. *Let R be a UFD and $D \subset R$ be a multiplicative set not containing 0. Then, $D^{-1}R$ is a UFD.* ⌟

Proof. Consider any arbitrary $\frac{x}{y} \in D^{-1}R$. Now, if $\frac{x}{y}$ is a unit or irreducible, we are done. So, we suppose that this is neither irreducible nor unit. Then, there exists the following decomposition:

$$\frac{x}{y} = \frac{a_1}{b_1} \cdot \frac{a_2}{b_2}$$

where neither a_1/b_1 nor a_2/b_2 is a unit. Now, we can write this as

$$\frac{x}{y} = \frac{1}{b_1 b_2} \frac{a_1}{1} \frac{a_2}{1}.$$

Since $a_1 \in R$, and R is a UFD, there exists a prime/irreducible factorization: $a_1 = p_1 \cdots p_n$ and similarly, there exists one for $a_2 = q_1 \cdots q_m$ where p_i and q_j are prime/irreducible. Plugging this in, our decomposition becomes:

$$\frac{x}{y} = \frac{1}{b_1 b_2} p_1 \cdots p_n q_1 \cdots q_m.$$

Note that $\frac{1}{b_1 b_2}$ is a unit (with its inverse being $\frac{b_1 b_2}{1}$). We claim that in $D^{-1}R$, each $p_i = \frac{p_i}{1}$ and each $q_i = \frac{q_i}{1}$ are irreducibles. To see this, suppose $\frac{p_1}{1} = \frac{\alpha}{\beta} \cdot \frac{\alpha'}{\beta'}$. Then, there exists some $d \in D$ such that $d(p_1 \beta\beta' - \alpha\alpha') = 0$. Since $d \neq 0$ by definition of our multiplicative set and R is an integral domain, $p_1\beta\beta' - \alpha\alpha' = 0 \implies p_1\beta\beta' = \alpha\alpha'$. Now, p_1 divides $\alpha\alpha'$ and p_1 is prime, so p_1 divides either α or α'. WLOG, suppose p_1 divides α. Then, $p_1 = w\alpha$. But p_1 is irreducible, so w is a unit. So, we see that

$$\frac{p_1}{1} \cdot \frac{w^{-1}}{\beta} = \frac{p_1 w^{-1}}{\beta} = \frac{\alpha}{\beta},$$

which means $\frac{p_1}{1}$ divides α/β, concluding our proof that each $p_i = \frac{p_i}{1}$ is prime. Similar strategy shows that each q_j is also prime. Now, we see that $\frac{x}{y}$ has a prime factorization. Now, we need to show that this factorization is unique. Suppose, we have another decomposition:

$$\frac{x}{y} = \frac{r_1}{d_1} \cdots \frac{r_t}{d_t} = \frac{1}{d_1 \cdots d_t} r_1 \cdots r_t.$$

To compare the equations, we have that:

$$\frac{x}{y} = \frac{1}{d_1 \cdots d_t} r_1 \cdots r_t = \frac{1}{b_1 b_2} p_1 \cdots p_n q_1 \cdots q_m$$

which implies

$$\frac{b_1 b_2}{d_1 \cdots d_t} r_1 \cdots r_t = p_1 \cdots p_n q_1 \cdots q_m.$$

We claim that $t = n + m$ and $r_i \sim p_i$ for all $i \in \{1, \cdots, n\}$ and $r_j \sim q_j$ for all $j \in \{1, \cdots, m\}$. We prove this by induction. Suppose $t = 0$. Then, $\frac{d_1 \cdots d_t}{b_1 b_2} p_1 \cdots p_n q_1 \cdots q_m = 1$. But this makes each p_i and q_j into units which is impossible since they are irreducible/prime in R. As such, $n = 0, m = 0$ and our statement is vacuously true. Now, suppose our statement is true for $t = k - 1$. Now,

$$\frac{b_1 b_2}{d_1 \cdots d_t} r_1 \cdots r_{k-1} r_k = p_1 \cdots p_n q_1 \cdots q_m.$$

We see that r_k divides $p_1 \cdots p_n q_1 \cdots q_m$. Since r_k is prime, it means r_k divides at least one of p_i or q_j. WLOG, suppose r_k divides q_m. Let $r_k = v q_m$. Since r_k is irreducible and q_m is irreducible, we must have that v is a unit. Then,

$$\begin{aligned}
\frac{b_1 b_2}{d_1 \cdots d_t} r_1 \cdots r_{k-1} r_k &= p_1 \cdots p_n q_1 \cdots q_m \\
\frac{v b_1 b_2}{d_1 \cdots d_t} r_1 \cdots r_{k-1} q_m &= p_1 \cdots p_n q_1 \cdots q_m \\
\frac{v b_1 b_2}{d_1 \cdots d_t} r_1 \cdots r_{k-1} &= p_1 \cdots p_n q_1 \cdots q_{m-1},
\end{aligned}$$

where the last line follows from the fact that we can cancel the q_m from both sides since $D^{-1}R$ is an integral domain (given D contains no zero and R is an integral domain). But, by inductive hypothesis, $k - 1 = n + m - 1$ which says $k = n + m$, and $r_i \sim p_i$ for all $i \in \{1, \cdots, n\}$ and $r_j \sim q_j$ for all $j \in \{n + 1, \cdots, n + m - 1\}$. Since, we already have that $r_t \sim q_m$, we are done. This shows that the factorization is unique in the sense as in UFD, so $D^{-1}R$ is a UFD. $\square$

2.19. **Noetherian Rings**

Proposition 2.153. *Let R be a ring. The following are equivalent:*

(1) *Every ideal of R is finitely generated.*
(2) *Any non-empty set X of ideals in R has a maximal element (i.e. there exists some $I \in X$ such that there is no $J \in X$ such that $I \subsetneq J$).*

⌟

Proof. (1) $\implies$ (2). Let X be a collection of ideals in R. Suppose, for contradiction, that X does not have a maximal element. Let $I_1 \in X$. Then, I_1 is not maximal by our assumption, so there exists some I_2 with $I_2 \subsetneq I_2$. Continuing this way, we get a chain of ideals:

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots .$$

Consider $I = \cup_i I_i$. Since this is the union of ideals in a chain, I is an ideal itself. Since I is finitely generated, $I = (a_1, \cdots, a_m)$. Then, each $a_j \in I$ implies $a_j \in I_{n_j}$ for some $n_j \in \mathbb{N}$. Let $N = \max\{n_1, \cdots, n_m\}$. Then, $a_1, \cdots, a_m \in I_N$. We have that $I_N = I$, and $I_{N+1} = I_{N+2} = \cdots = I$. But $I_N \subsetneq I_{N+1}$, which is a contradiction. So, there must be a maximal element.
(2) $\implies$ (1). Consider I fixed. Define X to be the set of ideals $J \trianglelefteq R$ such that $J \subset I$ and J is finitely generated. Then, Z has a maximal element J. We claim $J = I$. Suppose not; then there exists some $a \in I - J$. Then, the ideal $J + (a)$ such that $J \subsetneq J + (a) \subsetneq R$. This contradicts that J is maximal. □

Definition 2.154 (Noetherian Rings)**.** *A ring R is called Noetherian if every ideal of R is finitely generated.* ⌟

Proposition 2.155. *R is Noetherian if and and only if any ascending chain of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ stagnates (i.e. there exists some m such that $I_m = I_{m+1} = I_{m+2} = \cdots$).* ⌟

Proof. If R is Noetherian, then $\cup_i I_i$, as an ideal, must be finitely generated and therefore, the chain must stagnate. Conversely, suppose every such chain stagnates. Now, consider any fixed ideal I. We must show that I is finitely generated. Define X to be a chain of finitely generated ideals of R such that they are contained in I. This chain must stagnate at some ideal J. We claim $J = I$. We already know $J \subseteq I$. We show $I \subseteq J$. Suppose not: then there exists some $a \in I - J$. Then, $J + (a)$ is an ideal with $J \subsetneq J + (a) \subsetneq I$. This contradicts that J was the maximal such ideal, so $J = I$. □

Remark 2.156. While the ascending chain condition holds in Noetherian rings, the descending chain condition need not hold. For e.g., in the Noetherian ring $\mathbb{Z}$, we have infinitely long descending chain of ideals:

$$(2) \supset (4) \supset (8) \supset (16) \cdots .$$

However, a commutative ring that satisfies the descending chain condition is guaranteed to be a Noetherian ring; these rings are called Artinian rings. ⌟

Example 2.157. Every PID is a Noetherian ring. For e.g., $\mathbb{Z}$ is Noetherian. ⌟

Example 2.158. $\mathbb{C}$ is Noetherian. ⌟

Example 2.159. $\mathbb{C}[x_1, x_2, \cdots]$ and $\mathbb{Z}[x_1, x_2, \cdots]$ are *not* a Noetherian ring. The ideal $I = (x_1, x_2, \cdots)$ is not finitely generated. ⌟

Example 2.160. The ring of continuous real-valued functions on $[0, 1]$ is not Noetherian. ⌟

Proposition 2.161. *Let R be a Noetherian ring and let I be an ideal of R. Then, R/I is a Noetherian ring. Any homomorphic image of a Noetherian ring is also Noetherian.* ⌟

Proof. Suppose R is Noetherian. Now, consider any non-empty set, X, of ideals of R/I. These ideals correspond to ideals of R containing I. Then, if this set X does not have a maximal element, there is also no maximal element of the corresponding set in R, which is impossible since R is Noetherian. The second statement follows from the first isomorphism theorem. □

Example 2.162. For a field k, $k[x_1, x_2, \cdots, x_n]$ is a Noetherian ring. ⌟

Proposition 2.163. *Let R and S be Noetherian rings. Then, $R \times S$ is Noetherian.* ⌟

Proof. The ideals of $R \times S$ are of the form $I \times J$ where I is an ideal of R and J is an ideal of S. Now, consider some non-empty collection of ideals in $R \times S$:

$$\mathfrak{X} = \{I_i \times J_i\}_{i \in \mathcal{I}}.$$

Now, define

$$\mathfrak{X}^R := \{I_i \trianglelefteq R \mid I_i \times J \in \mathfrak{X} \text{ for some } J \trianglelefteq S, i \in \mathcal{I}\}.$$

Given R is a Noetherian ring, this non-empty collection of ideals has a maximal element, so let I be the maximal element of $\mathfrak{X}^R$. Now, define

$$\mathfrak{X}^S_{I\times\cdot} := \{J_i \trianglelefteq S \mid I \times J \in \mathfrak{X}, i \in \mathcal{I}\}.$$

Since S is Noetherian, this non-empty set also has a maximal element, which we denote by J. Now, we claim $I \times J$ is a maximal element of $\mathfrak{X}$. Note that $I \times J$, by definition, is in $\mathfrak{X}$. Now, suppose for contradiction, this $I \times J$ is not the maximal element. Then, there exists some $I' \times J' \in \mathfrak{X}$ such that $I \times J \subsetneq I' \times J'$. Then, $I \subseteq I'$ and $J \subseteq J'$. Since I is the maximal element of $\mathfrak{X}^R$ and $I' \in \mathfrak{X}^R$, we have that $I = I'$. This means, $J \subsetneq J'$. But J' is an element of $\mathfrak{X}^S_{I\times\cdot}$ that properly contains J. This is a contradiction, since J is the maximal element of $\mathfrak{X}^S_{I\times\cdot}$, and, therefore, $I \times J$ is indeed the maximal element. Since we have found a maximal element for any collection of ideals in $R \times S$, therefore, $R \times S$ is a Notherian ring. □

Proposition 2.164 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then $R[x]$ is a Noetherian ring.* ⌟

Proof. Suppose R is Noetherian. Now, let $I \trianglelefteq R[x]$. We must find a finite set of elements that generate I. First, consider the set

$$J := \{ \text{ leading coefficients of polynomials in } I\} \cup \{0\}.$$

Since I is an ideal, J is also an ideal in R. But R is Noetherian, so there exists a finite set of elements that generate J. But each element corresponds to the leading coefficient of some polynomial in I, so, there exists a finite set of polynomials $f_1(x), f_2(x), \cdots, f_r(x) \in I$ such that their leading coefficients generate J. Furthermore, consider $N \in \mathbb{Z}$ such that $N > \deg f_i$ for all $i \in \{1, \cdots, r\}$.
Now, for each $m < N$, define

$$J_m := (\{ \text{ leading coefficients of polynomials of degree } \leq m \text{ in } I\}) \cup \{0\}.$$

Again, since J_m is an ideal in R, it must be finitely generated, so there exists a finite set of polynomials $f_{m1}, \cdots, f_{ms_m} \in I$ whose degrees are $\leq m$ leading coefficients generate J_m.
Now, we claim that $I = I' = (f_{mi})_{m\in\{1,\cdots,N\}, i\in\{1,\cdots,s_m\}} + (f_1, \cdots, f_r)$. Suppose not. Then, there exists some $g \in I$ that is not in I'. Let g be the minimal such element.
Case 1: $\deg g > N$. Now, there exists some polynomials Q_i such that $\sum_i Q_i f_i$ which has the same leading term as g, so $\deg(g - \sum_i Q_i f_i) < \deg(g)$. Since g is the minimal element that is in I but not in I', then, $g - \sum_i Q_i f_i$ is in I'. But then, $g \in I'$.
Case 2: $n := \deg g \leq N$. Again, choose Q_j such that $\sum_j Q_j f_{nj}$ has the same leading term as g. Similar argument to case 1 shows that $g \in I'$. □

We now show that in any Noetherian integral domain, we can factorize elements. Note that, unlike in UFDs, this factorization is not necessarily unique in Noetherian IDs.

Proposition 2.165. *If R is a Noetherian integral domain, then any $a \in R$ such that $a \neq 0$ can be factorized as*

$$a = u\pi_1 \cdots \pi_n, \quad \textit{where } u \in R^\times, \pi_i \textit{ are irreducible.}$$

⌟

Proof. Consider the set

$$\mathfrak{X} := \{a \in R \mid a \neq 0, a \text{ cannot be factorized in the form mentioned}\}.$$

If this set is empty, then we are done. If not, since this is a Noetherian ring, this set has some maximal element $x \in X$. Now, since $x \in X$, x is not a unit: if x were a unit, we could write

$x = x$ (i.e. the number of irreducibles in the decomposition is $n = 0$). Furthermore, x is not irreducible, since otherwise we have the decomposition $x = 1 \cdot x$. Then, let $x = bc$ where neither b nor c is a unit. We have that $(x) \subsetneq (b)$: if $(x) = (b)$, then $b = xc'$ for some c', which leads to $b = bcc'$ implying $cc' = 1$ which implies c is a unit - a contradiction. Similarly, we have that $(x) \subsetneq (c)$. Now, since x is the maximal element and $(x) \subsetneq (b), (c)$, b and c are not in $\mathfrak{X}$ and, by definition, they are also not units. So they must have irreducible decompositions:

$$b = u\pi_1 \cdots \pi_n$$

and

$$c = v\pi'_1 \cdots \pi'_m.$$

But then we have the decomposition

$$x = uv\pi_1 \cdots \pi_n \pi'_1 \cdots \pi'_m.$$

□

2.20. **Polynomial Rings**

Definition 2.166. *If R is a ring and $\{x_i\}_{i\in I}$ is a collection of variables, then:*

(1) *A monomial is a $\prod_{i\in I} x_i^{n_i}$ where $a \in R$, $n_i \in \mathbb{Z}_{\geq 0}$, and n_i is 0 for all but finitely many $i \in I$.*
(2) *A polynomial is a sum of monomials.*
(3) *A monic polynomial is a term whose leading term's coefficient is 1.*
(4) *$R[x_i]_{i\in I}$ is called the polynomial ring where the operations, addition and multiplication, are defined naturally.*

⌟

Definition 2.167 (Power Series)**.** *A power series, $R[[x_i]]_{i\in I}$ is the ring consisting of infinite sums of monomials. In particular,*

$$R[[x]] = \{\sum_{n=0}^{\infty} a_n x^n \mid a_n \in R, n \in \mathbb{Z}_{\geq 0}\}.$$

⌟

Definition 2.168 (Degree of Polynomials)**.** *The polynomial $p(x) = a_0 + a_1 x + \cdots + a_d x^d$ is said to have degree $\deg p(x) = d$ if $a_d \neq 0$. We say that $\deg 0 = -\infty$.* ⌟

Proposition 2.169. *Let R be an integral domain. Then,*

(1) *$\deg p(x)q(x) = \deg p(x) + \deg q(x)$ if $p(x) \neq 0$ and $q(x) \neq 0$.*
(2) *$R[x]^\times = R^\times$*
(3) *$R[x]$ is also an integral domain.*

⌟

Proposition 2.170. *If R is an integral domain, show that $R[[x]]$ is an also an integral domain.* ⌟

Proof. Suppose R is an integral domain. Now, suppose for contradiction that $R[[x]]$ is not an integral domain. Then, there exists non-zero $a(x) = \sum_{m=0}^{\infty} a_m x^m$ and non-zero $b(x) = \sum_{n=0}^{\infty} b_m x^m$ such that $a(x)b(x) = 0$. Since $a(x) \neq 0$, there exists a minimal non-negative integer $p \in \mathbb{N}$ such that $a_p \neq 0$. Similarly, there exists a minimal non-negative integer $q \in \mathbb{N}$ such that $b_q \neq 0$.

Now, we consider $a(x)b(x) = \sum_{i=0}^{\infty} \sum_{m+n=i, 0\leq m,n\leq i} a_m b_n x^i$. In particular, consider the term of degree $p + q$. Then, since the product is 0, the coefficient of x^{p+q} is:

$$a_0 b_{p+q} + a_1 b_{p+q-1} + \cdots + a_{p+q} b_0 = 0.$$

Recall that p is the smallest degree term of $a(x)$ that is non-zero. As such, $a_0, \cdots, a_{p-1}$ are all 0. Therefore, we have that

$$a_p b_q + a_{p+1} b_{q-1} + \cdots + a_{p+q} b_0 = 0.$$

Now, $b_0, \cdots, b_{q-1}$ are all 0, so we see that

$$a_p b_q = 0.$$

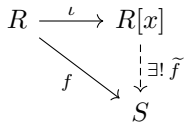
Given R is an integral domain, this means either $a_p = 0$ or $b_q = 0$. This contradicts our assumption that $a_p \neq 0$ and $b_q \neq 0$. Therefore, $R[[x]]$ is an integral domain. □

The universal property of the polynomial ring $R[x]$ says, roughly, that to define a ring homomorphism out of $R[x]$, it suffices to specify where the elements of R go and where x goes.

Proposition 2.171 (Universal Property of Polynomial Rings)**.** *Let R and S be commutative rings, and let $f : R \to S$ be a ring homomorphism. For any $s \in S$, there exists a unique ring homomorphism*

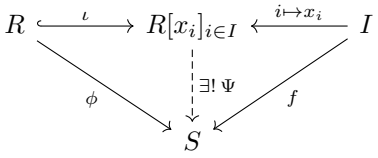
$$\widetilde{f} : R[x] \to S$$

such that $\widetilde{f}(r) = f(r)$ for every $r \in R$ and $\widetilde{f}(x) = s$. Equivalently, if $\iota : R \to R[x]$ is the natural inclusion, then $\widetilde{f}$ is the unique ring homomorphism satisfying $\widetilde{f} \circ \iota = f$ and $\widetilde{f}(x) = s$. This is expressed by the following commutative diagram:



⌟

Proposition 2.172 (Universal Property of Polynomial Rings)**.** *Let $\phi : R \to S$ be a ring morphism, let I be a set, and let $f : I \to S$ be a map of sets. Then there exists a unique ring morphism $\Psi : R[x_i]_{i\in I} \to S$ such that $\Psi|_R = \phi$ and $\Psi(x_i) = f(i)$ for every $i \in I$. Equivalently, there exists a unique Ψ making the following diagram commute:*



⌟

Proof. One can show this by defining $\Psi(a\prod_{i\in I} x_i^{n_i}) = \phi(a)\prod_{i\in I} f(i)^{n_i}$. Note that any polynomial can be defined using some map $\alpha : I \to \mathbb{Z}_{\geq 0}$:

$$x_\alpha = \prod_{i\in I} x_i^{\alpha(i)}.$$

Then, it is easy to show that $\Psi(\sum_\alpha r_\alpha x^\alpha)$ is well-defined (here, $r_\alpha \in R$), is a ring morphism, and is unique. □

The following is an extremely important proposition that we will repeatedly use in the future.

Proposition 2.173. *Let R be a commutative, unital ring. Let $f(x), g(x) \in R[x]$ and suppose g is a monic polynomial. Then, there exists unque polynomials $q(x), r(x) \in R[x]$ such that $f(x) = q(x)g(x) + r(x)$, where $\deg r(x) < \deg g(x)$.* ⌟

Proof. If $\deg f(x) < \deg g(x)$, then select $q(x) = 0$ and $r(x) = f(x)$ and we are done. Now, suppose $\deg f(x) \geq g(x)$. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$ and let $g(x) = \cdots + x^m$ since $g(x)$ is monic. Then,

$$f(x) = a_nx^{n-m}g(x) + f_1(x)$$

for some $f_1(x)$. Since $a_nx^{n-m}g(x)$ equals the leading term of f, $\deg f_1 < \deg f$. Now, continuing inductively, we can write $f_1(x) = q_1(x)g(x) + r(x)$ where $\deg r(x) < \deg g(x)$. So, $f(x) = a_nx^{n-m}g(x) + g(x)q_1(x) + r(x)$, which tells us

$$f(x) = g(x)(a_nx^{n-m} + q_1(x)) + r(x),$$

where $\deg r(x) < \deg g(x)$. This showed existence. To show uniqueness, suppose

$$f = gq_1 + r$$

and

$$f = gq_2 + r.$$

Then, $(q_1 - q_2)g = r_2 - r_1$. If $q_1 \neq q_2$, since g is monic, $\deg g(q_1 - q_2) = \deg g + \deg(q_1 - q_2) \geq \deg g$. But $\deg r_1 - r_2 < \deg g$. This gives us a contradiction. So, $q_1 = q_2$, but then $r_2 = r_1$, and we are done. □

Proposition 2.174. *If $I \trianglelefteq R$, then $R[x]/I[x] \cong (R/I)[x]$. In particular, if $I \trianglelefteq_{\text{prime}} R$, then $I[x] \trianglelefteq_{\text{prime}} R[x]$.* ⌟

Proof. The first part can be proven using the morphism: $\varphi : R[x] \to (R/I)[x]$ which sends $\varphi(a) = a + I$ and $\varphi(a_ix^i) = (a_i + I)x^i$. Then, its kernel is $I[x]$ □

Example 2.175. Let $R = \mathbb{Z}$ and consider the ideal $n\mathbb{Z} \trianglelefteq \mathbb{Z}$. Then, $\mathbb{Z}[x]/n\mathbb{Z}[x] \cong \mathbb{Z}/n\mathbb{Z}[x]$. In particular, if p is prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field and so $\mathbb{Z}/p\mathbb{Z}[x]$ is an integral domain. ⌟

Observation. *If $I \trianglelefteq_{\text{maximal}} R$, then $I[x]$ need not be a maximal ideal in $R[x]$. However, we do have the following guarantee:*

Proposition 2.176. *If $I \trianglelefteq_{\text{maximal}} R$, then $(I, x) \trianglelefteq_{\text{maximal}} R[x]$.* ⌟

Proof. Suppose $(I, x) \subsetneq J \subsetneq R[x]$. Now, suppose $p(x) \in J$ such that $p(x) \notin (I, x)$.
Case 1: $p(x) = p$ is constant. Since $p \notin I$, then, $I \subsetneq (I, p)$ in R. But I is maximal, so $(I, p) = R$. As such, $1 \in (I, p)$. Since $(I, p) \subseteq J$, $J = R[x]$. As such, all constant terms in J must be contained in I.
Case 2: $p(x) = f(x) + c$ where $f(x)$ consists of only non-constant terms. Now, $f(x) \in (x)$. Then, $f(x) + c + (-f(x)) = c \in J$ since $-f(x) \in (x) \in J$. But since c must be in I from case 1, $p(x) = f(x) + c \in (x) + I \subset (I, x)$. □

Proposition 2.177. *Given R is a commutative, unital ring,*

$$R[x]/(f(x), g(x)) \cong (R[x]/(f(x)))/(\overline{g(x)})$$

where $\overline{g(x)} = (g(x), f(x))/(f(x)) \in R[x]/(f(x))$. More generally,

$$R/(a, b) \cong (R/(a))/(\bar{b}).$$

⌟

Proof. This comes from the third isomorphism theorem. □

Definition 2.178 (Primitive Polynomial)**.** *A polynomial in which the gcd of its coefficients is 1 is called a primitive polynomial.* ⌟

Proposition 2.179. *If R is an integral domain, then $R[x]$ contains infinitely many irreducibles up to associates.* ⌟

Proof. Suppose R is infinite. Then, any degree 1 polynomial $(x - a)$ for $a \in R$ is irreducible and also pairwise non-associate (there cannot be any unit u such that $x - a = u(x - b)$, since comparing coefficients gives $u = 1$). This is because $(x - a)$ is monic, so it cannot be decomposed into two factors of degree > 0, and since f is monic, any constant factor must be 1. So we get infinitely many irreducibles.

Suppose R is finite. We assume there are finitely many irreducibles up to associates and come to a contradiction. A finite integral domain is a field, so we are looking for irreducibles in $F[x]$. Now, suppose $F[x]$ has finitely many irreducibles, $f_1(x), \cdots, f_n(x)$. Note that each $f_i(x)$ must have degree > 0, since irreducibles are not units. Now, $s(x) = f_1(x)f_2(x) \cdots f_n(x) + 1$ is a new polynomial of degree > 0. If $s(x)$ is irreducible, we have found a new irreducible polynomial. If not, $s(x)$ be decomposed into irreducibles since $F[x]$ is a UFD. Let $g(x)$ be an irreducible factor. We claim $g(x)$ does not divide any $f_i(x)$. Suppose, for contradiction, it divides $f_i(x)$. Then, as $f_i(x)$ is irreducible and $g(x)$ is a non-unit (irreducible) dividiing $f_i(x)$, they must be associates. As such, $f_i(x)$ also divides $g(x)$, which implies $f_i(x)$ divides $s(x)$. But this is impossible since this implies $f_i(x)$ divides 1, i.e. $f_i(x)$ is a unit which contradicts that $f_i(x)$ is irreducible. Since $g(x)$ does not divide any $f_i(x)$, Therefore, $g(x)$ is a new irreducible polynomial that is not associate to any of the f_i. □

2.21. **Polynomial Rings over Fields**

Recall that, any polynomial $a(x) \in R[x]$ and any monic polynomial $m(x) \in R[x]$, we can always find unique $q(x), r(x) \in R[x]$ such that $a(x) = m(x)q(x) + r(x)$ such that $\deg r(x) < \deg m(x)$. We will now look at polynomials over rings where this result holds for all polynomials, not only monic ones.

Consider a field $\mathbb{F}$. Define a norm on $\mathbb{F}[x]$ to be

$$N(p(x)) = \deg p(x), \text{ with } N(0) := 0.$$

Theorem 2.180. *Let $\mathbb{F}$ be a field. hen, $\mathbb{F}[x]$ is a Euclidean domain. Concretely, for any $a(x), b(x) \in \mathbb{F}[x]$, there exists $q(x), r(x) \in \mathbb{F}[x]$ such that*

$$a(x) = b(x)q(x) + r(x),$$

with

$$r(x) = 0 \text{ or } N(r(x)) < N(b(x)).$$

⌟

Proof. If $\deg b(x) > \deg a(x)$, then set $q(x) = 0$ and $r(x) = a(x)$ and we are done. If $a(x) = 0$, this same choice of $q(x)$ and $r(x)$ works again. So the case left for us to handle is when $\deg a(x) := n \geq m =: \deg b(x)$. We use induction on the degree n. Let

$$a(x) = a_0 + a_1 x + \cdots + a_n x^n$$

and

$$b(x) = b_0 + b_1 x + \cdots + b_m x^m.$$

We first find a quotient that gets rid of the highest degree term of $a(x)$. This is precisely the term $\frac{a_n}{b_m} x^{n-m}$, which is defined since $\mathbb{F}$ is a field. So, if we multiply $b(x)$ with this term, we get rid of the highest degree term in $a(x)$. Define

$$a'(x) = a(x) - \frac{a_n}{b_m} x^{n-m} b(x).$$

Then, note that $\deg a'(x) < b$. Now, by inductive hypothesis, there exists unique polynomials $q'(x)$ and $r'(x)$ such that

$$a'(x) = q'(x)b(x) + r'(x)$$

where $r'(x)$ has degree less than $b(x)$. Since $q'(x)$ takes care of dividing all the terms of degree $< n$ and $\frac{a_n}{b_m} x^{n-m}$, the $q(x)$ we are looking for is simply defined as:

$$q(x) = q'(x) + \frac{a_n}{b_m} x^{n-m}.$$

It is easy to check, then, that

$$a(x) = b(x)q(x) + r'(x)$$

satisfies the requirements. All that is left is to prove uniqueness. Suppose, we have two equations satisfying the requirements:

$$a(x) = q_1(x)b(x) + r_1(x)$$

and

$$a(x) = q_2(x)b(x) + r_2(x).$$

Then, $a(x) - q_1(x)b(x)$ and $a(x) - q_2(x)b(x)$ are both equations whose degree is less than m. Then, subtracting the second from the first, i.e. $a(x) - q_1(x)b(x) - (a(x) - q_2(x)b(x)) = b(x)(q_1(x) - q_2(x))$, we get a polynomial with degree less than m. This is impossible (since $\deg b(x) = m$) unless $q_1(x) - q_2(x) = 0$. So, $q_1(x) = q_2(x)$, which forces $r_1(x) = r_2(x)$. □

Corollary 2.181. *If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is a PID and a UFD.* ⌟

Proof. All Euclidean Domains are UFDs. □

Examples 2.182. *Here is a collection of examples of polynomials rings over fields.*

1. $\mathbb{Q}[x]$ *is a Euclidean domain, PID and a UFD.*
2. $\mathbb{Q}[x, y]$ *is not a PID; the ideal (x, y) is not generated by a single element.*
3. $\mathbb{Z}/p\mathbb{Z}[x]$*, where p is prime, is a Euclidean domain, PID, and a UFD.*
4. *Although $\mathbb{Z}[x]$ is not a PID, $\mathbb{Z}[x]/p\mathbb{Z}[x]$, for prime p, is a PID.*

2.22. **Polynomial Rings over UFDs**

This section is all about understanding when we can move between $R[x]$, where R is a UFD, and $F[x]$ where F is the field of fractions of R. Before that, we recall:

Fact 2.183. *In any UFD, gcd's exist and is unique up to associates.* ⌟

Definition 2.184 (Primitive Polynomial)**.** *Let R be a UFD, and let*

$$f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x]$$

be a nonzero polynomial. We say that $f(x)$ is primitive *if*

$$\gcd(a_0, a_1, \ldots, a_n) = 1.$$

Equivalently, $f(x)$ is primitive if every common divisor of its coefficients is a unit in R. In particular, 1 is associate to any unit u, and so primitive polynomials are polynomials whose coefficients' gcd is 1 or, equivalently, a unit. ⌟

Proposition 2.185 (Gauss's Lemma)**.** *Let R be a UFD and let F be its field of fractions. If $p(x)$ is reducible in $F[x]$, then it is reducible in $R[x]$. In particular, if $p(x) = A(x)B(x) \in F[x]$, then there exists $a(x), b(x) \in R[x]$ such that $p(x) = a(x)b(x) \in R[x]$ and $a(x) = rA(x), b(x) = sB(x)$ for $r, s \in F$.* ⌟

Proof. Suppose $p(x)$ is reducible in $F[x]$. Then, $p(x) = A(x)B(x)$. Now, take a common multiple of the denominators of the coefficients in $A(x)$ and in $B(x)$. Let this be d. Then, $dp(x) = a'(x)b'(x) \in R[x]$ where $a'(x) \in R[x]$ and $b'(x) \in R[x]$. If d is a unit, we are done: set $p(x) = d^{-1}a'(x)b'(x)$ and we have decomposed $p(x)$.

Now, since R is a UFD, d can be decomposed into irreducibles. Let $d = p_1p_2 \cdots p_n$ where each p_i is irreducible. Since R is a UFD, all irreducibles are primes, so (p_i) is a prime ideal for each i. We know, from Proposition 2.174, that if (p_i) is prime in R, then $(p_i)[x]$ is a prime ideal in $R[x]$. Therefore, $(R/(p_1))[x] \cong R[x]/((p_1)[x])$ is an integral domain. Mapping all coefficients to modulo p_1, we see then that

$$\overline{dp(x)} = \overline{0}.$$

So, $\overline{a'(x)b'(x)} = 0$. Since this is an integral domain, then one of these two factors must be 0. WLOG, suppose $\overline{a'(x)} = 0$. Then, in $R[x]$, $a'(x) \in (p_1)[x]$, i.e. all the coefficients of $a'(x)$ must be divisible by p_1. We can then cancel out p_1 from both sides of $dp(x) = a'(x)b'(x)$. Continuing this way, we can cancel out all p_i's, leaving us with a decomposition $p(x) = a(x)b(x)$ in $R[x]$, where neither $a(x)$ nor $b(x)$ are units, i.e. $p(x)$ is reducible in $R[x]$. □

Proposition 2.186 (Gauss's lemma - alternative version)**.** *Suppose R is a UFD. If $f(x) \in R[x]$ and $g(x) \in R[x]$ are primitive, then $f(x)g(x) \in R[x]$ is primitive.* ⌟

Proof. Suppose $f(x)$ and $g(x)$ are primitive. Suppose for contradiction that $f(x)g(x)$ is not primitive. In other words, $f(x)g(x) = \pi p(x)$, i.e. π divides every coefficient of $f(x)g(x)$. Now, since $f(x)g(x)$ is not primitive, π is not a unit and since R is a UFD, π must be prime. Therefore, (π) is a prime ideal, which means $R/(\pi)$ is an integral domain. Then, mapping to $R/(\pi)[x]$ (i.e. taking modulo π), we see that $\overline{f(x)} \cdot \overline{g(x)} = 0$. But since $f(x)$ is primitive, $\overline{f(x)} \neq 0$ and similarly $\overline{g(x)} \neq 0$. This is impossible in an integral domain since there are no zero divisors. As such, $f(x)g(x)$ must be primitive. □

Note that we have seen that reducible in $F[x]$ implies reducible in $R[x]$. The converse need not be true in general but can be shown for primitive polynomials and, therefore, monic polynomials:

Corollary 2.187. *Let R be a UFD and let F be its field of fractions. If $p(x)$ is a primitive polynomial, then $p(x)$ is irreducible in $R[x]$ if and only if $p(x)$ is irreducible in $F[x]$. In particular, a monic polynomial is irreducible in $R[x]$ if and only if it is irreducible in $F[x]$.* ⌟

Proof. By Gauss's lemma, $p(x)$ is irreducible in $F[x]$ implies it is reducible in $R[x]$. So, if $p(x)$ is irreducible in $R[x]$, then $p(x)$ is irreducible in $F[x]$. Conversely, suppose $p(x)$ is irreducible in $F[x]$. We show that it is irreducible in $R[x]$. We show this via a contrapositive. We show that if $p(x)$ is reducible in $R[x]$, then it is reducible in $F[x]$. Suppose $p(x) = a(x)b(x)$ in $R[x]$ where neither of the two factors is a unit. Since $p(x)$ is a unit, this means neither $a(x)$ nor $b(x)$ is a constant at all. But that means, this is a decomposition in $F[x]$ too where neither is a unit. So $p(x)$ is reducible in $F[x]$. □

Now, we prove the following:

Theorem 2.188. *R is a UFD if and only if $R[x]$ is a UFD.* ⌟

Corollary 2.189. *If R is a UFD, then $R[x_i]_{i\in\mathcal{I}}$ is a UFD.* ⌟

Proof. If $R[x]$ is a UFD, then it is easy to see that R, which is a subring of $R[x]$, must also be a UFD. Now, we prove the converse. Suppose R is a UFD. Suppose $p(x) \neq 0$ and $p(x)$ is not a constant. We will find a unique (up to associate) decomposition into irreducibles. Let $d =$ gcd of coefficients of $p(x)$. Then,

$$p(x) = dp'(x), \text{ where } p'(x) \text{ is primitive.}$$

Since R is a UFD, d can be factored into irreducibles already, so all that remains is for us to show that $p'(x)$ can also be decomposed. Note that if $p'(x)$ is irreducible in $F[x]$, then, since it is primitive, it is also irreducible in $R[x]$ and we are done. So, we now consider the case where it is reducible in $F[x]$.

So, for simplicity, we now study how to decompose a primitive, reducible polynomial $p(x)$ (that is not a constant) into irreducibles that are unique up to associates. Since F is a field, $F[x]$ is a PID and, therefore, a UFD, and so, $p(x)$ can be decomposed into irreducibles in $F[x]$:

$$p(x) = q_1(x)q_2(x)\cdots q_n(x).$$

Now, factor out $u \in F^\times$ to get

$$p(x) = up_1(x)\cdots p_n(x),$$

where each $p_i(x)$ is primitive. Now, $p(x)$ is primitive and $p_i(x)$ are primitive, which forces u to be a unit in R. Also, by the previous corollary, since each $p_i(x)$ is irreducible in $F[x]$, they must be irreducible in $R[x]$, so we now have a decomposition into irreducibles in $R[x]$.

All that remains is for us to show that this is a unique decomposition. Suppose, we have two decompositions:

$$p(x) = q_1(x)\cdots q_r(x) = q_1'(x)\cdots q_s'(x).$$

Since $p(x)$ is primitive, each q_i and q_i' is primitive (therefore, non-constant) and irreducible in $F[x]$. By unique factorization in $F[x]$ which is a UFD, $r = s$ and q_i is associate to q_i' (after some renumbering if needed). All we need to show is that $q_i(x)$ and $q_i'(x)$ are associate in $R[x]$. Suppose,

$$q_i(x) = \frac{a}{b}q_i'(x), \text{ where } a, b \in F.$$

Recall that $q_i(x)$ is primitive. We have

$$bq_i(x) = aq_i'(x) \in R[x].$$

Now the gcd of the coefficients on the left hand side is b and the gcd of the coefficients on the right hand side is a. But in R, which is a UFD, the gcd is unique up to a unit, so $a = bu$ for $u \in R^\times$. So,

$$q_i(x) = uq_i'(x) \in R[x].$$

□

Example 2.190. $\mathbb{Z}[x]$ is a UFD (but not a PID). ⌟

Example 2.191. $\mathbb{Z}[x, y]$ is a UFD. ⌟

Example 2.192. $\mathbb{Q}[x]$ and $\mathbb{Q}[x, y]$ are UFDs. ⌟

3. Categories

3.1. Definitions and Preliminary Examples

We will develop all our definitions under the following assumption: suppose there exists a universe, i.e. a set $\mathfrak{U}$, such that:

(1) if $X \in Y \in \mathfrak{U}$, then $X \in \mathfrak{U}$
(2) if $X, Y \in \mathfrak{U}$, then $\{X, Y\} \in \mathfrak{U}$
(3) if $X \in \mathfrak{U}$, then $\mathscr{P}(X) \in \mathfrak{U}$ where $\mathscr{P}(X)$ is the set of all subsets of X
(4) $\{0, 1, 2, 3, \cdots\} \in \mathfrak{U}$
(5) if $X \in \mathfrak{U}$ and $f : X \to \mathfrak{U}$ is a function, then $\operatorname{im} f \in \mathfrak{U}$.

The existence of such a universe is not guaranteed by ZFC, but we will put that aside for now because we will not really encounter most of the set theoretic issues.

Definition 3.1 (Small Set)**.** *If $X \in \mathfrak{U}$, we call X a small set.* ⌟

Definition 3.2 (Category)**.** *A category, $\mathscr{C}$, consists of a class of objects in the category, denoted by $\operatorname{Obj}(\mathscr{C})$, and for any pair of objects, $A, B \in \operatorname{Obj}(\mathscr{C})$, a se of morphisms between them, denoted by $\operatorname{Mor}_{\mathscr{C}}(A, B)$ or $\operatorname{Mor}_{\mathscr{C}}(A, B)$, that satisfies the following properties:*

(1) *(**Identity Morphisms**) For any object A, there is at least one morphism, $1_A \in \operatorname{Mor}(A, A)$, which we call the identity on A.*
(2) *(**Composition of Morphisms**) Morphisms can be composed: any morphisms $f \in \operatorname{Mor}(A, B)$ and $g \in \operatorname{Mor}(B, C)$ can be composed to get a morphism $gf \in \operatorname{Mor}(A, C)$. In other words, there is a function:*

$$\operatorname{Mor}_{\mathscr{C}}(A, B) \times \operatorname{Mor}_{\mathscr{C}}(B, C) \to \operatorname{Mor}_{\mathscr{C}}(A, C)$$

by

$$(f, g) \to gf$$

for all objects A, B and C.
(3) *(**Associativity of Composition**) The composition of morphisms is associative, i.e. $(hg)f = h(gf)$ for all morphisms f, g and h such that their compositions make sense.*
(4) *(**Identity Laws**) The identity morphisms are identity with respect to composition: for any $f \in \operatorname{Mor}_{\mathscr{C}}(A, B)$, $f1_A = f$ and $1_B f = f$.*
(5) *$\operatorname{Mor}_{\mathscr{C}}(A, B) \cap \operatorname{Mor}_{\mathscr{C}}(C, D) = \emptyset$ unless $A = B$ and $C = D$.*

⌟

Notation 3.3. *We may often simply say $A \in \mathscr{C}$ to mean $A \in \operatorname{Obj}(\mathscr{C})$. We may write $f : A \to B$ or $A \xrightarrow{f} B$ to mean $f \in \operatorname{Mor}_{\mathscr{C}}(A, B)$.*

Remark 3.4. The importance of viewing composition as a function is that the function must send equal inputs to equal outputs. In other words, if $f = g$ are morphisms and h is another morphism, then

$$f = g \implies h \circ f = h \circ g.$$

Similarly,

$$f = g \implies f \circ k = g \circ k.$$

Note that the converse is not true in general. The converse is only true for special morphisms, called monomorphisms and epimorphisms which we will study later on. ⌟

Definition 3.5 (Domain/Source and Codomain/Target)**.** *Given $f \in \operatorname{Mor}(A, B)$ is a morphism, we say A is the domain or source and B is the codomain or target.* ⌟

Definition 3.6 (Endomorphisms)**.** *A morphism from an object $A \in \operatorname{Obj}(\mathscr{C})$ to itself is called an endomorphism. We write $\operatorname{End}_{\mathscr{C}}(A) = \operatorname{Mor}_{\mathscr{C}}(A, A)$.* ⌟

Note that $1_A \in \operatorname{End}_{\mathscr{C}}(A)$.

Example 3.7. The category of groups, denoted by **Grp**, is a category where the objects are groups and the morphisms are group homomorphisms. The category of rings, denoted by **Ring**, is a category where the objects are rings and the morphisms are ring homomorphisms. ⌟

Example 3.8. The category of sets, denoted by **Sets**, where the objects are sets and the morphisms are set maps. ⌟

Example 3.9. Given a set S with some equivalence relation $\sim$ defined on its elements, we can encode the data into a category. Define a category whose objects are elements of S. The morphisms are

$$\mathrm{Mor}(x, y) = \{(x, y)\} \quad \text{if } x \sim y$$

and

$$\mathrm{Mor}(x, y) = \emptyset, \quad \text{if } x \not\sim y.$$

Since equivalence relations are reflective, we can define $1_x = (x, x)$. Since equicalence relations are transitive, we can compose morphisms in this category and the composition is associative. ⌟

Example 3.10. Given S is a set, we can define a category $\hat{S}$ such that $\mathrm{Obj}(\hat{S}) = \mathscr{P}(S)$ and for any objects A and B, we let $\mathrm{Mor}(A, B) = \{(A, B)\}$ if $A \subseteq B$ and $\mathrm{Mor}(A, B) = \emptyset$ otherwise. ⌟

Example 3.11. The category of vector spaces over a field k, denoted by $\mathbf{Vec}_k$, where the objects are vector spaces over k and the morphisms are linear transformations over k. ⌟

Example 3.12 (Preordered sets, Partially ordered sets). A preorder, denoted by $\leq$, is a reflexive (i.e. $a \leq a$), transitive (i.e. $a \leq b, b \leq c \implies a \leq c$) binary relation. A preordered set is a set S with a preorder, $\leq$. A preordered set can be viewed as a category: the objects in the category are the elements in the set, and we have a morphism $x \to y$ if $x \leq y$.

An order on a set is a preorder $\leq$ with the additional property that if $A \leq B$ and $B \leq A$, then $A = B$. A set with a preorder is often called a partially ordered set or poset. Given a partially ordered set (for example, $\mathbb{Z}$ with the relation $\leq$), we can define a category whose objects are elements of the set and for any $x \leq y$, we can define a morphism $x \to y$.

Note: all preorders are not orders. An example is the divisibility relation. Consider $\mathbb{Z}$ where $2 \mid -2$ and $-2 \mid 2$ but $2 \neq 2$. ⌟

In all these examples so far, it makes sense to look at elements within each object of each category. This is not a general feature of categories. In particular, the objects in a category need not be like sets of elements and the morphisms need not be like functions.

Definition 3.13 (Category $\mathscr{C}_A$)**.** *Let $\mathscr{C}$ be a category and let $A \in \mathrm{Obj}(\mathscr{C})$. Define the category $\mathscr{C}_A$ where:*

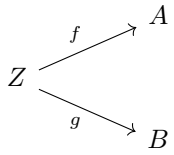
1. *The objects in $\mathscr{C}_A$ are morphisms $X \to A$ for objects $X \in \mathrm{Obj}(\mathscr{C})$ that have morphisms to A. In other words,*
$$\mathrm{Obj}(\mathscr{C}_A) := \{f : X \to A \mid f \text{ is a morphism in } \mathscr{C}\}.$$
2. *The morphisms are commutative diagrams in $\mathscr{C}$, i.e. there is a morphism $\pi : f_1 \to f_2$ for $f_1 : X \to A, f_2 : Y \to A$ if $f_1 = f_2 \circ \pi$ is a commutative diagram in $\mathscr{C}$.*
3. *The identity morphism is simply $1_f : f \to f$ which corresponds to $\mathrm{id}_X \in \mathrm{Mor}_{\mathscr{C}}(X, X)$.*
4. *In $\mathscr{C}_A$, morphisms can be composed since commutative diagrams in $\mathscr{C}$ can be composed. Concretely, for $\sigma : f_1 \to f_2$ and $\tau : f_2 \to f_3$, where $f_1 : X_1 \to A, f_2 : X_2 \to A, f_3 : X_3 \to A$, we can have $\tau\sigma : f_1 \to f_3$ since $f_1 = f_3 \circ \tau\sigma$.*

⌟

Definition 3.14 (Category $\mathscr{C}^A$)**.** *Given category $\mathscr{C}$ with an object A, we can also define the category $\mathscr{C}^A$ whose objects are morphisms $f : A \to X$ for $X \in \mathrm{Obj}(\mathscr{C})$.* ⌟

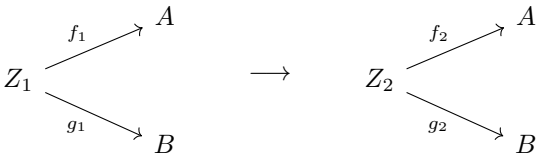
Definition 3.15 (Category $\mathscr{C}_{A,B}$)**.** *Given category $\mathscr{C}$, choose two fixed objects, $A, B \in \mathrm{Obj}(\mathscr{C})$. Then, we define the category $\mathscr{C}_{A,B}$ where:*

(1) *The objects are diagrams in $\mathscr{C}$ of the form:*

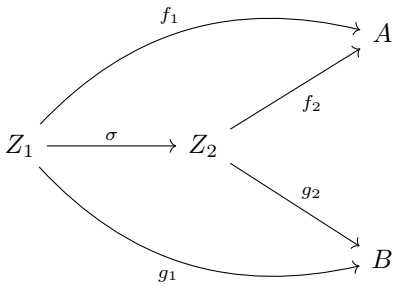


In other words, each object is defined by a pair of morphisms, $f : X \to A$ and $g : X \to B$, that start from some common source $X \in \mathrm{Obj}(\mathscr{C})$.

(2) *The morphisms are commutative diagrams. Concretely, we have a morphism connecting two objects*



that corresponds to some $\sigma : Z_1 \to Z_2$ in $\mathscr{C}$ such that the following diagram commutes:



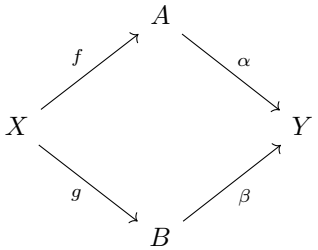
Equivalently, the commutativity conditions are

$$f_1 = f_2 \circ \sigma, \qquad g_1 = g_2 \circ \sigma.$$

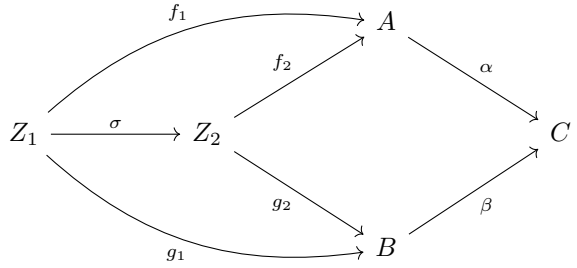
This is indeed a category. For any object in $\mathscr{C}_{A,B}$ (a diagram of a pair of morphisms starting from some X to A and B respectively), there is the identity morphism defined by $1_X \in \mathrm{Mor}(X, X)$. Morphisms in $\mathscr{C}_{A,B}$ can be composed since they can be composed in $\mathscr{C}$, and this composition will be associative. ⌟

Definition 3.16 (Category $\mathscr{C}_{\alpha,\beta}$)**.** *Given category $\mathscr{C}$, choose two fixed morphisms, $\alpha : A \to C$ and $\beta : B \to C$. Then, we define the fibred version of category $\mathscr{C}_{A,B}$, denoted by $\mathscr{C}_{\alpha,\beta}$, where:*

(1) *The objects are commutative diagrams starting from some common source X:*



(2) *The morphisms between objects correspond to morphisms that connect the different sources:*



It is easy to check this is also a category. The identity morphism in $\mathscr{C}_{\alpha,\beta}$ from the object in (1) to itself corresponds to 1_X in $\mathscr{C}$. Morphisms can be composed associatively since they can be composed associatively in commutative diagrams in $\mathscr{C}$. ⌟

Note that there are obvious analogues: $\mathscr{C}^{A,B}$ and $\mathscr{C}^{\alpha,\beta}$.

Finally, as an exercise, we look at how a single group can be viewed as a category:

Definition 3.17 (Group as a Category)**.** *Let $\mathscr{A}$ be a category with only one object, denoted by A, and morphisms in $\mathrm{Mor}(A, A)$. Note that the morphisms can be composed and there is $1_A \in \mathrm{Mor}(A, A)$. Finally, we require that every morphism $g \in \mathrm{Mor}(A, A)$ is an isomorphism, i.e. there exists some $g^{-1} \in \mathrm{Mor}(A, A)$ such that $gg^{-1} = g^{-1}g = 1_A$.*

Then, $\mathrm{Mor}(A, A)$ corresponds to a group. The maps in $g \in \mathrm{Mor}(A, A)$ correspond to elements in the group $g \in G$. The identity map $1_A \in \mathrm{Mor}(A, A)$ corresponds to the identity in the group, $1 \in G$. The composition of morphisms corresponds to the group operation. ⌟

Definition 3.18 (Monoids)**.** *Monoids are the same as groups, except elements are not required to have inverses. Concretely, a monoid is a set equipped with an associative binary operation and a two-sided identity.* ⌟

3.2. **Morphisms**

Definition 3.19 (Isomorphism)**.** *Let $\mathscr{C}$ be a category. We say $f \in \mathrm{Mor}(A, B)$ is an isomorphism if it has a two-sided inverse under composition, i.e. there exists $g \in \mathrm{Mor}(B, A)$ such that $gf = 1_A$ and $fg = 1_B$.* ⌟

Proposition 3.20. *The inverse of an isomorphism is unique. As such, we will denote the inverse of an isomorphism, f, by f^{-1}.* ⌟

Proof. Suppose $f \in \mathrm{Mor}(A, B)$ has two inverses, g and h. Then, $g = g \cdot 1_B = g(fh) = (gf)h = 1_A h = h$. □

Proposition 3.21. *1_A is an isomorphism and $(1_A)^{-1} = 1_A$ for each $A \in \mathscr{C}$. If f is an isomorphism, then f^{-1} is also an isomorphism, with $(f^{-1})^{-1} = f$. If $f \in \mathrm{Mor}(A, B)$ and $g \in \mathrm{Mor}(B, C)$ are isomorphisms, then gf is an isomorphism with $(gf)^{-1} = f^{-1}g^{-1}$.* ⌟

Proposition 3.22 (Automorphism group)**.** *If $A \in \mathrm{Obj}(\mathscr{C})$, then the invertible elements of $\mathrm{Mor}_{\mathscr{C}}(A, A)$ form a group under composition, called the* automorphism group *of A, denoted by $\mathrm{Aut}(A)$. Furthermore, the following are true:*

1. *If $f, g \in \mathrm{Aut}(A)$, then $gf \in \mathrm{Aut}(A)$, and composition of automorphisms is associative.*
2. *$1_A \in \mathrm{Aut}(A)$.*
3. *For any $f \in \mathrm{Aut}(A)$, $f^{-1} \in \mathrm{Aut}(A)$.*
4. *If $A \cong B$ (meaning there is some isomorphism $f : A \to B$), then $\mathrm{Aut}(A) \cong \mathrm{Aut}(B)$.*

⌟

Proposition 3.23 (Automorphism group)**.** *If $A \in \mathrm{Obj}(\mathscr{C})$, then the invertible elements of $\mathrm{Mor}_{\mathscr{C}}(A, A)$ form a group under composition, called the* automorphism group *of A, denoted by $\mathrm{Aut}(A)$. Furthermore, the following are true:*

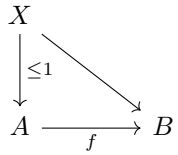
1. *If $f, g \in \mathrm{Aut}(A)$, then $gf \in \mathrm{Aut}(A)$, and composition of automorphisms is associative.*
2. *$1_A \in \mathrm{Aut}(A)$.*
3. *For any $f \in \mathrm{Aut}(A)$, $f^{-1} \in \mathrm{Aut}(A)$.*
4. *If $A \cong B$, then $\mathrm{Aut}(A) \cong \mathrm{Aut}(B)$.*

⌟

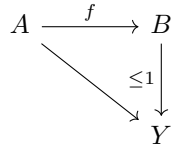
Definition 3.24 (Discrete Category)**.** *A category whose only morphisms are the identity morphisms is called a discrete category.* ⌟

Definition 3.25 (Monomorphisms/Mono and Epimorphisms/Epi)**.** *Let $\mathscr{C}$ be a category.*

1. *A morphism $f \in \mathrm{Mor}(A, B)$ is called a monomorphism (or mono) if the following holds: for all objects X and all morphisms $\alpha, \alpha' : X \to A$, $f \circ \alpha = f \circ \alpha \implies \alpha = \alpha'$. Pictorially:*



2. *A morphism $f \in \mathrm{Mor}(A, B)$ is called an epimorphism (or epi) if the following holds: for all objects Y and all morphisms $\beta, \beta' : B \to Y$, $\beta \circ f = \beta' \circ f \implies \beta = \beta'$. Pictorially,*



⌟

Proposition 3.26. *In the category* **Set***, epimorphisms are equivalent to surjections and monomorphisms are equivalent to injections.* ⌟

Proof. Suppose $f : A \to B$ is a surjection. Now, suppose $\beta, \beta' : B \to C$ such that $\beta' \circ f = \beta \circ f$. Then, for any $b \in B$, $\beta'(b) = \beta'(f(a)) = \beta(f(a)) = \beta(b)$, where a is the preimage of b under f (which exists since f is a surjection), implying f is an epimorphism. Conversely, suppose f is an epimorphism. Now, suppose f is not a surjection, i.e. there exists some $b \in B$ such that $b \notin f(A)$. Define $g : B \to \{0, 1\}$ via $g(x) = 0$ for all $x \in B$. Also define $h : B \to \{0, 1\}$ via $h(x) = 1$ if $x \notin f(A)$ and $h(x) = 0$ if $x \in f(A)$. Then, note that $g \circ f = h \circ f$, but since f is not surjective, $g \neq h$. This is a contradiction, so f must be surjective. Now, suppose $f : A \to B$ is a monomorphism. Suppose $f(a) = f(a')$ but $a \neq a'$. Then, we can construct two morphisms: $\iota_1, \iota_2 : \{0\} \to X$ by $\iota_1(0) = a$ and $\iota_2(0) = a'$. Clearly, $\iota_1 \neq \iota_2$, but $f \circ \iota_1 = f \circ \iota_2$ which contradicts that f is a monomorphism. So, $a = a'$ and, therefore, f is injective. Conversely, suppose f is injective. Then, for morphisms $g, h : X \to A$ such that $f \circ g = f \circ h$, we have that $f(g(x)) = f(h(x))$ for all $x \in X$. Since f is a injective, $g(x) = h(x)$ for all x, so $g = h$. □

Corollary 3.27. *In the category of groups,* **Grp***, we have that epimorphisms are surjections and monomorphisms are injections.* ⌟

Observation. *Surjections and epimorphisms are not necessarily equivalent. For example, in the category of commutative and unital rings, define $\iota : \mathbb{Z} \to \mathbb{Q}$ to be the inclusion morphism and let $g, h : \mathbb{Q} \to R$ be morphisms such that $g \circ \iota = h \circ \iota$. We claim that ι is an epimorphism. This is because for any $a/b \in \mathbb{Q}$, $g(\frac{a}{b}) = g(a)g(b)^{-1} = g(\iota(a))g(\iota(b))^{-1} = h(\iota(a))h(\iota(b))^{-1} = h(a)h(b)^{-1} = h(\frac{a}{b})$.*

Proposition 3.28. *Composition of monomorphisms is a monomorphism. Composition of epimorphisms is an epimorphism.* ⌟

Proof. Suppose $f \in \mathrm{Mor}(A, B)$ and $g \in \mathrm{Mor}(B, C)$ are epimorphisms. Consider $gf \in \mathrm{Mor}(A, C)$ with $\pi_1, \pi_2 : C \to Y$ such that $\pi_1 \circ gf = \pi_2 \circ gf$. Now, $(\pi_1 \circ g) \circ f = (\pi_2 \circ g) \circ f$ and since f is an epimorphism, $\pi_1 \circ g = \pi_2 \circ g$. Since f is an epimorphism, we have that $\pi_1 = \pi_2$, which shows gf is an epimorphism. Similarly logic shows the statement for monomorphisms. □

3.3. **Universal Properties**

We start by defining initial, final, and terminal objects in a category:

Definition 3.29 (Initial, Final, and Terminal Objects)**.** *Let $\mathscr{C}$ be a category. Then, $I \in \mathrm{Obj}(\mathscr{C})$ is called an initial object in $\mathscr{C}$ if for all $X \in \mathrm{Obj}(\mathscr{C})$, there exists only one unique morphism $f : I \to X$. An object $F \in \mathrm{Obj}(\mathscr{C})$ is called a final object if for all $X \in \mathrm{Obj}(\mathscr{C})$, there exists exactly one unique morphism $f : X \to F$. An object that is either initial or final is called a terminal object.* ⌟

Example 3.30 (Terminal objects in **Set**). In the category of sets, $\emptyset$ is an initial object and $\{x\}$ for any element x (or any singleton) is a final object. To see the former, we see that the following statement is vacuously true: for any set S, there exists a morphism $f : \emptyset \to S$ such that $f(x) \in X$ is well-defined for any $x \in \emptyset$. Its uniqueness is also vacuously true. To see the latter, there is a unique morphism from any set S to $\{x\}$ that maps everything to x. ⌟

Example 3.31. In the category **Ring** of commutative rings with identity and identity-preserving ring homomorphisms, the initial object is $\mathbb{Z}$ and the final object is the zero ring. ⌟

Example 3.32. In the category **Top**, the initial object is the empty topological space $\emptyset$ and a final object is any singleton topological space $\{x\}$. ⌟

Proposition 3.33. *The initial object in a category is unique up to unique isomorphism, i.e. if I_1 and I_2 are both initial objects, then $I_1 \cong I_2$ via a unique isomorphism. The final object in a category is also unique up to unique isomorphism.* ⌟

Proof. Let I_1 and I_2 both be initial. Then, there exists a unique morphism $\iota_1 : I_1 \to I_1$ and a unique morphism $\iota_1 : I_2 \to I_1$. Now, $\iota_1 \circ \iota_1 \in \mathrm{Mor}(I_1, I_2)$. However, since I_1 is unique, there is only one unique morphism from I_1 to itself and since 1_{I_1} is another such candidate, we must have that $\iota_2 \circ \iota_1 = 1_{I_1}$. Similarly, $\iota_1 \circ \iota_2 = 1_{I_1}$. Since ι_1 is a unique map, the isomorphism is unique. The proof for final objects is similar. □

At a high-level, when we say an object has a *universal property*, we mean that the object is a terminal object in some category defined with respect to the property. As an example, we look at the following:

Proposition 3.34. *The quotient $A/\sim$ is universal with respect to the property of mapping A to a set such that equivalent elements have the same image. This is equivalent to saying that in the category $\mathscr{C}^A$ (see 3.14), the morphism $\pi : A \to A/\sim$ is the initial object.* ⌟

Proof. We denote each object in the category $\mathscr{C}^A$ by (f, Z) if $f : A \to Z$ is a morphism in the original category $\mathscr{C}$ that A is an object of. Now, $(\pi, A/\sim)$ is an object in $\mathscr{C}^A$. For any other object (f, Z), we must find a unqiue morphism $\pi_f : \pi \to f$ (i.e. some morphism π' such that $f = \pi' \circ \pi$ in $\mathscr{C}$). To do so, define

$$\pi' : A/\sim \to Z$$

by

$$\pi'(\bar{a}) = f(a)$$

where $\bar{a}$ is some representative in the equivalence class of a. This is well-defined since if $\bar{a} = \bar{a}'$, then $a \sim a'$, so $f(a) = f(a')$, which means $\pi'(\bar{a}) = \pi(\bar{a})$. Clearly, $f = \pi' \circ \pi$. Uniqueness is straightforward. □

3.4. **Functors**

Definition 3.35 (Covariant Functors)**.** *A* covariant functor $F : \mathscr{A} \to \mathscr{B}$ *from a category* $\mathscr{A}$ *to a category* $\mathscr{B}$ *consists of the following data:*

(1) *A map on objects*
$$F : \mathrm{Obj}(\mathscr{A}) \longrightarrow \mathrm{Obj}(\mathscr{B}).$$
(2) *For every* $A_1, A_2 \in \mathrm{Obj}(\mathscr{A})$ *and every morphism* $m : A_1 \to A_2$ *in* $\mathscr{A}$*, there is a morphism*
$$F(m) : F(A_1) \longrightarrow F(A_2)$$
in $\mathscr{B}$*.*

Furthermore, F *preserves identities:*
$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$$
for every $A \in \mathrm{Obj}(\mathscr{A})$*, and* F *preserves composition:*
$$F(m_1 \circ m_2) = F(m_1) \circ F(m_2)$$
whenever the composition $m_1 \circ m_2$ *is defined.* ⌟

Diagrammatically, a covariant functor sends
$$A_1 \xrightarrow{m} A_2 \qquad \xrightarrow{F} \qquad F(A_1) \xrightarrow{F(m)} F(A_2).$$

Definition 3.36 (Contravariant Functors)**.** *A* contravariant functor $F : \mathscr{A} \to \mathscr{B}$ *consists of the following data:*

(1) *A map on objects*
$$F : \mathrm{Obj}(\mathscr{A}) \longrightarrow \mathrm{Obj}(\mathscr{B}).$$
(2) *For every* $A_1, A_2 \in \mathrm{Obj}(\mathscr{A})$ *and every morphism* $m : A_1 \to A_2$ *in* $\mathscr{A}$*, there is a morphism*
$$F(m) : F(A_2) \longrightarrow F(A_1)$$
in $\mathscr{B}$*.*

Again, F *preserves identities:*
$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}\,.$$
However, F *reverses composition:*
$$F(m_1 \circ m_2) = F(m_2) \circ F(m_1)$$
whenever the composition $m_1 \circ m_2$ *is defined.* ⌟

Examples 3.37. *The following are examples of covariant functors.*

(1) ***Identity functor.*** *For any category* $\mathscr{A}$*, the identity functor*
$$\mathrm{id}_{\mathscr{A}} : \mathscr{A} \longrightarrow \mathscr{A}$$
sends each object and each morphism to itself.
(2) ***Forgetful functors.*** *Forgetful functors are functors that map objects in some category to objects in a different category where some of its internal structure is forgotten. An example of a forgetful functor is*
$$F : \mathbf{Vec}_k \longrightarrow \mathbf{Sets},$$
which sends each vector space to its underlying set and each linear transformation to its underlying set map.
(3) *Another example of a forgetful functor is*
$$\mathbf{Rings} \to \mathbf{Mon},$$
from rings to monoids which forgets the additive structure but not the multiplicative structure of rings.

Example 3.38. Consider the functor

$$\mathrm{GL}_n : \mathbf{Rings} \to \mathbf{Grps}$$

which maps as follows: for every ring A, define

$$\mathrm{GL}_n(A) = \{M \in M_n(A) \mid M \text{ is invertible}\}.$$

This is the general linear group of degree n over A, with group operation given by matrix multiplication. Given a ring homomorphism $f : A \to B$, define

$$\mathrm{GL}_n(f) : \mathrm{GL}_n(A) \to \mathrm{GL}_n(B)$$

by applying f entrywise. For example, when $n = 2$,

$$\mathrm{GL}_2(f) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} f(a) & f(b) \\ f(c) & f(d) \end{pmatrix}.$$

This is indeed a functor. Firstly, it is well-defined: If $M \in \mathrm{GL}_n(A)$, then $\mathrm{GL}_n(f)(M) \in \mathrm{GL}_n(B)$. Secondly, Secondly, it preserves identities: for every ring A, $\mathrm{GL}_n(1_A) = 1_{\mathrm{GL}_n(A)}$. Finally, it preserves composition: For ring homomorphisms $f : A \to B$ and $g : B \to C$, $\mathrm{GL}_n(g \circ f) = \mathrm{GL}_n(g) \circ \mathrm{GL}_n(f)$. ⌟

The following is a particularly important example of a covariant functor:

Definition 3.39 (Category of categories)**.** *The category of categories is a category where the objects are categories and the morphisms are functors. The identity morphisms on each object (i.e. category), $\mathscr{A}$, is the identity functor, $1_{\mathscr{A}}$. This category is denoted by* **CAT**. ⌟

Diagrammatically, a contravariant functor sends

$$A_1 \xrightarrow{\ m\ } A_2 \qquad \xrightarrow{\ F\ } \qquad F(A_1) \xleftarrow{\ F(m)\ } F(A_2)\ .$$

Remark 3.40. We can regard a contravariant functor F as a covariant functor

$$F : \mathscr{A}^{\mathrm{op}} \longrightarrow \mathscr{B},$$

where $\mathscr{A}^{\mathrm{op}}$ is the *opposite category* of $\mathscr{A}$, obtained by reversing the direction of every morphism. ⌟

Definition 3.41 (Presheaf)**.** *Let $\mathscr{A}$ be a category. Then, a presheaf on $\mathscr{A}$ is a functor $\mathscr{A}^{\mathrm{op}} \to$* **Set** ⌟

Recall dual vector spaces (see B.1).

Definition 3.42 (Dual Functor)**.** *We now construct the contravariant functor called the dual functor from the category of vector spaces over field k to the category of vector spaces over field k:*

$$(-)^* : k\text{-}\mathbf{Vect} \to k\text{-}\mathbf{Vect}.$$

*Recall that the objects of k-*Vect *are vector spaces over k, and the morphisms are k-linear maps. The* dual functor $(-)^*$ *is defined as follows:*

(1) *On objects: For every vector space V over k,*

$$V \longmapsto V^* := \mathrm{Hom}_k(V, k).$$

(2) *On morphisms: Given a linear map $f : V \to W$, define its* dual map

$$f^* : W^* \to V^*$$

by

$$f^*(\lambda) := \lambda \circ f$$

for every $\lambda \in W^*$. As such, for every $v \in V$,

$$f^*(\lambda)(v) = \lambda(f(v)).$$

⌟

We verify the two functor axioms. Firstly, this functor takes identity to identity. Let V be a vector space. Then, $(1_V)^* = 1_{V^*}$ because, for every $\varphi \in V^*$, $(1_V)^*(\varphi) = \varphi \circ 1_V = \varphi$. Secondly, it preserves composition: Suppose we have linear maps $V \xrightarrow{f} W \xrightarrow{g} U$. The corresponding dual maps are $U^* \xrightarrow{g^*} W^* \xrightarrow{f^*} V^*$. Let $\varphi \in U^*$. Then

$$\begin{aligned}(g \circ f)^*(\varphi) &= \varphi \circ (g \circ f) \\ &= (\varphi \circ g) \circ f \quad \text{(by associativity)} \\ &= f^*(g^*(\varphi)) \\ &= (f^* \circ g^*)(\varphi).\end{aligned}$$

Therefore, $(g \circ f)^* = f^* \circ g^*$. Thus both functor axioms hold, and $(-)^*$ is a contravariant functor.

Definition 3.43 (Functor h^A or $\mathrm{Hom}_{\mathscr{C}}(A, _)$)**.** *Suppose $A \in \mathrm{Obj}(\mathscr{C})$. We define the covariant functor*

$$h^A : \mathscr{C} \longrightarrow \mathbf{Set}$$

by

$$h^A(B) = \mathrm{Mor}_{\mathscr{C}}(A, B).$$

Furthermore, for any morphism $f : X \to Y$ in $\mathscr{C}$, define

$$h^A(f) : \mathrm{Mor}_{\mathscr{C}}(A, X) \longrightarrow \mathrm{Mor}_{\mathscr{C}}(A, Y)$$

by

$$h^A(f)(g) = f \circ g,$$

where $g : A \to X$.

Thus, h^A acts on morphisms by postcomposition. ⌟

Definition 3.44 (Functor h_A or $\mathrm{Hom}_{\mathscr{C}}(_, A)$)**.** *Let $A \in \mathrm{Obj}(\mathscr{C})$. Define*

$$h_A : \mathscr{C} \longrightarrow \mathscr{S}$$

by

$$h_A(X) = \mathrm{Mor}_{\mathscr{C}}(X, A).$$

For a morphism $f : X \to Y$ in $\mathscr{C}$, define

$$h_A(f) : \mathrm{Mor}_{\mathscr{C}}(Y, A) \longrightarrow \mathrm{Mor}_{\mathscr{C}}(X, A)$$

by

$$h_A(f)(g) = g \circ f,$$

where $g : Y \to A$.

Thus, h_A acts on morphisms by precomposition. ⌟

Example 3.45 (Free Groups as Functors). Suppose S is a set. Recall that the free group on S, denoted by $F(S)$, is the collection of formal expressions, or words, such as $x^{-4}y^2zy^{-3}$ where $x, y, z \in S$. Two words are equal if one can be obtained from the other by cancellation; for example, $x^3xy = x^4y = x^2x^{-1}yx^2y$. Words can be multiplies: $x^3y \cdot x^2y^{-1} = x^y x^2 y^{-1}$. Now, we can view F as a functor instead:

$$F : \mathbf{Set} \to \mathbf{Grp}$$

via

$$S \to F(S)$$

and

$$(s \xrightarrow{f} s') \to (F(S) \xrightarrow{F(f)} F(S'))$$

where we are mapping each set to the free group on the set. ⌟

Definition 3.46 (Faithful and Full Functors)**.** *A covariant functor $F : \mathscr{A} \to \mathscr{B}$ is called* faithful *if, for every pair of objects $A, A' \in \mathrm{Obj}(\mathscr{A})$, the induced map*

$$\mathrm{Mor}_{\mathscr{A}}(A, A') \longrightarrow \mathrm{Mor}_{\mathscr{B}}(F(A), F(A'))$$

is injective. The functor F is called full *if this map is surjective. If F is both faithful and full, then it is called* fully faithful. ⌟

Example 3.47. Consider the functor: **Ab** $\to$ **Grps**. Then, this functor is both faithful and full: this is because group homomorphisms between any pair $F(A)$ and $F(A')$ also respect the abelian structure, so if $F(f) = F(g)$ as group morphisms, then $f = g$ as abelian group morphisms. ⌟

Proposition 3.48. *The contravariant dual functor*

$$(-)^* : k\text{-}\mathsf{Vect} \to k\text{-}\mathsf{Vect}$$

is faithful, but not full. ⌟

Proof. Recall that for any pair of vector spaces V, W, the dual functor induces a map

$$\Phi_{V,W} : \mathrm{Hom}_k(V, W) \to \mathrm{Hom}_k(W^*, V^*), \qquad f \mapsto f^*.$$

We first show that this map is injective for all V, W. Suppose $f, g : V \to W$ satisfy $f^* = g^*$. So, for every $\varphi \in W^*$,

$$f^*(\varphi) = g^*(\varphi) \implies \varphi \circ f = \varphi \circ g.$$

So, for every $v \in V$ and $\varphi \in W^*$, $\varphi(f(v)) = \varphi(g(v))$, which means

$$\varphi(f(v) - g(v)) = 0.$$

Now, recall the following fact about linear functionals: if $w \in W$ satisfies $\varphi(w) = 0$ for every $\varphi \in W^*$, then $w = 0$. This is because, if $w \neq 0$, we may extend $\{w\}$ to a basis of W and define a linear functional $\varphi : W \to k$ that sends w to 1 and all other basis vectors to 0. This contradicts the assumption that every linear functional vanishes on w. We apply this to our case now. For a given fixed v, set $w = f(v) - g(v) \in W$. Then, $\varphi \in W^*$ satisfies that $\varphi(w) = 0$ for every $\varphi \in W^*$. As such, $w = 0 \implies f(v) = g(v)$. Since this is true for all v, we see that $f = g$, and therefore, $\Phi_{V,W}$ is injective.

To show that the dual functor is not full, it suffices to find vector spaces V, W and a linear map $T : W^* \to V^*$ that is not equal to f^* for any linear map $f : V \to W$. Let $V = k$ and let

$$W = \bigoplus_{n=1}^{\infty} ke_n,$$

which is the vector space of infinite sequences of elements of k in which only finitely many entries are nonzero. Thus, every element of W is a finite linear combination of the basis vectors $e_1, e_2, \ldots$, where $e_1 = (1, 0, 0, 0, \cdots)$ and $e_2 = (0, 1, 0, 0, \cdots)$ and so on. For each $n \geq 1$, define $\delta_n \in W^*$ by

$$\delta_n(e_m) = \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$$

Now, we define $\sigma \in W^*$ by

$$\sigma(e_n) = 1$$

for every $n \geq 1$. Note that

$$\sigma \notin \mathrm{span}_k\{\delta_1, \delta_2, \ldots\},$$

since every finite linear combination of the δ_n vanishes on all but finitely many basis vectors, whereas σ does not. So,

$$\{\delta_1, \delta_2, \ldots, \sigma\}$$

is a linearly independent subset of W^*. Extending this set to a basis of W^*, we can define a linear functional

$$L : W^* \to k$$

satisfying

$$L(\delta_n) = 0 \quad \text{for all } n, \qquad L(\sigma) = 1.$$

Now define the linear map

$$T : W^* \to k^*$$

by

$$T(\varphi)(a) = aL(\varphi)$$

for every $\varphi \in W^*$ and $a \in k$. We claim that T is not the dual of any linear map $f : k \to W$. Suppose, for contradiction, that $T = f^*$ for some $f : k \to W$. Let $w = f(1) \in W$. Then, for every n,

$$\begin{aligned}\delta_n(w) &= \delta_n(f(1)) \\ &= f^*(\delta_n)(1) \\ &= T(\delta_n)(1) \\ &= L(\delta_n) \\ &= 0.\end{aligned}$$

Since all coordinates of w are zero, we conclude that $w = 0$. However,

$$f^*(\sigma)(1) = \sigma(f(1)) = \sigma(0) = 0,$$

whereas

$$T(\sigma)(1) = L(\sigma) = 1.$$

This contradicts $T = f^*$. Therefore T is not the dual of any linear map $f : k \to W$, so $\Phi_{k,W}$ is not surjective. □

Definition 3.49 (Identity Functor)**.** *Let $\mathscr{C}$ be a category. Then, we have the identity function:*

$$1_{\mathscr{C}} : \mathscr{C} \to \mathscr{C}$$

which maps

$$1_{\mathscr{C}}(A) = A, \forall A \in \text{obj}(\mathscr{C})$$

and

$$1_{\mathscr{C}}(f) = f, \forall f \in \text{Mor}(\mathscr{C}).$$

The identity functor is full and faithful. ⌟

Definition 3.50 (Constant Functor)**.** *Let $\mathscr{C}$ and $\mathscr{D}$ be categories with $Y \in \text{obj}(\mathscr{D})$ a fixed object. Then, we have the constant functor $F_Y : \mathscr{C} \to \mathscr{D}$ defined by*

$$F_Y(X) = Y, \forall X \in \text{Obj}(\mathscr{C})$$

and

$$F_Y(f) = 1_Y, \forall f \in \text{Mor}(\mathscr{C}).$$

⌟

Definition 3.51 (Opposite/Dual Category)**.** *Given $\mathscr{C}$ is a category, there is an opposite (or dual) category, denoted by $\mathscr{C}^{\text{op}}$, where the objects are the same as the objects in $\mathscr{C}$ but the morphisms are reverses (i.e. if $f : A \to B$ in $\mathscr{C}$, then $f : B \to A$ in $\mathscr{C}^{\text{op}}$).* ⌟

The opposite category's construction speaks to the central notion that, in categories, we treat objects and morphisms without paying attention to how they act on "elements" so to speak. For example, we define if $f : A \to B$ in $\mathscr{C}$, the one might expect this to be a function such that for all "elements" $a \in A$, we get that $f(a) = b \in B$. The problem with such a treatment is that then it is impossible to view f as a morphism $f : B \to A$ because $f(b)$ is not even defined for "element" $b \in B$. Instead, the correct way to treat f is as simply some morphism, $f \in \text{Hom}_{\mathscr{C}}(A, B)$. Then, we construct a new set of morphisms $\text{Hom}_{\mathscr{C}^{\text{op}}}(B, A)$ where each $f \in \text{Hom}_{\mathscr{C}^{\text{op}}}(B, A)$ corresponds to $f \in \text{Hom}_{\mathscr{C}}(A, B)$. In particular, for the definition of the category and functors to hold, this correspondence is sufficient; we do not need these morphisms to be defined on "elements".

Proposition 3.52. *If $\mathscr{C}$ is a category, then there is a category, $\mathscr{C}^{\text{op}}$, with the same objects as $\mathscr{C}$ and such that*

$$\text{Hom}_{\mathscr{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathscr{C}}(Y, X).$$

Then, to give a contravariant functor $\mathscr{C} \to \mathscr{D}$ is the same as giving a covariant functor $\mathscr{C}^{\text{op}} \to \mathscr{D}$, which is also the same as giving a covariant functor $\mathscr{C} \to \mathscr{D}^{\text{op}}$. ⌟

Proof. First, we construct the category $\mathscr{C}^{\mathrm{op}}$ such that $\mathrm{Obj}(\mathscr{C}^{\mathrm{op}}) = \mathrm{Obj}(\mathscr{C})$. Now, if $X, Y \in \mathrm{Obj}(\mathscr{C}^{\mathrm{op}})$, we define

$$\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,Y) := \{f^{\mathrm{op}} \mid X \xrightarrow{f^{\mathrm{op}}} Y \text{ in } \mathscr{C}^{\mathrm{op}}\} = \{f \mid Y \xrightarrow{f} X \text{ in } \mathscr{C}\} =: \mathrm{Hom}_{\mathscr{C}}(Y,X).$$

Note that this is a correspondence: this is because $(\mathscr{C}^{\mathrm{op}})^{\mathrm{op}}$ has morphisms $\mathrm{Hom}_{(\mathscr{C}^{\mathrm{op}})^{\mathrm{op}}}(X,Y) = \{(f^{\mathrm{op}})^{\mathrm{op}} \mid X \xrightarrow{(f^{\mathrm{op}})^{\mathrm{op}}} Y \text{ in } (\mathscr{C}^{\mathrm{op}})^{\mathrm{op}}\} = \{f^{\mathrm{op}} \mid Y \xrightarrow{f^{\mathrm{op}}} X \text{ in } \mathscr{C}^{\mathrm{op}}\} = \{f \mid X \xrightarrow{f} Y \text{ in } \mathscr{C}\} = \mathrm{Hom}_{\mathscr{C}}(X,Y)$. As such, $Y \xrightarrow{f} X$ in $\mathscr{C}$ corresponds to $X \xrightarrow{f^{\mathrm{op}}} Y$ in $\mathscr{C}^{\mathrm{op}}$.

We define compositions of morphisms as follows: if $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,Y)$ and $g^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(Y,Z)$, then we define $g^{\mathrm{op}} \circ f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,Z)$ to be the morphism $f \circ g \in \mathrm{Hom}_{\mathscr{C}}(Z,X)$. This exists since in $\mathscr{C}$, $f \in \mathrm{Hom}_{\mathscr{C}}(Y,X)$ and $g \in \mathrm{Hom}_{\mathscr{C}}(Z,Y)$, so we get the composition $f \circ g \in \mathrm{Hom}_{\mathscr{C}}(Z,X)$. In summary,

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} \text{ in } \mathscr{C}^{\mathrm{op}} \leftrightarrow f \circ g \text{ in } \mathscr{C}.$$

Note in particular that $(f \circ g)^{\mathrm{op}} = g^{\mathrm{op}} \circ f^{\mathrm{op}}$. This is a category since:

(1) Identity morphisms exist: for any object $X \in \mathscr{C}^{\mathrm{op}}$, there is an identity morphism $\mathrm{id}_X^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,X)$ which is the same as the identity morphism $\mathrm{id}_X \in \mathrm{Hom}_{\mathscr{C}}(X,X)$ in category $\mathscr{C}$. This morphism satisfies that for any $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,Y)$, we have that $f^{\mathrm{op}} \circ \mathrm{id}_X^{\mathrm{op}} = f^{\mathrm{op}}$ because $f \in \mathrm{Hom}_{\mathscr{C}}(Y,X)$ in $\mathscr{C}$, so, in $\mathscr{C}$, we have that $\mathrm{id}_X \circ f = f$. Remembering our definition of composition, we get the desired result. A similar argument shows that if $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(Y,X)$, then $\mathrm{id}_X^{\mathrm{op}} \circ f^{\mathrm{op}} = f^{\mathrm{op}}$ in $\mathscr{C}^{\mathrm{op}}$.

(2) Composition of morphisms is associative: suppose $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(X,Y), g^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(Y,Z)$ and $h^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(Z,W)$. Then, $h^{\mathrm{op}} \circ (g^{\mathrm{op}} \circ f^{\mathrm{op}})$ in $\mathscr{C}^{\mathrm{op}}$ corresponds to $(f \circ g) \circ h$ in $\mathscr{C}$. Since composition is associative in $\mathscr{C}$, this is equal to $f \circ (g \circ h)$, which corresponds to the morphism $(h^{\mathrm{op}} \circ g^{\mathrm{op}}) \circ f^{\mathrm{op}}$ in $\mathscr{C}^{\mathrm{op}}$. As such

$$h^{\mathrm{op}} \circ (g^{\mathrm{op}} \circ f^{\mathrm{op}}) = (h^{\mathrm{op}} \circ g^{\mathrm{op}}) \circ f^{\mathrm{op}} \quad \text{in } \mathscr{C}^{\mathrm{op}}.$$

Now, suppose $F : \mathscr{C} \to \mathscr{D}$ is a contravariant functor. Then, for all $X \in \mathrm{Obj}(\mathscr{C})$, $F(X) \in \mathrm{Obj}(\mathscr{D})$ and for any morphism $X \xrightarrow{f} Y$, we have that $F(Y) \xrightarrow{F(f)} F(X)$ in $\mathscr{D}$. For any morphism $Y \xrightarrow{g} Z$, we have that

$$F(g \circ f) = F(f) \circ F(g).$$

Now, we define the functor $\Psi : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$ as follows:

$$\Psi(X) = F(X), \forall X \in \mathrm{Obj}(\mathscr{C}^{\mathrm{op}}) = \mathrm{Obj}(\mathscr{C}),$$

and if

$$Y \xrightarrow{f^{\mathrm{op}}} X \quad \text{in } \mathscr{C}^{\mathrm{op}},$$

then we know that

$$X \xrightarrow{f} Y \quad \text{in } \mathscr{C},$$

so we define

$$\Psi(f^{\mathrm{op}}) = F(f) \in \mathrm{Hom}_{\mathscr{D}}(F(Y), F(X)), \forall f^{\mathrm{op}} \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(Y,X).$$

In other words, Ψ sends a morphism in $\mathscr{C}^{\mathrm{op}}$ to the same image that F sends the corresponding morphism in $\mathscr{C}$ to. We now check that this is indeed a functor:

(1) Ψ sends identity to identity: consider $\mathrm{id}_X^{\mathrm{op}} : X \to X$ in $\mathscr{C}^{\mathrm{op}}$. This corresponds to $\mathrm{id}_X : X \to X$ in $\mathscr{C}$. Then, $\Psi(\mathrm{id}_X^{\mathrm{op}}) = F(\mathrm{id}_X) = \mathrm{id}_{F(X)} = \mathrm{id}_{\Psi(X)}$ since F is a contravariant functor that must preserve identities and $F(X) = \Psi(X)$ for any object X.

(2) Ψ preserves composition. Suppose $X \xrightarrow{f^{\mathrm{op}}} Y \xrightarrow{g^{\mathrm{op}}} Z$ in $\mathscr{C}^{\mathrm{op}}$. Then, $g^{\mathrm{op}} \circ f^{\mathrm{op}}$ in $\mathscr{C}^{\mathrm{op}}$ corresponds to the morphism $f \circ g$ in $\mathscr{C}$. Now, $F(f \circ g) = F(g) \circ F(f)$ since F is a contravariant functor. Using the definition of our new functor, we get that:

$$\Psi(g^{\mathrm{op}} \circ f^{\mathrm{op}}) = F(f \circ g) = F(g) \circ F(f) = \Psi(g^{\mathrm{op}}) \circ \Psi(f^{\mathrm{op}}).$$

As such, we have found a covariant functor $\Psi : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$. Since this functor agrees with F on every object and morphism (through corresondence between $\mathrm{Mor}(\mathscr{C}^{\mathrm{op}})$ and $\mathrm{Mor}(\mathscr{C})$), we see that F uniquely determines a covariant functor Ψ.

Now, we show the converse: suppose $\Psi : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$ is a covariant functor. Now, we construct a contravariant functor $F : \mathscr{C} \to \mathscr{D}$. Let

$$F(X) = \Psi(X), \forall X \in \mathrm{Obj}(\mathscr{C}) = \mathrm{Obj}(\mathscr{C}^{\mathrm{op}}).$$

Let $Y \xrightarrow{f^{\text{op}}} X$ in $\mathscr{C}^{\text{op}}$. Then, in $\mathscr{C}$, we have that $X \xrightarrow{f} Y$. Note that, since Ψ is a covariant functor and since Ψ and F agree on objects, we have that

$$\Psi(Y) \xrightarrow{\Psi(f^{\text{op}})} \Psi(X) \implies F(Y) \xrightarrow{\Psi(f^{\text{op}})} F(X).$$

So, we define $F(f) = \Psi(f^{\text{op}}) \in \text{Hom}_{\mathscr{D}}(F(Y), F(X))$ for all $f \in \text{Hom}_{\mathscr{C}}(X, Y)$. Note that, then, F takes identity to identity: $\text{id}_X \in \text{Hom}_{\mathscr{C}}(X, X)$ corresponds to $\text{id}_X^{\text{op}} \in \text{Hom}_{\mathscr{C}^{\text{op}}}(X, X)$, so $F(\text{id}_X) = \Psi(\text{id}_X^{\text{op}}) = \text{id}_{\Psi(X)} = \text{id}_{F(X)}$. Finally, we check composition of morphisms: let $f \in \text{Hom}_{\mathscr{C}}(X, Y)$ and $g \in \text{Hom}_{\mathscr{C}}(Y, Z)$. Then, $g \circ f \in \text{Hom}_{\mathscr{C}}(X, Z)$. Then, in $\mathscr{C}^{\text{op}}$, we have that $f^{\text{op}} \in \text{Hom}_{\mathscr{C}^{\text{op}}}(Y, X)$ and $g^{\text{op}} \in \text{Hom}_{\mathscr{C}^{\text{op}}}(Z, Y)$ and $f^{\text{op}} \circ g^{\text{op}} \in \text{Hom}_{\mathscr{C}^{\text{op}}}(Z, X)$. Now, given the definition of morphisms in the opposite category, we can define

$$F(g \circ f) = \Psi(f^{\text{op}} \circ g^{\text{op}}) = \Psi(f^{\text{op}}) \circ \Psi(g^{\text{op}}) = F(f) \circ F(g).$$

As such F reverses composition of morphisms. This gives us a contravariant functor $F : \mathscr{C} \to \mathscr{D}$. As such $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$ determines a contravariant functor $F : \mathscr{C} \to \mathscr{D}$.

Finally, we check that this is a correspondence. Suppose $F : \mathscr{C} \to \mathscr{D}$. Now, we get a functor $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$. This determines a functor $G : \mathscr{C} \to \mathscr{D}$. Now, note that $G(X) = \Psi(X) = F(X)$ so, G and F agree on objects. Furthermore, $G(f) = \Psi(f^{\text{op}}) = F(f)$, so G and F agree on morphisms. Altogether, we see that G and F are the same functors. Conversely, suppose $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$. This determines a functor $G : \mathscr{C} \to \mathscr{D}$. This then determines a functor $\varphi : \mathscr{C}^{\text{op}} \to \mathscr{D}$. Now, $\varphi(X) = G(X) = \Psi(X)$ and $\varphi(f^{\text{op}}) = G(f) = \Psi(f^{\text{op}})$, so $\Psi = \varphi$.

To prove the last part, suppose $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$ is a covariant functor. Then, we define a covariant functor $S : \mathscr{C} \to \mathscr{D}^{\text{op}}$ as follows: define

$$S(X) = \Psi(X), \forall X \in \text{Obj}(\mathscr{C}) = \text{Obj}(\mathscr{C}^{\text{op}}).$$

Recall that if $D \xrightarrow{m} D'$ in $\mathscr{D}$, then $D' \xrightarrow{m^{\text{op}}} D$ in $\mathscr{D}^{\text{op}}$. Now, let $Y \xrightarrow{f^{\text{op}}} X$ in $\mathscr{C}^{\text{op}}$. Then, $X \xrightarrow{f} Y$ in $\mathscr{C}$. Since Ψ is covariant, we have that $\Psi(Y) \xrightarrow{\Psi(f^{\text{op}})} \Psi(X) \implies S(Y) \xrightarrow{\Psi(f^{\text{op}})} S(X)$ in $\mathscr{D}$. Now, we define the morphism $S(X) \xrightarrow{S(f)} S(Y)$ in $\mathscr{D}^{\text{op}}$ to be the corresponding morphism as $S(Y) \xrightarrow{\Psi(f^{\text{op}})} S(X)$ in $\mathscr{D}$. In other words, we define $S(f) = (\Psi(f^{\text{op}}))^{\text{op}}$, for any $f \in \text{Hom}_{\mathscr{C}}(X, Y)$ to be the corresponding morphism $\Psi(f^{\text{op}}) \in \text{Hom}_{\mathscr{D}}(S(Y), S(X))$.

Note that $S(\text{id}_X) = (\Psi(\text{id}_X^{\text{op}}))^{\text{op}} = (\text{id}_{\Psi(X)})^{\text{op}} = (\text{id}_{S(X)})^{\text{op}} = \text{id}_{S(X)}$. Furthermore, if $X \xrightarrow{f} Y \xrightarrow{g} Z$ in $\mathscr{C}$, then $Z \xrightarrow{g^{\text{op}}} Y \xrightarrow{f^{\text{op}}} X$ in $\mathscr{C}^{\text{op}}$. Now, $g \circ f \in \text{Hom}_{\mathscr{C}}(X, Z)$ corresponds to $f^{\text{op}} \circ g^{\text{op}} \in \text{Hom}_{\mathscr{C}^{\text{op}}}(Z, X)$. We then have that

$$S(g \circ f) = (\Psi(f^{\text{op}} \circ g^{\text{op}}))^{\text{op}}.$$

Now, in $\mathscr{D}$, $\Psi(f^{\text{op}} \circ g^{\text{op}}) = \Psi(f^{\text{op}}) \circ \Psi(g^{\text{op}})$. Then, in $\mathscr{D}^{\text{op}}$, $(\Psi(f^{\text{op}} \circ g^{\text{op}}))^{\text{op}} = (\Psi(f^{\text{op}}) \circ \Psi(g^{\text{op}}))^{\text{op}} = (\Psi(g^{\text{op}}))^{\text{op}} \circ (\Psi(f^{\text{op}}))^{\text{op}}$, which is in turn equal to $S(g) \circ S(f)$, so we get that

$$S(g \circ f) = S(g) \circ S(f).$$

As such, S is a covariant functor from $\mathscr{C}$ to $\mathscr{D}^{\text{op}}$. So, we have that a covariant functor $\mathscr{C}^{\text{op}} \to \mathscr{D}$ determines a covariant functor $\mathscr{C} \to \mathscr{D}^{\text{op}}$.

Convesely, suppose $S : \mathscr{C} \to \mathscr{D}^{\text{op}}$ is a covariant functor. Then, define $\varphi : \mathscr{C}^{\text{op}} \to \mathscr{D}$ to be a functor as follows: for all $X \in \text{Obj}(\mathscr{C}) = \text{Obj}(\mathscr{C}^{\text{op}})$, $\varphi(X) = S(X)$. Now, let $X \xrightarrow{f} Y$ in $\mathscr{C}$. Then, $S(X) \xrightarrow{S(f)} S(Y)$ in $\mathscr{D}^{\text{op}}$. This corresponds to $S(X) \xrightarrow{(S(f))^{\text{op}}} S(Y)$ in $\mathscr{D}$. We define $\varphi(f^{\text{op}}) = (S(f))^{\text{op}}$. This preserves identity: $\varphi(\text{id}_X^{\text{op}}) = (S(\text{id}_X))^{\text{op}} = (\text{id}_{S(X)})^{\text{op}} = \text{id}_{S(X)} = \text{id}_{\varphi(X)}$. Furthermore, $\varphi(g^{\text{op}} \circ f^{\text{op}}) = (S(f \circ g))^{\text{op}} = (S(f) \circ S(g))^{\text{op}} = (S(g))^{\text{op}} \circ (S(f))^{\text{op}} = \varphi(g^{\text{op}}) \circ \varphi(f^{\text{op}})$. As such, φ is a covariant functor.

A similar argument to the second part shows that this is indeed a correspondence. If $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$ is a covariant functor, we get a functor $S : \mathscr{C} \to \mathscr{D}^{\text{op}}$ which in turn gives us $\varphi : \mathscr{C}^{\text{op}} \to \mathscr{D}$. Since Ψ and φ agree on all objects and all morphisms (as $\Psi(f^{\text{op}}) = (S(f))^{\text{op}} = \varphi(f^{\text{op}})$), we have that $\Psi = \varphi$. Similarly, if $S : \mathscr{C} \to \mathscr{D}^{\text{op}}$, we get a functor $\Psi : \mathscr{C}^{\text{op}} \to \mathscr{D}$ which gives us a functor $T : \mathscr{C} \to \mathscr{D}^{\text{op}}$. Again, S and T agree on objects and morphisms, so $S = T$. □

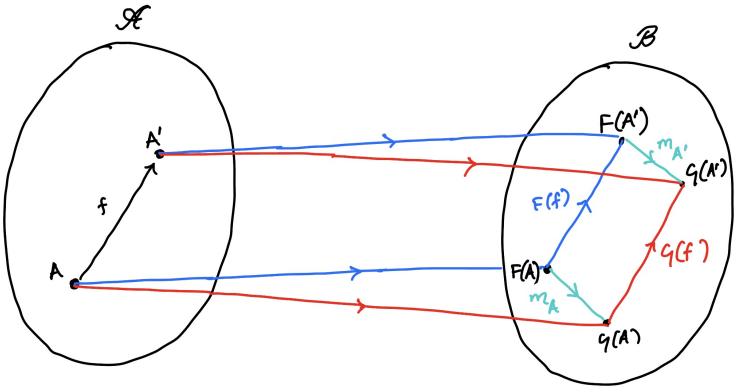


FIGURE 1. An illustration of natural transformations.

3.5. **Natural Transformations and Equivalence of Categories**

Definition 3.53 (Natural Transformations)**.** *Let $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{A} \to \mathscr{B}$ be two covariant functors.*

A natural transformation $m : F \to G$ *is the data of a morphism*

$$m_A : F(A) \longrightarrow G(A)$$

for each $A \in \mathrm{Obj}(\mathscr{A})$, such that for every morphism $f : A \to B$ in $\mathscr{A}$, the following diagram commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ {\scriptstyle m_A}\downarrow & & \downarrow{\scriptstyle m_B} \\ G(A) & \xrightarrow[G(f)]{} & G(B). \end{array}$$

Equivalently, for every morphism $f : A \to B$, we have

$$m_B \circ F(f) = G(f) \circ m_A.$$

⌟

Example 3.54. We already saw that we have the functor

$$\mathrm{GL}_n : \mathbf{Rings} \to \mathbf{Grps}$$

defined on objects by

$$\mathrm{GL}_n(A) = \{M \in M_n(A) \mid M \text{ is invertible}\},$$

and on morphisms by applying ring homomorphisms $f : A \to A'$ entrywise to matrices. Then, we have a natural transformation

$$\det : \mathrm{GL}_n \to \mathrm{GL}_1,$$

whose component at each ring A is the group homomorphism

$$\det_A : \mathrm{GL}_n(A) \to \mathrm{GL}_1(A) = A^\times,$$

given by

$$M \longmapsto \det(M).$$

Since M is invertible, $\det(M)$ is a unit in A. Now we verify naturality: consider a ring homomorphism $f : A \to A'$. We must show that

$$\mathrm{GL}_1(f) \circ \det_A = \det_{A'} \circ \mathrm{GL}_n(f).$$

For example, when $n = 2$, we have

$$\det_A \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

whereas

$$\det_{A'} \begin{pmatrix} f(a) & f(b) \\ f(c) & f(d) \end{pmatrix} = f(a)f(d) - f(b)f(c).$$

Since f is a ring homomorphism,

$$f(ad - bc) = f(a)f(d) - f(b)f(c).$$

More generally, since the determinant is a polynomial with integer coefficients in the matrix entries,

$$\det(\mathrm{GL}_n(f)(M)) = f(\det(M))$$

for every $M \in \mathrm{GL}_n(A)$. Therefore,

$$\mathrm{GL}_1(f) \circ \det_A = \det_{A'} \circ \mathrm{GL}_n(f),$$

proving that det is a natural transformation. ⌟

As with functors, we have identity and constant natural transformations too:

Definition 3.55 (Identity natural transformation)**.** *Let F be a functor: $\mathscr{C} \xrightarrow{F} \mathscr{D}$. Then, the identity natural transformation's components are defined as*

$$\mathrm{id}_X = \mathrm{id}_{F(X)} : F(X) \to F(X), \forall X \in \mathrm{Obj}(\mathscr{C}).$$

⌟

Recall the constant functor: Let $\mathscr{C}$ and $\mathscr{D}$ be categories. Fix an object $Y \in \mathrm{Obj}(\mathscr{D})$. Then, we have the constant functor of Y defined as:

$$C_Y : \mathscr{C} \to \mathscr{D}$$

which sends

$$X \to Y, \forall X \in \mathrm{Obj}(\mathscr{C})$$

and

$$(X \xrightarrow{f} X') \to (Y \xrightarrow{1_Y} Y).$$

Now, we define the constant natural transformation:

Definition 3.56 (Constant natural transformation)**.** *Let $\mathscr{C}$ and $\mathscr{D}$ be categories and $Y_1, Y_2 \in \mathrm{Obj}(\mathscr{D})$ with $Y_1 \xrightarrow{g} Y_2$. Now, we can define a natural transformation*

$$\eta : C_{Y_1} \to C_{Y_2}$$

whose components

$$\eta_X : C_{Y_1}(X) \to C_{Y_2}(X)$$

are defined by, for each $X \in \mathrm{Obj}(\mathscr{C})$,

$$\eta_X = g.$$

⌟

Definition 3.57 (Natural Isomorphisms)**.** *A* natural isomorphism *is a natural transformation $m : F \to G$ such that each*

$$m_A : F(A) \longrightarrow G(A)$$

is an isomorphism for every $A \in \mathrm{Obj}(\mathscr{A})$. ⌟

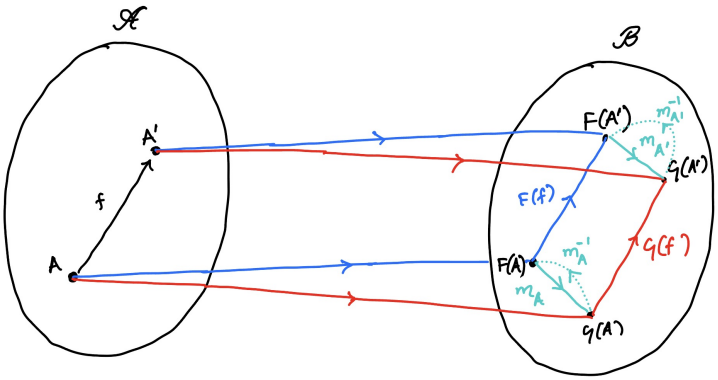


FIGURE 2. An illustration of natural isomorphisms.

Recall that for a vector space V over a field k, its dual space is $V^* := \mathrm{Hom}_k(V, k)$. An element $\varphi \in V^*$ is therefore a linear functional $\varphi : V \to k$. Now, we can take the dual once again to obtain the *double dual* of V:

$$V^{**} := (V^*)^* = \mathrm{Hom}_k(V^*, k).$$

Thus, an element $\Phi \in V^{**}$ is a linear functional

$$\Phi : V^* \to k.$$

In other words, elements of V^* are linear functionals on V, whereas elements of V^{**} are linear functionals on V^*. Now, we can define the canonical evaluation map. Let V be a vector space over k. We define the *canonical evaluation map*: for every $v \in V$, we define the linear functional $D_V(v) \in V^{**}$ defined by

$$D_V(v)(\varphi) := \varphi(v) \in k$$

for every $\varphi \in V^*$.

Proposition 3.58. *Let V be a finite-dimensional vector space over k. Then the canonical evaluation map*

$$D_V : V \xrightarrow{\sim} V^{**}$$

is an isomorphism of vector spaces. ⌟

Proof. We first show that D_V is injective. Suppose $v \in \ker(D_V)$. Then $D_V(v) = 0$, which means that $D_V(v)(\varphi) = 0$ for every $\varphi \in V^*$. By definition of the evaluation map, this implies

$$\varphi(v) = 0, \qquad \forall \varphi \in V^*.$$

Recall that linear functionals separate points. In particular, if $v \neq 0$, we can extend $\{v\}$ to a basis of V and construct a linear functional $\varphi : V \to k$ such that $\varphi(v) = 1$. This contradicts the assertion that $\varphi(v) = 0$ for every $\varphi \in V^*$. Thus $v = 0$, so

$$\ker(D_V) = \{0\}.$$

Therefore, D_V is injective. Now, since V is finite-dimensional, we know that

$$\dim_k V = \dim_k V^* = \dim_k V^{**}.$$

Hence D_V is an injective linear map between finite-dimensional vector spaces of equal dimension, and is therefore surjective. Consequently, D_V is an isomorphism. □

Observation. *The canonical evaluation map $D_V : V \to V^{**}$ is injective for every vector space V, even when V is infinite-dimensional. However, when V is infinite-dimensional, D_V is not surjective in general.*

Definition 3.59 (Double-dual functor)**.** *Recall that the dual functor*

$$(-)^* : \textbf{\textit{Fin-k-vect}} \to \textbf{\textit{Fin-k-vect}}$$

is contravariant. Given a linear map $f : V \to W$, the dual functor produces the linear map $f^* : W^* \to V^*$. Applying the dual functor once more produces

$$f^{**} : V^{**} \to W^{**}.$$

This allows us to define the double-dual functor

$$(-)^* \circ (-)^* : \textbf{\textit{Fin-k-vect}} \to \textbf{\textit{Fin-k-vect}}$$

which is covariant. It acts on objects by

$$V \longmapsto V^{**}$$

and on morphisms by

$$f \longmapsto f^{**}.$$

⌟

Observation. *Given a linear map $f : V \to W$ and an element $\Phi \in V^{**}$, the double-dual map*

$$f^{**} : V^{**} \to W^{**}$$

is defined by

$$f^{**}(\Phi) = \Phi \circ f^*.$$

Equivalently, for every $\varphi \in W^*$,

$$\begin{aligned} f^{**}(\Phi)(\varphi) &= \Phi(f^*(\varphi)) \\ &= \Phi(\varphi \circ f). \end{aligned}$$

3.5.1. *The Natural Isomorphism.* We now have two covariant functors from **Fin-k-vect** to itself:

(1) The identity functor

$$1_{\textbf{Fin-k-vect}} : \textbf{Fin-k-vect} \to \textbf{Fin-k-vect},$$

which sends $V \mapsto V$ and $f \mapsto f$.

(2) The double-dual functor

$$(-)^* \circ (-)^* : \textbf{Fin-k-vect} \to \textbf{Fin-k-vect},$$

which sends $V \mapsto V^{**}$ and $f \mapsto f^{**}$.

We will show that these two functors are naturally isomorphic.

Theorem 3.60. *There exists a natural isomorphism*

$$D : 1_{\textbf{\textit{Fin-k-vect}}} \xrightarrow{\sim} (-)^* \circ (-)^*$$

whose component at each finite-dimensional vector space V is the canonical evaluation map

$$D_V : V \xrightarrow{\sim} V^{**}.$$

⌟

Proof. We have already shown that, for every finite-dimensional vector space V, the canonical evaluation map

$$D_V : V \to V^{**}$$

is an isomorphism. It remains to show that the family

$$D = \{D_V\}_{V \in \mathrm{Obj}(\textbf{Fin-k-vect})}$$

defines a natural transformation.

Let $f : V \to W$ be any linear map. We must show that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\quad f \quad} & W \\ {\scriptstyle D_V}\downarrow & & \downarrow{\scriptstyle D_W} \\ V^{**} & \xrightarrow[\quad f^{**} \quad]{} & W^{**} \end{array}$$

So, we must verify that

$$f^{**} \circ D_V = D_W \circ f.$$

Let $v \in V$. We need to show that

$$f^{**}(D_V(v)) = D_W(f(v)).$$

Notice that both sides are elements of W^{**}, meaning that they are linear maps from W^* to k. Therefore, it suffices to show that they agree on every element $\varphi \in W^*$. First, consider the left-hand side:

$$\begin{aligned}\big(f^{**}(D_V(v))\big)(\varphi) &= D_V(v)(f^*(\varphi)) \\ &= f^*(\varphi)(v) \\ &= (\varphi \circ f)(v) \\ &= \varphi(f(v)).\end{aligned}$$

Now consider the right-hand side:

$$\big(D_W(f(v))\big)(\varphi) = \varphi(f(v)).$$

Therefore,

$$\big(f^{**}(D_V(v))\big)(\varphi) = \big(D_W(f(v))\big)(\varphi)$$

for every $\varphi \in W^*$. So,

$$f^{**}(D_V(v)) = D_W(f(v))$$

for every $v \in V$. So,

$$f^{**} \circ D_V = D_W \circ f,$$

proving that the naturality diagram commutes. Therefore, D is a natural transformation. Moreover, since every component

$$D_V : V \to V^{**}$$

is an isomorphism, D is a natural isomorphism. Consequently,

$$D : 1_{\textbf{Fin-k-vect}} \xrightarrow{\sim} (-)^* \circ (-)^*.$$

□

Definition 3.61 (Equivalence between Categories)**.** *An equivalence between categories $\mathscr{A}$ and $\mathscr{B}$ consists of a pair of functions, $\mathscr{A} \xrightarrow{F} \mathscr{B}$ and $\mathscr{B} \xrightarrow{G} \mathscr{A}$ together with natural isomorphisms:*

$$\eta : 1_{\mathscr{A}} \to G \circ F$$

and

$$\epsilon : 1_{\mathscr{B}} \to F \circ G.$$

If there exists an equivalence, we say that $\mathscr{A}$ and $\mathscr{B}$ are equivalence, which is denoted by

$$\mathscr{A} \simeq \mathscr{B}.$$

⌟

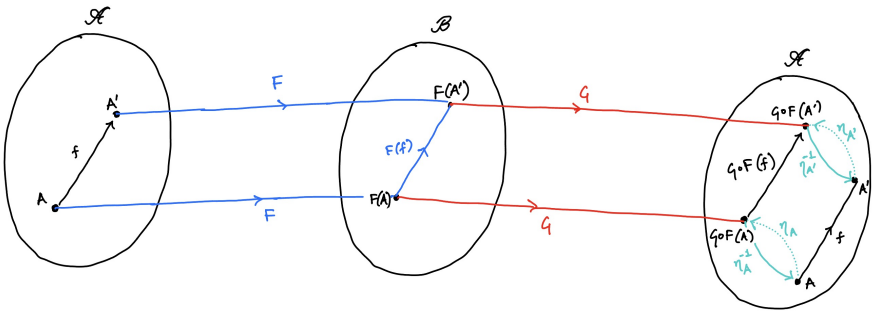


FIGURE 3. An illustration of one of the two requirements of an equivalence between categories.

Definition 3.62 (Essentially surjective on objects functors)**.** *A functor $F : \mathscr{A} \to \mathscr{B}$ is essentially surjective on objects if for all objects $B \in \mathrm{Obj}(\mathscr{B})$, there exists an object $A \in \mathrm{Obj}(\mathscr{A})$ such that $F(A) \cong B$.* ⌟

Proposition 3.63. *A functor $F : \mathscr{A} \to \mathscr{B}$ is an equivalence if and only if it is full, faith, and essentially surjective on objects.* ⌟

Proof. Suppose $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{A}$ constitute an equivalence between $\mathscr{A}$ and $\mathscr{B}$. We first prove the forward direction.

(1) F is faithful: Let $A \xrightarrow{f} A'$ be a morphism in $\mathscr{A}$. Now, suppose $F(f) = F(g)$ as morphisms. Then, $G(F(f)) = G(F(g))$. Now, since η is a natural transformation between $G \circ F$ and $1_{\mathscr{A}}$, we have that
$$\eta_{A'} \circ f = GF(f) \circ \eta_A = GF(g) \circ \eta_A = \eta_{A'} \circ g.$$
Since $\eta_{A'}$ is an isomorphism, we get that $f = g$. Therefore, F is faithful.

(2) F is full: we already know that G is faithful. Now, suppose we have a morphism
$$F(A) \xrightarrow{h} F(A'), \text{ in } \mathscr{B}.$$
Let
$$f = \eta_{A'}^{-1} \circ G(h) \circ \eta_A.$$
Then, we have the following by the definition of f itself:
$$\eta_{A'} \circ f = G(h) \circ \eta_A.$$
On the other hand, we also also have by the definition of the natural isomorphisms, η, (see Figure 3):
$$\eta_{A'} \circ f = G(F(f)) \circ \eta_A.$$
So, we get that $G(F(f)) \circ \eta_A = G(h) \circ \eta_A$. Given η_A is an isomorphism, we have $G(F(f)) = G(h)$. Since G is faithful, $F(f) = h$. As such, F is full.

(3) F is essentially surjective on objects: consider $B \in \mathscr{B}$ fixed. Now, consider $G(B) \in \mathrm{Obj}(\mathscr{A})$. We have the morphism:
$$\epsilon_B^{-1} : F(G(B)) \to B.$$
This is an isomorphism, so
$$F(G(B)) \cong B.$$

Conversely, suppose F is faithful, full, and essentially surjective on objects. We will construct the explicit natural isomorphisms, $\eta : 1_{\mathscr{A}} \to G \circ F$ and $\epsilon : 1_{\mathscr{B}} \to F \circ G$. Let $B \in \mathrm{Obj}(\mathscr{B})$. Since F is essentially surjective on objects, there exists some $A \in \mathrm{Obj}(\mathscr{A})$ such that $F(A) \cong B$. Now, let us assign $G(B) = A$ (note that we are not defining G to be a functor yet, this is just an assignment or label so to speak), so $F(G(B)) \cong B$. Let this isomorphism be
$$\epsilon_B : F(G(B)) \xrightarrow{\sim} B.$$
Since this can be done for all objects in B, G assigns each object in $\mathscr{B}$ to some object in $\mathscr{A}$. To turn this into a functor, we must make G map morphisms in $\mathscr{B}$ to morphisms in $\mathscr{A}$. To do so, suppose $B \xrightarrow{h} B'$. Then,
$$F(G(B)) \xrightarrow{\epsilon_B} B \xrightarrow{h} B' \xrightarrow{\epsilon_{B'}^{-1}} F(G(B')).$$
This defined the morphism:
$$(\epsilon_{B'}^{-1} \circ h \circ \epsilon_B) : F(G(B)) \to F(G(B')).$$
Now, F is full, so there exists some morphism $G(B) \xrightarrow{g} G(B')$ such that $F(g) = \epsilon_{B'}^{-1} \circ h \circ \epsilon_B$ and since F is faithful, this morphism g is unique. As such,
$$G(h) = g, \quad \text{s.t. } F(g) = \epsilon_{B'}^{-1} \circ h \circ \epsilon_B = F(G(h)).$$
We now verify that G is indeed a functor:

(1) $G(1_B) = 1_{G(B)}$: we know that $F(G(1_B)) = \epsilon_B^{-1} \circ 1_B \circ \epsilon_B = 1_{F(G(B))} = F(1_{G(B)})$. Now, F is faithful, so $G(1_B) = 1_{G(B)}$.

(2) $G(h' \circ h) = G(h') \circ G(h)$: suppose $B \xrightarrow{h} B' \xrightarrow{h'} B''$. By definition,

$$F(G(h' \circ h)) = \epsilon_{B''}^{-1} \circ h' \circ h \circ \epsilon_B.$$

Now, proceed as:

$$\begin{aligned} F(G(h' \circ h)) &= \epsilon_{B''}^{-1} \circ h' \circ h \circ \epsilon_B \\ &= \epsilon_{B''}^{-1} \circ h' \circ \epsilon_{B'} \circ \epsilon_{B'}^{-1} \circ h \circ \epsilon_B \\ &= F(G(h')) \circ F(G(h)) \\ &= F(G(h') \circ G(h)). \end{aligned}$$

Since F is faithful, we conclude.

Finally, we show that ϵ is a natural transformation: for each $B \xrightarrow{h} B'$, we have that $F(G(h)) = \epsilon_{B'}^{-1} \circ h \circ \epsilon_B$, which shows that

$$\epsilon_{B'} \circ F(G(h)) = h \circ \epsilon_B.$$

As such, we have attained the natural isomorphism:

$$\epsilon : F \circ G \xrightarrow{\sim} 1_{\mathscr{B}}.$$

A similar strategy to this (and is simpler since we have already constructed the functor G). □

Remark 3.64. One reason why Proposition 3.63 is powerful is that we can prove that a functor F is an equivalence without constructing the "inverse" functor G or without constructing the natural isomorphisms η and ϵ; we can simply show that by proving that the functor is full, faithful and essentially surjective. ⌟

Corollary 3.65. *Let $F : \mathscr{C} \to \mathscr{D}$ be a full and faithful functor. Then, $\mathscr{C}$ is equivalent to the full subcategory $\mathscr{C}'$ of $\mathscr{D}$ whose objects are of the form $F(C)$ for some $C \in \mathscr{C}$.* ⌟

Definition 3.66 (Representable functors)**.** *Let $F : \mathscr{C} \to \mathbf{Sets}$ be a covariant functor. We say F is representable by $U \in \mathrm{Obj}(\mathscr{C})$ if if there is a natural isomorphism $i : \mathrm{Hom}_{\mathscr{C}}(U, -) \xrightarrow{\sim} F$.* ⌟

Proposition 3.67. *F is representable by U if and only if there is a distinguished element $u \in F(U)$ and for every $X \in \mathrm{Obj}(\mathscr{C})$ and every $x \in F(X)$, there is a unique morphism $f : U \to X$ such that $F(f)(u) = x$. Furthermore, the pair (U, u) is unique up to unique isomorphism.* ⌟

Proof. Suppose F is representable by U. Let the isomorphism be $i_X : \mathrm{Hom}_{\mathscr{C}}(U, X) \xrightarrow{\sim} F(X)$ for each $X \in \mathrm{Obj}(\mathscr{C})$. Consider any fixed object, X, in $\mathscr{C}$ now. If $F(X)$ is empty, the statement is vacuously true. Otherwise, let $x \in F(X)$ be an arbitrary element. Then, via the isomorphism, we have $i_X^{-1}(x) \in \mathrm{Hom}_{\mathscr{C}}(U, X)$. In particular, define

$$f := i_X^{-1}(x) \in \mathrm{Hom}_{\mathscr{C}}(U, X).$$

Note that $i_X(f) = x$. Now, we leverage the fact that i is a natural transformation. We have that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{C}}(U,U) & \xrightarrow{f\circ(-)} & \mathrm{Hom}_{\mathscr{C}}(U,X) \\ \downarrow{\scriptstyle i_U} & & \downarrow{\scriptstyle i_X} \\ F(U) & \xrightarrow[F(f)]{} & F(X). \end{array}$$

Given $\mathrm{id}_U \in \mathrm{Hom}_{\mathscr{C}}(U, U)$, by the commutativity of the diagram, we have:

$$\begin{aligned} (F(f) \circ i_U)(\mathrm{id}_U) &= i_X(f \circ \mathrm{id}_U) \\ F(f)(i_U(\mathrm{id}_U)) &= i_X(f) \\ &= x \quad \text{(by definition of } x\text{)}. \end{aligned}$$

Define $i_U(\mathrm{id}_U)$ to be $u \in F(U)$. Then, we see that $F(f)(u) = x$. Since the definition of u is irrespective of the specific object X and the specific element $x \in F(X)$, this is a distinguished element that works for every object X and every $x \in F(X)$ with the construction that $f : U \to X$ is defined via $f = i_X^{-1}(x)$. Finally, we claim that the choice of f is unique. Suppose there existed another morphism $g : U \to X$ that satisfies this requirement. Then, we have that $F(g)(u) = x$. Then, we would have the following diagram commute:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{C}}(U,U) & \xrightarrow{g\circ(-)} & \mathrm{Hom}_{\mathscr{C}}(U,X) \\ \Big\downarrow{\scriptstyle i_U} & & \Big\downarrow{\scriptstyle i_X} \\ F(U) & \xrightarrow[F(g)]{} & F(X). \end{array}$$

This tells us that $i_X \circ g \circ \mathrm{id}_U = F(g) \circ i_U(\mathrm{id}_U)$. Now, recalling that i_X is an isomorphism, we can write:

$$\begin{aligned} i_X \circ g \circ \mathrm{id}_U &= F(g) \circ i_U(\mathrm{id}_U) \\ i_X \circ g &= F(g)(i_U(\mathrm{id}_U)) \\ g &= i_X^{-1} \circ F(g)(i_U(\mathrm{id}_U)) \\ &= i_X^{-1} \circ F(g)(u) \\ &= i_X^{-1} \circ F(g)(u) \\ &= i_X^{-1}(x) \\ &= f. \end{aligned}$$

As such, $g = f$ and we have shown that f is unique.

Now, we prove the converse. Suppose there exists an element $u \in F(U)$ such that for all $X \in \mathrm{Obj}(\mathscr{C})$ and all $x \in F(X)$, there exists a unique morphism $f : U \to X$ satisfying $F(f)(u) = x$. Note that then, we get two arrows in **Sets** for free:

$$\mathrm{Hom}_{\mathscr{C}}(U,U) \xrightarrow{f\circ(-)} \mathrm{Hom}_{\mathscr{C}}(U,X)$$

and

$$F(U) \xrightarrow{F(f)} F(X).$$

Now, we construct the map $i_X : \mathrm{Hom}_{\mathscr{C}}(U,X) \to F(X)$ by first setting $i_X(g) = F(g)(u)$ for any $g \in \mathrm{Hom}_{\mathscr{C}}(U,X)$. We claim this is an injective map. Suppose $i_X(g) = i_X(h) \implies F(g)(u) = F(h)(u)$. Recall that, by hypothesis, u is a distinguished element such that for all objects X and all elements $x \in F(X)$, there is one *unique* morphism $f : U \to X$ such that $F(f)(u) = x$. Let $i_X(g) = i_X(h) = x \in F(X)$. Then, since both $F(g)(u) = x$ and $F(h)(u) = x$, by the uniqueness of the morphism, we have that $g = h$. We also claim that i_X is surjective. This follows from the definition, since for every $x \in F(X)$, we have a unique morphism $f : U \to X$ such that $F(f)(u) = x$, and as such we have a $i_X(f) = F(f)(u) = x \in F(X)$. Since i_X is bijective, this is an isomorphism. Next, we claim this is a natural transformation. Consider any $m : X \to Y$ morphism in in $\mathscr{C}$. Now, observe that, for any $\Psi \in \mathrm{Hom}_{\mathscr{C}}(U,X)$, we have that

$$\begin{aligned} (F(m) \circ i_X)(\Psi) &= F(m)(F(\psi)(u)) \\ &= F(m \circ \Psi)(u). \end{aligned}$$

On the other hand,

$$\begin{aligned} i_Y \circ m \circ \Psi &= i_Y(m \circ \Psi) \\ &= F(m \circ \Psi)(u) \end{aligned}$$

Since Ψ was chosen arbitrarily, we have that the following square commutes:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{C}}(U,X) & \xrightarrow{m\circ(-)} & \mathrm{Hom}_{\mathscr{C}}(U,Y) \\ \downarrow{\scriptstyle i_X} & & \downarrow{\scriptstyle i_Y} \\ F(X) & \xrightarrow[F(m)]{} & F(Y). \end{array}$$

As such, $i : \mathrm{Hom}_{\mathscr{C}}(U,-) \to F$ is a natural isomorphism. Finally, we show that (U,u) is unique up to unique isomorphism. Suppose, (V,v) is another pair that satisfies the same constraints. In other words, $v \in F(V)$ and for every $X \in \mathrm{Obj}(\mathscr{C})$ and every $x \in F(X)$, there is a unique morphism $f' : V \to X$ such that $F(f')(v) = x$. Then, given $U \in \mathrm{Obj}(\mathscr{C})$ and $u \in F(U)$, there is a unique morphism $\Psi_{VU} : V \to U$ such that $F(\Psi_{VU})(v) = u$. On the other hand, we have a unique morphism $\Psi_{UV} : U \to V$ such that $F(\Psi_{UV})(u) = v$. Now, consider

$$\begin{aligned} F(\Psi_{VU} \circ \Psi_{UV})(u) &= (F(\Psi_{VU}) \circ F(\Psi_{UV}))(u) \\ &= F(\Psi_{VU})(v) \\ &= u. \end{aligned}$$

Now, note that given $U \in \mathrm{Obj}(\mathscr{C})$ and $u \in F(U)$, $F(\mathrm{id}_U)(u) = \mathrm{id}_{F(U)}(u) = u$. So, both $\Psi_{VU} \circ \Psi_{UV}$ and id_U satisfy the requirement, and since the morphism is unique, we must have that

$$\Psi_{VU} \circ \Psi_{UV} = \mathrm{id}_U.$$

A similar argument shows that

$$\Psi_{UV} \circ \Psi_{VU} = \mathrm{id}_V.$$

As such $V \cong U$ and, if the isomorphism is $\Psi_{UV} : U \to V$, then $F(\Psi)(u) = v$. So, (U,u) is unique up to unique isomorphism. □

Proposition 3.68. *The forgetful functor from the category of rings to the category of sets is representable.* ⌟

Proof. Let F be the forgetful functor from **Rings** to **Sets**, that maps each ring to its underlying set of elements. We claim that F is representable by $\mathbb{Z}[x]$. Now, consider any ring R with any fixed element $r \in R$. Then, consider the morphism: $f_{R,r} : \mathbb{Z}[x] \to R$ defined by

$$f_{R,r}(n) = n \cdot 1_R, \quad \forall n \in \mathbb{Z},$$
$$f_{R,r}(x) = r$$

and

$$f_{R,r}(\sum_{i=0}^{d} a_i x^i) = \sum_{i=0}^{d} a_i r^i.$$

We claim this is a ring morphism:

(1) $f_{R,r}(0) = 0 \cdot 1_R = 0$
(2) $f_{R,r}(1) = 1 \cdot 1_R = 1_R$
(3) $f_{R,r}(\sum_{i=0}^{d} a_i x^i + \sum_{j=0}^{d'} b_j x^j) = f_{R,r}(\sum_{i=0}^{d}(a_i + b_i)x^i) = \sum_{i=0}^{d}(a_i + b_i)r^i = \sum_{i=0}^{d} a_i r^i + \sum_{j=0}^{d'} b_j r^j = f_{R,r}(\sum_{i=0}^{d} a_i x^i) + f_{R,r}(\sum_{j=0}^{d'} b_j x^j)$, and
(4) $f_{R,r}(\sum_{i=0}^{d} a_i x^i \cdot \sum_{j=0}^{d'} b_j x^j) = f_{R,r}(\sum_{k=0}^{d+d'} \sum_{0\le i\le k} a_i b_{k-i} x^k) = \sum_{k=0}^{d+d'} \sum_{0\le i\le k} a_i b_{k-i} r^k = (\sum_{i=0}^{d} a_i r^i)(\sum_{j=0}^{d'} b_j r^j) = f_{R,r}(\sum_{i=0}^{d} a_i x^i) f_{R,r}(\sum_{j=0}^{d'} b_j x^j)$.

Note that in the third and fourth part of this calculation, there are some harmless assumptions (for e.g., WLOG, we assume $d \ge d'$ and we then pad the lower degree polynomial with zeros so that the addition and the multiplication, as written, makes sense). Finally, we observe that $F(f_{R,r})(x) = r$. This is straightforward since F is the forgetful functor so it simply maps $f_{R,r}$ to itself as a set function and, by definition, we set that $f_{R,r}(x) = r$. All that remains to show is that this is, indeed, the unique morphism. Suppose have another morphism $g : \mathbb{Z}[x] \to R$ such that $g(x) = r$. Since ring morphisms are required to send 1 to 1 in the category of commutative,

unital rings, this means g and $f_{R,r}$ agree on both x and integers, so they agree on every element of $\mathbb{Z}[x]$. As such, $g = f_{R,r}$. Then, by the previous proposition, F is representable by $\mathbb{Z}[x]$. $\square$

3.6. **Yoneda's Lemma**

Definition 3.69 (Natural transformation $\mathrm{Hom}_{\mathscr{C}}(f,-)$)**.** *Given $X \xrightarrow{f} Y$ is an arrow in $\mathscr{C}$, recall that we have functors $\mathrm{Hom}_{\mathscr{C}}(X,-)$ and $\mathrm{Hom}_{\mathscr{C}}(Y,-)$. Then, we have the natural transformation:*

$$\mathrm{Hom}_{\mathscr{C}}(f,-) : \mathrm{Hom}_{\mathscr{C}}(Y,-) \implies \mathrm{Hom}_{\mathscr{C}}(X,-)$$

defined as follows: for any object Z,

$$\mathrm{Hom}_{\mathscr{C}}(f,Z) : \mathrm{Hom}_{\mathscr{C}}(Y,Z) \to \mathrm{Hom}_{\mathscr{C}}(X,Z)$$

by

$$\mathrm{Hom}_{\mathscr{C}}(f,Z)(g) = g \circ f.$$

⌟

Proposition 3.70 (Yoneda's Lemma, Part I)**.** *Suppose that $\mathscr{C}$ is a category with small Hom-sets and that $F : \mathscr{C} \to \mathbf{Sets}$ is a covariant functor. For any object $X \in \mathscr{C}$, there is a bijection:*

$$\Phi_{F,X} : \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), F) \xrightarrow{\sim} F(X)$$

given by

$$\Phi_{F,X}(\eta) = \eta_X(1_X)$$

and

$$\Phi^{-1}_{F,X}(x) = \eta, \ \textit{s.t.}\ \eta_Y(g) = F(g)(x), \quad \forall x \in F(X), \forall Y \in \mathrm{Obj}(\mathscr{C}).$$

Here, $\mathrm{Nat}(G,F)$ is the set of natural transformations from G to F. ⌟

Proof. We define the map $\Phi_{F,X}(\Psi) = \Psi_X(1_X)$ for any Ψ a natural transformation from $\mathrm{Hom}_{\mathscr{C}}(X,-)$ to F. We first show that this well-defined. Suppose $\Psi = \Psi'$. Then, $\Psi_X = \Psi'_X$. Then, for the identity morphism $1_X : X \to X$ in $\mathscr{C}$, we have that $\Psi_X(1_X) = \Psi'_X(1_X)$ and, so, $\Phi_{F,X}(\Psi) = \Phi_{F,X}(\Psi')$.

Now, we construct a map in the other direction. For any object $Y \in \mathrm{Obj}(\mathscr{C})$ and any morphism $g : X \to Y$, we define $\varphi_Y(g) = F(g)(x)$. First, we claim this is well-defined. This is because if $X \xrightarrow{g} Y$ is the same arrow as $X \xrightarrow{h} Y$, then, $g = h \implies F(g) = F(h)$ and so $F(g)(x) = F(h)(x)$. Next, we claim this is a natural transformation. First, we show that the following square commutes:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{C}}(X,X) & \xrightarrow{g\circ(-)} & \mathrm{Hom}_{\mathscr{C}}(X,Y) \\ \Big\downarrow{\scriptstyle \varphi_X} & & \Big\downarrow{\scriptstyle \varphi_Y} \\ F(X) & \xrightarrow[F(g)]{} & F(Y) \end{array}$$

Indeed, we have that:

$$\begin{aligned} F(g)(\varphi_X(f)) &= F(g)(F(f)(x)) \\ &= (F(g) \circ F(f))(x) \\ &= F(g \circ f)(x) \\ &= \varphi_Y(g \circ f) \end{aligned}$$

Generalizing this, we get that, for $m : Y \to Z$ an arrow in $\mathscr{C}$, the following commutes:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{C}}(X,Y) & \xrightarrow{m\circ(-)} & \mathrm{Hom}_{\mathscr{C}}(X,Z) \\ \Big\downarrow{\scriptstyle \varphi_Y} & & \Big\downarrow{\scriptstyle \varphi_Z} \\ F(Y) & \xrightarrow[F(m)]{} & F(Z) \end{array}$$

We show this as follows:

$$\begin{aligned} F(m)(\varphi_Y(f)) &= F(m)(F(f)(x)) \quad \text{where } f \in \mathrm{Hom}_{\mathscr{C}}(X,Y) \\ &= F(m \circ f)(x) \\ &= \varphi_Z(m \circ f) \end{aligned}$$

Having shown that φ is indeed a natural transformation, we now notice that for $x \in F(X)$, $\phi_{F,X}(\varphi) = \varphi_X(1_X) = F(1_X)(x) = 1_{F(X)}(x) = x$, so we define

$$\Xi : F(X) \to \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), F)$$

by

$$\Xi(x) = \varphi_x$$

where φ_x is our new notation for the natural transformation we constructed.

All that remains is for us to show that this constitutes a bijection. For any natural transformation $\Psi \in \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), F)$, we observe that:

$$\begin{aligned} \Xi(\Phi_{F,X}(\Psi)) &= \Xi(\Psi_X(1_X)) \\ &= \varphi_{\Psi_X(1_X)}. \end{aligned}$$

Now, observe that for a given object Y and a morphism $g \in \mathrm{Hom}_{\mathscr{C}}(X,Y)$, we have that

$$\begin{aligned} (\varphi_{\Psi_X(1_X)})_Y(g) &= F(g)(\Psi_X(1_X)) \\ &= \Psi_Y(g \circ 1_X) \\ &= \Psi_Y(g) \end{aligned}$$

Since this is true for any object Y and any morphism $g \in \mathrm{Hom}_{\mathscr{C}}(X,Y)$, we have that $\varphi_{\Psi_X(1_X)}) = \Psi$. For the other direction, consider a fixerd $x \in F(X)$. Then,

$$\begin{aligned} \Phi_{F,X}(\Xi(x)) &= \Phi_{F,X}(\varphi_x) \\ &= (\varphi_x)_X(1_X) \\ &= x. \end{aligned}$$

As such, $\phi_{F,X}$ is indeed a bijection. □

Yoneda's lemma shows that the set of elements $F(X)$ corresponds to the set of natural transformations from $\mathrm{Hom}_{\mathscr{C}}(X,-)$ to F.

Proposition 3.71 (Yoneda's Lemma, Part II: bijection is natural in X and Y)**.** *If $f : X \to Y$, then*

$$F(f) \circ \Phi_{F,X} = \Phi_{F,Y} \circ (\circ \mathrm{Hom}_{\mathscr{C}}(f,-))$$

and that, if $\alpha : F \to G$ is a natural transformation, then

$$\alpha_X \circ \Phi_{F,X} = \Phi_{G,X} \circ (\alpha \circ).$$

⌟

Proof. Let $\psi \in \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), F)$. Then,

$$\begin{aligned} (\Phi_{F,Y} \circ (\circ \mathrm{Hom}_{\mathscr{C}}(f,-)))(\psi) &= \Phi_{F,Y}(\psi \circ \mathrm{Hom}_{\mathscr{C}}(f,-)) \\ &= (\psi \circ \mathrm{Hom}_{\mathscr{C}}(f,-))_Y(1_Y) \\ &= \psi_Y(1_Y \circ f) \\ &= \psi_Y(f) \end{aligned}$$

On the other hand,

$$\begin{aligned} (F(f) \circ \Phi_{F,X})(\psi) &= F(f)(\psi_X(1_X)) \\ &= \psi_Y(f \circ 1_X) \quad \text{(by naturality of } \psi) \\ &= \psi_Y(f). \end{aligned}$$

As ψ was chosen arbitrarily, we have proven that

$$F(f) \circ \Phi_{F,X} = \Phi_{F,Y} \circ (\circ \operatorname{Hom}_{\mathscr{C}}(f, -)).$$

Next, suppose $\alpha : F \to G$ is a natural transformation. Let $\psi \in \operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X, -), F)$ like before. Then,

$$(\alpha_X \circ \Phi_{F,X})(\psi) = \alpha_X(\psi_X(1_X))$$

On the other hand,

$$\begin{aligned} \Phi_{G,X} \circ (\alpha\circ)(\psi) &= \phi_{G,X} \circ (\alpha \circ \psi) \\ &= (\alpha \circ \psi)_X(1_X) \\ &= \alpha(\psi_X(1_X)). \end{aligned}$$

Since ψ was chosen arbitrarily, we have that

$$\alpha_X \circ \Phi_{F,X} = \Phi_{G,X} \circ (\alpha\circ).$$

□

Corollary 3.72 (Corollary of Yoneda's Lemma)**.** *There are isomorphisms*

$$\operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X, -), \operatorname{Hom}_{\mathscr{C}}(Y, -)) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{C}}(Y, X)$$

which are natural in X and Y. In particular, the isomorphism is given by

$$\Xi_{XY}(\eta) = \eta_X(1_X), \quad \forall \eta \in \operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X, -), \operatorname{Hom}_{\mathscr{C}}(Y, -))$$

and

$$\Xi^{-1}_{XY}(f) = \operatorname{Hom}_{\mathscr{C}}(f, -), \quad \forall f \in \operatorname{Hom}_{\mathscr{C}}(Y, X).$$

⌟

Proof. Observe that setting the functor F to $\operatorname{Hom}_{\mathscr{C}}(Y, -)$, we get that

$$\Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),X} : \operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X, -), \operatorname{Hom}_{\mathscr{C}}(Y, -)) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{C}}(Y, X).$$

All that remains is for us to show that this is natural in X and Y. First, we claim that this is natural in X. Let $f : X \to Z$ be an arbitrary morphism. Then, we have already established that

$$F(f) \circ \Phi_{F,X} = \Phi_{F,Z} \circ (\circ \operatorname{Hom}_{\mathscr{C}}(f, -)).$$

Applying this to our case, we get that

$$(f \circ (-)) \circ \Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),X} = \Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),Z} \circ (\circ \operatorname{Hom}_{\mathscr{C}}(f, -)).$$

On the other hand, we claim that this is natural in Y. Consider the natural transformation $\alpha : \operatorname{Hom}_{\mathscr{C}}(Y, -) \to \operatorname{Hom}_{\mathscr{C}}(Z, -)$. Then, we get that

$$\alpha_X \circ \Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),X} = \Phi_{\operatorname{Hom}_{\mathscr{C}}(Z,-),X} \circ (\alpha\circ).$$

As such, given an arbitrary morphism $g : Z \to Y$, we can define α to be the natural transformation $\operatorname{Hom}_{\mathscr{C}}(g, -) : \operatorname{Hom}_{\mathscr{C}}(Y, -) \to \operatorname{Hom}_{\mathscr{C}}(Z, -)$. Plugging this definition of α into our equation, we have proven the naturality in Y too. □

Remark 3.73. We discuss the naturality in X and Y as follows. Suppose $f : X \to X'$ is a morphism in $\mathscr{C}$. Then, keeping Y fixed, the following diagram commutes:

$$\begin{array}{ccc} \operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X, -), \operatorname{Hom}_{\mathscr{C}}(Y, -)) & \xrightarrow{\Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),X}} & \operatorname{Hom}_{\mathscr{C}}(Y, X) \\ \Big\downarrow{\scriptstyle (-)\circ\eta_f} & & \Big\downarrow{\scriptstyle f\circ(-)} \\ \operatorname{Nat}(\operatorname{Hom}_{\mathscr{C}}(X', -), \operatorname{Hom}_{\mathscr{C}}(Y, -)) & \xrightarrow[\Phi_{\operatorname{Hom}_{\mathscr{C}}(Y,-),X'}]{} & \operatorname{Hom}_{\mathscr{C}}(Y, X') \end{array}$$

Here, $\eta_f = \mathrm{Hom}_{\mathscr{C}}(f, -)$. On the other hand, naturality in Y means that, for any arbitrary morphism $g : Y' \to Y$, keeping X fixed, the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), \mathrm{Hom}_{\mathscr{C}}(Y,-)) & \xrightarrow{\Phi_{\mathrm{Hom}_{\mathscr{C}}(Y,-),X}} & \mathrm{Hom}_{\mathscr{C}}(Y, X) \\ \Big\downarrow{\scriptstyle \eta_g \circ (-)} & & \Big\downarrow{\scriptstyle (-)\circ g} \\ \mathrm{Nat}(\mathrm{Hom}_{\mathscr{C}}(X,-), \mathrm{Hom}_{\mathscr{C}}(Y',-)) & \xrightarrow[\Phi_{\mathrm{Hom}_{\mathscr{C}}(Y',-),X}]{} & \mathrm{Hom}_{\mathscr{C}}(Y', X) \end{array}$$

Here, $\eta_g = \mathrm{Hom}_{\mathscr{C}}(g, -)$. ⌟

3.7. **Limits and Colimits**

Definition 3.74 (Limit of a functor)**.** *Let $\mathscr{J}$ be a small category, let $\mathscr{C}$ be a category, and let $F : \mathscr{J} \to \mathscr{C}$ be a functor. A* limit *of F is an object $\varprojlim F \in \mathrm{Obj}(\mathscr{C})$, together with a natural transformation*

$$\phi : C_{\varprojlim F} \to F,$$

(where $C_{\varprojlim F} : \mathscr{J} \to \mathscr{C}$ is the constant functor at L), satisfying the following universal property:

For every object $X \in \mathrm{Obj}(\mathscr{C})$ and every natural transformation

$$\Psi : C_X \to F,$$

there exists a unique morphism

$$\alpha : X \to L$$

such that

$$\phi \circ C_\alpha = \Psi.$$

Here, $C_\alpha : C_X \to C_L$ is the natural transformation whose component at every $j \in \mathrm{Obj}(\mathscr{J})$ is α. ⌟

Observation. *A natural transformation $\phi : C_L \to F$ consists of morphisms*

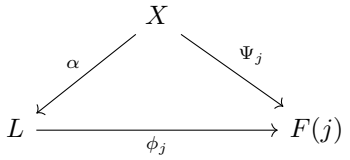
$$\phi_j : L \to F(j)$$

for every $j \in \mathrm{Obj}(\mathscr{J})$, satisfying

$$F(u) \circ \phi_j = \phi_{j'}$$

for every morphism $u : j \to j'$ in $\mathscr{J}$.

Such a pair (L, ϕ) is called a cone over F*. The universal property of the limit says that every other cone (X, Ψ) factors uniquely through (L, ϕ). Equivalently, for every $j \in \mathrm{Obj}(\mathscr{J})$, the following diagram commutes:*



This definition uses natural transformations, but since they are constant transformations, the definition could be made somewhat simpler:

Definition 3.75 (Limit of functor)**.** *Let $\mathscr{J}$ be a small category, let $\mathscr{C}$ be a category, and let*

$$F : \mathscr{J} \to \mathscr{C}$$

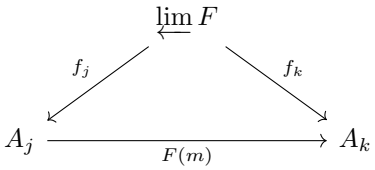
be a functor. Write $A_j := F(j)$ for every $j \in \mathrm{Obj}(\mathscr{J})$.

A limit *of F is an object $\varprojlim F$ of $\mathscr{C}$, together with morphisms*

$$f_j : \varprojlim F \to A_j =: F(j)$$

for every $j \in \mathrm{Obj}(\mathscr{J})$, satisfying the following two conditions:

(1) *Compatibility. For every morphism $m : j \to k$ in $\mathscr{J}$, the following diagram commutes:*



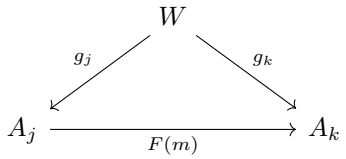
In other words,

$$F(m) \circ f_j = f_k.$$

(2) *Universal property. Suppose* $W \in \mathrm{Obj}(\mathscr{C})$ *is another object, together with morphisms*

$$g_j : W \to A_j =: F(j)$$

for every $j \in \mathrm{Obj}(\mathscr{J})$*, such that for every morphism* $m : j \to k$ *in* $\mathscr{J}$*, the following diagram commutes:*



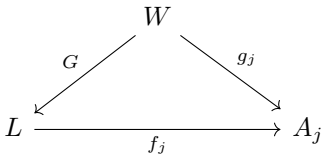
Then there exists a unique morphism

$$G : W \to \varprojlim F$$

such that

$$g_j = f_j \circ G$$

for every $j \in \mathrm{Obj}(\mathscr{J})$*. Equivalently, for every* $j \in \mathrm{Obj}(\mathscr{J})$*, the following diagram commutes:*



If the limit exists, it is unique up to a unique isomorphism compatible with the maps f_j*.* ⌟

Example 3.76. Suppose $\mathscr{J}$ is a category with two objects, j_1 and j_2, with morphisms that are $1_{j_1}, 1_{j_2}, j_1 \xrightarrow{u} j_2$ and $j_1 \xrightarrow{v} j_2$. Consider a functor $\mathscr{J} \xrightarrow{F}$ **k-vect**. This functor must identify two k-vector spaces: let's call them V and W. It must also identify two corresponding morphisms: let's call them $V \xrightarrow{f} W$ and $V \xrightarrow{g} W$. Then, the limit of F, $\varprojlim F$, has a morphism to V, let's call it

$$\varprojlim F \xrightarrow{\phi} V.$$

Now, by definition of the limit, there must be a morphism to W which has to be equal to both $f \circ \phi$ and $g \circ \phi$:

$$\varprojlim F \xrightarrow{\phi \circ f = \phi \circ g} W.$$

This means $\operatorname{im}\phi \subset \ker(f-g)$. Now, if there is another object T with morphism $\Psi : T \to V$ with $f \circ \Psi = g \circ \Psi$, then $\operatorname{im}\Psi \subset \ker(f-g)$. As such,

$$\varprojlim F = \ker(f-g).$$

⌟

Definition 3.77 (Colimit of functor)**.** *Let* $\mathscr{J}$ *be a small category, let* $\mathscr{C}$ *be a category, and let*

$$F : \mathscr{J} \to \mathscr{C}$$

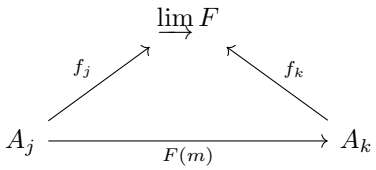
be a functor. Write $A_j := F(j)$ *for every* $j \in \mathrm{Obj}(\mathscr{J})$*.*

A colimit *of the functor* F *is an object* $\varinjlim F$ *of* $\mathscr{C}$*, together with morphisms*

$$f_j : A_j = F(j) \to \varinjlim F$$

for every $j \in \mathrm{Obj}(\mathscr{J})$*, satisfying the following two conditions:*

(1) *Compatibility. For every morphism* $m : j \to k$ *in* $\mathscr{J}$*, the following diagram commutes:*



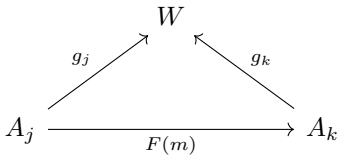
In other words,

$$f_k \circ F(m) = f_j.$$

(2) *Universal property. Suppose* $W \in \mathrm{Obj}(\mathscr{C})$ *is another object, together with morphisms*

$$g_j : A_j = F(j) \to W$$

for every $j \in \mathrm{Obj}(\mathscr{J})$*, such that for every morphism* $m : j \to k$ *in* $\mathscr{J}$*, the following diagram commutes:*



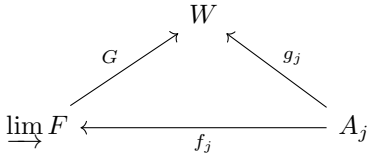
Then there exists a unique morphism

$$G : \varinjlim F \to W$$

such that

$$g_j = G \circ f_j$$

for every $j \in \mathrm{Obj}(\mathscr{J})$*. Equivalently, for every* $j \in \mathrm{Obj}(\mathscr{J})$*, the following diagram commutes:*



If the colimit exists, it is unique up to a unique isomorphism compatible with the maps f_j*.* ⌟

Example 3.78. Let $\mathscr{J}$ be a category of three 3 objects: j_1, j_2 and j_3. Apart from the identity morphisms, suppose we have two more morphisms: $j_1 \to j_2$ and $j_1 \to j_3$. Then, define the functor $F : \mathscr{J} \to \mathbf{Rings}$. This functor specifies three rings A, B and C with morphisms $A \to B$ and $A \to C$. Then, the colimit is $\varinjlim F = B \otimes_A C$. ⌟

Example 3.79. In the ⌟

Proposition 3.80. *The category* **Set** *has all small limits and colimits. In particular, the limit is defined as follows: the object is*

$$L := \left\{ (x_j)_{j \in \mathrm{Obj}(\mathscr{J})} \in \prod_{j \in \mathrm{Obj}(\mathscr{J})} A_j \;\middle|\; F(m)(x_j) = x_k \text{ for every } m : j \to k \right\},$$

and the morphisms are defined as, for each $j \in \mathrm{Obj}(\mathscr{J})$*,*

$$f_j : L \to A_j$$

by

$$f_j((x_i)_i) = x_j.$$

(The exact construction of the colimit is provided in the proof). ⌟

Proof. Let $\mathscr{J}$ be a small category, and let

$$F : \mathscr{J} \to \mathbf{Set}$$

be a functor. Write $A_j := F(j)$ for each $j \in \mathrm{Obj}(\mathscr{J})$. We first show that there exists a limit. Define the set

$$L := \left\{ (x_j)_{j \in \mathrm{Obj}(\mathscr{J})} \in \prod_{j \in \mathrm{Obj}(\mathscr{J})} A_j \;\middle|\; F(m)(x_j) = x_k \text{ for every } m : j \to k \right\}.$$

For each $j \in \mathrm{Obj}(\mathscr{J})$, define

$$f_j : L \to A_j$$

by

$$f_j((x_i)_i) = x_j.$$

We claim that $(L, \{f_j\})$ is a limit of F. We verify the two conditions. Let $m : j \to k$ be any morphism in $\mathscr{J}$. For every $(x_i)_i \in L$, we have

$$\begin{aligned}(F(m) \circ f_j)((x_i)_i) &= F(m)(x_j) \\ &= x_k \\ &= f_k((x_i)_i).\end{aligned}$$

Therefore,

$$F(m) \circ f_j = f_k.$$

Now, we verify the universal property. Suppose W is another set, together with maps

$$g_j : W \to A_j$$

for every $j \in \mathrm{Obj}(\mathscr{J})$, satisfying

$$F(m) \circ g_j = g_k$$

for every morphism $m : j \to k$. Define

$$G(w) := (g_j(w))_{j \in \mathrm{Obj}(\mathscr{J})}.$$

This is well-defined, since for every morphism $m : j \to k$,

$$F(m)(g_j(w)) = g_k(w),$$

so the tuple $(g_j(w))_j$ belongs to L. Furthermore, for every $w \in W$,

$$\begin{aligned}(f_j \circ G)(w) &= f_j((g_i(w))_i) \\ &= g_j(w).\end{aligned}$$

So, $f_j \circ G = g_j$ for every j. To prove uniqueness, suppose $H : W \to L$ also satisfies

$$f_j \circ H = g_j$$

for every j. Since elements of L are tuples determined by their coordinates, for every $w \in W$,

$$H(w) = (g_j(w))_j = G(w).$$

Therefore $H = G$, proving uniqueness. Next, we show colimit exists. Consider the disjoint union

$$S := \bigsqcup_{j \in \mathrm{Obj}(\mathscr{J})} A_j.$$

Elements of S are pairs (j, x), where $j \in \mathrm{Obj}(\mathscr{J})$ and $x \in A_j$. Now, define $\sim$ to be the smallest equivalence relation on S such that

$$(j, x) \sim (k, F(m)(x))$$

for every morphism $m : j \to k$ in $\mathscr{J}$ and every $x \in A_j$. Let

$$L := S/\!\sim.$$

For each $j \in \mathrm{Obj}(\mathscr{J})$, define

$$f_j : A_j \to L$$

by

$$f_j(x) = [(j, x)],$$

where $[(j, x)]$ denotes the equivalence class of (j, x). We claim that $(L, \{f_j\})$ is a colimit of F. First, we check compatibility. Let $m : j \to k$ be any morphism in $\mathscr{J}$. For every $x \in A_j$,

$$\begin{aligned}(f_k \circ F(m))(x) &= f_k(F(m)(x)) \\ &= [(k, F(m)(x))] \\ &= [(j, x)] \\ &= f_j(x).\end{aligned}$$

Therefore,

$$f_k \circ F(m) = f_j.$$

Next, we check the universal property. Suppose W is another set, together with maps

$$g_j : A_j \to W$$

for every $j \in \mathrm{Obj}(\mathscr{J})$, satisfying

$$g_k \circ F(m) = g_j$$

for every morphism $m : j \to k$. Define

$$G([(j, x)]) := g_j(x).$$

We first verify that G is well-defined. By construction, the equivalence relation $\sim$ is generated by identifications

$$(j, x) \sim (k, F(m)(x)).$$

For each such identification, we have

$$g_j(x) = g_k(F(m)(x))$$

by the compatibility of the maps g_j. So, the value of G is unchanged under every generating identification, and hence under the equivalence relation $\sim$. Therefore G is well-defined. Also, for every $x \in A_j$,

$$\begin{aligned}(G \circ f_j)(x) &= G([(j, x)]) \\ &= g_j(x).\end{aligned}$$

So, $G \circ f_j = g_j$ for every j. To prove uniqueness, suppose $H : L \to W$ also satisfies

$$H \circ f_j = g_j$$

for every j. Every element of L is of the form $[(j, x)]$ for some j and $x \in A_j$. So,

$$\begin{aligned}H([(j, x)]) &= H(f_j(x)) \\ &= g_j(x) \\ &= G([(j, x)]).\end{aligned}$$

Therefore $H = G$, proving uniqueness. So,

$$\varinjlim F = L.$$

□

Now, we show that colimits and limits can be viewed as initial and terminal objects respectively. To do so, we will require the following lemma:

Lemma 3.81. *Let $\emptyset$ denote the empty category, i.e. the category with no objects and no morphisms. Then, there is a functor F from $\emptyset$ to any other category $\mathscr{C}$.* ⌟

Proof. Let $\mathscr{C}$ be an arbitrary category. Now, we construct a functor $F : \emptyset \to \mathscr{C}$, as follows:

$$F : \mathrm{Obj}(\emptyset) \to \mathrm{Obj}(\mathscr{C}), \quad F(X) \in \mathrm{Obj}(\mathscr{C}), \forall X \in \mathrm{Obj}(\emptyset).$$

Since the empty category has no object, F is the unique empty function. Similarly, if $X \xrightarrow{f} Y$ in $\emptyset$, we define $F : \mathrm{Hom}_{\emptyset}(X, Y) \to \mathrm{Hom}_{\mathscr{C}}(F(X), F(Y))$ where $F(X) \xrightarrow{F(f)} F(Y)$ in $\mathscr{C}$. Since the empty category has no morphisms, this function is also the unique empty function. From this, it follows that $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ for any $X \xrightarrow{\mathrm{id}_X} X$ in $\emptyset$ - a statement that is again vacuously true. Finally, for morphisms g and f in $\emptyset$ such that $g \circ f$ makes sense, we have that

$$F(g \circ f) = F(g) \circ F(f).$$

This statement is also vacuously true since $\emptyset$ has no morphisms in the first place. This gives us a functor from $\emptyset$ to $\mathscr{C}$. □

Proposition 3.82. *An object I of category $\mathscr{C}$ is a colimit for functor $F : \emptyset \to \mathscr{C}$ if and only if there is a unique morphism from I to every object of $\mathscr{C}$, i.e. I is an initial object in $\mathscr{C}$.* ⌟

Proof. Suppose $I \in \mathrm{Obj}(\mathscr{C})$ is a colimit for functor $F : \emptyset \to \mathscr{C}$. Now, we will use the definition of a colimit. Let $A_j = F(j)$ for any $j \in \mathrm{Obj}(\emptyset)$ and let $f_j : A_j \to I$ be the morphisms that are compatible with morphisms $F(m) : A_j \to A_k$ in $\mathscr{C}$ for $j \xrightarrow{m} k$ in $\emptyset$, i.e. $f_j = f_k \circ F(m)$. Now, consider X to be an arbitrary object in $\mathscr{C}$. We make the following claims:

(1) There exists a set of morphisms $x_j : A_j \to X$ for all $j \in \text{Obj}(\emptyset)$. Recall that $A_j = F(j)$ for object $j \in \text{Obj}(\emptyset)$. But the empty category has no object, so there is no object A_j either. As such, the family $\{x_j : A_j \to X\}_{j \in \text{Obj}(\emptyset)}$ is the unique empty set.
(2) The morphisms x_j are compatible in the sense that if $j \xrightarrow{m} k$ is a morphism in $\emptyset$, then $x_j = x_k \circ F(m)$. This is also vacuously true since there is no such morphism m in the empty category in the first place.

Due these two claims, by the universal property of the colimit, we have a unique morphism $\Psi_X : I \to X$ such that $\Psi_X \circ f_j = x_j$ for all $j \in \text{Obj}(\emptyset)$. Note that f_j is a morphism from $A_j = F(j)$ and x_j is also a morphism from $A_j = F(j)$. But since there is no object j in the empty category, A_j does not exist and, as such, these morphisms, x_j and f_j, do not exist. As such, the statement reduces to there is a unique morphism $\Psi_X : I \to X$ for all objects $X \in \text{Obj}(\mathscr{C})$.

Conversely, suppose I is an object in $\mathscr{C}$ such that there is a unique morphism $\Psi_X : I \to X$ for all $X \in \text{Obj}(\mathscr{C})$. Now, we show that I is a colimit for the functor $F : \emptyset \to \mathscr{C}$. Let $A_j = F(j)$ for each object j in $\emptyset$. Now, we construct the set $\{f_j : A_j \to I\}_{j \in \text{Obj}(\emptyset)}$ such that $f_j = f_k \circ F(m)$ for all $j \xrightarrow{m} k$ in the empty category. Since $\text{Obj}(\emptyset)$ is empty, this is the empty set, and the compatibilities are vacuously guaranteed. Furthermore, suppose Y is another object with morphisms $y_j : A_j \to Y$ for all $j \in \text{Obj}(\emptyset)$ satisfies the same property, i.e. $y_j = y_k \circ F(m)$ for any $j \xrightarrow{m} k$ in the empty category. Then, we claim that there exists a unique morphism $\Psi : I \to Y$ such that $\Psi \circ f_j = y_j$ for all $j \in \text{Obj}(\emptyset)$. To see this, recall that I is the initial object, and we already have the unique morphism $\Psi_Y : I \to Y$. Setting $\Psi = \Psi_Y$, we see that the rest of the constraints are automatically vacuously satisfied since $\{f_j : A_j \to I\}_{j \in \text{Obj}(\emptyset)}$ is the empty set. □

Proposition 3.83. *An object T of category $\mathscr{C}$ is a limit for functor $F : \emptyset \to \mathscr{C}$ if and only if there is a unique morphism from every object of $\mathscr{C}$ to T, i.e. T is a terminal object in $\mathscr{C}$.* ⌟

Proof. Suppose $T \in \text{Obj}(\mathscr{C})$ is a limit for the functor $F : \emptyset \to \mathscr{C}$. Let $A_j = F(j)$ for each $j \in \text{Obj}(\emptyset)$. Then, we have a collection of morphisms $f_j : T \to A_j$ for each $j \in \text{Obj}(\emptyset)$ such that if $j \xrightarrow{m} k$ in the empty category, then $f_k = F(m) \circ f_j$. Now, let $X \in \text{Obj}(\mathscr{C})$. We now make two claims:

(1) There exists a family of morphisms $x_j : X \to A_j$ for each $j \in \text{Obj}(\emptyset)$. This is vacuously true since $\text{Obj}(\emptyset)$ is empty, so $\{x_j : X \to A_j\}_{j \in \text{Obj}(\emptyset)}$ is the empty set.
(2) The morphisms x_j are compatible with the morphisms between A_j in the sense that if $j \xrightarrow{m} k$ is a morphism in the empty category, then $x_k = F(m) \circ x_j$. This is, again, vacuous true since there are no morphisms in the empty category in the first place.

Now, by the universal property of the limit, there exists a unique morphism $\Psi_X : X \to T$ with $f_j \circ \Psi_X = x_j$ for all $j \in \text{Obj}(\emptyset)$. Since $\text{Obj}(\emptyset)$ is an empty set, any morphism from $X \to T$ would satisfy this constraint, so the statement reduces to simply that we have one unique morphism $\Psi_X : X \to T$. This is true for all objects $X \in \text{Obj}(\mathscr{C})$, so T is an object in $\mathscr{C}$ such that there is a unique morphism from every object in $\mathscr{C}$ to T.

Conversely, suppose T is an object in $\mathscr{C}$ such that there is a unique morphism from every object of $\mathscr{C}$ to T. We now prove that T is a limit for the functor F. We first claim that there is a family of morphisms $\{f_j : T \to A_j\}_{j \in \text{Obj}(\emptyset)}$ that are compatible with morphisms between A_j in the sense that if $j \xrightarrow{m} k$ is a morphism in the empty category, then $f_k = F(m) \circ F_j$. This is vacuously true since the empty category has no objects or morphisms in the first place. Now, suppose L is another object with morphisms $\varphi_j : L \to A_j$ for every $j \in \text{Obj}(\emptyset)$ with the same compatibility, i.e. $\varphi_k = F(m) \circ \varphi_j$ for all morphisms $m : j \to k$ in $\text{Obj}(\emptyset)$. Then, we claim that there is a unique morphism $\Psi_L : L \to T$ such that $\varphi_j = f_j \circ \Psi_L$. To see this, since T is a terminal object, we have a unique morphism $\Psi : L \to T$. Now, we claim that $\Psi_L = \Psi$ satisfies this. This is true since the empty category has no object or morphism, so the compatibility constraint is automatically and vacuously satisfied. As such, T is a limit for the functor F. □

We use this two answer two questions as follows:

Problem 3.84. *Do the categories of sets, groups and rings have initial objects? If so, what are they?*

Proof. In the category of sets, the empty set, $\emptyset$, is the initial object. Indeed, there is a unique morphism (i.e. set function) $\varphi_S : \emptyset \to S$ for any set S, namely the unique empty function.

In the category of groups, the trivial group, $\{1\}$, is the initial object. For any other group G, we have the unique group morphism: $\varphi : (1) \to G$. Note that $\varphi(1) = \mathrm{id}_G$ where id_G is the identity element of group G. Since $1 \cdot 1 = 1$, we have that $\varphi(1 \cdot 1) = \varphi(1) = \mathrm{id}_G = \mathrm{id}_G \cdot \mathrm{id}_G = \varphi(1) \cdot \varphi(1)$, so this is a group morphism. This is obviously unique since any other group morphism must send identity to identity.

In the category of rings, $\mathbb{Z}$ is the initial object. For any other commutative, ring R, we have that $\varphi : \mathbb{Z} \to R$ by $\varphi(n) = n1_R$. This is a ring morphism: $\varphi(0) = 0 \cdot 1_R = 0$, $\varphi(1) = 1 \cdot 1_R = 1_R$, $\varphi(n + m) = (n + m)1_R = n1_R + m1_R = \varphi(n) + \varphi(m)$, and $\varphi(nm) = nm1_R = n1_R \cdot m1_R = \varphi(n) \cdot \varphi(m)$. This is the unique such morphism since if $\Psi : \mathbb{Z} \to R$, then $\Psi(1) = 1_R$, so $\Psi(n) = \Psi(n \cdot 1) = \Psi(1 + \cdots + 1) = \Psi(1) + \cdots + \Psi(1) = n1_R$. As such, $\varphi = \Psi$. □

Problem 3.85. *Do the categories of sets, groups and rings have terminal objects? If so, what are they?*

Answer. In the category of sets, any singleton set $\{x\}$ is a terminal object. This is because for any set S, we have a unique morphism $f_S : S \to \{x\}$ that maps every element of S to x. Any other morphism from S to $\{x\}$ would be equal to f_S since there is only one element that such a set function can map to.

In the category of groups, the trivial group, $\{1\}$, is the terminal object. For any group G, we have the morphism $\varphi_G : G \to \{1\}$ that sends everything to 1. This is a group homomorphism since $\varphi(1_G) = 1$ sends identity to identity and for all $\varphi(g \cdot h) = 1 = 1 \cdot 1 = \varphi(g) \cdot \varphi(h)$. Any other morphism from G to the trivial group must be the same as φ_G since there is only one element to map all inputs into.

In the category of rings, the terminal object is the ring $\{0\}$. For any ring R, we have the morphism $\varphi : R \to \{0\}$ that sends everything to 0. This is a ring morphism since $\varphi(0) = 0$, $\varphi(1) = 0 = 1$ (since $0 = 1$ in the zero ring), $\varphi(r + s) = 0 = 0 + 0 = \varphi(r) + \varphi(s)$ and $\varphi(rs) = 0 = 0 \cdot 0 = \varphi(r) \cdot \varphi(s)$. All other ring morphisms from R to $\{0\}$ must be the same as this since there is only one element to map every input into.

Proposition 3.86. *Suppose that $\mathscr{I}$ is a small category and that F is a functor from $\mathscr{I}$ to the category* **Rings**. *Let G denote the forgetful functor from* **Rings** *to* **Sets**. *Then, the following objects are isomorphic as sets:*

$$G(\varprojlim F) \xrightarrow{\sim} \varprojlim (G \circ F).$$

⌟

Proof. First, we focus on the category of rings. Let $A_j := F(j)$ for each $j \in \mathrm{Obj}(\mathscr{I})$. Note that $A_j \in \mathrm{Obj}(\mathbf{Rings})$. By the definition of limits, there exists morphisms $f_j : \varprojlim F \to A_j$ for all $j \in \mathrm{Obj}(\mathscr{I})$ such that if $j \xrightarrow{m} k$ is an arrow in $\mathscr{I}$, then, $f_k = F(m) \circ f_j$.

Now, in the category of sets, let $S_j := (G \circ F)(j)$ for all $j \in \mathrm{Obj}(\mathscr{I})$ and let the morphisms be $p_j : \varprojlim (G \circ F) \to S_j$. Now, observe that $G(\varprojlim F)$ is an object in **Sets** with morphisms $G(f_j) : G(\varprojlim F) \to S_j$. Furthermore, for any arrow $j \xrightarrow{m} k$ in $\mathscr{I}$, we get that $G(f_k) = (G \circ F)(m) \circ G(f_j)$ because

$$\begin{aligned} F(m) \circ f_j &= f_k \\ G(F(m) \circ f_j) &= G(f_k) \\ G(F(m)) \circ G(f_j) &= G(f_k) \\ (G \circ F)(m) \circ G(f_j) &= G(f_k). \end{aligned}$$

We will refer to these conditions as the collection $\mathfrak{X}$ for future reference. As such, by the universal property of limits, we have a unique morphism

$$\Psi : G(\varprojlim F) \to \varprojlim (G \circ F)$$

satisfying

$$p_j \circ \Psi = G(f_j), \forall j \in \mathrm{Obj}(\mathscr{I}).$$

Next, we make use of the fact that we know what the limit in **Sets** for $G \circ F$ is:

$$\varprojlim(G \circ F) = \{(x_j)_{j \in \mathrm{Obj}(\mathscr{I})} \in \prod_{j \in \mathrm{Obj}(\mathscr{I})} (G \circ F)(j) \mid (G \circ F)(m)(x_j) = x_k, \forall j \xrightarrow{m} k\}$$

Now, we observe that this already comes with a ring structure: $(0_{F(j)})_{j \in \mathrm{Obj}(\mathscr{I})} \in \prod_{j \in \mathrm{Obj}(\mathscr{I})} G(F(j))$, since $F(j)$ is a ring which must have one and G is simply the forgetful functor which maps the ring to its underlying set, so the zeros remain. Similarly, $(1_{F(j)})_{j \in \mathrm{Obj}(\mathscr{I})} \in \prod_{j \in \mathrm{Obj}(\mathscr{I})} G(F(j))$. Suppose $(x_j)_{j \in \mathrm{Obj}(\mathscr{I})}$ and $(x'_j)_{j \in \mathrm{Obj}(\mathscr{I})}$ are elements in this set, then we can add them to get $(x_j + x'_j)_{j \in \mathscr{I}}$ which is also in the set since $(G \circ F)(m)(x_j + x'_j) = G(F(m)(x_j) + F(m)(x'_j)) = x_k + x'_k$. Finally, the product is $(x_j \cdot x'_j)_{j \in \mathscr{I}}$ is in the set since $(G \circ F)(m)(x_j \cdot x'_j) = x_k \cdot x'_k$. We identify this ring in the category **Rings** by S, i.e.

$$G(S) = \varprojlim(G \circ F).$$

Now, notice that we have morphisms $\tilde{p}_j : S \to F(j)$ defined by

$$\tilde{p}_j((x_i)_{i \in \mathrm{Obj}(\mathscr{I})}) = x_j.$$

This is also known to be a ring morphism since this is the projection morphism that direct product comes with. Finally, these morphisms are compatible with the morphisms between objects in $\mathscr{I}$: for $j \xrightarrow{m} k$ a morphism in $\mathscr{I}$, we have that

$$\begin{aligned}(F(m) \circ \tilde{p}_j)((x_i)_{i \in \mathrm{Obj}(\mathscr{I})}) &= F(m)(x_j) \\ &= x_k \quad \text{(since } (G \circ F)(m)(x_j) = x_k \text{ by def. of } S \text{ and } G \text{ is forgetful)} \\ &= \tilde{p}_k((x_i)_{i \in \mathrm{Obj}(\mathscr{I})}).\end{aligned}$$

As such

$$F(m) \circ \tilde{p}_j = \tilde{p}_k.$$

Therefore, by the universal property of the limit $\varprojlim F$, we must have a unique morphism

$$\Phi : S \to \varprojlim F$$

such that the compatibilities are preserved, i.e.

$$f_j \circ \Phi = \tilde{p}_j$$

Now, we move back to the category of sets. Note that

$$G(S) = \varprojlim(G \circ F)$$

and we get the morphism

$$G(\Phi) : \varprojlim(G \circ F) \to G(\varprojlim F).$$

In particular, $G(\Phi)$ preserves the compatibilities since G is a functor and it preserves compositions.

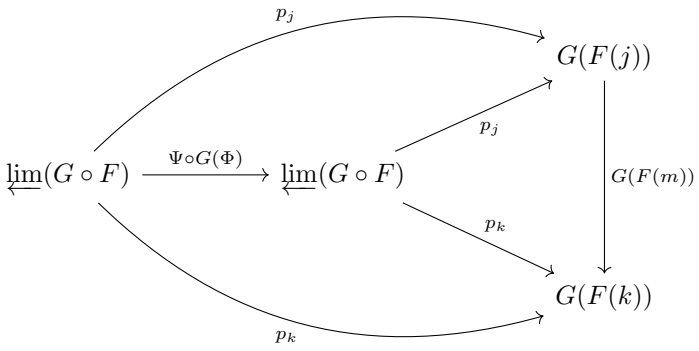
Now, observe that

$$\Psi \circ G(\Phi) : \varprojlim(G \circ F) \to \varprojlim(G \circ F).$$

In particular,

$$\begin{aligned}p_j \circ \Psi \circ G(\Phi) &= G(f_j) \circ G(\Phi) \\ &= G(f_j \circ \Phi) \\ &= G(\tilde{p}_j) \\ &= p_j.\end{aligned}$$

So we get that the following commutes:



But, by the universal property of the limit, the morphism from $\varprojlim(G \circ F)$ to $\varprojlim(G \circ F)$ must be unique and since $1_{\varprojlim(G \circ F)}$ is another such morphism, we must have that

$$\Psi \circ G(\Phi) = 1_{\varprojlim(G \circ F)}.$$

Finally, we show that $G(\Phi) \circ \Psi = 1_{G(\varprojlim F)}$. First, notice that we have some morphism

$$\tilde{\Psi} : \varprojlim F \to S$$

in **Rings** such that

$$G(\tilde{\Psi}) = \Psi.$$

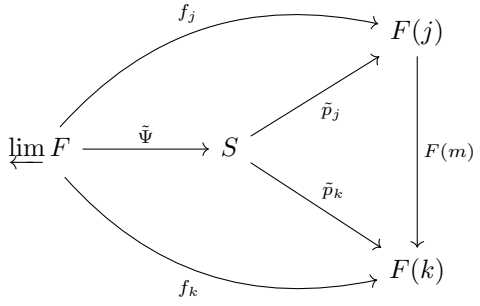
We construct this morphism as, for any $r \in \varprojlim F$, we define

$$\tilde{\Psi}(r) = (f_j(r))_{j \in \mathrm{Obj}(\mathscr{I})}.$$

First, note that this is well-defined. This is because $(G \circ F)(m)(f_j(r)) = F(m)(f_j(r)) = f_k(r)$ since $F(m) \circ f_j = f_k$. As such, $\tilde{\Psi}(r)$ is indeed in S. Since each f_j send 0 to 0, 1 to 1, respects additive and multiplicative structures, $\tilde{\Psi}$ is also a ring morphism. Finally, we check that

$$\begin{aligned}(\tilde{p}_j \circ \tilde{\Psi})(r) &= \tilde{p}_j((f_i(r))_{i \in \mathrm{Obj}(\mathscr{I})}) \\ &= f_j(r) \\ \implies \tilde{p}_j \circ \tilde{\Psi} &= f_j.\end{aligned}$$

We have that the following commutes:



Mapping this diagram, under G, to the category of sets, we see that $G(\tilde{\Psi})$ is another morphism that satisfies the conditions in $\mathfrak{X}$ (see the start of the proof) and since Ψ was the unique morphism that satisfies these conditions, we must have that $G(\tilde{\Psi}) = \Psi$. Now, in the category of rings, we have that $\Phi \circ \tilde{\Psi} : \varprojlim F \to \varprojlim L$ which satisfies that $f_j = f_j \circ \Phi \circ \tilde{\Psi}$. Since $1_{\varprojlim F}$ is another such morphism, we must have that

$$\Phi \circ \tilde{\Psi} = 1_{\varprojlim F}.$$

Then,

$$\Phi \circ \tilde{\Psi} = 1_{\varprojlim F}$$
$$G(\Phi \circ \tilde{\Psi}) = 1_{G(\varprojlim F)}$$
$$G(\Phi) \circ \Psi = 1_{G(\varprojlim F)}.$$

With this, we have construct the isomorphism between $G(\varprojlim F)$ and $\varprojlim(G \circ F)$, and we are done. □

Remark 3.87. Technically, we should be first proving the existence of the limit for F in **Rings**. Indeed, our construction of the ring S midway would satisfy that requirement. ⌟

Remark 3.88. The corresponding statement is not true for colimits. We prove this as follows. Let $\mathscr{I}$ be the empty category. Then, we have already shown the colimit for the functor $F : \emptyset \to$ **Rings** is the initial object, i.e. $\mathbb{Z}$. However, after mapping to **Sets** under the forgetful functor G, we get $\mathbb{Z} \in \text{Obj}(\textbf{Sets})$ which is now no longer an initial object (since the initial object in **Sets** is the emtpyset, $\emptyset$ and since $\mathbb{Z}$ is not isomorphic to $\emptyset$ (they have different cardinalities), we get that the statement is false. (Note that here we are using the results shown in problem 2). ⌟

3.8. **Adjoints**

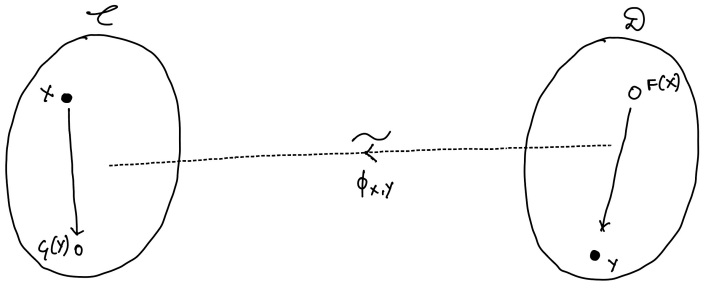


FIGURE 4. Illustration of adjunction

Definition 3.89 (Adjoints)**.** *Suppose, we have the following categories and functors:*

$$F : \mathscr{C} \to \mathscr{D}, \quad G : \mathscr{D} \to \mathscr{C}.$$

We say F is left-adjoint to G (or, equivalently, G is right-adjoint to F), if: for every pair of objects $X \in \mathrm{Obj}(\mathscr{C})$ and $Y \in \mathrm{Obj}(\mathscr{D})$, we have a bijection:

$$\phi_{X,Y} : \mathrm{Hom}_{\mathscr{D}}(F(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{C}}(X, G(Y))$$

that is natural in both X and Y, i.e.

- *if $X \xrightarrow{f} X'$ in $\mathscr{C}$, then*

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{D}}(F(X'), Y) & \xrightarrow{\phi_{X',Y}} & \mathrm{Hom}_{\mathscr{C}}(X', G(Y)) \\ \downarrow{\scriptstyle (-)\circ F(f)} & & \downarrow{\scriptstyle (-)\circ f} \\ \mathrm{Hom}_{\mathscr{D}}(F(X), Y) & \xrightarrow[\phi_{X,Y}]{} & \mathrm{Hom}_{\mathscr{C}}(X, G(Y)) \end{array}$$

- *if $Y \xrightarrow{g} Y'$ in $\mathscr{D}$, then*

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{D}}(F(X), Y) & \xrightarrow{\phi_{X,Y}} & \mathrm{Hom}_{\mathscr{C}}(X, G(Y)) \\ \downarrow{\scriptstyle g\circ(-)} & & \downarrow{\scriptstyle G(g)\circ(-)} \\ \mathrm{Hom}_{\mathscr{D}}(F(X), Y') & \xrightarrow[\phi_{X,Y'}]{} & \mathrm{Hom}_{\mathscr{C}}(X, G(Y')) \end{array}$$

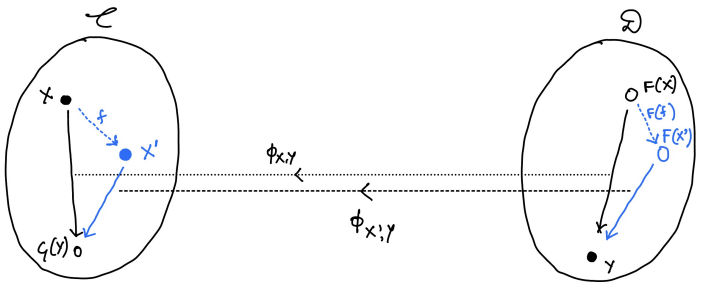


FIGURE 5. Illustration of naturality (in X) of adjunction, for a given morphism $f : X \to X'$ in category $\mathscr{C}$.

Definition 3.90 (Units/Counits)**.** *Given that F is left adjoint to G:*

- *For every $X \in \mathrm{Obj}(\mathscr{C})$, we have the identity morphism, $1_{F(X)} : F(X) \to F(X)$. This allows us to define, the morphisms:*

$$\eta_X = \phi_{X,F(X)}(1_{F(X)}) : X \to G(F(X)).$$

The collection of these morphisms define a natural transformation, called the unit:

$$\eta : 1_{\mathscr{C}} \to G \circ F, \quad \text{where } \eta_X = \phi_{X,F(X)}(1_{F(X)}).$$

- *For every $Y \in \mathrm{Obj}(\mathscr{D})$, we have the identity morphism, $1_{G(Y)} : G(Y) \to G(Y)$. This allows us to define the morphisms:*

$$\mu_Y = \phi^{-1}_{G(Y),Y}(1_{G(Y)}) : F(G(Y)) \to Y.$$

The collection of these morphisms define a natural transformation, called the counit:

$$\mu : F \circ G \to 1_{\mathscr{D}}, \quad \text{where } \mu_Y := \phi^{-1}_{G(Y),Y}(1_{G(Y)}).$$

⌟

Proposition 3.91. *The unit and counit are natural transformations. In other words, for $f : X \to X'$ in $\mathscr{C}$, we have*

$$G(F(f)) \circ \eta_X = \eta_{X'} \circ f,$$

and for $g : Y \to Y'$ in $\mathscr{D}$, we have

$$g \circ \mu_Y = \mu_{Y'} \circ F(G(g)).$$

⌟

Proof. Recall that the adjunction, $\phi_{X,Y}$, is natural in Y. Then, setting $Y = F(X)$, we get that

$$\begin{aligned} G(F(f)) \circ \eta_X &= G(F(f)) \circ \phi_{X,F(X)}(1_{F(X)}) \\ &= \phi_{X,F(X')}(F(f) \circ 1_{F(X)}) \\ &= \phi_{X,F(X')}(F(f)) \end{aligned}$$

On the other hand, we consider the naturality in X, after setting $Y = F(X')$: :

$$\begin{aligned} \eta_{X'} \circ f &= \phi_{X',F(X')}(1_{F(X')}) \circ f \\ &= \phi_{X,F(X')}(1_{F(X')} \circ F(f)) \\ &= \phi_{X,F(X')}(F(f)) \end{aligned}$$

As such, we have that

$$G(F(f)) \circ \eta_X = \eta_{X'} \circ f.$$

The proof for the other statement is similar. □

Now, we show how the adjunction, $\phi_{X,Y}$ and $\phi^{-1}_{X,Y}$, can be recovered using the unit and counit:

Proposition 3.92. *Given $p : F(X) \to Y$ in $\mathscr{D}$ and $\phi_{X,Y}$ is an adjunction between F and G, we have that*

$$\phi_{X,Y}(p) = G(p) \circ \eta_X.$$

Given $q : X \to G(Y)$ in $\mathscr{C}$, we have that

$$\phi^{-1}_{X,Y}(q) = \mu_Y \circ F(q).$$

⌟

Proof. Use naturality in Y: set Y to be $F(X)$ and Y' to be Y. Then, $F(X) \xrightarrow{p} Y$ gives us:

$$\phi_{X,Y}(p \circ 1_{F(X)}) = G(p) \circ \phi_{X,F(X)}(1_{F(X)})$$
$$\phi_{X,Y}(p) = G(p) \circ \eta_X$$

Considering naturality in X gives us the second identity. Let $X \xrightarrow{q} G(Y)$. Then, with X' set to be $G(Y)$ in the naturality of Y, we observe that since $\mu_Y = \phi^{-1}_{G(Y),Y}(1_{G(Y)}) \in \mathrm{Hom}_{\mathscr{D}}(F(G(Y)), Y)$, we have:

$$\phi_{X,Y}(\mu_Y \circ F(q)) = \phi_{G(Y),Y}(\mu_Y) \circ q$$
$$\phi_{X,Y}(\mu_Y \circ F(q)) = q$$
$$\phi^{-1}_{X,Y}(q) = \mu_Y \circ F(q).$$

□

Remark 3.93. Note that all pairs of natural transformations $1_{\mathscr{C}} \implies G(F)$ and $F(G) \implies 1_{\mathscr{D}}$ do not define an adjunction. In other words, there is no "converse" of the last proposition. ⌟

Most of the interesting examples of adjoints will come up when we look at modules and other categories. For now, we look at some basic examples.

Example 3.94. Let $G : \mathbf{Rings} \to \mathbf{Sets}$ be the forgetful functor. Then,

$$F : \mathbf{Sets} : \mathbf{Rings}$$

by

$$F(\Omega) = \mathbb{Z}[x_\omega]_{\omega \in \Omega}$$

is left adjoint to G. In this case, we have the unit:

$$\eta_\Omega : \Omega \to \mathbb{Z}[x_\omega]_{\omega \in \Omega}$$

by

$$\omega \to x_\omega, \forall \omega \in \Omega.$$

We have the counit:

$$\mu_R : \mathbb{Z}[x_r]_{r \in R} \to R$$

by

$$x_r \to r$$

⌟

Example 3.95. Let $G : \mathbf{Fields} \to \mathbf{IDs}$ be the forgetful functor. Note that in **IDs**, the morphisms are injective ring morphisms. Then, define the functor $F : \mathbf{IDs} \to \mathbf{Fields}$ by

$$R \to Q(R).$$

Then, F is left adjoint to G, allowing us to write:

$$\mathrm{Hom}(Q(R), k) \cong \mathrm{Hom}(R, K)$$

for any integral domain R and any field k. In this case, the unit is $\eta_R : R \to Q(R)$ and the counit is $\mu_k : k \to k$. ⌟

Proposition 3.96. *Suppose $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$ are functors such that F is left adjoint to G. Then,*

(1) *G preserves small limits.*
(2) *F preserves small colimits.*

⌟

Proof. We show that G preserves small limits. Let $\mathcal{J}$ be a small category and let $H : \mathcal{J} \to \mathcal{D}$ be a functor. Suppose that H has a limit L, with projection morphisms

$$\lambda_j : L \to H(j)$$

for each $j \in \mathrm{Ob}(\mathcal{J})$. We show that $G(L)$ has morphisms $G(\lambda_j) : G(L) \to G(H(j))$, making it a limit of $G \circ H$. First, for every morphism $a : j \to k$ in $\mathcal{J}$, we have that

$$H(a) \circ \lambda_j = \lambda_k.$$

Applying G gives

$$G(H(a)) \circ G(\lambda_j) = G(\lambda_k).$$

Now let $X \in \mathrm{Ob}(\mathcal{C})$ has morphisms

$$\beta_j : X \to G(H(j)).$$

satisfying, for every morphism $a : j \to k$,

$$G(H(a)) \circ \beta_j = \beta_k.$$

By the adjunction, each β_j corresponds uniquely to a morphism

$$\alpha_j := \phi^{-1}_{X,H(j)}(\beta_j) : F(X) \to H(j).$$

We claim that $(\alpha_j)_j$ is a cone over H. Indeed, by naturality of ϕ,

$$\begin{aligned}
\phi_{X,H(k)}(H(a) \circ \alpha_j) &= G(H(a)) \circ \phi_{X,H(j)}(\alpha_j) \\
&= G(H(a)) \circ \beta_j \\
&= \beta_k \\
&= \phi_{X,H(k)}(\alpha_k).
\end{aligned}$$

Since $\phi_{X,H(k)}$ is injective, we obtain

$$H(a) \circ \alpha_j = \alpha_k.$$

Therefore $(\alpha_j)_j$ is a cone over H. By the universal property of limits and since L is a limit of H, there exists a unique morphism

$$u : F(X) \to L$$

such that

$$\lambda_j \circ u = \alpha_j$$

for every j. Define

$$v := \phi_{X,L}(u) : X \to G(L).$$

By naturality,

$$\begin{aligned}
G(\lambda_j) \circ v &= G(\lambda_j) \circ \phi_{X,L}(u) \\
&= \phi_{X,H(j)}(\lambda_j \circ u) \\
&= \phi_{X,H(j)}(\alpha_j) \\
&= \beta_j.
\end{aligned}$$

Therefore, v satisfies the required compatibility conditions. To show uniqueness, suppose another morphism $v' : X \to G(L)$ satisfies

$$G(\lambda_j) \circ v' = \beta_j$$

for every j. Let

$$u' := \phi^{-1}_{X,L}(v') : F(X) \to L.$$

Then,

$$\begin{aligned}
\phi_{X,H(j)}(\lambda_j \circ u') &= G(\lambda_j) \circ v' \\
&= \beta_j \\
&= \phi_{X,H(j)}(\alpha_j).
\end{aligned}$$

So,

$$\lambda_j \circ u' = \alpha_j.$$

By uniqueness in the universal property of L, we have $u' = u$. As such,

$$v' = \phi_{X,L}(u') = \phi_{X,L}(u) = v.$$

Therefore $G(L)$ is a limit of $G \circ H$, and G preserves small limits. The proof for the F preserving colimits uses similar techniques. □

Example 3.97. The forgetful functor $G : \mathbf{Rings} \to \mathbf{Sets}$ preserves small limits: $G(R \times S) = G(R) \times G(S)$. But G does not preserve small colimits. For example, the colimit of the empty functor in rings is the initial object $\mathbb{Z}$ and the colimit in **Sets** is the initial object $\emptyset$, which are clearly not isomorphic. ⌟

4. Module Theory

References

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[DF04] David S. Dummit and Richard M. Foote. *Abstract Algebra*. John Wiley & Sons, 3rd edition, 2004.

Appendix A. Euclidean Algorithm, Bézout's Identity

Euclidean Algorithm. The Euclidean Algorithm is an algorithm for finding the greatest common divisor of two non-zero integers a and b.

$$a = q_0 b + r_0 \tag{9}$$
$$b = q_1 r_0 + r_1 \tag{10}$$
$$r_0 = q_2 r_1 + r_2 \tag{11}$$
$$r_1 = q_3 r_2 + r_3 \tag{12}$$
$$\vdots$$
$$r_{n-2} = q_n r_{n-1} + r_n \tag{13}$$
$$r_{n-1} = q_{n+1} r_n. \tag{14}$$

Here r_n is the last nonzero remainder. Such an r_n must exist since

$$|b| > |r_0| > |r_1| > \cdots > |r_n|$$

is a strictly decreasing sequence of positive integers whenever the remainders are nonzero, and therefore the process cannot continue indefinitely. The final nonzero remainder r_n is the greatest common divisor of a and b, i.e.,

$$\gcd(a, b) = r_n =: (a, b).$$

Bézout's Identity. Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers x and y such that

$$ax + by = \gcd(a, b).$$

In other words, the greatest common divisor of a and b can be written as an integer linear combination of a and b.

Proof. Suppose $d = \gcd(a, b)$. Then, $a = pd$ and $b = qd$ for some $p, q \in \mathbb{Z}$. Now, p and q must be relatively prime since, otherwise, their greatest common divisor could be absorbed into d. So, all we need to do is prove the statement for when a and b are relatively prime now.

Suppose a and b are relatively prime. Consider the set $S = \{ax + by \mid x, y \in \mathbb{Z}\}$. By the well-ordering principle, S has a smallest element. Suppose the smallest element is $m = ax_0 + by_0$. Then, we must show that $m \mid a$ adn $m \mid b$. Using the division algorithm on a with divisor m, we get that $a = qm + r$ where $0 \leq r < m$. However, $r = a - qm = a - q(ax_0 - by_0) = a(1 - qx_0) + b(-qy_0) \in S$ and is smaller than m, so r must be 0. Similarly, m divides b. Therefore, m must be 1. □

Appendix B. Linear Algebra Review

B.1. Dual Vector Spaces

Definition B.1 (Dual vector spaces)**.** *Let k be a field and V a vector space over k. A* linear functional *on V is a k-linear map*

$$\varphi : V \to k.$$

The dual vector space *of V, denoted by V^*, is the set of all linear functionals on V:*

$$V^* := \mathrm{Hom}_k(V, k) = \{\varphi : V \to k \mid \varphi \text{ is } k\text{-linear}\}.$$

⌟

Observation. *Note that the dual space V^* is itself a vector space over k: for $\varphi, \psi \in V^*$ and $a \in k$, define addition and scalar multiplication pointwise:*

$$(\varphi + \psi)(v) := \varphi(v) + \psi(v), \qquad (a\varphi)(v) := a\varphi(v).$$

These operations satisfy the vector space axioms.

Example B.2. Consider $V = \mathbb{R}^2$, with standard basis

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Define two linear functionals $e_1^*, e_2^* \in V^*$ by

$$e_1^*(x, y) = x, \qquad e_2^*(x, y) = y.$$

Every linear functional $\varphi \in V^*$ can be written uniquely as

$$\varphi = ae_1^* + be_2^*$$

for some $a, b \in \mathbb{R}$. To see this, note that since φ is linear, for any $(x, y) \in \mathbb{R}^2$,

$$\begin{aligned}\varphi(x, y) &= \varphi(xe_1 + ye_2) \\ &= x\varphi(e_1) + y\varphi(e_2).\end{aligned}$$

Setting $a = \varphi(e_1)$ and $b = \varphi(e_2)$ gives

$$\varphi(x, y) = ax + by = ae_1^* + be_2^*.$$

Therefore,

$$V^* = \operatorname{span}_{\mathbb{R}}\{e_1^*, e_2^*\},$$

and $\{e_1^*, e_2^*\}$ is a basis of V^*. In particular,

$$\dim_{\mathbb{R}} V^* = 2 = \dim_{\mathbb{R}} V.$$

⌟

Definition B.3. *Let V be a finite-dimensional vector space over k with basis*

$$\{e_1, \ldots, e_n\}.$$

The dual basis *is the collection*

$$\{e_1^*, \ldots, e_n^*\} \subseteq V^*$$

defined by

$$e_i^*(e_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

⌟

We summarize the previous discussion as the following proposition:

Proposition B.4. *The dual basis is a basis of V^*. Therefore,*

$$\dim_k V^* = \dim_k V.$$

⌟

Observation. *Elements of V are vectors, whereas elements of V^* are linear maps from V to k. Thus, although V and V^* have the same dimension when V is finite-dimensional, they are conceptually different objects.*

Lemma B.5. *If $w \in W$ satisfies $\varphi(w) = 0$ for every $\varphi \in W^*$, then $w = 0$.* ⌟

Proof. If $w \neq 0$, we may extend $\{w\}$ to a basis of W and define a linear functional $\varphi : W \to k$ that sends w to 1 and all other basis vectors to 0. This contradicts the assumption that every linear functional vanishes on w. □

Appendix C. Additional Results on Prime Ideals

C.1. Prime ideals and Nilpotents

We will show that the set of nilpotent elements of a ring equals to the intersection of all ideals in the spectrum of a ring. We will require the following lemma:

Lemma C.1. *Suppose that I is an ideal of a ring R and that $S \subset R - I$ is a subset. Then, R contains a maximal ideal J containing I but not containing any element of S.* ⌟

Proof. Consider the set $X = \{K \trianglelefteq R \mid I \subseteq K, K \cap S = \emptyset\}$. This set is partially ordered and non-empty since $I \in X$. Now, consider a chain of ideals $\mathcal{C} \subset X$. Then, consider the element

$$J = \cup_{K \in \mathcal{C}} K.$$

J is clearly an ideal (the interesting part is to show that if, $a, b \in J$, then $a + b \in J$; this follows because $a \in K_n$ and $b \in K_m$, then, $a + b \in K_n$ if $K_m \subseteq K_n$, otherwise $a + b \in K_m$). Next, we can see that $I \subset J$ since $I \subseteq K$ for each K in the chain. Furthermore, none of the ideals in the chain intersect with S, so S does not intersect with J. Therefore, $J \in X$. So, we have an upper bound for any chain in X that is inside X itself. So, by Zorn's lemma, X has a maximal element J that contains I and does not intersect with S. □

We will also need the following lemma:

Lemma C.2. *Suppose now that $x \in R$ is not nilpotent and $I = (0)$ and $S = \{1, x, x^2, \cdots\}$. Let J be a maximal ideal not intersecting with S, which exists by the previous lemma. Show that J is a prime ideal.* ⌟

Proof. Let J be our maximal ideal not intersecting with I. Now, suppose $ab \in J$. We show that either a or b is in J. Suppose, for contradiction, neither a nor b is in J. Then, $J \subsetneq J + (a)$. Since $J + (a)$ is a larger ideal containing J, it must intersect with S. Then, there exists some n such that $x^n \in S$ such that $x^n \in J + (a)$, i.e. $x^n = j + ra$ for some $j \in J, r \in R$ and $n \in \mathbb{Z}_{\geq 0}$. Similarly, there exists some $m \in \mathbb{Z}_{\geq 0}$ such that $x^m = j' + sb$ for some $j' \in J$ and $s \in R$. Now, one can show that $x^{n+m} \in J$, which would show that $J \cap S \neq \emptyset$, which is a contradiction, so J is a prime ideal. □

Finally, we come to the main result of this section:

Proposition C.3. *The intersection of all prime ideals of R equals the set of nilpotent elements of R. In other words, $\cap_{P \in \mathrm{Spec}\,(R)} P = \{r \in R \mid r \text{ nilpotent}\}$.* ⌟

Proof. Consider a nilpotent r. Then, r is nilpotent shows that $r^n = 0$ for some $n \in \mathbb{Z}_{\geq 0}$. But 0 is an element in every prime ideal, so $0 \in \cap_{P \in \mathrm{Spec}\,(R)} P$, so $r^n \in \cap_{P \in \mathrm{Spec}\,(R)} P$. But then, $r^n \in P$ for each prime ideal P, which shows that $r \in P$ for each prime ideal P. So, $r \in \cap_{P \in \mathrm{Spec}\,(R)} P$. On the other hand, suppose $r \in \cap_{P \in \mathrm{Spec}\,(R)} P$. We show that r is nilpotent. We show this through a contrapositive: we will prove that if r is not nilpotent, then r is not in $\cap_{P \in \mathrm{Spec}\,(R)} P$. Given r is not nilpotent, we have that $J \cap S = \emptyset$ where $S = \{1, r, r^2, \cdots\}$. So J does not contain r, and so, we have a prime ideal that does not contain r. Therefore, $r \notin \cap_{P \in \mathrm{Spec}\,(R)} P$. □

C.2. Minimal Prime Ideals

Definition C.4 (Minimal Prime Ideal)**.** *A minimal prime ideal is a prime ideal that strictly contains no other prime ideal.* ⌟

Proposition C.5. *Any prime ideal of a ring R contains a minimal prime ideal.* ⌟

Proof. If R is an integral domain, then (0) is a prime ideal and contains no other prime ideal and is contained in every prime ideal, so (0) is the minimal prime ideal in every prime ideal. Suppose, then, that R is not an integral domain. Suppose P is a prime ideal of ring R. Now, consider the set $\mathfrak{X} = \{Q \trianglelefteq_{\text{prime}} R \mid Q \subseteq P\}$. Note that $\mathfrak{X} \neq \emptyset$ since P is in the set. Now, define a partial ordering as

$$Q_1 \leq Q_2$$

if

$$Q_2 \subseteq Q_1.$$

Then, consider a chain in this set: $Q_1 \leq Q_2 \leq Q_3 \leq \cdots$. Consider

$$X = \cap_i Q_i.$$

Firstly, we claim $X \in \mathfrak{X}$. Firstly, X is an ideal, since $0 \in Q_i$ for all i (as they are all ideals). Furthermore, if $a, b \in X$. Then, a and b are in every ideal Q_i, and so, $a+b, ab$ and ra (for any $r \in R$) are in every ideal Q_i, which implies $a+b, ab$ and ra are in X. Next, we show that X is a prime ideal. Suppose $xy \in X$. Then, $xy \in Q_i$ for each i. Since each Q_i is prime, either $x \in Q_i$ or $y \in Q_I$. Of course, if $x \in Q_i$ for all Q_i (or if $y \in Q_i$ for all i), we are done. So suppose that $x \notin Q_n$ for some n. Then, $y \in Q_n$ (since $xy \in Q_n$ and Q_n is a prime ideal).

Case 1: $Q_i \leq Q_n$. In this case, $Q_n \subseteq Q_i$, so $y \in Q_i$. **Case 2:** $Q_n \leq Q_i$. In this case, $Q_i \subseteq Q_n$. Since $x \notin Q_n$, we have that $x \notin Q_i$. But $xy \in Q_i$, so $y \in Q_i$.

In either case, we see that $y \in Q_i$ for all i. As such, $y \in X$. Therefore, X is a prime ideal. Now, X is an upper bound of the chain and X is a prime ideal. Furthermore, since every $Q_i \subseteq P$ and $X = \cap_i Q_i$, therefore, $X \subseteq P$. Finally, $X \subseteq Q_i$ for all i, which means $Q_i \leq X$ for all i. Therefore, X is an upper bound of the chain that is in $\mathfrak{X}$ itself. Therefore, by Zorn's lemma, $\mathfrak{X}$ has a maximal element, M. Since this is maximal with respect to our partial ordering, this means M is minimal with respect to inclusion and so, $M \subseteq P$ is a minimal prime ideal.

Furthermore, $X \in \mathfrak{X}$, since $X = \cap_i Q_i$ and $Q_i \subseteq P$. □

Proposition C.6 (Every element of a minimal prime ideal is a zero divisor)**.** *Suppose P is a minimal prime ideal. Then, every element of $(R-P)^{-1}P = P_P \trianglelefteq R_P$ is nilpotent. Therefore, every element of P is either zero or a zero divisor.* ⌟

Proof. Suppose P is a minimal prime ideal. Now, we know

$$\text{Spec}(D^{-1}R) \leftrightarrow \{Q \in \text{Spec}(R) \mid D \cap Q = \emptyset\}.$$

As such,

$$\text{Spec}(R_P) \leftrightarrow \{Q \in \text{Spec}(R) \mid Q \cap (R-P) = \emptyset\},$$

which is equivalent to

$$\text{Spec}(R_P) \leftrightarrow \{Q \in \text{Spec}(R) \mid Q \subseteq P\}.$$

Since P is minimal, the right hand side only has P. herefore, the only prime ideal of R_P is P_P. Now, let $\frac{a}{b} \in P_P$. Then, consider the ring

$$(R_P)_{\frac{a}{b}}.$$

Now,

$$\text{Spec}((R_P)_{\frac{a}{b}}) \leftrightarrow \{Q \in \text{Spec}(R_P) \mid Q \cap \{1, \frac{a}{b}, \frac{a^2}{b^2}, \cdots\} = \emptyset\}.$$

But the right hand side has no element, so $\text{Spec}((R_P)_{\frac{a}{b}}) = \emptyset$. However, every non-zero ring has at least one prime ideal, so this implies $(R_P)_{\frac{a}{b}} \cong \{0\}$. This means $\frac{a}{b}$ is nilpotent in R_P, since $D^{-1}R = 0$ if and only if $0 \in D$.

Now, if $(\frac{a}{b})^n = \frac{0}{1}$ for $a \in P, b \notin P$, there exists some $r \in R - P$ such that $r(a^n) = 0 \implies ra^{n-1}a = 0$. Now, since $a/1$ is nilpotent, there exists some minimal $n \in \mathbb{Z}_{\geq 1}$ such that $ra^n = 0$ for $r \notin P$. Then, $ra^{n-1} \neq 0$. As such, $ra^{n-1}a = 0$ shows that a is a zero divisor. □

Proposition C.7. *Consider the set $\mathcal{Z}$ of ideals of R in which every element is a zero divisor. Then, every element of $\mathcal{Z}$ is contained in a maximal element. Every maximal element of $\mathcal{Z}$ is a prime. The set of zero divisors of R is a union of prime ideals.* ⌟

Proof. Let $\mathcal{Z} := \{I \trianglelefteq R \mid \text{all elements of } I \text{ are zero divisors}\}$. Now, $\mathcal{Z} \neq \emptyset$ as $(0) \in \mathcal{Z}$. Furthermore, $\mathcal{Z}$ is partially ordered with inclusion. Now, consider a chain $I_1 \subseteq I_2 \subseteq I_3 \cdots$. Then, let $I = \cup_i I_i$. We claim that I is an ideal in R. This is true since $0 \in I_i$ for each i. Suppose $a, b \in I$. Then, $a \in I_n$ for some n and $b \in I_m$ for some m. Then, $a, b \in I_{\max(m,n)}$. Since $I_{\max(m,n)}$ is an ideal, $a+b, ab, ra$ (for any $r \in R$) are all in $I_{\max(m,n)}$, and so they are all in I. Furthermore, $I \in \mathcal{Z}$. This is because for any $x \in I$, $x \in I_n$ for some n which makes it a zero divisor and so all elements of I are zero divisors. Clearly, every $I_n \subseteq I$, so we have found an upper bound of this chain that is in $\mathcal{Z}$. As such, by Zorn's lemma, the set $\mathcal{Z}$ has a maximal element.

Now, suppose I is a maximal element of this set $\mathcal{Z}$. Suppose $xy \in I$. Now, suppose $x \notin I$ and $y \notin I$. Then,, $I \subsetneq I + (x)$. Because I is maximal in $\mathcal{Z}$, $I + (x) \notin \mathcal{Z}$. As such, $I + (x)$ contains

some element that is not a zero divisor: let that be $i + rx$. Similarly, $I \subsetneq I + (y)$ implies there exists some element in the right hand side that is not a zero divisor: let that be $j + sy$. Now, consider their product:

$$(i + rx)(j + sy) = ij + isy + rjx + rsxy \in I.$$

However, since $i + rx$ and $j + sy$ are not zero divisors, their product is not a zero divisor. This is a contradiction since they are in I and every element of I is a zero divisor. As such, either $x \in I$ or $y \in I$, which means I is a prime ideal.

Finally, we show that the set of zero divisors is a union of prime ideals. Clearly, the union of all maximal elements of $\mathcal{Z}$ is contained in the set of zero divisors because the maximal elements are themselves in $\mathcal{Z}$ and, therefore, consists only of zero divisors. Conversely, if x is a zero divisor, then there exists some $a \in R$ such that $ax = 0$. Now, consider the ideal (x). Since $a(rx) = r(ax) = r(0) = 0$, therefore, every element of (x) is a zero divisor. As such, $(x) \in \mathcal{Z}$. Since every element of $\mathcal{Z}$ is contained in some maximal element of $\mathcal{Z}$, therefore, (x) is contained in some ideal M that is maximal in $\mathcal{Z}$, so $x \in M$. Since the maximal element is prime, every zero divisor is contained in some prime ideal. As such, the set of zero divisors is contained in the union of maximal elements of $\mathcal{Z}$, which is a union of prime ideals. □

Proposition C.8. *Suppose now that R is reduced. Then, every zero divisor in R lies in a minimal prime ideal.* ⌟

Proof. Suppose R is reduced. Now, suppose $s \neq 0$ and $rs = 0$. As such, r is a zero divisor. We claim that there exists a minimal prime ideal of R not containing s. Suppose s were in every prime ideal. Then, s would be in their intersection. We proved already that the intersection of all prime ideals is the set of all nilpotents in R, so s becomes a nilpotent in R. But $s \neq 0$ and R is reduced, so this is impossible. As such, there exists a prime ideal $P' \trianglelefteq R$ such that $s \notin P'$. We know that P' contains a minimal prime ideal P, and $s \notin P$. Since r is a zero divisor, $rs = 0 \in P$ with $s \notin P$ implies that $r \in P$, so every zero divisor lies in a minimal prime ideal. □

C.3. **Minimal Prime Ideals in Noetherian Rings**

Lemma C.9. *Suppose I and J are ideals and $IJ \subseteq P$ for some prime ideal P, then either $I \subseteq P$ or $J \subseteq P$.* ⌟

Proof. Consider any arbitary $y \in J$. Now, if $I \subseteq P$, we are done. So, suppose $I \nsubseteq P$. Then, there exists $x \in I$ such that $x \notin P$. Now, for any $y \in J$, $xy \in IJ \subset P$. Since $x \notin P$, therefore, $y \in P$. □

Proposition C.10. *Suppose $R \neq 0$ and R is Noetherian. Suppose I is an ideal of R. Then, there exists a finite number of prime ideals, $P_1, \cdots, P_n$ such that*

$$P_1 \cdots P_n \subset I.$$

Furthermore, R has finitely many minimal prime ideals. ⌟

Proof. If $I = R$: since $R \neq 0$, there is a maximal prime ideal P and every maximal ideal is prime. So, $P \subseteq R = I$ and we are done.

If I is prime itself, we know that this contains some minimal prime ideal P, so $P \subseteq I$, and we are done.

Now, suppose $I \neq R$ and I is also not prime. Define the following set: $\mathfrak{X} := \{J \trianglelefteq R \mid / \exists \text{ prime ideals } p_1, \cdots, p_n \text{ s.t. } p_1 \cdots p_n \subset I\}$. We know that $\mathfrak{X}$ has a maximal element since R is Noetherian. so let J be a maximal element of $\mathfrak{X}$. Suppose J is not prime. Then there exist $a, b \in R$ such that $ab \in J$ but $a \notin J$ and $b \notin J$. Now, $J \subsetneq J + (a), J \subsetneq J + (b)$. Since J is maximal in $\mathfrak{X}$, neither $J + (a)$ nor $J + (b)$ belongs to $\mathfrak{X}$. Hence there exist prime ideals $P_1, \ldots, P_m$ and $Q_1, \ldots, Q_n$ such that

$$P_1 \cdots P_m \subseteq J + (a)$$

and

$$Q_1 \cdots Q_n \subseteq J + (b).$$

Therefore,

$$\begin{aligned} P_1 \cdots P_m Q_1 \cdots Q_n &\subseteq (J + (a))(J + (b)) \\ &\subseteq J. \end{aligned}$$

The last step follows from:

$$(J + (a))(J + (b)) \subseteq J^2 + aJ + bJ + (ab) \subseteq J,$$

since J is an ideal and $ab \in J$. This means a finite product of prime ideals is contained in J, contradicting the fact that $J \in \mathfrak{X}$. Therefore, $\mathfrak{X} = \emptyset$, and so for every ideal $I \trianglelefteq R$, there exist finitely many prime ideals $P_1, \ldots, P_n$ such that

$$P_1 \cdots P_n \subseteq I.$$

Note that $(0) \subseteq q$ for any minimal prime ideal q. Then, $p_1 \cdots p_n \subseteq (0) \subseteq q$. Then, at least one of p_i is in q, so $(0) \subseteq p_i \subseteq q$. Now, p_i contains some minimal prime ideal m so,

$$(0) \subseteq m \subseteq p_i \subseteq q.$$

Now, m is minimal and q is minimal, so this forces $m = p_i = q$. As such, q equals to some minimal prime ideal in $\{p_1, \cdots, p_n\}$. □